

Coordinate Geometry in the IMO Shortlist

Version 2.0

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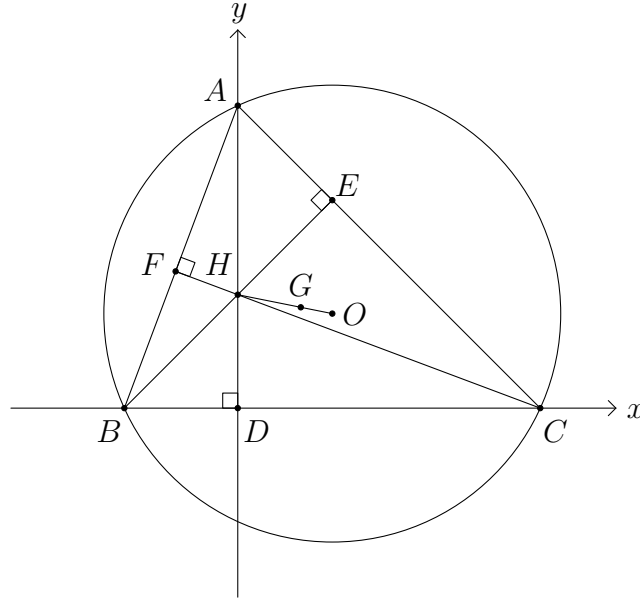
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Preliminary Settings

We first provide 3 common settings which will be used repeatedly in this note. We will quote these results directly whenever they are used. But one should learn how to derive these.

Setting 1: $A(0, a)$, $B(b, 0)$, $C(c, 0)$

Remarks. This setting is used for triangle geometry, where the figure can be constructed from $\triangle ABC$ step by step. It is especially useful if there is a symmetry between B and C . In that case the coordinates of the symmetric points can be found easily by swapping the variables b and c , and perhaps other related variables. For convenience, we usually impose the condition $a > 0$. In case $\triangle ABC$ is acute, we can further assume $b < 0 < c$.



From $m_{AB} = -\frac{a}{b}$, we easily get

$$AB: \quad y = -\frac{a}{b}x + a. \quad (1.1)$$

By symmetry, we have

$$AC: \quad y = -\frac{a}{c}x + a. \quad (1.2)$$

Let H be the orthocentre of $\triangle ABC$. Since $CH \perp AB$, we have $m_{CH} = \frac{-1}{m_{AB}} = \frac{b}{a}$. Therefore, the altitude from C has equation

$$CH: \quad y = \frac{b}{a}x - \frac{bc}{a}. \quad (1.3)$$

By symmetry, we have

$$BH: \quad y = \frac{c}{a}x - \frac{bc}{a}. \quad (1.4)$$

Note that the altitude from A is simply

$$AH: \quad x = 0. \quad (1.5)$$

By solving any two of these, we obtain

$$H\left(0, -\frac{bc}{a}\right). \quad (1.6)$$

Let D, E, F be the feet of altitudes from A, B, C respectively. Obviously, we have

$$D(0, 0). \quad (1.7)$$

Also, by solving (1.2) and (1.4), we get

$$E\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right). \quad (1.8)$$

By symmetry, we have

$$F\left(\frac{b(a^2 + bc)}{a^2 + b^2}, \frac{ab(b - c)}{a^2 + b^2}\right). \quad (1.9)$$

Let G be the centroid of $\triangle ABC$. By computing the arithmetic mean, we obtain

$$G = \left(\frac{b + c}{3}, \frac{a}{3}\right). \quad (1.10)$$

Let O be the circumcentre of $\triangle ABC$. Since $OG : GH = 1 : 2$ and O, G, H are collinear, we find that

$$O = \left(\frac{3 \cdot \frac{b + c}{3} + (-1) \cdot 0}{3 + (-1)}, \frac{3 \cdot \frac{a}{3} + (-1) \cdot \left(-\frac{bc}{a}\right)}{3 + (-1)}\right) = \left(\frac{b + c}{2}, \frac{a^2 + bc}{2a}\right). \quad (1.11)$$

From this, the equation of the circumcircle of $\triangle ABC$ must be of the form

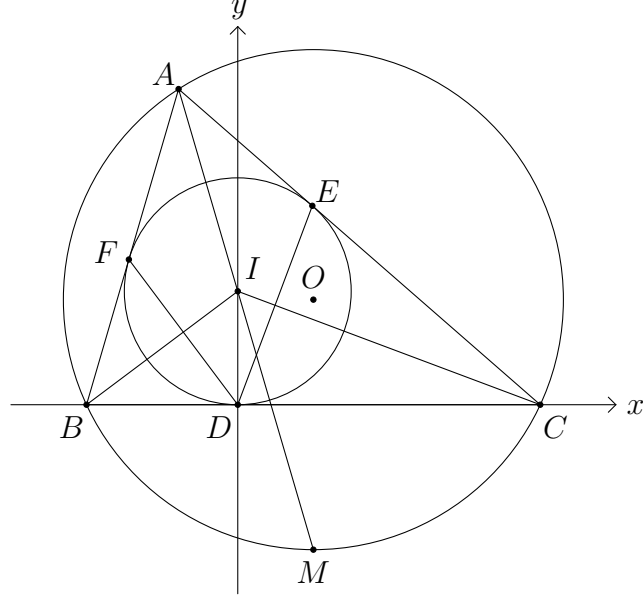
$$x^2 + y^2 - (b + c)x - \frac{a^2 + bc}{a}y + d = 0.$$

Putting $x = b$ and $y = 0$, we obtain $d = bc$. So

$$(ABC): \quad x^2 + y^2 - (b + c)x - \frac{a^2 + bc}{a}y + bc = 0. \quad (1.12)$$

Setting 2: $I(0, 1), B(b, 0), C(c, 0)$

Remarks. This setting is used for triangle geometry involving the incentre I of $\triangle ABC$. Setting 1 is not suitable for this purpose since the coordinates of I would involve radicals in that case. This setting can also be used if I is the excentre instead of the incentre. The only difference is that the inequalities mentioned below involving b and c would change accordingly. In particular, for incentre I , we may assume $b < 0 < c$.



Since $\angle CBA = 2\angle CBI$, by the double angle formula for tangent, we have

$$m_{AB} = \frac{2m_{BI}}{1 - m_{BI}^2} = \frac{2\left(-\frac{1}{b}\right)}{1 - \left(-\frac{1}{b}\right)^2} = -\frac{2b}{b^2 - 1}.$$

This implies

$$AB: \quad y = -\frac{2b}{b^2 - 1}x + \frac{2b^2}{b^2 - 1}. \quad (2.1)$$

By symmetry, we have

$$AC: \quad y = -\frac{2c}{c^2 - 1}x + \frac{2c^2}{c^2 - 1}. \quad (2.2)$$

Solving (2.1) and (2.2), we have

$$x = \frac{2b^2(c^2 - 1) - 2c^2(b^2 - 1)}{2b(c^2 - 1) - 2c(b^2 - 1)} = \frac{2(c - b)(c + b)}{2(c - b)(bc + 1)} = \frac{b + c}{bc + 1}$$

and

$$y = \frac{2b}{b^2 - 1} \cdot \left(b - \frac{b + c}{bc + 1}\right) = \frac{2b}{b^2 - 1} \cdot \frac{c(b^2 - 1)}{bc + 1} = \frac{2bc}{bc + 1}.$$

This means

$$A\left(\frac{b + c}{bc + 1}, \frac{2bc}{bc + 1}\right). \quad (2.3)$$

From the configuration, A has positive y -coordinates. Therefore, we may assume

$$bc + 1 < 0. \quad (2.4)$$

Let D, E, F be the contact points of the incircle $\odot I$ with the sides BC, CA, AB respectively. It is obvious that

$$D(0, 0). \quad (2.5)$$

From this, the equation of the incircle is

$$\odot I: \quad x^2 + y^2 - 2y = 0. \quad (2.6)$$

Also, we easily get

$$BI: \quad y = -\frac{1}{b}x + 1. \quad (2.7)$$

Since $DF \perp BI$, we know that $m_{DF} = \frac{-1}{m_{BI}} = b$. Thus, we find that

$$DF: \quad y = bx. \quad (2.8)$$

Solving (2.1) and (2.8), we have

$$x = \frac{2b^2}{b(b^2 - 1) + 2b} = \frac{2b}{b^2 + 1}$$

and

$$y = b \cdot \frac{2b}{b^2 + 1} = \frac{2b^2}{b^2 + 1}.$$

Therefore, we obtain

$$F\left(\frac{2b}{b^2 + 1}, \frac{2b^2}{b^2 + 1}\right). \quad (2.9)$$

By symmetry, we have

$$CI: \quad y = -\frac{1}{c}x + 1 \quad (2.10)$$

and

$$E\left(\frac{2c}{c^2 + 1}, \frac{2c^2}{c^2 + 1}\right). \quad (2.11)$$

The side lengths of $\triangle ABC$ are given by

$$BC = c - b, \quad (2.12)$$

$$CA = \sqrt{\left(\frac{b+c}{bc+1} - c\right)^2 + \left(\frac{2bc}{bc+1} - 0\right)^2} = \sqrt{\frac{b^2(1+c^2)^2}{(bc+1)^2}} = \frac{b(1+c^2)}{bc+1} \quad (2.13)$$

(since $b < 0$ and $bc + 1 < 0$), and by symmetry

$$AB = \left| \frac{c(1+b^2)}{bc+1} \right| = -\frac{c(1+b^2)}{bc+1} \quad (2.14)$$

(since $c > 0$ and $bc + 1 < 0$).

Let O be the circumcentre of $\triangle ABC$. Firstly, O lies on the perpendicular bisector $x = \frac{b+c}{2}$ of BC . Secondly, from

$$\begin{aligned}
& OA = OB \\
\Rightarrow & \left(\frac{b+c}{2} - \frac{b+c}{bc+1} \right)^2 + \left(y - \frac{2bc}{bc+1} \right)^2 = \left(\frac{b+c}{2} - b \right)^2 + y^2 \\
\Rightarrow & \frac{(b+c)^2}{(bc+1)^2} - \frac{(b+c)^2}{bc+1} + \frac{4b^2c^2}{(bc+1)^2} - \frac{4bc}{bc+1}y = b^2 - b(b+c) \\
\Rightarrow & \frac{(b+c)^2 - (b+c)^2(bc+1) + 4b^2c^2 + bc(bc+1)^2}{(bc+1)^2} = \frac{4bc}{bc+1}y \\
\Rightarrow & y = \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{4(bc+1)},
\end{aligned}$$

we obtain

$$O \left(\frac{b+c}{2}, \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{4(bc+1)} \right). \quad (2.15)$$

It follows that

$$(ABC): \quad x^2 + y^2 - (b+c)x - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc+1)}y + bc = 0. \quad (2.16)$$

Let AI meet (ABC) again at M . Recall that M is the midpoint of $\widehat{BC}$ on (ABC) . From (2.3), we can find that

$$AI: \quad y = \frac{bc-1}{b+c}x + 1. \quad (2.17)$$

Putting $x = \frac{b+c}{2}$, we obtain

$$M \left(\frac{b+c}{2}, \frac{bc+1}{2} \right). \quad (2.18)$$

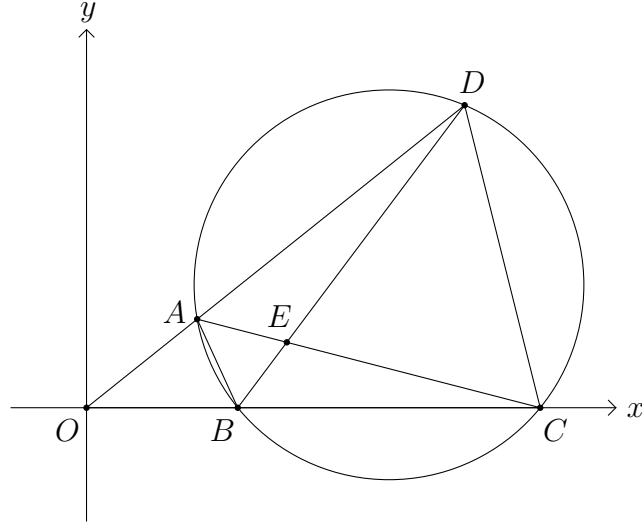
Setting 3: $O(0,0)$, $A(a,ma)$, $B(b,0)$, $C(c,0)$, $D(d,md)$

Remarks. This setting is used for a cyclic quadrilateral $ABCD$, where O is the intersection of the opposite sidelines AD and BC . As O lies outside $(ABCD)$, we may assume b, c have the same sign, and a, d have the same sign. The same setting applies if O is the intersection of the diagonals of a cyclic quadrilateral.

The key property of this construction is the following relation. We have

$$\begin{aligned}
& A, B, C, D \text{ are concyclic} \\
\Leftrightarrow & OA \times OD = OB \times OC \\
\Leftrightarrow & \sqrt{a^2 + (ma)^2} \times \sqrt{d^2 + (md)^2} = bc \\
\Leftrightarrow & ad(m^2 + 1) = bc.
\end{aligned} \quad (3.1)$$

Note that we have used the fact that $bc > 0$ and $ad > 0$.



Let E be the intersection of AC and BD . We easily obtain

$$AC: \quad y = \frac{ma}{a-c}x - \frac{mac}{a-c} \quad (3.2)$$

and similarly,

$$BD: \quad y = \frac{md}{d-b}x - \frac{mdb}{d-b}. \quad (3.3)$$

Solving these, we get

$$x = \frac{mdb(a-c) - mac(d-b)}{md(a-c) - ma(d-b)} = \frac{abc + abd - acd - bcd}{ab - cd}$$

and

$$y = \frac{ma}{a-c} \cdot \left(\frac{abc + abd - acd - bcd}{ab - cd} - c \right) = \frac{ma}{a-c} \cdot \frac{abd - acd - bcd + c^2d}{ab - cd} = \frac{mad(b-c)}{ab - cd}.$$

This means

$$E \left(\frac{abc + abd - acd - bcd}{ab - cd}, \frac{mad(b-c)}{ab - cd} \right). \quad (3.4)$$

Note that we also have

$$AB: \quad y = \frac{ma}{a-b}x - \frac{mab}{a-b} \quad (3.5)$$

and

$$CD: \quad y = \frac{md}{d-c}x - \frac{mdc}{d-c} \quad (3.6)$$

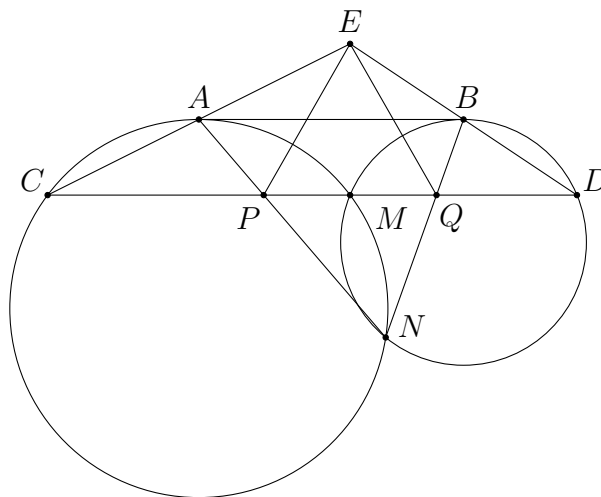
by symmetry.

In this note, our main goal is to show how coordinate geometry can be applied to solve the general case of the shortlist problems. Therefore, we will omit the consideration on some limiting cases. For example, we may use $y = mx + c$ to represent the equation

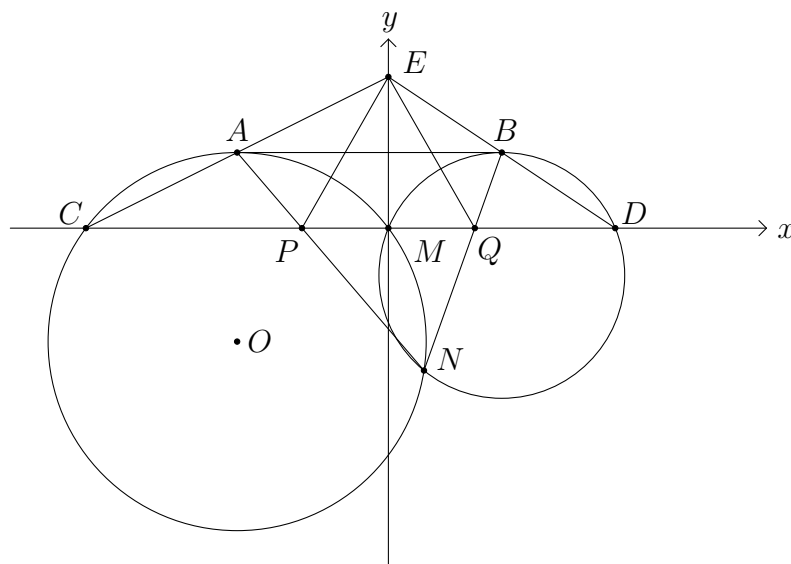
of a straight line even though there is a possibility that it is a vertical line, or we may define an intersection point even though it may be the ‘intersection’ of two parallel lines. The limiting cases can usually be handled by continuity or other arguments. But we will not do so throughout this note.

IMO Shortlist Problems

Problem 1. (IMO Shortlist 2000 G2/Problem 1) AB is tangent to the circles $CAMN$ and $NMBD$. M lies between C and D on the line CD , and CD is parallel to AB . The chords NA and CM meet at P ; the chords NB and MD meet at Q . The rays CA and DB meet at E . Prove that $PE = QE$.



Solution.



Let $M(0,0)$, $C(2c,0)$, $D(2d,0)$. Since the tangent AB is parallel to the chord CM , the point A lies on the perpendicular bisector of CM (for example, one can deduce this from $\angle AMC = \angle MAB = \angle ACM$). Let $A(c,a)$ and hence $B(d,a)$. Let $O(c,y)$

be the circumcentre of $\triangle ACM$. From

$$\begin{aligned} OC &= OA \\ \Rightarrow c^2 + y^2 &= (a - y)^2 \\ \Rightarrow y &= \frac{a^2 - c^2}{2a}, \end{aligned}$$

we get $O\left(c, \frac{a^2 - c^2}{2a}\right)$. So

$$(ACM): \quad x^2 + y^2 - 2cx - \frac{a^2 - c^2}{a}y = 0. \quad (1)$$

Similarly, we have

$$(BDM): \quad x^2 + y^2 - 2dx - \frac{a^2 - d^2}{a}y = 0. \quad (2)$$

The difference of (1) and (2) is

$$2(d - c)x + \frac{c^2 - d^2}{a}y = 0,$$

i.e. $y = \frac{2a}{c + d}x$. This is the equation of the radical axis of the two circles. Putting this in (1), we solve

$$\left(1 + \frac{4a^2}{(c + d)^2}\right)x^2 - \left(2c + \frac{2(a^2 - c^2)}{c + d}\right)x = 0.$$

The root $x = 0$ is rejected. The other root $x = \frac{2(c + d)(a^2 + cd)}{4a^2 + (c + d)^2}$ is the x -coordinate of N . So we have $N\left(\frac{2(c + d)(a^2 + cd)}{4a^2 + (c + d)^2}, \frac{4a(a^2 + cd)}{4a^2 + (c + d)^2}\right)$. Thus,

$$m_{AN} = \frac{4a(a^2 + cd) - a(4a^2 + (c + d)^2)}{2(c + d)(a^2 + cd) - c(4a^2 + (c + d)^2)} = \frac{-a(d - c)^2}{2(d - c)a^2 + c(d^2 - c^2)} = -\frac{a(d - c)}{2a^2 + c^2 + cd}.$$

This yields

$$AN \quad y = -\frac{a(d - c)}{2a^2 + c^2 + cd}x + \frac{2a(a^2 + cd)}{2a^2 + c^2 + cd}.$$

It follows that $P\left(\frac{2(a^2 + cd)}{d - c}, 0\right)$. By symmetry, we have $Q\left(\frac{2(a^2 + cd)}{c - d}, 0\right)$.

Next, we solve

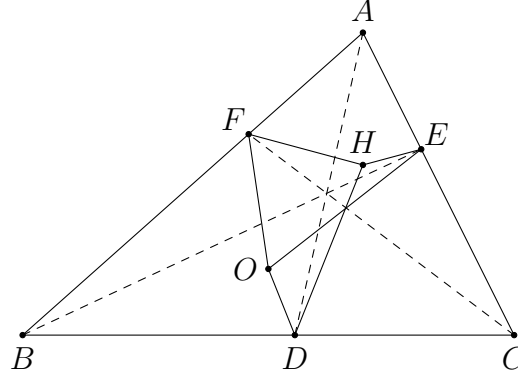
$$\begin{aligned} AC: \quad y &= -\frac{a}{c}x + 2a, \\ BD: \quad y &= -\frac{a}{d}x + 2a \end{aligned}$$

to get $E(0, 2a)$. From above, P and Q are symmetric in the y -axis. So $PE = QE$ as desired. $\square$

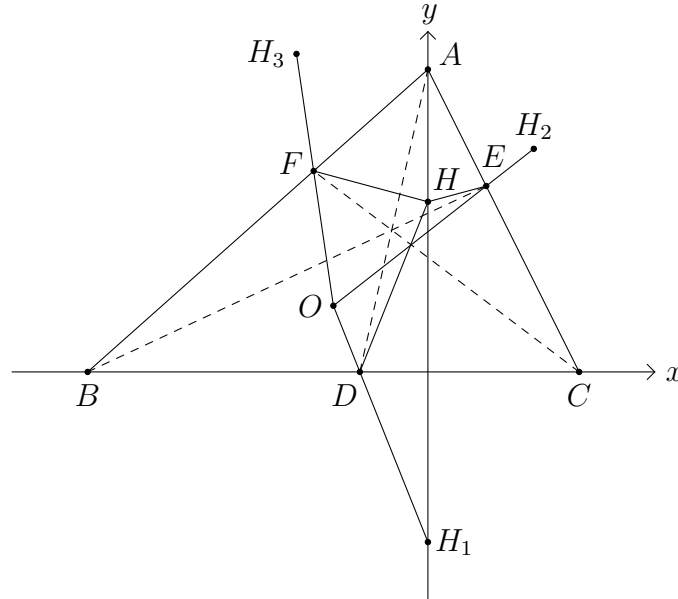
Problem 2. (IMO Shortlist 2000 G3) Let O be the circumcentre and H the orthocentre of an acute triangle ABC . Show that there exist points D, E , and F on sides BC, CA , and AB respectively such that

$$OD + DH = OE + EH = OF + FH$$

and the lines AD, BE , and CF are concurrent.



Solution.



Guessing the positions of D, E, F is important. To make the computation of the sum $OD + DH$, etc. easy, one way is to take the point D such that OD and DH are symmetric in the line BC . It turns out this is exactly the correct choice of D .

Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $H\left(0, -\frac{bc}{a}\right)$ and $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$. Let H_1, H_2, H_3 be the reflection of H in BC, CA, AB respectively. Suppose OH_1 meets BC at D ,

OH_2 meets CA at E , and OH_3 meets AB at F . We claim that these are the desired points D, E, F .

It is obvious that $H_1\left(0, \frac{bc}{a}\right)$. So

$$OH_1: \quad y = \frac{a^2 - bc}{a(b + c)}x + \frac{bc}{a}.$$

This yields $D\left(-\frac{bc(b + c)}{a^2 - bc}, 0\right)$.

Next, recall that the intersection point of AC and BH has coordinates

$$\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right).$$

This is the midpoint of HH_2 . So we easily get

$$H_2 = 2\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right) - \left(0, -\frac{bc}{a}\right) = \left(\frac{2c(a^2 + bc)}{a^2 + c^2}, \frac{c(2a^2c - a^2b + bc^2)}{a(a^2 + c^2)}\right).$$

This yields

$$m_{OH_2} = \frac{2c(2a^2c - a^2b + bc^2) - (a^2 + bc)(a^2 + c^2)}{a(4c(a^2 + bc) - (b + c)(a^2 + c^2))} = \frac{-a^4 - 3a^2bc + 3a^2c^2 + bc^3}{a(-a^2b + 3a^2c + 3bc^2 - c^3)}.$$

Thus, the equation of OH_2 is

$$y = \frac{-a^4 - 3a^2bc + 3a^2c^2 + bc^3}{a(-a^2b + 3a^2c + 3bc^2 - c^3)}x - \frac{-a^4 - 3a^2bc + 3a^2c^2 + bc^3}{a(-a^2b + 3a^2c + 3bc^2 - c^3)} \cdot \frac{b + c}{2} + \frac{a^2 + bc}{2a},$$

which can be simplified to

$$OH_2: \quad y = \frac{-a^4 - 3a^2bc + 3a^2c^2 + bc^3}{a(-a^2b + 3a^2c + 3bc^2 - c^3)}x + \frac{2a^4c + a^2b^2c + 3a^2bc^2 - 2a^2c^3 + b^2c^3 - bc^4}{a(-a^2b + 3a^2c + 3bc^2 - c^3)}.$$

Together with

$$AC: \quad y = -\frac{a}{c}x + a,$$

we combine and solve them to get

$$\begin{aligned} x &= \frac{c(a^2(-a^2b + 3a^2c + 3bc^2 - c^3) - (2a^4c + a^2b^2c + 3a^2bc^2 - 2a^2c^3 + b^2c^3 - bc^4))}{a^2(-a^2b + 3a^2c + 3bc^2 - c^3) + c(-a^4 - 3a^2bc + 3a^2c^2 + bc^3)} \\ &= \frac{c(-a^4b + a^4c - a^2b^2c + a^2c^3 - b^2c^3 + bc^4)}{-a^4b + 2a^4c + 2a^2c^3 + bc^4} \\ &= \frac{c(c - b)(a^2 + c^2)(a^2 + bc)}{(a^2 + c^2)(2a^2c - a^2b + bc^2)} \\ &= \frac{c(c - b)(a^2 + bc)}{2a^2c - a^2b + bc^2} \end{aligned}$$

and

$$y = -\frac{a}{c} \cdot \frac{c(c-b)(a^2+bc)}{2a^2c - a^2b + bc^2} + a = \frac{ac(a^2+b^2)}{2a^2c - a^2b + bc^2}.$$

Hence, $E\left(\frac{c(c-b)(a^2+bc)}{2a^2c - a^2b + bc^2}, \frac{ac(a^2+b^2)}{2a^2c - a^2b + bc^2}\right)$.

We first check that $OD + DH = OE + EH$. By the property of reflection, we have $OD + DH = OD + DH_1 = OH_1$. Similarly, we have $OE + EH = OH_2$. So it suffices to check $OH_1 = OH_2$. (This is a well-known property of the orthocentre.) Indeed, we have

$$\begin{aligned} OH_1^2 &= \left(\frac{b+c}{2}\right)^2 + \left(\frac{a^2+bc}{2a} - \frac{bc}{a}\right)^2 \\ &= \frac{(b+c)^2}{4} + \frac{(a^2-bc)^2}{4a^2} \\ &= \frac{(a^2+b^2)(a^2+c^2)}{4a^2} \end{aligned}$$

and

$$\begin{aligned} OH_2^2 &= \left(\frac{b+c}{2} - \frac{2c(a^2+bc)}{a^2+c^2}\right)^2 + \left(\frac{a^2+bc}{2a} - \frac{c(2a^2c - a^2b + bc^2)}{a(a^2+c^2)}\right)^2 \\ &= \left(\frac{a^2b - 3a^2c - 3bc^2 + c^3}{2(a^2+c^2)}\right)^2 + \left(\frac{a^4 + 3a^2bc - 3a^2c^2 - bc^3}{2a(a^2+c^2)}\right)^2 \\ &= \frac{a^8 + a^6b^2 + 3a^6c^2 + 3a^4b^2c^2 + 3a^4c^4 + 3a^2b^2c^4 + a^2c^6 + b^2c^6}{4a^2(a^2+c^2)^2} \\ &= \frac{(a^2+b^2)(a^2+c^2)^3}{4a^2(a^2+c^2)^2} \\ &= \frac{(a^2+b^2)(a^2+c^2)}{4a^2} \\ &= OH_1^2. \end{aligned}$$

This shows $OH_1 = OH_2$. Similarly, we also have $OD + DH = OF + FH$. It remains to prove AD, BE, CF are concurrent.

From $A(0, a)$ and $D\left(-\frac{bc(b+c)}{a^2-bc}, 0\right)$, we easily obtain

$$AD: \quad y = \frac{a(a^2-bc)}{bc(b+c)}x + a.$$

From $B(b, 0)$ and $E\left(\frac{c(c-b)(a^2+bc)}{2a^2c - a^2b + bc^2}, \frac{ac(a^2+b^2)}{2a^2c - a^2b + bc^2}\right)$, we find that

$$BE: \quad y = \frac{ac(a^2+b^2)}{a^2b^2 - 3a^2bc + a^2c^2 - 2b^2c^2 + bc^3}x - \frac{abc(a^2+b^2)}{a^2b^2 - 3a^2bc + a^2c^2 - 2b^2c^2 + bc^3}.$$

Solving these, we have

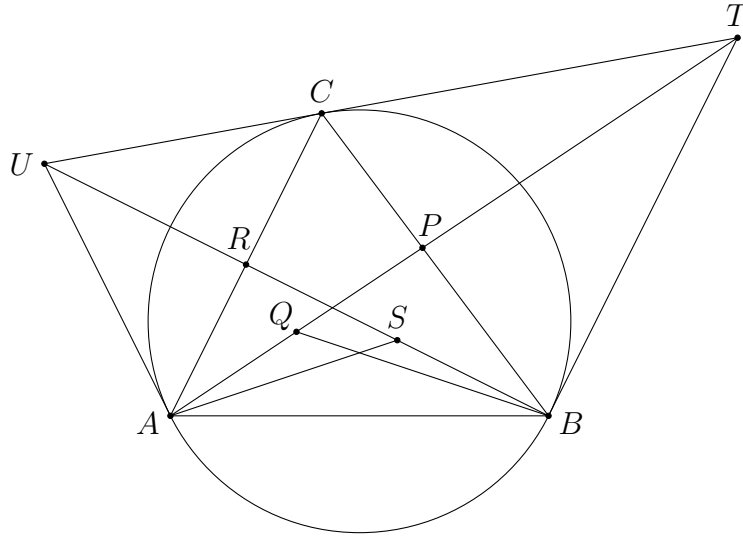
$$\begin{aligned}
x &= \frac{bc(b+c)(a^2b^2 - 3a^2bc + a^2c^2 - 2b^2c^2 + bc^3 + bc(a^2 + b^2))}{bc^2(a^2 + b^2)(b+c) - (a^2 - bc)(a^2b^2 - 3a^2bc + a^2c^2 - 2b^2c^2 + bc^3)} \\
&= \frac{bc(b+c)(a^2b^2 - 2a^2bc + a^2c^2 + b^3c - 2b^2c^2 + bc^3)}{-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4} \\
&= \frac{bc(b+c)(b-c)^2(a^2 + bc)}{-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4}
\end{aligned}$$

and

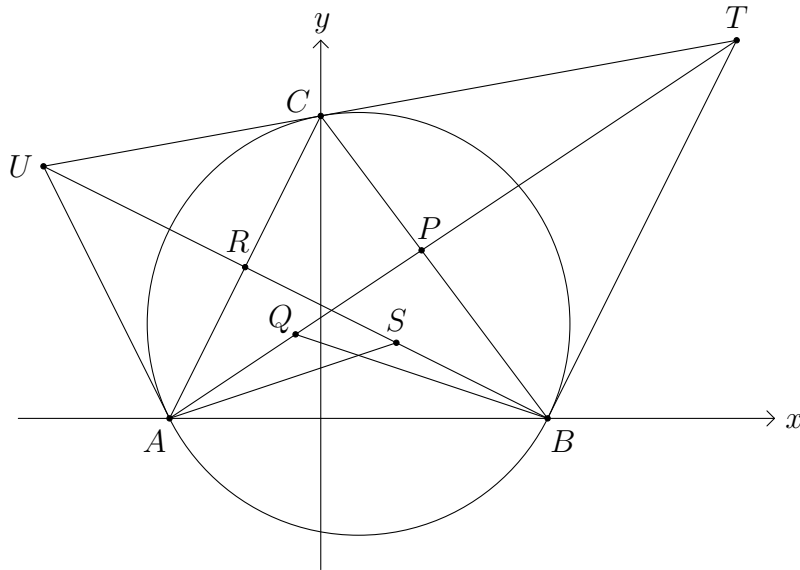
$$\begin{aligned}
y &= \frac{a(a^2 - bc)}{bc(b+c)} \cdot \frac{bc(b+c)(b-c)^2(a^2 + bc)}{-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4} + a \\
&= \frac{a((a^2 - bc)(b-c)^2(a^2 + bc) + (-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4))}{-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4} \\
&= \frac{abc(a^4 + a^2b^2 + a^2c^2 + b^2c^2)}{-a^4b^2 + 3a^4bc - a^4c^2 + a^2b^3c + a^2bc^3 + b^4c^2 - b^3c^3 + b^2c^4}.
\end{aligned}$$

Since both expressions are symmetric in b and c , this is also the intersection of AD and CF . Therefore, AD, BE, CF are concurrent. The proof is complete. $\square$

Problem 3. (IMO Shortlist 2000 G5) The tangents at B and A to the circumcircle of an acute angled triangle ABC meet the tangent at C at T and U respectively. AT meets BC at P , and Q is the midpoint of AP ; BU meets CA at R , and S is the midpoint of BR . Prove that $\angle ABQ = \angle BAS$. Determine, in terms of ratios of side lengths, the triangles for which this angle is a maximum.



Solution.



Let $A(a, 0)$, $B(b, 0)$, $C(0, c)$. The circumcentre O has coordinates $\left(\frac{a+b}{2}, \frac{c^2+ab}{2c}\right)$.
From

$$m_{BT} = \frac{-1}{m_{BO}} = -\frac{c(a-b)}{c^2 + ab},$$

we get

$$BT: \quad y = -\frac{c(a-b)}{c^2+ab}x + \frac{bc(a-b)}{c^2+ab}. \quad (1)$$

Similarly, from

$$m_{CT} = \frac{-1}{m_{CO}} = -\frac{c(a+b)}{ab-c^2},$$

we get

$$CT: \quad y = -\frac{c(a+b)}{ab-c^2}x + c. \quad (2)$$

Solving (1) and (2), we obtain

$$x = \frac{(ab - c^2)(c(c^2 + ab) - bc(a - b))}{c(a + b)(c^2 + ab) - c(a - b)(ab - c^2)} = \frac{(ab - c^2)c(b^2 + c^2)}{2ac(b^2 + c^2)} = \frac{ab - c^2}{2a}$$

and

$$y = -\frac{c(a+b)}{ab-c^2} \cdot \frac{ab-c^2}{2a} + c = \frac{c(a-b)}{2a}.$$

This means $T\left(\frac{ab-c^2}{2a}, \frac{c(a-b)}{2a}\right)$. It is easy to get

$$AT: \quad y = \frac{c(a-b)}{ab-c^2-2a^2}x - \frac{ac(a-b)}{ab-c^2-2a^2}.$$

Together with

$$BC: \quad y = -\frac{c}{b}x + c,$$

we find that

$$x = \frac{b((ab - c^2 - 2a^2) + a(a - b))}{b(a - b) + (ab - c^2 - 2a^2)} = \frac{b(a^2 + c^2)}{2a^2 - 2ab + b^2 + c^2}$$

and

$$y = -\frac{c}{b} \cdot \frac{b(a^2 + c^2)}{2a^2 - 2ab + b^2 + c^2} + c = \frac{c(a - b)^2}{2a^2 - 2ab + b^2 + c^2}.$$

So $P\left(\frac{b(a^2 + c^2)}{2a^2 - 2ab + b^2 + c^2}, \frac{c(a - b)^2}{2a^2 - 2ab + b^2 + c^2}\right)$. Thus,

$$Q = \frac{A + P}{2} = \left(\frac{2a^3 - a^2b + ab^2 + ac^2 + bc^2}{2(2a^2 - 2ab + b^2 + c^2)}, \frac{c(a - b)^2}{2(2a^2 - 2ab + b^2 + c^2)} \right).$$

It follows that

$$m_{BQ} = \frac{c(a - b)^2}{(2a^3 - a^2b + ab^2 + ac^2 + bc^2) - 2b(2a^2 - 2ab + b^2 + c^2)} = \frac{c(a - b)}{2a^2 - 3ab + 2b^2 + c^2}.$$

By symmetry, we have $m_{AS} = \frac{c(b - a)}{2b^2 - 3ba + 2a^2 + c^2}$. Hence,

$$\tan \angle QBA = -m_{BQ} = m_{AS} = \tan \angle BAS.$$

This implies $\angle QBA = \angle BAS$.

Next, to maximize $\angle QBA$, it means we need to maximize $\tan \angle QBA$ since the angle is acute. We may assume $b > 0 > a$. Also, as $2a^2 - 3ab + 2b^2 > 0$, we can apply the AM-GM inequality to get

$$\tan \angle QBA = \frac{c(b - a)}{2a^2 - 3ab + 2b^2 + c^2} \leq \frac{c(b - a)}{2\sqrt{(2a^2 - 3ab + 2b^2)(c^2)}} = \frac{b - a}{2\sqrt{2a^2 - 3ab + 2b^2}}.$$

Equality holds when $c = \sqrt{2a^2 - 3ab + 2b^2}$. For the remaining term, we have

$$\frac{b - a}{2\sqrt{2a^2 - 3ab + 2b^2}} = \frac{1}{2} \sqrt{\frac{a^2 - 2ab + b^2}{2a^2 - 3ab + 2b^2}} = \frac{1}{2} \sqrt{\frac{1}{2} + \frac{(-a)b}{2(2(-a)^2 + 3(-a)b + 2b^2)}}.$$

Again applying the AM-GM inequality, this is bounded above by

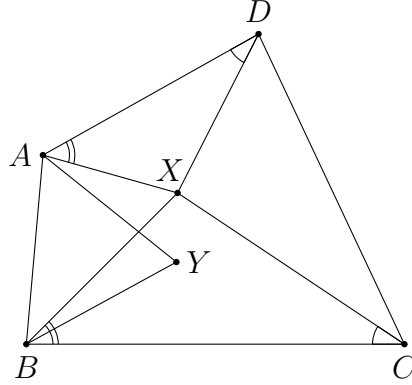
$$\frac{1}{2} \sqrt{\frac{1}{2} + \frac{(-a)b}{2(7(-a)b)}} = \frac{1}{2} \sqrt{\frac{4}{7}}.$$

Equality holds when $-a = b$. In that case, $c = \sqrt{2a^2 - 3ab + 2b^2} = \sqrt{7}b$. Therefore, the maximum value is attained when $AB = 2b$, $AC = BC = \sqrt{b^2 + c^2} = 2\sqrt{2}b$, which means $AB : BC : CA = 1 : \sqrt{2} : \sqrt{2}$. $\square$

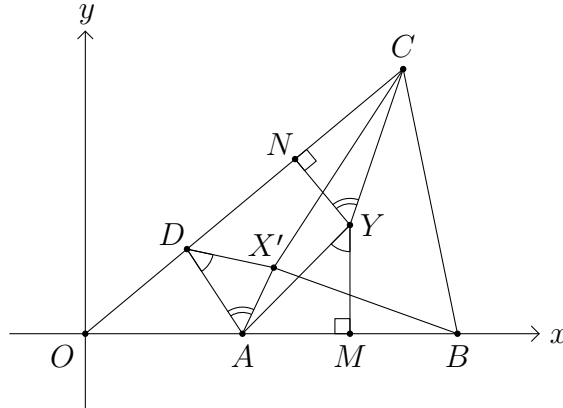
Problem 4. (IMO Shortlist 2000 G6) Let $ABCD$ be a convex quadrilateral. The perpendicular bisectors of its sides AB and CD meet at Y . Denote by X a point inside the quadrilateral $ABCD$ such that

$$\angle ADX = \angle BCX < 90^\circ \quad \text{and} \quad \angle DAX = \angle CBX < 90^\circ.$$

Show that $\angle AYB = 2\angle ADX$.



Solution.



We set $AB \cap CD = O$ as the origin, and let $A(a, 0)$, $B(b, 0)$, $C(c, mc)$ and $D(d, md)$ (thus, the equation of CD is $y = mx$). Let $M\left(\frac{a+b}{2}, 0\right)$ and $N\left(\frac{c+d}{2}, \frac{m(c+d)}{2}\right)$ be the midpoints of AB and CD respectively. Clearly, $m_{NY} = \frac{-1}{m_{CD}} = -\frac{1}{m}$, so that

$$NY : \quad y = -\frac{1}{m}x + \frac{(m^2 + 1)(c + d)}{2m}.$$

As the x -coordinate of Y is $\frac{a+b}{2}$, we obtain $Y\left(\frac{a+b}{2}, \frac{(m^2 + 1)(c + d) - (a + b)}{2m}\right)$.

Note that there is only one point X inside $ABCD$ such that $\angle ADX = \angle BCX$ and $\angle DAX = \angle CBX$. Here is a brief proof of this fact. Indeed, the two conditions

together mean $\triangle XAD \sim \triangle XBC$. So we need $\frac{XA}{XB} = \frac{XD}{XC} = \frac{AD}{BC}$. The locus of X satisfying each of $\frac{XA}{XB} = \frac{AD}{BC}$ and $\frac{XD}{XC} = \frac{AD}{BC}$ is a circle (Apollonius circle). The two circles have at most two intersection points. But it is not possible that both intersection points lie inside $ABCD$ since the centres of the two circles lie on the rays CD and BA beyond D and A (if $AD < BC$), or on the rays DC and AB beyond C and B (if $AD > BC$) respectively.

In view of the final result and the symmetry, we define X' to be a point inside $ABCD$ such that $\angle ADX' = \angle AYM$ and $\angle DAX' = \angle CYN$. If we can show that $\angle X'AD = \angle X'BC$ and $\angle X'DA = \angle X'CB$, then we must have $X' = X$ from above, and hence $\angle AYB = 2\angle AYM = 2\angle ADX$.

Firstly, we have

$$\begin{aligned}\tan \angle CYN &= \frac{m_{NY} - m_{CY}}{1 + m_{NY}m_{CY}} = \frac{-\frac{1}{m} - \frac{(d-c)m^2 + (c+d-a-b)}{m(a+b-2c)}}{1 - \frac{(d-c)m^2 + (c+d-a-b)}{m^2(a+b-2c)}} \\ &= \frac{m((c-d)m^2 + (c-d))}{(a+b-c-d)m^2 + (a+b-c-d)} = \frac{(c-d)m}{a+b-c-d}.\end{aligned}$$

Using $\frac{m_{AD} - m_{AX'}}{1 + m_{AD}m_{AX'}} = \tan \angle DAX' = \tan \angle CYN$, we obtain

$$\begin{aligned}m_{AX'} &= \frac{m_{AD} - \tan \angle CYN}{1 + m_{AD} \tan \angle CYN} = \frac{\frac{md}{d-a} - \frac{(c-d)m}{a+b-c-d}}{1 + \frac{d(c-d)m^2}{(d-a)(a+b-c-d)}} \\ &= \frac{(ac+bd-2cd)m}{d(c-d)m^2 + (d-a)(a+b-c-d)}.\end{aligned}$$

This implies

$$AX': \quad y = \frac{(ac+bd-2cd)m}{d(c-d)m^2 + (d-a)(a+b-c-d)}x - \frac{(ac+bd-2cd)am}{d(c-d)m^2 + (d-a)(a+b-c-d)}.$$

Similarly, we have

$$\tan \angle AYM = \frac{1}{\tan \angle YAM} = \frac{1}{m_{AY}} = \frac{(b-a)m}{(m^2+1)(c+d) - (a+b)}.$$

Using $\frac{m_{DX'} - m_{AD}}{1 + m_{DX'}m_{AD}} = \tan \angle ADX' = \tan \angle AYM$, we obtain

$$m_{DX'} = \frac{m_{AD} + \tan \angle AYM}{1 - m_{AD} \tan \angle AYM} = \frac{\frac{md}{d-a} + \frac{(b-a)m}{(m^2+1)(c+d) - (a+b)}}{1 - \frac{(b-a)dm^2}{(d-a)((m^2+1)(c+d) - (a+b))}}$$

$$= \frac{(d(c+d)m^2 + a^2 - 2ad + d^2 - ab + cd)m}{(cd + d^2 - ac - bd)m^2 + (d-a)(c+d-a-b)}.$$

This implies

$$DX': \quad y = \frac{(d(c+d)m^2 + a^2 - 2ad + d^2 - ab + cd)m}{(cd + d^2 - ac - bd)m^2 + (d-a)(c+d-a-b)}x + \frac{(-(ac+bd)m^2 + 2ab - ac - bd)dm}{(cd + d^2 - ac - bd)m^2 + (d-a)(c+d-a-b)}.$$

Solving the equations of AX' and DX' , we find that

$$\begin{aligned} x &= \frac{d(-(ac+bd)m^2 + (2ab - ac - bd))(d(c-d)m^2 + (d-a)(a+b-c-d)) + a(ac+bd-2cd)((cd + d^2 - ac - bd)m^2 + (d-a)(c+d-a-b))}{(ac+bd-2cd)((cd + d^2 - ac - bd)m^2 + (d-a)(c+d-a-b)) - (d(c+d)m^2 + (a^2 - 2ad + d^2 - ab + cd))(d(c-d)m^2 + (d-a)(a+b-c-d))} \\ &= \frac{d^2(d-c)(ac+bd)m^4 + (-a^3c^2 + a^3cd - a^2bcd + a^2bd^2 + 2a^2c^2d - a^2cd^2 + 3abcd^2 - 3abd^3 - 2ac^2d^2 - b^2d^3 + 2bd^4)m^2 + (d-a)^2(a+b-c-d)(ac-bd)}{d^2(d^2-c^2)m^4 + (-a^2c^2 + 2a^2d^2 + 2ac^2d - 4ad^3 - b^2d^2 + 2bcd^2 - 2c^2d^2 + 2d^4)m^2 + (d-a)^2(a+b-c-d)(a-b+c-d)} \\ &= \frac{(d^2m^2 + (d-a)^2)((d-c)(ac+bd)m^2 + (a+b-c-d)(ac-bd))}{(d^2m^2 + (d-a)^2)((d^2-c^2)m^2 + (a+b-c-d)(a-b+c-d))} \\ &= \frac{(d-c)(ac+bd)m^2 + (a+b-c-d)(ac-bd)}{(d^2-c^2)m^2 + (a+b-c-d)(a-b+c-d)}. \end{aligned}$$

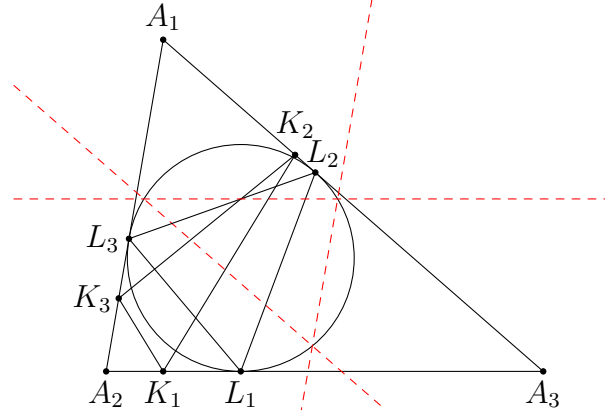
(The factorization can be guessed since the expression must be symmetric by swapping a and b and swapping c and d .) Then y is equal to

$$\begin{aligned} & \frac{(ac+bd-2cd)m}{d(c-d)m^2 + (d-a)(a+b-c-d)} \cdot (x-a) \\ &= \frac{(ac+bd-2cd)m}{d(c-d)m^2 + (d-a)(a+b-c-d)} \cdot \frac{d(c-d)(a-b)m^2 + (a+b-c-d)(d-a)(a-b)}{(d^2-c^2)m^2 + (a+b-c-d)(a-b+c-d)} \\ &= \frac{(a-b)(ac+bd-2cd)m}{(d^2-c^2)m^2 + (a+b-c-d)(a-b+c-d)}. \end{aligned}$$

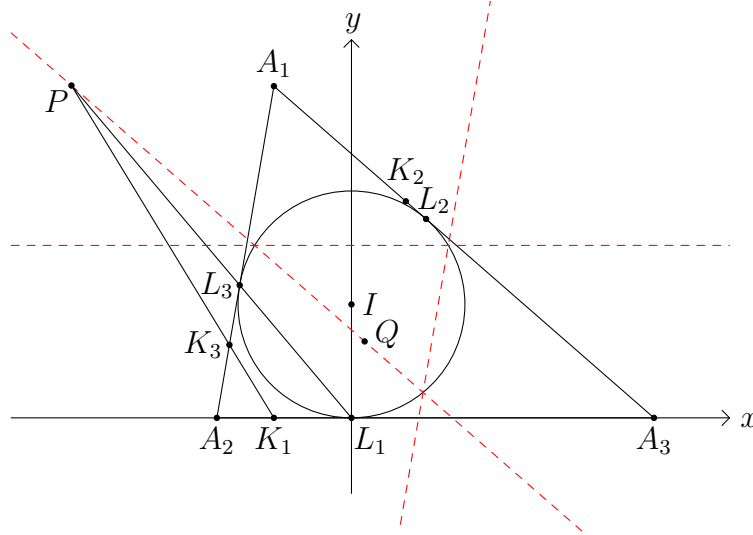
Note that both coordinates of X' remain unchanged if we swap a and b and swap c and d . Therefore, by symmetry, X' satisfies $\angle X'CB = \angle BYM$ and $\angle X'BC = \angle DYN$ as well. This implies $\angle X'CB = \angle X'DA$ and $\angle X'BC = \angle X'AD$. The proof is complete. $\square$

Problem 5. (IMO Shortlist 2000 G8/Problem 6) $A_1A_2A_3$ is an acute-angled triangle. The foot of the altitude from A_i is K_i and the incircle touches the side opposite A_i at L_i . The line K_1K_2 is reflected in the line L_1L_2 . Similarly, the line K_2K_3 is reflected

in L_2L_3 and K_3K_1 is reflected in L_3L_1 . Show that the three new lines form a triangle with vertices on the incircle.



Solution.



Let $I(0,1)$ be the incentre. Let $L_1(0,0)$, $A_2(b,0)$, $A_3(c,0)$. Then $A_1\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$, $L_3\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right)$. Clearly, $K_1\left(\frac{b+c}{bc+1}, 0\right)$. Then

$$A_1A_2: \quad y = -\frac{2b}{b^2-1}x + \frac{2b^2}{b^2-1}. \quad (1)$$

The slope of A_3K_3 is $\frac{-1}{m_{A_1A_2}} = \frac{b^2-1}{2b}$. So

$$A_3K_3: \quad y = \frac{b^2-1}{2b}x - \frac{(b^2-1)c}{2b}. \quad (2)$$

We solve (1) and (2). This yields

$$x = \frac{4b^3 + (b^2-1)^2c}{(b^2-1)^2 + (2b)^2} = \frac{4b^3 + (b^2-1)^2c}{(b^2+1)^2}$$

and

$$y = \frac{b^2 - 1}{2b} \cdot \left(\frac{4b^3 + (b^2 - 1)^2 c}{(b^2 + 1)^2} - c \right) = \frac{b^2 - 1}{2b} \cdot \frac{4b^3 - 4b^2 c}{(b^2 + 1)^2} = \frac{2b(b - c)(b^2 - 1)}{(b^2 + 1)^2}.$$

Therefore, we obtain $K_3 \left(\frac{4b^3 + (b^2 - 1)^2 c}{(b^2 + 1)^2}, \frac{2b(b - c)(b^2 - 1)}{(b^2 + 1)^2} \right)$.

Let ℓ be the reflection of $K_1 K_3$ in the line $L_1 L_3$. Suppose P is the intersection of $L_1 L_3$ and $K_1 K_3$. Then P lies on ℓ . Recall that

$$L_1 L_3 : \quad y = bx. \quad (3)$$

Also, from

$$\begin{aligned} m_{K_1 K_3} &= \frac{2b(b - c)(b^2 - 1)(bc + 1)}{(4b^3 + (b^2 - 1)^2 c)(bc + 1) - (b + c)(b^2 + 1)^2} \\ &= \frac{2b(b - c)(b^2 - 1)(bc + 1)}{b(b^2 - 1)^2 c^2 + 4b^2(b^2 - 1)c - b(b^2 - 1)^2} \\ &= \frac{2(b - c)(bc + 1)}{b^2 c^2 - b^2 + 4bc - c^2 + 1}, \end{aligned}$$

we have

$$K_1 K_3 : \quad y = \frac{2(b - c)(bc + 1)}{b^2 c^2 - b^2 + 4bc - c^2 + 1} x - \frac{2(b - c)(b + c)}{b^2 c^2 - b^2 + 4bc - c^2 + 1}. \quad (4)$$

We now solve (3) and (4). This yields

$$x = \frac{2(b - c)(b + c)}{2(b - c)(bc + 1) - b(b^2 c^2 - b^2 + 4bc - c^2 + 1)} = -\frac{2(b - c)(b + c)}{(b^2 + 1)(bc^2 + 2c - b)}$$

and

$$y = bx = -\frac{2b(b - c)(b + c)}{(b^2 + 1)(bc^2 + 2c - b)}.$$

Thus, $P \left(-\frac{2(b - c)(b + c)}{(b^2 + 1)(bc^2 + 2c - b)}, -\frac{2b(b - c)(b + c)}{(b^2 + 1)(bc^2 + 2c - b)} \right)$.

Let Q be the reflection of K_1 in the line $L_1 L_3$. Then Q also lies on ℓ . Firstly, the slope of $K_1 Q$ is $\frac{-1}{m_{L_1 L_3}} = -\frac{1}{b}$. So

$$K_1 Q : \quad y = -\frac{1}{b}x + \frac{b + c}{b(bc + 1)}.$$

Combining with (3), we get

$$x = \frac{b + c}{(bc + 1)(b^2 + 1)}$$

and

$$y = bx = \frac{b(b+c)}{(bc+1)(b^2+1)}.$$

So the intersection of K_1Q and L_1L_3 is $\left(\frac{b+c}{(bc+1)(b^2+1)}, \frac{b(b+c)}{(bc+1)(b^2+1)}\right)$. Note that this is the midpoint of K_1Q . So the coordinates of Q are

$$\left(\frac{2(b+c)}{(bc+1)(b^2+1)} - \frac{b+c}{bc+1}, \frac{2b(b+c)}{(bc+1)(b^2+1)}\right) = \left(\frac{(b+c)(1-b^2)}{(bc+1)(b^2+1)}, \frac{2b(b+c)}{(bc+1)(b^2+1)}\right).$$

We use P and Q to find ℓ . We have

$$\begin{aligned} m_{PQ} &= \frac{2b(b+c)(bc^2+2c-b) + 2b(b-c)(b+c)(bc+1)}{(b+c)(1-b^2)(bc^2+2c-b) + 2(b-c)(b+c)(bc+1)} \\ &= \frac{2bc(b^2+1)}{-b(b^2+1)c^2 + b(b^2+1)} = \frac{2c}{1-c^2}. \end{aligned}$$

So the equation of ℓ is

$$y = \frac{2c}{1-c^2}x + \frac{2c}{1-c^2} \cdot \frac{2(b-c)(b+c)}{(b^2+1)(bc^2+2c-b)} - \frac{2b(b-c)(b+c)}{(b^2+1)(bc^2+2c-b)},$$

i.e.

$$\ell: \quad y = \frac{2c}{1-c^2}x + \frac{2(b+c)(b-c)}{(b^2+1)(1-c^2)}. \quad (5)$$

Let ℓ' be the reflection of K_1K_2 in the line L_1L_2 . By symmetry,

$$\ell': \quad y = \frac{2b}{1-b^2}x + \frac{2(b+c)(c-b)}{(c^2+1)(1-b^2)}. \quad (6)$$

We solve (5) and (6). We get

$$\begin{aligned} x &= \frac{2(b+c)(c-b)(b^2+1)(1-c^2) - 2(b+c)(b-c)(c^2+1)(1-b^2)}{(b^2+1)(c^2+1)(2c(1-b^2) - 2b(1-c^2))} \\ &= \frac{2(b+c)(c-b)(2-2b^2c^2)}{2(b^2+1)(c^2+1)(c-b)(bc+1)} = \frac{2(b+c)(1-bc)}{(b^2+1)(c^2+1)} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{2c}{1-c^2} \cdot \frac{2(b+c)(1-bc)}{(b^2+1)(c^2+1)} + \frac{2(b+c)(b-c)}{(b^2+1)(1-c^2)} \\ &= \frac{4c(b+c)(1-bc) + 2(b+c)(b-c)(c^2+1)}{(b^2+1)(c^2+1)(1-c^2)} \\ &= \frac{2(b+c)^2(1-c^2)}{(b^2+1)(c^2+1)(1-c^2)} = \frac{2(b+c)^2}{(b^2+1)(c^2+1)}. \end{aligned}$$

Therefore, the intersection of ℓ and ℓ' has coordinates

$$\left(\frac{2(b+c)(1-bc)}{(b^2+1)(c^2+1)}, \frac{2(b+c)^2}{(b^2+1)(c^2+1)} \right).$$

We check that it lies on the incircle, whose equation is

$$x^2 + (y-1)^2 = 1.$$

We first compute

$$\begin{aligned} x^2 + (y-1)^2 &= \left(\frac{2(b+c)(1-bc)}{(b^2+1)(c^2+1)} \right)^2 + \left(\frac{2(b+c)^2}{(b^2+1)(c^2+1)} - 1 \right)^2 \\ &= \left(\frac{2(b+c)(1-bc)}{(b^2+1)(c^2+1)} \right)^2 + \left(\frac{-b^2c^2 + b^2 + 4bc + c^2 - 1}{(b^2+1)(c^2+1)} \right)^2. \end{aligned}$$

So we have to prove

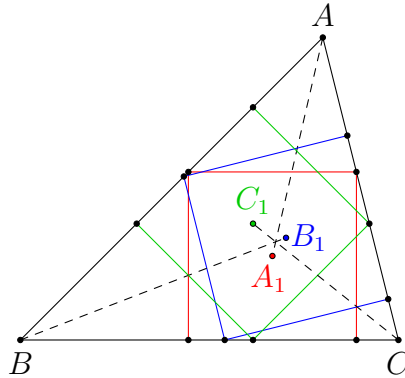
$$(2(b+c)(1-bc))^2 + (-b^2c^2 + b^2 + 4bc + c^2 - 1)^2 = ((b^2+1)(c^2+1))^2.$$

Instead of expanding everything, we use factorization. Indeed, we have

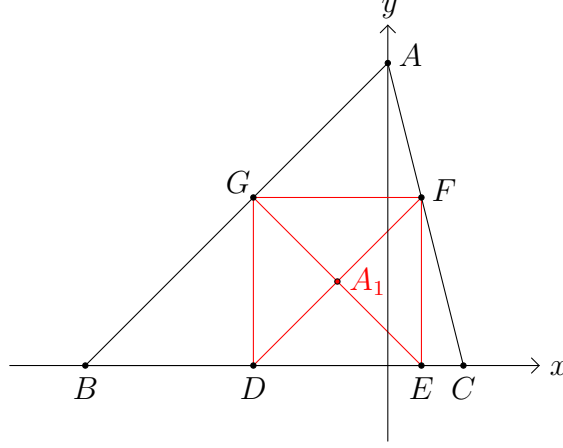
$$\begin{aligned} &((b^2+1)(c^2+1))^2 - (-b^2c^2 + b^2 + 4bc + c^2 - 1)^2 \\ &= ((b^2+1)(c^2+1) - (-b^2c^2 + b^2 + 4bc + c^2 - 1)) \\ &\quad \times ((b^2+1)(c^2+1) + (-b^2c^2 + b^2 + 4bc + c^2 - 1)) \\ &= (2b^2c^2 - 4bc + 2)(2b^2 + 4bc + 2c^2) \\ &= 4(1-bc)^2(b+c)^2. \end{aligned}$$

This shows ℓ and ℓ' intersect on the incircle. By symmetry, the three lines form a triangle with vertices on the incircle. $\square$

Problem 6. (IMO Shortlist 2001 G1) Let A_1 be the centre of the square inscribed in acute triangle ABC with two vertices of the square on side BC . Thus one of the two remaining vertices of the square is on side AB and the other is on AC . Points B_1, C_1 are defined in a similar way for inscribed squares with two vertices on sides AC and AB , respectively. Prove that lines AA_1, BB_1, CC_1 are concurrent.



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$ where $a > 0$ and $b < 0 < c$. Let $DEFG$ be the square with centre A_1 such that D, E lie on BC , F lies on CA and G lies on AB . Let $D(d, 0)$. Since

$$AB: \quad y = -\frac{a}{b}x + a, \quad (1)$$

we put $x = d$ to get $G\left(d, \frac{a(b-d)}{b}\right)$. Next, since

$$AC: \quad y = -\frac{a}{c}x + a, \quad (2)$$

by putting $y = \frac{a(b-d)}{b}$, we get $F\left(\frac{cd}{b}, \frac{a(b-d)}{b}\right)$. It follows that $E\left(\frac{cd}{b}, 0\right)$. As we have $DE = DG$, we need

$$\frac{cd}{b} - d = \frac{a(b-d)}{b}.$$

Solving this, we obtain $d = \frac{ab}{a+c-b}$. From this, we easily get

$$G\left(\frac{ab}{a+c-b}, \frac{a(c-b)}{a+c-b}\right) \quad \text{and} \quad E\left(\frac{ac}{a+c-b}, 0\right).$$

Note that A_1 is the midpoint of the diagonal EG . So

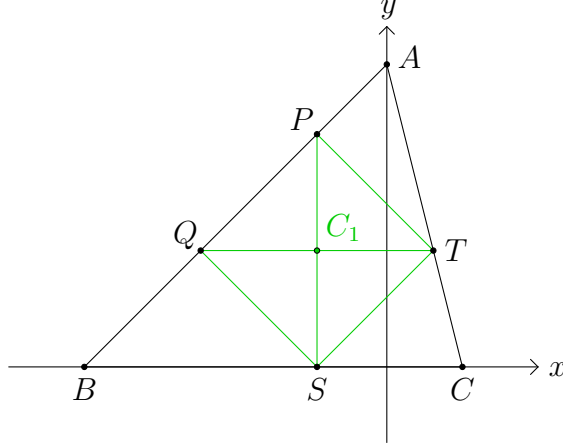
$$A_1\left(\frac{a(b+c)}{2(a+c-b)}, \frac{a(c-b)}{2(a+c-b)}\right).$$

Thus,

$$m_{AA_1} = \frac{a(c-b) - 2a(a+c-b)}{a(b+c)} = \frac{b-c-2a}{b+c}$$

and hence

$$AA_1: \quad y = \frac{b-c-2a}{b+c}x + a. \quad (3)$$



Next, let $PQST$ be the square with centre C_1 such that P, Q lie on AB , S lies on BC and T lies on CA . As P lies on AB , from (1), we may let $P\left(p, \frac{a(b-p)}{b}\right)$. Since

$$m_{PT} = \frac{-1}{m_{AB}} = \frac{b}{a},$$

the equation of PT is

$$PT: \quad y = \frac{b}{a}x - \frac{(a^2 + b^2)p - a^2b}{ab}.$$

Solving this with (2), we obtain

$$x = \frac{a + \frac{(a^2 + b^2)p - a^2b}{ab}}{\frac{b}{a} + \frac{a}{c}} = \frac{c(a^2 + b^2)p}{b(a^2 + bc)}$$

and

$$y = -\frac{a}{c} \cdot \frac{c(a^2 + b^2)p}{b(a^2 + bc)} + a = \frac{a(-(a^2 + b^2)p + b(a^2 + bc))}{b(a^2 + bc)}.$$

This means $T\left(\frac{c(a^2 + b^2)p}{b(a^2 + bc)}, \frac{a(-(a^2 + b^2)p + b(a^2 + bc))}{b(a^2 + bc)}\right)$.

From $m_{ST} = m_{AB} = -\frac{a}{b}$, the equation of ST is

$$y = -\frac{a}{b}x + \frac{a}{b} \cdot \frac{c(a^2 + b^2)p}{b(a^2 + bc)} + \frac{a(-(a^2 + b^2)p + b(a^2 + bc))}{b(a^2 + bc)}.$$

After simplification, this becomes

$$ST: \quad y = -\frac{a}{b}x + \frac{a(c-b)(a^2 + b^2)p + ab^2(a^2 + bc)}{b^2(a^2 + bc)}.$$

Putting $y = 0$, we deduce $S\left(\frac{(c-b)(a^2+b^2)p+b^2(a^2+bc)}{b(a^2+bc)}, 0\right)$.

From $m_{QS} = m_{PT} = \frac{b}{a}$, we get

$$QS: \quad y = \frac{b}{a}x - \frac{(c-b)(a^2+b^2)p+b^2(a^2+bc)}{a(a^2+bc)}.$$

We solve this with (1). This yields

$$x = \frac{a + \frac{(c-b)(a^2+b^2)p+b^2(a^2+bc)}{a(a^2+bc)}}{\frac{b}{a} + \frac{a}{b}} = \frac{b((c-b)p + (a^2+bc))}{a^2+bc}$$

and

$$y = -\frac{a}{b} \cdot \frac{b((c-b)p + (a^2+bc))}{a^2+bc} + a = -\frac{a(c-b)p}{a^2+bc}.$$

This means $Q\left(\frac{b((c-b)p + (a^2+bc))}{a^2+bc}, -\frac{a(c-b)p}{a^2+bc}\right)$.

We now use the constraint $PQ = QS$. We have

$$\begin{aligned} PQ^2 &= \left(p - \frac{b((c-b)p + (a^2+bc))}{a^2+bc}\right)^2 + \left(\frac{a(b-p)}{b} + \frac{a(c-b)p}{a^2+bc}\right)^2 \\ &= \left(\frac{(a^2+b^2)p - b(a^2+bc)}{a^2+bc}\right)^2 + \left(\frac{-a((a^2+b^2)p - b(a^2+bc))}{b(a^2+bc)}\right)^2 \\ &= \left(\frac{(a^2+b^2)p - b(a^2+bc)}{b(a^2+bc)}\right)^2 (a^2+b^2). \end{aligned}$$

Since the y -coordinate of T is $\frac{a(-(a^2+b^2)p + b(a^2+bc))}{b(a^2+bc)}$, which is positive, we know that

$$PQ = -\frac{(a^2+b^2)p - b(a^2+bc)}{b(a^2+bc)}\sqrt{a^2+b^2}. \quad (4)$$

We also have

$$\begin{aligned} QS^2 &= \left(\frac{b((c-b)p + (a^2+bc))}{a^2+bc} - \frac{(c-b)(a^2+b^2)p+b^2(a^2+bc)}{b(a^2+bc)}\right)^2 + \left(\frac{a(c-b)p}{a^2+bc}\right)^2 \\ &= \left(\frac{-a^2(c-b)p}{b(a^2+bc)}\right)^2 + \left(\frac{a(c-b)p}{a^2+bc}\right)^2 \\ &= \left(\frac{a(c-b)p}{b(a^2+bc)}\right)^2 (a^2+b^2). \end{aligned}$$

Since the y -coordinate of Q is $-\frac{a(c-b)p}{a^2+bc}$, which is positive, we know that

$$QS = \frac{a(c-b)p}{b(a^2+bc)}\sqrt{a^2+b^2}. \quad (5)$$

Equating (4) and (5), we need

$$-(a^2+b^2)p + b(a^2+bc) = a(c-b)p.$$

This yields $p = \frac{b(a^2+bc)}{a^2+b^2-ab+ac}$. Thus, we deduce

$$P\left(\frac{b(a^2+bc)}{a^2+b^2-ab+ac}, \frac{a(b-a)(b-c)}{a^2+b^2-ab+ac}\right) \quad \text{and} \quad S\left(\frac{a^2c+b^2c+abc-ab^2}{a^2+b^2-ab+ac}, 0\right).$$

Again, C_1 is the midpoint of PS . So its coordinates are

$$\left(\frac{a^2b+a^2c+abc+2b^2c-ab^2}{2(a^2+b^2-ab+ac)}, \frac{a(b-a)(b-c)}{2(a^2+b^2-ab+ac)}\right).$$

It follows that

$$m_{CC_1} = \frac{a(b-a)(b-c)}{(a^2b+a^2c+abc+2b^2c-ab^2) - 2c(a^2+b^2-ab+ac)} = \frac{b-a}{a-b+2c}.$$

Therefore, we have

$$CC_1: \quad y = \frac{b-a}{a-b+2c}x - \frac{c(b-a)}{a-b+2c}. \quad (6)$$

The equation of BB_1 is not obtained by swapping b and c in (6), since the orientation of the squares are reversed. Indeed, we should do a reflection in the y -axis, meaning to replace x by $-x$, and then swap b and $-c$. After these replacements, we obtain the equation

$$BB_1: \quad y = \frac{c+a}{a+c-2b}x - \frac{b(c+a)}{a+c-2b}. \quad (7)$$

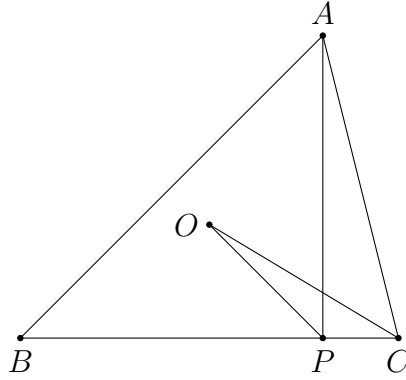
The equations (3), (6), (7) can be rewritten as

$$\begin{aligned} (b-c-2a)x - (b+c)y &= -a(b+c), \\ (b-a)x - (a-b+2c)y &= c(b-a), \\ (c+a)x - (a+c-2b)y &= b(c+a). \end{aligned}$$

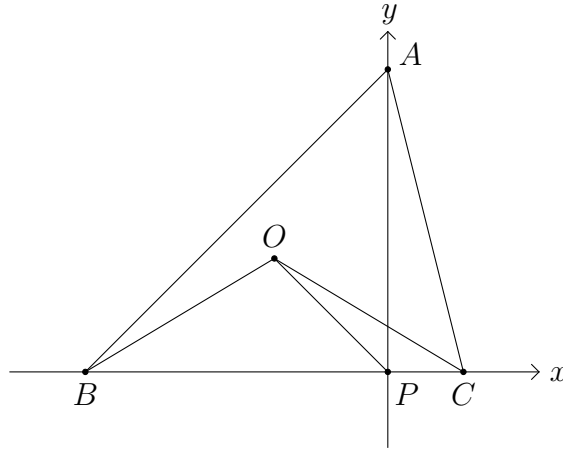
Since the second equation minus the third equation gives the first equation, the three straight lines are concurrent as desired (it is obvious that they are not parallel). $\square$

Problem 7. (IMO Shortlist 2001 G2/Problem 1) Let ABC be an acute-angled triangle with circumcentre O . Let P on BC be the foot of the altitude from A . Suppose that $\angle BCA \geq \angle ABC + 30^\circ$.

Prove that $\angle CAB + \angle COP < 90^\circ$.



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$, $P(0, 0)$ where $b < 0 < a, c$. Then $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$. Note that $\angle OCP = 90^\circ - \angle BAC$. Therefore,

$$\begin{aligned}
 & \angle CAB + \angle COP < 90^\circ \\
 \Leftrightarrow & \angle COP < \angle OCP \\
 \Leftrightarrow & CP < OP \\
 \Leftrightarrow & c^2 < \left(\frac{b+c}{2}\right)^2 + \left(\frac{a^2+bc}{2a}\right)^2 \\
 \Leftrightarrow & a^4 + (b^2 + 4bc - 3c^2)a^2 + b^2c^2 > 0. \tag{1}
 \end{aligned}$$

We use the condition $\angle BCA - \angle ABC \geq 30^\circ$. We have

$$\begin{aligned}
 & \angle BCA - \angle ABC \geq 30^\circ \\
 \Rightarrow & \tan(\angle BCA - \angle ABC) \geq \tan 30^\circ \\
 \Rightarrow & \frac{\tan \angle BCA - \tan \angle ABC}{1 + \tan \angle BCA \tan \angle ABC} \geq \frac{1}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{\frac{a}{c} - \left(-\frac{a}{b}\right)}{1 - \frac{a^2}{bc}} \geq \frac{1}{\sqrt{3}} \\
&\Rightarrow \frac{a(b+c)}{bc-a^2} \geq \frac{1}{\sqrt{3}}.
\end{aligned}$$

Since $\angle BCA > \angle ABC$, we have $|b| > |c|$. This implies $b+c < 0$. Therefore, from the above inequality, we know that $bc - a^2 < 0$. It follows that

$$\sqrt{3}a(b+c) \leq bc - a^2,$$

i.e.

$$a^2 + \sqrt{3}(b+c)a - bc \leq 0.$$

This implies the discriminant of the quadratic polynomial in a on the left is nonnegative. In other words, we have

$$\Delta = \left(\sqrt{3}(b+c)\right)^2 + 4bc = 3b^2 + 10bc + 3c^2 \geq 0.$$

Since $3b^2 + 10bc + 3c^2 = (b+3c)(3b+c)$ and $3b+c = 2b+(b+c) < 0$, this yields $b+3c \leq 0$.

Recall that we need to prove (1). This is obvious if $b^2 + 4bc - 3c^2 \geq 0$. Thus, it suffices to consider $b^2 + 4bc - 3c^2 < 0$. In that case, we have

$$\begin{aligned}
&a^4 + (b^2 + 4bc - 3c^2)a^2 + b^2c^2 \\
&= \left(a^2 + \frac{b^2 + 4bc - 3c^2}{2}\right)^2 - \frac{b^4 + 8b^3c + 6b^2c^2 - 24bc^3 + 9c^4}{4} \\
&\geq -\frac{b^4 + 8b^3c + 6b^2c^2 - 24bc^3 + 9c^4}{4} \\
&= -\frac{1}{4}(b-c)(b+3c)(b^2 + 6bc - 3c^2).
\end{aligned}$$

We have $b-c < 0$ and $b+3c \leq 0$ from above. Also, we have

$$b^2 + 6bc - 3c^2 = (b^2 + 4bc - 3c^2) + 2bc < 0 + 0 = 0$$

from the assumption. These imply

$$a^4 + (b^2 + 4bc - 3c^2)a^2 + b^2c^2 \geq 0.$$

Equality holds when $b = -3c$ and $a = \sqrt{3}c$. In that case, we check that

$$AB^2 + AC^2 = 12c^2 + 4c^2 = 16c^2 = BC^2,$$

contradicting $\triangle ABC$ being an acute triangle. Thus, the inequality is strict as desired. $\square$

Problem 8. (IMO Shortlist 2001 G3) Let ABC be a triangle with centroid G . Determine, with proof, the position of the point P in the plane of ABC such that $AP \cdot AG + BP \cdot BG + CP \cdot CG$ is a minimum, and express this minimum value in terms of the side lengths of ABC .

Solution. Let $A(a, b)$, $B(c, d)$, $C(e, f)$ so that $G\left(\frac{a+c+e}{3}, \frac{b+d+f}{3}\right)$. For any $P(x, y)$, we have

$$\begin{aligned}
& AP \cdot AG + BP \cdot BG + CP \cdot CG \\
&= \sum_{\text{cyc}} \sqrt{(x-a)^2 + (y-b)^2} \cdot \sqrt{\left(\frac{c+e-2a}{3}\right)^2 + \left(\frac{d+f-2b}{3}\right)^2} \\
&\geq \sum_{\text{cyc}} \left((x-a) \left(\frac{c+e-2a}{3}\right) + (y-b) \left(\frac{d+f-2b}{3}\right) \right) \\
&= \frac{(c+e-2a) + (e+a-2c) + (a+c-2e)}{3} x \\
&\quad + \frac{(d+f-2b) + (f+b-2d) + (b+d-2f)}{3} y \\
&\quad - \frac{a(c+e-2a) + c(e+a-2c) + e(a+c-2e)}{3} \\
&\quad - \frac{b(d+f-2b) + d(f+b-2d) + f(b+d-2f)}{3} \\
&= \frac{2}{3} (a^2 + c^2 + e^2 - 2ac - 2ce - 2ea + b^2 + d^2 + f^2 - 2bd - 2df - 2fb), \tag{1}
\end{aligned}$$

where the Cauchy-Schwarz inequality is used. Note that the last expression is independent of P , and hence should be the minimum if equality can hold. Indeed, equality holds if and only if (iff)

$$\frac{y-b}{x-a} = \frac{\frac{d+f-2b}{3}}{\frac{c+e-2a}{3}} \tag{2}$$

together with two other similar equations obtained by changing the variables cyclically. Equation (2) can be rewritten as

$$\frac{y-b}{x-a} = \frac{\frac{b+d+f}{3} - b}{\frac{a+c+e}{3} - a}, \tag{3}$$

which holds iff $m_{AP} = m_{AG}$, i.e. P lies on AG . Similarly, equality holds iff P lies on all of AG, BG, CG . This means the minimum value is attained when $P = G$. (If one cannot spot the identity (3), one may regard (2) as the equation of a straight line and find the concurrent point of it and the other two lines.)

Now we have to express the minimum value (1) in terms of the side lengths, which are

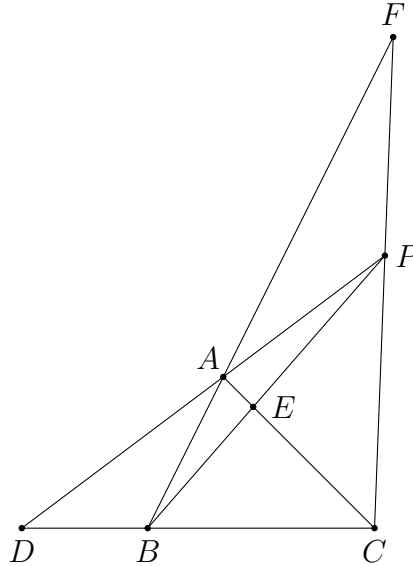
$$\begin{aligned} AB &= \sqrt{(a-c)^2 + (b-d)^2}, \\ BC &= \sqrt{(c-e)^2 + (d-f)^2}, \\ CA &= \sqrt{(e-a)^2 + (f-b)^2}. \end{aligned}$$

Indeed, (1) can be rewritten as

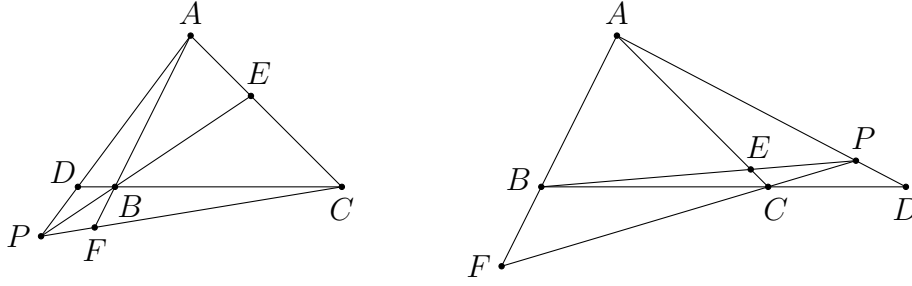
$$\begin{aligned} & \frac{1}{3}((a-c)^2 + (c-e)^2 + (e-a)^2 + (b-d)^2 + (d-f)^2 + (f-b)^2) \\ &= \frac{1}{3}(AB^2 + BC^2 + CA^2). \end{aligned}$$

□

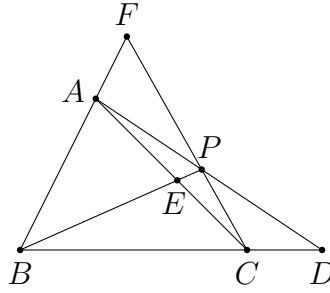
Problem 9. (IMO Shortlist 2001 G6) Let ABC be a triangle and P an exterior point in the plane of the triangle. Suppose the lines AP, BP, CP meet the sides BC, CA, AB (or extensions thereof) in D, E, F , respectively. Suppose further that the areas of triangles PBD, PCE, PAF are all equal. Prove that each of these areas is equal to the area of triangle ABC itself.



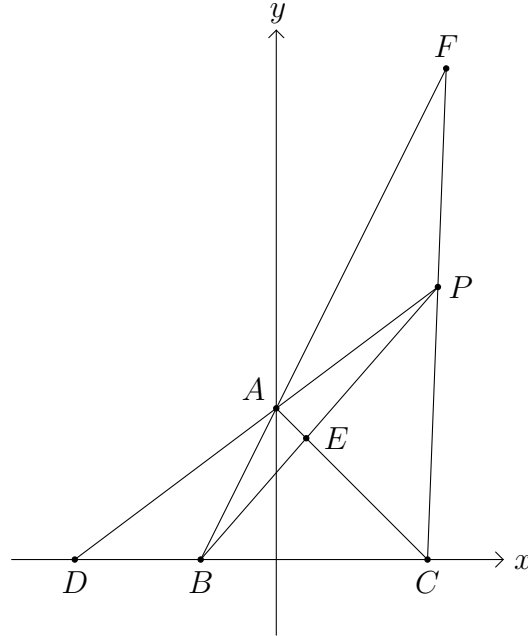
Solution. We first handle the configuration issue using some simple geometric arguments. Without loss of generality (WLOG), assume A and C lie on different sides of the line BP . If P and B lie on the same side of AC , then $\triangle PBD$ lies strictly inside $\triangle PAF$, which is impossible. Thus, P and B lie on different sides of AC . If B lies between A and F , then $\triangle PCE$ lies strictly inside $\triangle PAF$, which is impossible. Thus, A lies between B and F .



If C lies between B and D , then $\triangle PCE$ lies strictly inside $\triangle PBD$, which is impossible. Thus, B lies between C and D . It follows that the configuration must be the one shown in the question.



Now, we use coordinate geometry.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$ and $P(s, t)$. We have

$$PA: \quad y = \frac{t-a}{s}x + a.$$

Putting $y = 0$, we obtain $D\left(\frac{as}{a-t}, 0\right)$. Solving

$$\begin{aligned} PB: \quad y &= \frac{t}{s-b}x - \frac{bt}{s-b}, \\ AC: \quad y &= -\frac{a}{c}x + a, \end{aligned}$$

we deduce $E\left(\frac{c(as+bt-ab)}{as+ct-ab}, \frac{a(c-b)t}{as+ct-ab}\right)$. By symmetry, we swap b and c to deduce $F\left(\frac{b(as+ct-ac)}{as+bt-ac}, \frac{a(b-c)t}{as+bt-ac}\right)$. Firstly, using BD as the base, we have

$$[PBD] = \frac{1}{2} \left(b - \frac{as}{a-t} \right) t = \frac{(-as-bt+ab)t}{2(a-t)}. \quad (1)$$

Secondly, by the shoelace formula and the configuration, we have

$$\begin{aligned} [PCE] &= \frac{1}{2} (x_P y_E + x_E y_C + x_C y_P - x_E y_P - x_C y_E - x_P y_C) \\ &= \frac{1}{2} \left(-\frac{c(as+bt-ab)}{as+ct-ab}t + \frac{a(c-b)t}{as+ct-ab}(s-c) + ct \right) \\ &= \frac{c(c-b)t^2 + a(c-b)(s-c)t}{2(as+ct-ab)} \\ &= \frac{(c-b)(as+ct-ac)t}{2(as+ct-ab)}. \end{aligned} \quad (2)$$

Similarly, we have

$$\begin{aligned} [PAF] &= \frac{1}{2} (x_P y_F + x_F y_A + x_A y_P - x_F y_P - x_A y_F - x_P y_A) \\ &= \frac{1}{2} \left(\frac{b(as+ct-ac)}{as+bt-ac}(a-t) + \frac{a(b-c)t}{as+bt-ac}s - as \right) \\ &= \frac{b(as+ct-ac)(a-t) + a(-as-ct+ac)s}{2(as+bt-ac)} \\ &= \frac{(as+ct-ac)(-as-bt+ab)}{2(as+bt-ac)}. \end{aligned} \quad (3)$$

Equating (1) and (2), we get

$$(-as-bt+ab)(as+ct-ab) = (c-b)(as+ct-ac)(a-t).$$

After expansion, this gives

$$a^2 s^2 + 2abst + c(2b-c)t^2 - a^2(3b-c)s - a(b^2 + 3bc - 2c^2)t + a^2(b^2 + bc - c^2) = 0. \quad (4)$$

Equating (1) and (3), we get

$$(as+bt-ac)t = (as+ct-ac)(a-t).$$

After expansion, this gives

$$2ast + (b+c)t^2 - a^2s - 3act + a^2c = 0,$$

which means

$$s = \frac{-(b+c)t^2 + 3act - a^2c}{a(2t-a)}. \quad (5)$$

Putting this into (4) and multiplying both sides by $(2t-a)^2$, we have

$$\begin{aligned} 0 &= \left(-(b+c)t^2 + 3act - a^2c \right)^2 + 2b(2t-a) \left(-(b+c)t^2 + 3act - a^2c \right)t \\ &\quad + c(2b-c)(2t-a)^2t^2 - a(3b-c)(2t-a) \left(-(b+c)t^2 + 3act - a^2c \right) \\ &\quad - a(b^2 + 3bc - 2c^2)(2t-a)^2t + a^2(b^2 + bc - c^2)(2t-a)^2 \\ &= -3(b-c)^2t^4 + 4a(b-c)^2t^3 + 5a^2(b-c)^2t^2 - 5a^3(b-c)^2t + (b-c)^2a^4 \\ &= -(b-c)^2(3t-a)(t^3 - at^2 - 2a^2t + a^3). \end{aligned}$$

Note that $b-c \neq 0$ and $3t-a \neq 0$ (because $t > a$ from the configuration). This leads to

$$t^3 - at^2 - 2a^2t + a^3 = 0. \quad (6)$$

Next, we easily find that

$$[ABC] = \frac{a(c-b)}{2}.$$

Comparing this with (2), our goal is to prove

$$(as + ct - ac)t = a(as + ct - ab),$$

i.e. $ast + ct^2 - a^2s - 2act + a^2b = 0$, or

$$s = \frac{-ct^2 + 2act - a^2b}{a(t-a)}.$$

Comparing with (5), we need to prove

$$\left(-(b+c)t^2 + 3act - a^2c \right)(t-a) = (-ct^2 + 2act - a^2b)(2t-a).$$

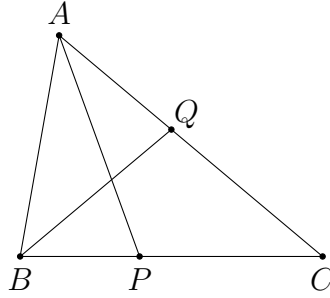
After expansion, we see that this is the same as

$$(b-c)t^3 - a(b-c)t^2 - 2a^2(b-c)t + a^3(b-c) = 0.$$

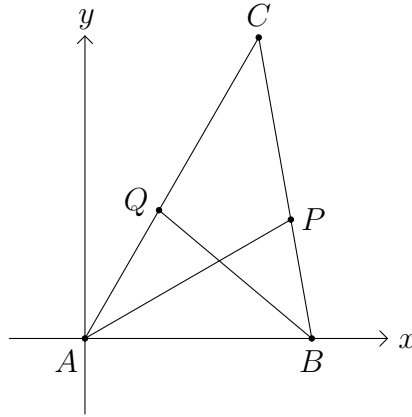
This follows from (6), and so the proof is complete. $\square$

Problem 10. (IMO Shortlist 2001 G8/Problem 5) In a triangle ABC , let AP bisect $\angle BAC$, with P on BC , and let BQ bisect $\angle ABC$, with Q on CA . It is known that $\angle BAC = 60^\circ$ and that $AB + BP = AQ + QB$.

What are the possible angles of triangle ABC ?



Solution.



Let $A(0,0)$ and $B(1,0)$. Since $\angle BAP = 30^\circ$ and $\angle BAC = 60^\circ$, we have $m_{AP} = \frac{1}{\sqrt{3}}$ and $m_{AC} = \sqrt{3}$. So we have

$$AP: \quad y = \frac{1}{\sqrt{3}}x \quad (1)$$

and

$$AQ: \quad y = \sqrt{3}x. \quad (2)$$

Let $\angle QBA = \angle CBQ = \theta$, and let $m = \tan \theta$. Then we have

$$BQ: \quad y = -mx + m. \quad (3)$$

Note that $m_{BC} = \frac{2m_{BQ}}{1 - m_{BQ}^2} = -\frac{2m}{1 - m^2}$. So

$$BP: \quad y = -\frac{2m}{1 - m^2}x + \frac{2m}{1 - m^2}. \quad (4)$$

We need to find P and Q (while C is not needed). From (1) and (4), we get

$$x = \frac{2\sqrt{3}m}{(1 - m^2) + 2\sqrt{3}m}$$

and

$$y = \frac{1}{\sqrt{3}}x = \frac{2m}{(1-m^2) + 2\sqrt{3}m}.$$

This means $P\left(\frac{2\sqrt{3}m}{(1-m^2) + 2\sqrt{3}m}, \frac{2m}{(1-m^2) + 2\sqrt{3}m}\right)$. Similarly, from (2) and (3), we get

$$x = \frac{m}{m + \sqrt{3}}$$

and

$$y = \sqrt{3}x = \frac{\sqrt{3}m}{m + \sqrt{3}}.$$

This means $Q\left(\frac{m}{m + \sqrt{3}}, \frac{\sqrt{3}m}{m + \sqrt{3}}\right)$. We now compute the lengths one by one. Clearly, $AB = 1$. Next, we have

$$\begin{aligned} BP^2 &= \left(\frac{2\sqrt{3}m}{(1-m^2) + 2\sqrt{3}m} - 1\right)^2 + \left(\frac{2m}{(1-m^2) + 2\sqrt{3}m}\right)^2 \\ &= \left(\frac{-(1-m^2)}{(1-m^2) + 2\sqrt{3}m}\right)^2 + \left(\frac{2m}{(1-m^2) + 2\sqrt{3}m}\right)^2 \\ &= \frac{(m^2 + 1)^2}{((1-m^2) + 2\sqrt{3}m)^2}. \end{aligned}$$

Since $2\theta < 120^\circ$, the angle θ is acute. So we know that $m > 0$. Also, note that the y -coordinate $\frac{2m}{(1-m^2) + 2\sqrt{3}m}$ of P is positive. Therefore, we deduce

$$BP = \frac{m^2 + 1}{(1-m^2) + 2\sqrt{3}m}. \quad (5)$$

Similarly, we have

$$AQ^2 = \left(\frac{m}{m + \sqrt{3}}\right)^2 + \left(\frac{\sqrt{3}m}{m + \sqrt{3}}\right)^2 = \frac{4m^2}{(m + \sqrt{3})^2}.$$

As the x -coordinate $\frac{m}{m + \sqrt{3}}$ of Q is positive, we have

$$AQ = \frac{2m}{m + \sqrt{3}}. \quad (6)$$

For QB , we have

$$\begin{aligned} QB^2 &= \left(\frac{m}{m + \sqrt{3}} - 1\right)^2 + \left(\frac{\sqrt{3}m}{m + \sqrt{3}}\right)^2 \\ &= \left(\frac{-\sqrt{3}}{m + \sqrt{3}}\right)^2 + \left(\frac{\sqrt{3}m}{m + \sqrt{3}}\right)^2 = \frac{3(m^2 + 1)}{(m + \sqrt{3})^2}. \end{aligned}$$

This implies

$$QB = \frac{\sqrt{3(m^2 + 1)}}{m + \sqrt{3}}. \quad (7)$$

(One can also deduce these using trigonometry.) We combine the results (5), (6), (7) to get

$$1 + \frac{m^2 + 1}{(1 - m^2) + 2\sqrt{3}m} = \frac{2m}{m + \sqrt{3}} + \frac{\sqrt{3(m^2 + 1)}}{m + \sqrt{3}}. \quad (8)$$

Now,

$$\begin{aligned} 1 + \frac{m^2 + 1}{(1 - m^2) + 2\sqrt{3}m} &= \frac{2m}{m + \sqrt{3}} + \frac{\sqrt{3(m^2 + 1)}}{m + \sqrt{3}} \\ \Rightarrow (2 + 2\sqrt{3}m)(m + \sqrt{3}) &= (2m + \sqrt{3(m^2 + 1)})(1 - m^2 + 2\sqrt{3}m) \\ \Rightarrow 2m^3 - 2\sqrt{3}m^2 + 6m + 2\sqrt{3} &= (1 - m^2 + 2\sqrt{3}m)\sqrt{3(m^2 + 1)} \\ \Rightarrow 4m^6 - 8\sqrt{3}m^5 + 36m^4 - 16\sqrt{3}m^3 + 12m^2 + 24\sqrt{3}m + 12 \\ &= 3m^6 - 12\sqrt{3}m^5 + 33m^4 + 33m^2 + 12\sqrt{3}m + 3 \\ \Rightarrow m^6 + 4\sqrt{3}m^5 + 3m^4 - 16\sqrt{3}m^3 - 21m^2 + 12\sqrt{3}m + 9 &= 0. \end{aligned}$$

Since the radicals only occur in the coefficients of odd degree terms, we look for roots which are rational multiples of $\sqrt{3}$. Two of the roots are $m = \pm\sqrt{3}$. So we get the factorization

$$(m^2 - 3)(m^4 + 4\sqrt{3}m^3 + 6m^2 - 4\sqrt{3}m - 3) = 0.$$

Indeed, the root $m = -\sqrt{3}$ is a double root. This implies

$$(m - \sqrt{3})(m + \sqrt{3})^2(m^3 + 3\sqrt{3}m^2 - 3m - \sqrt{3}) = 0.$$

However, both $m = \pm\sqrt{3}$ should be rejected as $0 < m = \tan \theta < \tan 60^\circ = \sqrt{3}$. So we have

$$m^3 + 3\sqrt{3}m^2 - 3m - \sqrt{3} = 0. \quad (9)$$

A cubic equation of $\tan \theta$ is closely related to the triple angle $\tan 3\theta$. In fact, we have

$$\tan 3\theta = \frac{m + \tan 2\theta}{1 - m \tan 2\theta} = \frac{m + \frac{2m}{1 - m^2}}{1 - m \cdot \frac{2m}{1 - m^2}} = \frac{3m - m^3}{1 - 3m^2}.$$

From (9), we have

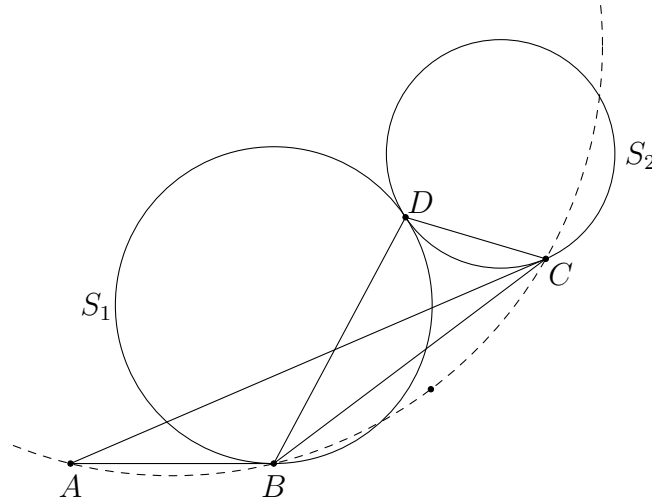
$$\frac{3m - m^3}{1 - 3m^2} = -\sqrt{3}.$$

Therefore, we obtain $\tan 3\theta = -\sqrt{3} = \tan 120^\circ$. Thus, $3\theta = 120^\circ, 300^\circ, 480^\circ$, which gives $\theta = 40^\circ, 100^\circ, 160^\circ$. From $0 < \theta < 60^\circ$, the only possible value is $\theta = 40^\circ$.

We have to show that $m = \tan 40^\circ$ does satisfy (8). Instead of checking whether the above steps are reversible, we use an indirect approach. Firstly, when $\theta = 0$, i.e.

$m = 0$, the left-hand side of (8) is 2 while the right-hand side is 1. Secondly, when $m = \frac{\sqrt{3}}{2}$ (we need a value less than $\sqrt{3}$, and it is simpler to have a rational multiple of $\sqrt{3}$), the left-hand side is $\frac{20}{13}$ while the right-hand side is $\frac{2 + \sqrt{7}}{3} > \frac{20}{13}$. By the intermediate value theorem, there must be an angle θ strictly between 0° and 60° such that $m = \tan \theta$ satisfies (8). This must be $\theta = 40^\circ$ from above. So this gives the only possible $\triangle ABC$ where $\angle B = 2\theta = 80^\circ$ and $\angle C = 40^\circ$. $\square$

Problem 11. (IMO Shortlist 2002 G1) Let B be a point on a circle S_1 , and let A be a point distinct from B on the tangent at B to S_1 . Let C be a point not on S_1 such that the line segment AC meets S_1 at two distinct points. Let S_2 be the circle touching AC at C and touching S_1 at a point D on the opposite side of AC from B . Prove that the circumcentre of triangle BCD lies on the circumcircle of triangle ABC .



Solution. Let O_1 and O_2 be the centres of S_1 and S_2 respectively. Note that O_1, D, O_2 are collinear. Let $D(0, 0)$, $O_1(a, 0)$, $O_2(b, 0)$ so that

$$S_1 : x^2 + y^2 - 2ax = 0$$

and

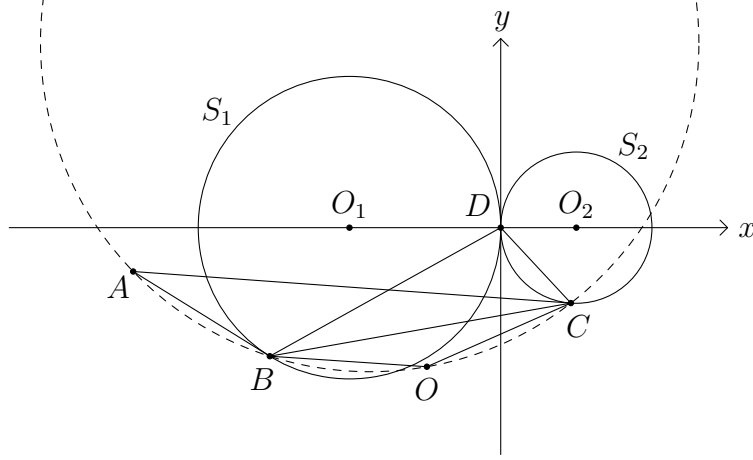
$$S_2 : x^2 + y^2 - 2bx = 0.$$

Let $B(c, d)$, $C(e, f)$ so that

$$c^2 + d^2 - 2ac = 0, \tag{1}$$

$$e^2 + f^2 - 2be = 0. \tag{2}$$

As AB is tangent to S_1 , we have $AB \perp BO_1$. So $m_{AB} = \frac{-1}{m_{BO_1}} = -\frac{c-a}{d}$.



Similarly, we have $m_{AC} = -\frac{e-b}{f}$. Then

$$\tan \angle BAC = \frac{m_{AC} - m_{AB}}{1 + m_{AC}m_{AB}} = \frac{-(e-b)d + (c-a)f}{fd + (e-b)(c-a)}. \quad (3)$$

Next, we easily get

$$\tan \angle BDC = \frac{m_{CD} - m_{BD}}{1 + m_{CD}m_{BD}} = \frac{cf - ed}{ec + fd}.$$

Let O be the circumcentre of $\triangle BCD$. Note that $\angle BOC = 2\angle BDC$ (which always holds using directed angles). So

$$\tan \angle BOC = \frac{2 \tan \angle BDC}{1 - \tan^2 \angle BDC} = \frac{2(cf - ed)(ec + fd)}{(ec + fd)^2 - (cf - ed)^2}. \quad (4)$$

(Alternatively, one can obtain this by finding the coordinates of O and then the slopes of OB and OC .) As

$$\begin{aligned} & A, B, C, O \text{ are concyclic} \\ \Leftrightarrow & \angle BAC = \angle BOC \\ \Leftrightarrow & \tan \angle BAC = \tan \angle BOC, \end{aligned}$$

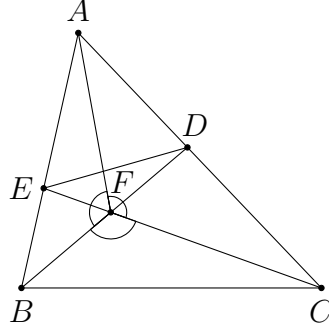
we need to show (3) and (4) are equal. Thus, we use (1) and (2) to remove a and b in (3). Indeed, since $a = \frac{c^2 + d^2}{2c}$ and $b = \frac{e^2 + f^2}{2e}$, we have

$$\tan \angle BAC = \frac{-\frac{e^2 - f^2}{2e} \cdot d + \frac{c^2 - d^2}{2c} \cdot f}{fd + \frac{e^2 - f^2}{2e} \cdot \frac{c^2 - d^2}{2c}} = \frac{2(-e^2cd + f^2cd + c^2ef - d^2ef)}{4cdef + (e^2 - f^2)(c^2 - d^2)}.$$

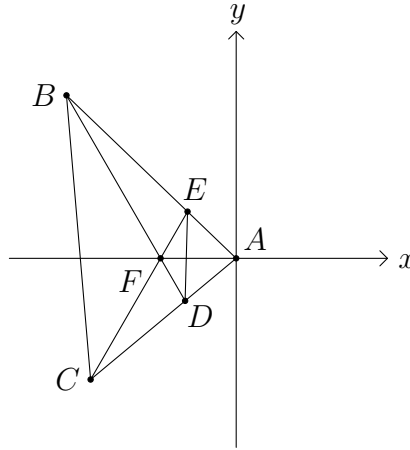
It is easy to check that this is just the expansion of (4). So we are done. $\square$

Problem 12. (IMO Shortlist 2002 G2) Let ABC be a triangle for which there exists an interior point F such that $\angle AFB = \angle BFC = \angle CFA$. Let the lines BF and CF meet the sides AC and AB at D and E respectively. Prove that

$$AB + AC \geq 4DE.$$



Solution.



Let $A(0,0)$ and $F(-1,0)$. As $\angle AFB = \angle AFC = 120^\circ$, we have $m_{BF} = -\sqrt{3}$ and $m_{CF} = \sqrt{3}$. This gives

$$BF: \quad y = -\sqrt{3}x - \sqrt{3}, \quad (1)$$

$$CF: \quad y = \sqrt{3}x + \sqrt{3}. \quad (2)$$

Thus, we let $B(b, -\sqrt{3}(b+1))$ and $C(c, \sqrt{3}(c+1))$ where $b, c \leq -1$. We deduce

$$AB: \quad y = -\frac{\sqrt{3}(b+1)}{b}x, \quad (3)$$

$$AC: \quad y = \frac{\sqrt{3}(c+1)}{c}x. \quad (4)$$

Solving (1) and (4), we get $D\left(-\frac{c}{2c+1}, -\frac{\sqrt{3}(c+1)}{2c+1}\right)$. Solving (2) and (3), we get $E\left(-\frac{b}{2b+1}, \frac{\sqrt{3}(b+1)}{2b+1}\right)$. Then we have $AB = \sqrt{4b^2 + 6b + 3}$, $AC = \sqrt{4c^2 + 6c + 3}$ and

$$\begin{aligned} DE &= \sqrt{\left(-\frac{b}{2b+1} + \frac{c}{2c+1}\right)^2 + \left(\frac{\sqrt{3}(b+1)}{2b+1} + \frac{\sqrt{3}(c+1)}{2c+1}\right)^2} \\ &= \sqrt{\left(\frac{c-b}{(2b+1)(2c+1)}\right)^2 + 3\left(\frac{4bc+3b+3c+2}{(2b+1)(2c+1)}\right)^2} \\ &= \sqrt{\frac{48b^2c^2 + 72bc(b+c) + 4(7b^2 + 25bc + 7c^2) + 36(b+c) + 12}{(2b+1)^2(2c+1)^2}}. \end{aligned}$$

For simplicity, we write $s = 2b+1 \leq -1$ and $t = 2c+1 \leq -1$. Then $AB = \sqrt{s^2 + s + 1}$, $AC = \sqrt{t^2 + t + 1}$ and

$$\begin{aligned} DE &= \sqrt{\frac{3(s-1)^2(t-1)^2 + 9(s-1)(t-1)(s+t-2) + 7(s-1)^2 + 25(s-1)(t-1) + 7(t-1)^2 + 18(s+t-2) + 12}{s^2t^2}} \\ &= \frac{\sqrt{3s^2t^2 + 3st(s+t) + (s^2 + st + t^2)}}{st}. \end{aligned}$$

Firstly, by the triangle inequality, we have

$$\begin{aligned} AB + AC &= \sqrt{\left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} + \sqrt{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &\geq \sqrt{\left(s + \frac{1}{2} + t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right)^2} \\ &= \sqrt{(s+t+1)^2 + 3}. \end{aligned}$$

Therefore, it suffices to prove

$$s^2t^2((s+t+1)^2 + 3) \geq 16(3s^2t^2 + 3st(s+t) + (s^2 + st + t^2)).$$

Let $\alpha = s+t$ and $\beta = st$. Using $s, t \leq -1$, we easily deduce $\alpha \leq -2$ and $\beta \geq -\alpha - 1$ (which is the same as $(s+1)(t+1) \geq 0$). Also, we have $\beta \leq \frac{\alpha^2}{4}$ since this is the same as $(s-t)^2 \geq 0$. Now,

$$\begin{aligned} &s^2t^2((s+t+1)^2 + 3) \geq 16(3s^2t^2 + 3st(s+t) + (s^2 + st + t^2)) \\ \Leftrightarrow &\beta^2(\alpha^2 + 2\alpha + 4) \geq 16(3\beta^2 + 3\alpha\beta + \alpha^2 - \beta) \\ \Leftrightarrow &(\alpha^2 + 2\alpha - 44)\beta^2 + 16(1 - 3\alpha)\beta - 16\alpha^2 \geq 0. \end{aligned} \tag{5}$$

Case 1. $\alpha^2 + 2\alpha - 44 > 0$

In this case, we must have $\alpha < -6$ as $\alpha \leq -2$. The axis of symmetry of the graph of the quadratic polynomial in (5) is $\frac{8(3\alpha - 1)}{\alpha^2 + 2\alpha - 44}$. We show that this is less than the lower bound $-\alpha - 1$ for β . Indeed,

$$\begin{aligned} & \frac{8(3\alpha - 1)}{\alpha^2 + 2\alpha - 44} < -\alpha - 1 \\ \Leftrightarrow & \alpha^3 + 3\alpha^2 - 18\alpha - 52 < 0 \\ \Leftrightarrow & \alpha^2(\alpha + 6) - 3\alpha(\alpha + 6) - 52 < 0. \end{aligned}$$

This holds since $\alpha < -6$. Thus, to prove (5), it suffices to consider $\beta = -\alpha - 1$. We have

$$\begin{aligned} & (\alpha^2 + 2\alpha - 44)(-\alpha - 1)^2 + 16(1 - 3\alpha)(-\alpha - 1) - 16\alpha^2 \geq 0 \\ \Leftrightarrow & \alpha^4 + 4\alpha^3 - 7\alpha^2 - 54\alpha - 60 \geq 0 \\ \Leftrightarrow & \alpha^3(\alpha + 6) - 2\alpha^2(\alpha + 6) + 5\alpha(\alpha + 6) - 84(\alpha + 1) + 24 \geq 0. \end{aligned}$$

This is true.

Case 2. $\alpha^2 + 2\alpha - 44 = 0$

This means $\alpha = -1 - \sqrt{45}$. Equation (5) is the same as $(1 - 3\alpha)\beta \geq \alpha^2 = -2\alpha + 44$. As it is linear in β and $1 - 3\alpha > 0$, it suffices to check the inequality when $\beta = -\alpha - 1$. Indeed,

$$\begin{aligned} & (1 - 3\alpha)(-\alpha - 1) \geq -2\alpha + 44 \\ \Leftrightarrow & 3\alpha^2 + 4\alpha - 45 \geq 0. \end{aligned}$$

This holds since $3\alpha^2 + 4\alpha - 45 = -2\alpha + 87 \geq 0$.

Case 3. $\alpha^2 + 2\alpha - 44 < 0$

In this case, in order to prove (5), we need to consider the endpoints $\beta = -\alpha - 1$ and $\beta = \frac{\alpha^2}{4}$ respectively. When $\beta = -\alpha - 1$, we have

$$\begin{aligned} & (\alpha^2 + 2\alpha - 44)(-\alpha - 1)^2 + 16(1 - 3\alpha)(-\alpha - 1) - 16\alpha^2 \geq 0 \\ \Leftrightarrow & \alpha^4 + 4\alpha^3 - 7\alpha^2 - 54\alpha - 60 \geq 0 \\ \Leftrightarrow & (\alpha + 2)^2(\alpha^2 - 11) - 10\alpha - 16 \geq 0. \end{aligned}$$

Recall that $\alpha \leq -2$. Therefore, we have $-10\alpha - 16 \geq 4$. It suffices to prove

$$(\alpha + 2)^2(\alpha^2 - 11) \geq -4$$

for $\alpha \leq -2$.

- If $-\sqrt{7} \leq \alpha \leq -2$, then $(\alpha + 2)^2(\alpha^2 - 11) \geq (-\sqrt{7} + 2)^2((-2)^2 - 11) > -4$.
- If $-3 \leq \alpha \leq -\sqrt{7}$, then $(\alpha + 2)^2(\alpha^2 - 11) \geq (-3 + 2)^2((-\sqrt{7})^2 - 11) = -4$.
- If $-\sqrt{11} \leq \alpha \leq -3$, then $(\alpha + 2)^2(\alpha^2 - 11) \geq (-\sqrt{11} + 2)^2((-3)^2 - 11) > -4$.
- If $\alpha \leq -\sqrt{11}$, then $(\alpha + 2)^2(\alpha^2 - 11) \geq 0 > -4$.

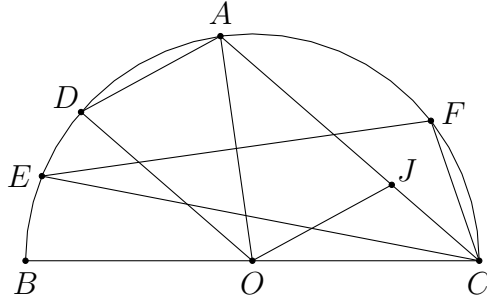
Next, when $\beta = \frac{\alpha^2}{4}$, we have

$$\begin{aligned}
 & (\alpha^2 + 2\alpha - 44) \left(\frac{\alpha^2}{4} \right)^2 + 16(1 - 3\alpha) \left(\frac{\alpha^2}{4} \right) - 16\alpha^2 \geq 0 \\
 \Leftrightarrow & \alpha^6 + 2\alpha^5 - 44\alpha^4 - 192\alpha^3 - 192\alpha^2 \geq 0 \\
 \Leftrightarrow & \alpha^2(\alpha + 4)^2(\alpha^2 - 6\alpha - 12) \geq 0.
 \end{aligned}$$

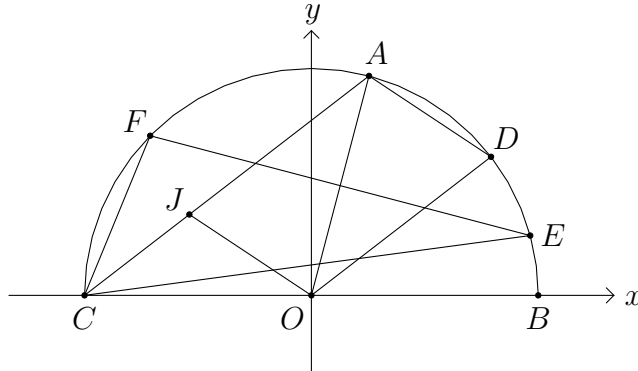
As $\alpha \leq -2$, we have $\alpha^2 - 6\alpha - 12 = \alpha^2 - 6(\alpha + 2) \geq 0$. So the inequality still holds.

Combining all cases, we see (5) always holds. $\square$

Problem 13. (IMO Shortlist 2002 G3/Problem 2) BC is a diameter of a circle with centre O . A is any point on the circle with $\angle AOC > 60^\circ$. EF is the chord which is the perpendicular bisector of AO . D is the midpoint of the minor arc AB . The line through O parallel to AD meets AC at J . Show that J is the incentre of triangle CEF .



Solution.



Let $O(0,0)$, $B(1,0)$, $C(-1,0)$ and the equation of the semicircle be $x^2 + y^2 = 1$. Let $A(a,b)$ so that

$$a^2 + b^2 = 1.$$

Since $m_{AO} = \frac{b}{a}$, we have $m_{EF} = -\frac{a}{b}$ and hence

$$EF: \quad y = -\frac{a}{b}x + \frac{a^2}{2b} + \frac{b}{2}.$$

Using $a^2 + b^2 = 1$, this becomes

$$EF: \quad y = -\frac{a}{b}x + \frac{1}{2b}.$$

Putting this in $x^2 + y^2 = 1$, we solve

$$\left(1 + \frac{a^2}{b^2}\right)x^2 - \frac{a}{b^2}x + \left(\frac{1}{4b^2} - 1\right) = 0.$$

This is the same as

$$x^2 - ax + \left(\frac{1}{4} - b^2\right) = 0.$$

The roots are $x = \frac{a \pm \sqrt{3}b}{2}$, which implies

$$y = -\frac{a}{b} \cdot \frac{a \pm \sqrt{3}b}{2} + \frac{1}{2b} = \frac{(1 - a^2) \mp \sqrt{3}ab}{2b} = \frac{b \mp \sqrt{3}a}{2}.$$

WLOG assume

$$E\left(\frac{a + \sqrt{3}b}{2}, \frac{-\sqrt{3}a + b}{2}\right), \quad F\left(\frac{a - \sqrt{3}b}{2}, \frac{\sqrt{3}a + b}{2}\right).$$

Next, the perpendicular bisector of AB has equation

$$y = -\frac{a-1}{b}x.$$

Putting this in $x^2 + y^2 = 1$, we solve

$$\left(1 + \frac{(a-1)^2}{b^2}\right)x^2 = 1.$$

The larger root is $x = \sqrt{\frac{b^2}{b^2 + (a-1)^2}} = \sqrt{\frac{1+a}{2}}$. This yields $D\left(d, -\frac{(a-1)d}{b}\right)$ where

$$d = \sqrt{\frac{1+a}{2}}.$$

From

$$m_{AD} = \frac{b + \frac{(a-1)d}{b}}{a - d} = \frac{b^2 + (a-1)d}{b(a-d)},$$

We get

$$OJ: \quad y = \frac{b^2 + (a-1)d}{b(a-d)}x. \quad (1)$$

We also have

$$AC: \quad y = \frac{b}{a+1}x + \frac{b}{a+1}. \quad (2)$$

Solving (1) and (2), we get

$$x = \frac{b^2(a-d)}{(b^2 + (a-1)d)(a+1) - b^2(a-d)} = \frac{b^2(a-d)}{(a^2 + b^2 - 1)d + b^2} = a - d$$

since $a^2 + b^2 = 1$. Then we easily get $J\left(a-d, \frac{b^2 + (a-1)d}{b}\right)$.

We now check the results. Firstly,

$$\begin{aligned} & \angle ECJ = \angle JCF \\ \Leftrightarrow & \quad \tan \angle ECJ = \tan \angle JCF \\ \Leftrightarrow & \quad \frac{m_{CJ} - m_{CE}}{1 + m_{CJ}m_{CE}} = \frac{m_{CF} - m_{CJ}}{1 + m_{CF}m_{CJ}} \\ \Leftrightarrow & \quad \frac{\frac{b}{a+1} - \frac{-\sqrt{3}a+b}{a+\sqrt{3}b+2}}{1 + \frac{b}{a+1} \cdot \frac{-\sqrt{3}a+b}{a+\sqrt{3}b+2}} = \frac{\frac{\sqrt{3}a+b}{a-\sqrt{3}b+2} - \frac{b}{a+1}}{1 + \frac{\sqrt{3}a+b}{a-\sqrt{3}b+2} \cdot \frac{b}{a+1}} \\ \Leftrightarrow & \quad \frac{\sqrt{3}a^2 + \sqrt{3}b^2 + \sqrt{3}a+b}{a^2 + b^2 + 3a + \sqrt{3}b+2} = \frac{\sqrt{3}a^2 + \sqrt{3}b^2 + \sqrt{3}a-b}{a^2 + b^2 + 3a - \sqrt{3}b+2} \\ \Leftrightarrow & \quad \frac{\sqrt{3}a+b+\sqrt{3}}{3a+\sqrt{3}b+3} = \frac{\sqrt{3}a-b+\sqrt{3}}{3a-\sqrt{3}b+3} \\ \Leftrightarrow & \quad \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}. \end{aligned}$$

This is true. Secondly,

$$\begin{aligned} & \angle CFJ = \angle JFE \\ \Leftrightarrow & \quad \tan \angle CFJ = \tan \angle JFE \\ \Leftrightarrow & \quad \frac{m_{FJ} - m_{CF}}{1 + m_{FJ}m_{CF}} = \frac{m_{EF} - m_{FJ}}{1 + m_{EF}m_{FJ}} \\ \Leftrightarrow & \quad \frac{\frac{\sqrt{3}ab - b^2 - 2(a-1)d}{b(-a-\sqrt{3}b+2d)} - \frac{\sqrt{3}a+b}{a-\sqrt{3}b+2}}{1 + \frac{\sqrt{3}ab - b^2 - 2(a-1)d}{b(-a-\sqrt{3}b+2d)} \cdot \frac{\sqrt{3}a+b}{a-\sqrt{3}b+2}} = \frac{\frac{a}{b} - \frac{\sqrt{3}ab - b^2 - 2(a-1)d}{b(-a-\sqrt{3}b+2d)}}{1 - \frac{a}{b} \cdot \frac{\sqrt{3}ab - b^2 - 2(a-1)d}{b(-a-\sqrt{3}b+2d)}}. \end{aligned}$$

This is the same as

$$\begin{aligned}
& \frac{2b(\sqrt{3}a^2 + \sqrt{3}b^2 + \sqrt{3}a - b) - 2(a^2 + b^2 + a + \sqrt{3}b - 2)d}{2b(a^2 + b^2 - a - \sqrt{3}b) + 2(-\sqrt{3}a^2 - \sqrt{3}b^2 + \sqrt{3}a + 3b)d} \\
&= \frac{b(a^2 + b^2) - 2bd}{-\sqrt{3}b(a^2 + b^2) + 2(a^2 + b^2 - a)d},
\end{aligned}$$

i.e.

$$\begin{aligned}
& (b(\sqrt{3}a - b + \sqrt{3}) - (a + \sqrt{3}b - 1)d)(-\sqrt{3}b + 2(-a + 1)d) \\
&= (b - 2bd)(b(-a - \sqrt{3}b + 1) + (\sqrt{3}a + 3b - \sqrt{3}))d.
\end{aligned}$$

After simplification, this becomes

$$b^2(-a + \sqrt{3}b - 2) + 2(\sqrt{3}ab + b^2 - a - \sqrt{3}b + 1)d^2 = 0.$$

Recall that d is defined by $d = \sqrt{\frac{1+a}{2}}$. So we need to check

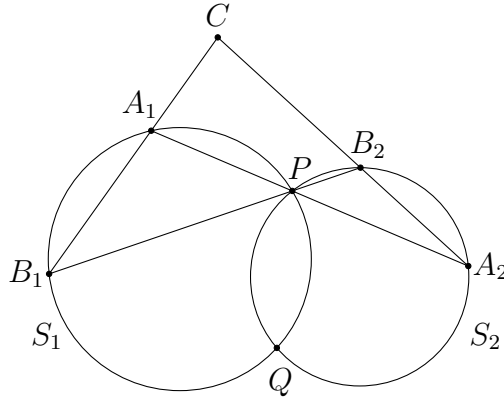
$$b^2(-a + \sqrt{3}b - 2) + (\sqrt{3}ab + b^2 - a - \sqrt{3}b + 1)(1 + a) = 0.$$

Indeed,

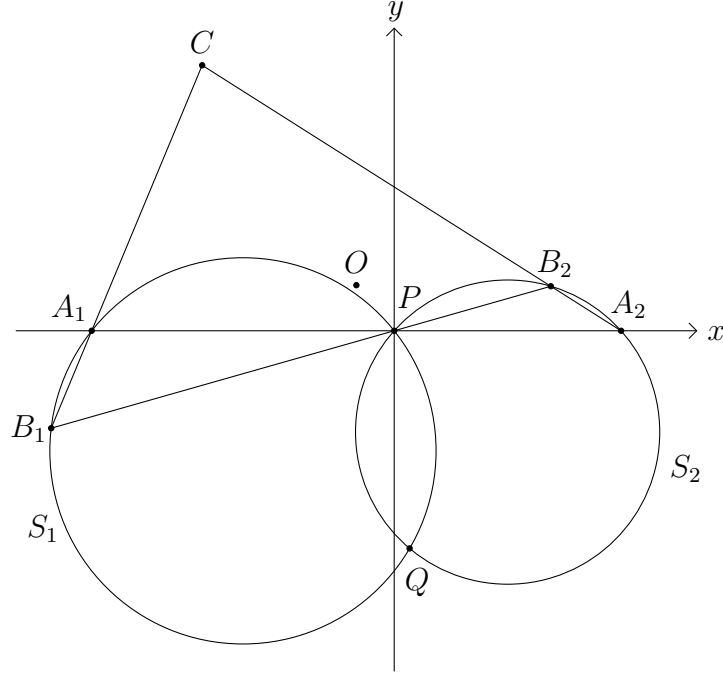
$$\begin{aligned}
& b^2(-a + \sqrt{3}b - 2) + (\sqrt{3}ab + b^2 - a - \sqrt{3}b + 1)(1 + a) \\
&= \sqrt{3}b(a^2 + b^2 - 1) - a^2 - b^2 + 1 = 0.
\end{aligned}$$

So both CJ and FJ are angle bisectors of $\triangle CEF$. From the configuration, J must be the incentre of $\triangle CEF$. $\square$

Problem 14. (IMO Shortlist 2002 G4) Circles S_1 and S_2 intersect at points P and Q . Distinct points A_1 and B_1 (not at P or Q) are selected on S_1 . The lines A_1P and B_1P meet S_2 again at A_2 and B_2 respectively, and the lines A_1B_1 and A_2B_2 meet at C . Prove that, as A_1 and B_1 vary, the circumcentres of triangles A_1A_2C all lie on one fixed circle.



Solution.



Let $P(0,0)$, $A_1(a,0)$, $A_2(b,0)$, and let

$$\begin{aligned} S_1: \quad x^2 + y^2 - ax + cy &= 0, \\ S_2: \quad x^2 + y^2 - bx + dy &= 0. \end{aligned} \tag{1}$$

Let $B_1(s, t)$ be an arbitrary point such that

$$s^2 + t^2 - as + ct = 0. \tag{2}$$

We first show that when B_1 varies, the circumcentre O of $\triangle A_1A_2C$ is fixed. Putting

$$B_1P: \quad y = \frac{t}{s}x$$

into (1), we solve

$$\left(1 + \frac{t^2}{s^2}\right)x^2 + \left(\frac{dt}{s} - b\right)x = 0.$$

The solution $x = 0$ corresponds to the point P (rejected), and the other solution is $x = \frac{s(bs - dt)}{s^2 + t^2}$. This easily yields $B_2\left(\frac{s(bs - dt)}{s^2 + t^2}, \frac{t(bs - dt)}{s^2 + t^2}\right)$. Then we obtain

$$\begin{aligned} A_1B_1: \quad y &= \frac{t}{s-a}x - \frac{at}{s-a}, \\ A_2B_2: \quad y &= -\frac{bs - dt}{bt + ds}x + \frac{b(bs - dt)}{bt + ds}. \end{aligned}$$

We solve these equations and obtain

$$x = \frac{\frac{at}{s-a} + \frac{b(bs-dt)}{bt+ds}}{\frac{t}{s-a} + \frac{bs-dt}{bt+ds}} = \frac{at(bt+ds) + b(s-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)}$$

and

$$y = \frac{t}{s-a} \left(\frac{at(bt+ds) + b(s-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)} - a \right) = \frac{t(b-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)}.$$

Thus, $C \left(\frac{at(bt+ds) + b(s-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)}, \frac{t(b-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)} \right)$. Note that we must have $O \left(\frac{a+b}{2}, y \right)$. Also,

$$\begin{aligned} OA_1 &= OC \\ \Leftrightarrow \left(\frac{a-b}{2} \right)^2 + y^2 &= \left(\frac{a+b}{2} - x_C \right)^2 + (y - y_C)^2 \\ \Leftrightarrow y^2 - (y - y_C)^2 &= \left(\frac{a+b}{2} - x_C \right)^2 - \left(\frac{a-b}{2} \right)^2 \\ \Leftrightarrow 2y_C y - y_C^2 &= (a - x_C)(b - x_C). \end{aligned}$$

It follows that

$$\begin{aligned} y &= \frac{(a - x_C)(b - x_C) + y_C^2}{2y_C} \\ &= \frac{(a-b)(s-a)(bs-dt)(b-a)t(bt+ds) + t^2(b-a)^2(bs-dt)^2}{(t(bt+ds) + (s-a)(bs-dt))^2} \\ &= \frac{2t(b-a)(bs-dt)}{t(bt+ds) + (s-a)(bs-dt)} \\ &= \frac{(b-a)(-ds^2 - dt^2 + abt + ads)}{2(bs^2 + bt^2 - abs + adt)} \\ &= \frac{(b-a)(ab+cd)}{2(ad-bc)} \end{aligned}$$

by using (2). Since the coordinates of O are independent of s and t , we prove that O is independent of B_1 and B_2 .

Next, we interchange the roles of A_i and B_i in the question and set up the coordinates again. Suppose $P(0,0)$ and S_1, S_2 are two circles satisfying

$$\begin{aligned} S_1 : \quad x^2 + y^2 - ax + cy &= 0, \\ S_2 : \quad x^2 + y^2 - bx + cy &= 0. \end{aligned}$$

(This time we assume $c = d$ so that PQ is the y -axis.) Consider an arbitrary point $B_1(s, t)$ on S_1 . We pick A_1 to be the point $(a, 0)$ on S_1 and construct the other points accordingly. From above, this choice of A_1 does not affect the position of the circumcentre O' of $\triangle B_1B_2C$. Therefore, it remains to prove that O' lies on a fixed circle when B_1 varies.

We first rewrite

$$B_2 \left(\frac{s(bs - dt)}{s^2 + t^2}, \frac{t(bs - dt)}{s^2 + t^2} \right) = \left(\frac{s(bs - ct)}{as - ct}, \frac{t(bs - ct)}{as - ct} \right)$$

using $d = c$ and (2), and similarly

$$\begin{aligned} & C \left(\frac{at(bt + ds) + b(s - a)(bs - dt)}{t(bt + ds) + (s - a)(bs - dt)}, \frac{t(b - a)(bs - dt)}{t(bt + ds) + (s - a)(bs - dt)} \right) \\ &= \left(\frac{at(bt + cs) + b(s - a)(bs - ct)}{\textcolor{red}{bs^2 + bt^2 - abs + act}}, \frac{t(b - a)(bs - ct)}{\textcolor{red}{bs^2 + bt^2 - abs + act}} \right) \\ &= \left(\frac{abt^2 + acst + b^2s^2 - bcst - ab^2s + abct}{\textcolor{red}{ct(a - b)}}, \frac{t(b - a)(bs - ct)}{\textcolor{red}{ct(a - b)}} \right) \\ &= \left(\frac{ab(as - ct - s^2) + acst + b^2s^2 - bcst - ab^2s + abct}{ct(a - b)}, \frac{ct - bs}{c} \right) \\ &= \left(\frac{a^2bs - abs^2 + acst + b^2s^2 - bcst - ab^2s}{ct(a - b)}, \frac{ct - bs}{c} \right) \\ &= \left(\frac{(ab - bs + ct)s}{ct}, \frac{ct - bs}{c} \right). \end{aligned}$$

As $m_{B_1B_2} = m_{B_1P} = \frac{t}{s}$, the perpendicular bisector of B_1B_2 has equation

$$y = -\frac{s}{t}x + \frac{s^2(as + bs - 2ct)}{2t(as - ct)} + \frac{t(as + bs - 2ct)}{2(as - ct)},$$

i.e.

$$y = -\frac{s}{t}x + \frac{as + bs - 2ct}{2t}. \quad (3)$$

As $m_{B_1C} = m_{A_1B_1} = \frac{t}{s - a}$, the perpendicular bisector of B_1C has equation

$$y = \frac{a - s}{t}x - \frac{(a - s)(ab - bs + 2ct)s}{2ct^2} + \frac{2ct - bs}{2c},$$

i.e.

$$y = \frac{a - s}{t}x + \frac{bcs - abt - 2c^2t - abc}{2ct}. \quad (4)$$

We now solve (3) and (4). We have

$$x = \frac{t}{a} \left(\frac{as + bs - 2ct}{2t} - \frac{bcs - abt - 2c^2t - abc}{2ct} \right) = \frac{cs + bt + bc}{2c}$$

and

$$y = -\frac{s}{t} \cdot \frac{cs + bt + bc}{2c} + \frac{as + bs - 2ct}{2t} = \frac{-cs^2 - bst + acs - 2c^2t}{2ct} = \frac{-bs + ct - c^2}{2c}.$$

This gives $O' \left(\frac{cs + bt + bc}{2c}, \frac{-bs + ct - c^2}{2c} \right)$.

When $B_1 = P$, we have $C = A_2$, and so O' is the centre $\left(\frac{b}{2}, -\frac{c}{2} \right)$ of S_2 . Similarly, when $B_2 = P$, O' is the centre $\left(\frac{a}{2}, -\frac{c}{2} \right)$ of S_1 . When $B_1 = Q$, $O' = Q = (0, -c)$. (These can also be obtained by considering various choices of s, t .) The desired locus circle should pass through these 3 points. It is clear that the x -coordinate of its centre is $\frac{a+b}{4}$. It is then not hard to deduce the equation of the circle

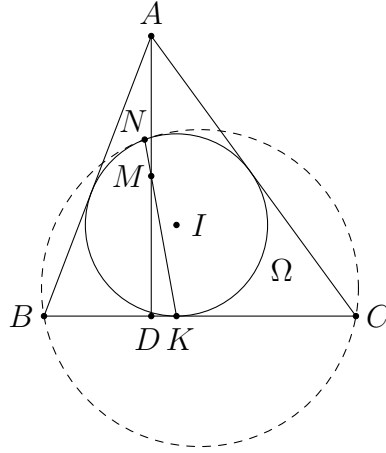
$$x^2 + y^2 - \frac{a+b}{2}x + \frac{ab+3c^2}{2c}y + \frac{ab+c^2}{2} = 0.$$

It remains to show that O' always lies on this circle. Indeed,

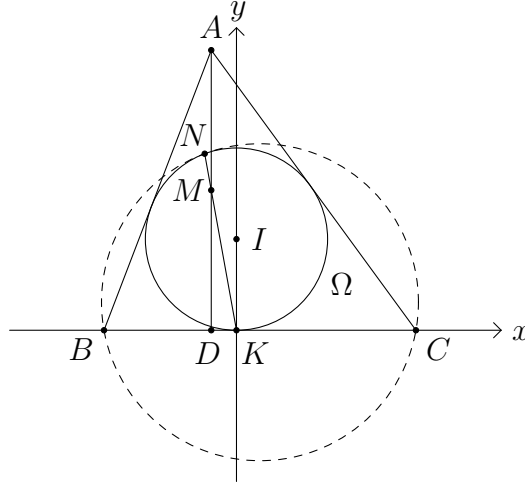
$$\begin{aligned} & \left(\frac{cs + bt + bc}{2c} \right)^2 + \left(\frac{-bs + ct - c^2}{2c} \right)^2 - \frac{a+b}{2} \left(\frac{cs + bt + bc}{2c} \right) + \frac{ab+3c^2}{2c} \left(\frac{-bs + ct - c^2}{2c} \right) \\ & + \frac{ab+c^2}{2} \\ &= \frac{1}{4c^2} (c^2s^2 + 2bcst + b^2t^2 + 2bc^2s + 2b^2ct + b^2c^2 + b^2s^2 - 2bcst + c^2t^2 + 2bc^2s - 2c^3t + c^4 \\ & - ac^2s - bc^2s - abct - b^2ct - abc^2 - b^2c^2 - ab^2s - 3bc^2s + abct + 3c^3t - abc^2 - 3c^4 \\ & + 2abc^2 + 2c^4) \\ &= \frac{1}{4c^2} (c^2s^2 + b^2t^2 + b^2ct + b^2s^2 + c^2t^2 + c^3t - ac^2s - ab^2s) \\ &= \frac{1}{4c^2} (s^2 + t^2 - as + ct)(b^2 + c^2) \\ &= 0. \end{aligned}$$

This proves the locus is a circle. □

Problem 15. (IMO Shortlist 2002 G7) The incircle Ω of the acute-angled triangle ABC is tangent to its side BC at a point K . Let AD be an altitude of triangle ABC , and let M be the midpoint of the segment AD . If N is the common point of the circle Ω and the line KM (distinct from K), then prove that the incircle Ω and the circumcircle of triangle BCN are tangent to each other at the point N .



Solution.



Let $K(0,0)$, $I(0,1)$, $B(b,0)$, $C(c,0)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$. Thus, $D\left(\frac{b+c}{bc+1}, 0\right)$ and $M\left(\frac{b+c}{bc+1}, \frac{bc}{bc+1}\right)$. From this, we find that

$$KM: \quad y = \frac{bc}{b+c}x.$$

Putting this in

$$\Omega: \quad x^2 + y^2 - 2y = 0,$$

we need to solve

$$x^2 + \frac{b^2c^2}{(b+c)^2}x^2 - \frac{2bc}{b+c}x = 0.$$

The solutions correspond to the points K and N . So the root $x = 0$ is rejected. The other root is $x = \frac{2bc(b+c)}{b^2c^2 + (b+c)^2}$. So we obtain $N\left(\frac{2bc(b+c)}{b^2c^2 + (b+c)^2}, \frac{2b^2c^2}{b^2c^2 + (b+c)^2}\right)$. The

x -coordinate of the centre O of (BCN) is $\frac{b+c}{2}$. From

$$\begin{aligned}
OB &= ON \\
\Rightarrow \left(\frac{b+c}{2} - b\right)^2 + y^2 &= \left(\frac{b+c}{2} - \frac{2bc(b+c)}{b^2c^2 + (b+c)^2}\right)^2 + \left(y - \frac{2b^2c^2}{b^2c^2 + (b+c)^2}\right)^2 \\
\Rightarrow b^2 - b(b+c) &= \frac{4b^2c^2(b+c)^2 + 4b^4c^4}{(b^2c^2 + (b+c)^2)^2} - \frac{2bc(b+c)^2}{b^2c^2 + (b+c)^2} - \frac{4b^2c^2}{b^2c^2 + (b+c)^2}y \\
\Rightarrow -bc &= -\frac{2bc(b^2 + c^2)}{b^2c^2 + (b+c)^2} - \frac{4b^2c^2}{b^2c^2 + (b+c)^2}y \\
\Rightarrow y &= \frac{b^2c^2 - b^2 + 2bc - c^2}{4bc},
\end{aligned}$$

we get $O\left(\frac{b+c}{2}, \frac{b^2c^2 - b^2 + 2bc - c^2}{4bc}\right)$. To show that (BCN) is tangent to Ω , it suffices to show N, I, O are collinear. It is easy to find that

$$NI: y = \frac{b^2c^2 - (b+c)^2}{2bc(b+c)}x + 1.$$

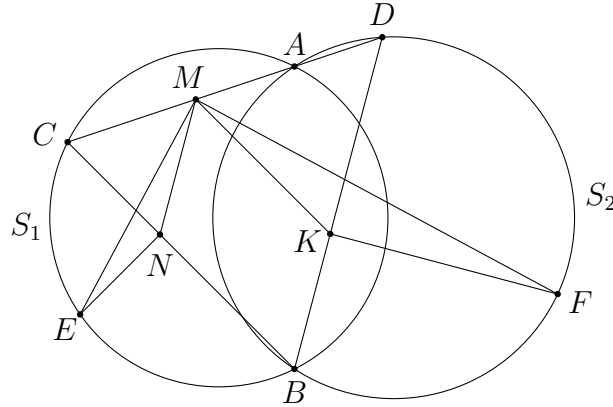
We now check that the coordinates of O satisfy this equation. Indeed,

$$\frac{b^2c^2 - (b+c)^2}{2bc(b+c)} \cdot \frac{b+c}{2} + 1 = \frac{b^2c^2 - (b+c)^2}{4bc} + 1 = \frac{b^2c^2 - b^2 + 2bc - c^2}{4bc}.$$

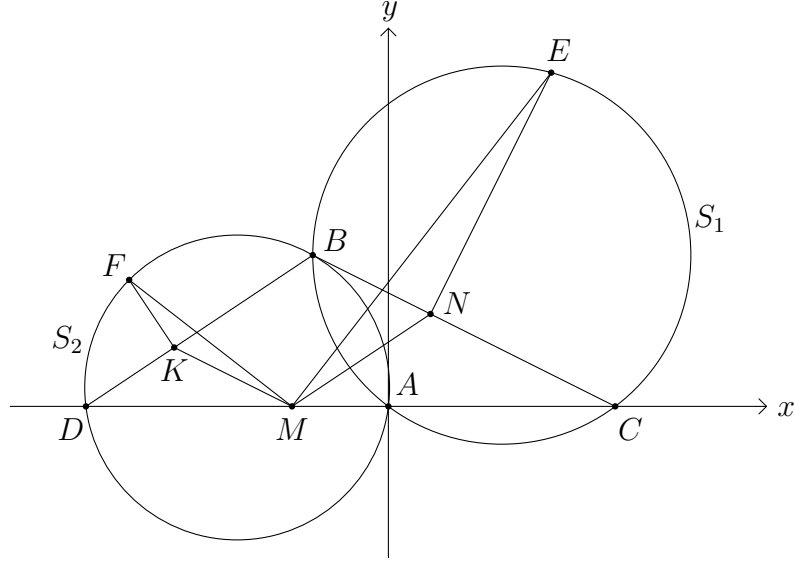
This completes the proof. $\square$

Problem 16. (IMO Shortlist 2002 G8) Let two circles S_1 and S_2 meet at the points A and B . A line through A meets S_1 again at C and S_2 again at D . Let M, N, K be three points on the line segments CD, BC, BD respectively, with MN parallel to BD and MK parallel to BC . Let E and F be points on those arcs BC of S_1 and BD of S_2 respectively that do not contain A . Given that EN is perpendicular to BC and FK is perpendicular to BD , prove that $\angle EMF = 90^\circ$.

(Remark: A should lie between C and D or otherwise the result may not hold.)



Solution.



Let $A(0,0)$, $B(a,b)$, $C(c,0)$, $D(d,0)$, $M(m,0)$ where $c > 0 > d$ and $b > 0$. As

$$m_{MN} = m_{BD} = \frac{b}{a-d},$$

we have

$$MN: \quad y = \frac{b}{a-d}x - \frac{bm}{a-d}.$$

Also, it is easy to get

$$BC: \quad y = \frac{b}{a-c}x - \frac{bc}{a-c}.$$

Solving these equations, we obtain

$$x = \frac{bm(a-c) - bc(a-d)}{b(a-c) - b(a-d)} = \frac{(a-c)m - c(a-d)}{d-c}$$

and

$$y = \frac{b}{a-c} \cdot \left(\frac{(a-c)m - c(a-d)}{d-c} - c \right) = \frac{b}{a-c} \cdot \frac{(a-c)m - c(a-c)}{d-c} = \frac{b(m-c)}{d-c}.$$

This yields $N \left(\frac{(a-c)m - c(a-d)}{d-c}, \frac{b(m-c)}{d-c} \right)$.

Next, from $m_{NE} = -\frac{1}{m_{BC}} = -\frac{a-c}{b}$, the equation of NE is

$$y = -\frac{a-c}{b}x + \frac{a-c}{b} \cdot \frac{(a-c)m - c(a-d)}{d-c} + \frac{b(m-c)}{d-c},$$

i.e.

$$NE: \quad y = -\frac{a-c}{b}x + \frac{b^2(m-c) + (a-c)^2m - c(a-c)(a-d)}{b(d-c)}. \quad (1)$$

The x -coordinate of the circumcentre O_1 of S_1 is $\frac{c}{2}$. Also, as

$$\begin{aligned} O_1A &= O_1B \\ \Rightarrow \quad \left(\frac{c}{2}\right)^2 + y^2 &= \left(\frac{c}{2} - a\right)^2 + (y-b)^2 \\ \Rightarrow \quad 0 &= a^2 - ac + b^2 - 2by \\ \Rightarrow \quad y &= \frac{b^2 + a^2 - ac}{2b}, \end{aligned}$$

the coordinates of O_1 are $\left(\frac{c}{2}, \frac{b^2 + a^2 - ac}{2b}\right)$. Thus, we deduce

$$S_1: \quad x^2 + y^2 - cx - \frac{b^2 + a^2 - ac}{b}y = 0. \quad (2)$$

Putting (1) in (2), we solve

$$\begin{aligned} \left(1 + \frac{(a-c)^2}{b^2}\right)x^2 + \left(-\frac{2(a-c)(b^2(m-c) + (a-c)^2m - c(a-c)(a-d))}{b^2(d-c)}\right. \\ \left.-c + \frac{(a-c)(b^2 + a^2 - ac)}{b^2}\right)x + \left(\frac{(b^2(m-c) + (a-c)^2m - c(a-c)(a-d))^2}{b^2(d-c)^2}\right. \\ \left.-\frac{(b^2 + a^2 - ac)(b^2(m-c) + (a-c)^2m - c(a-c)(a-d))}{b^2(d-c)}\right) = 0. \end{aligned}$$

After simplification and removing the common factor $b^2 + (a-c)^2$ from each coefficient, this becomes

$$\begin{aligned} (d-c)^2x^2 + (d-c)(-2(a-c)m + (ac + ad - 2cd))x + \left((b^2 + (a-c)^2)m^2\right. \\ \left.- (b^2(c+d) + (a-c)(ac + ad - 2cd))m + cd(b^2 + (a-c)(a-d))\right) = 0. \end{aligned}$$

The discriminant of this quadratic polynomial in x is

$$\begin{aligned} (d-c)^2(-2(a-c)m + (ac + ad - 2cd))^2 - 4(d-c)^2\left((b^2 + (a-c)^2)m^2\right. \\ \left.- (b^2(c+d) + (a-c)(ac + ad - 2cd))m + cd(b^2 + (a-c)(a-d))\right) \\ = (d-c)^2(-4b^2m^2 + 4b^2(c+d)m - 4b^2cd + a^2(d-c)^2). \end{aligned}$$

Let

$$\Delta = -4b^2m^2 + 4b^2(c+d)m - 4b^2cd + a^2(d-c)^2.$$

From the configuration, we should take the larger root. Therefore,

$$\begin{aligned} x &= \frac{(d-c)(2(a-c)m - (ac + ad - 2cd)) + |d-c|\sqrt{\Delta}}{2(d-c)^2} \\ &= \frac{2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta}}{2(d-c)} \end{aligned}$$

since $c > d$. Then we have

$$\begin{aligned} y &= -\frac{a-c}{b} \cdot \frac{2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta}}{2(d-c)} \\ &\quad + \frac{b^2(m-c) + (a-c)^2m - c(a-c)(a-d)}{b(d-c)} \\ &= \frac{2b^2(m-c) + a(a-c)(d-c) + (a-c)\sqrt{\Delta}}{2b(d-c)}. \end{aligned}$$

This means the coordinates of E are

$$\left(\frac{2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta}}{2(d-c)}, \frac{2b^2(m-c) + a(a-c)(d-c) + (a-c)\sqrt{\Delta}}{2b(d-c)} \right).$$

We swap the variables c and d to obtain the coordinates of F . Although we should choose the smaller root this time when we solve the quadratic equation, the sign change is cancelled by the effect when we remove the absolute value of $|c-d|$ (previously we have $|d-c| = -(d-c)$). Note that Δ is unchanged. Therefore, we have

$$F \left(\frac{2(a-d)m - (ac + ad - 2cd) - \sqrt{\Delta}}{2(c-d)}, \frac{2b^2(m-d) + a(a-d)(c-d) + (a-d)\sqrt{\Delta}}{2b(c-d)} \right).$$

To show that $ME \perp MF$, it suffices to check $m_{ME} \times m_{MF} = -1$. Note that

$$\begin{aligned} m_{ME} &= \frac{2b^2(m-c) + a(a-c)(d-c) + (a-c)\sqrt{\Delta}}{b(2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta} - 2(d-c)m)} \\ &= \frac{2b^2(m-c) + a(a-c)(d-c) + (a-c)\sqrt{\Delta}}{b(2(a-d)m - (ac + ad - 2cd) - \sqrt{\Delta})} \end{aligned}$$

and similarly

$$m_{MF} = \frac{2b^2(m-d) + a(a-d)(c-d) + (a-d)\sqrt{\Delta}}{b(2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta})}.$$

Thus, it remains to check

$$\begin{aligned} &(2b^2(m-c) + a(a-c)(d-c) + (a-c)\sqrt{\Delta}) \\ &\quad \times (2b^2(m-d) + a(a-d)(c-d) + (a-d)\sqrt{\Delta}) \\ &\quad + b^2(2(a-d)m - (ac + ad - 2cd) - \sqrt{\Delta})(2(a-c)m - (ac + ad - 2cd) - \sqrt{\Delta}) = 0. \end{aligned}$$

In the expansion of the left-hand side, the terms not involving $\sqrt{\Delta}$ are

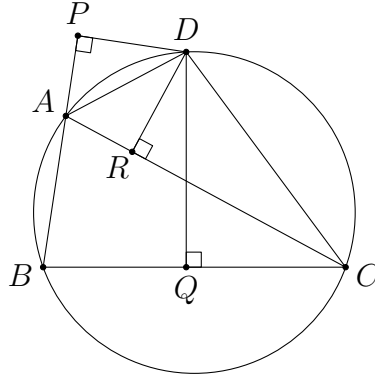
$$\begin{aligned}
& 4b^4(m-c)(m-d) + 2ab^2(c-d)((m-c)(a-d) - (m-d)(a-c)) \\
& - a^2(a-c)(a-d)(c-d)^2 + (a-c)(a-d)\Delta + 4b^2(a-c)(a-d)m^2 \\
& - 2b^2(ac+ad-2cd)(a-c+a-d)m + b^2(ac+ad-2cd)^2 + b^2\Delta \\
& = 4b^4(m-c)(m-d) + 4b^2(a-c)(a-d)m^2 - 4b^2(c+d)(a-c)(a-d)m \\
& + b^2(a^2(-c^2+6cd-d^2) - 4acd(c+d) + 4c^2d^2) - a^2(a-c)(a-d)(c-d)^2 \\
& + (b^2 + (a-c)(a-d))\Delta \\
& = 0
\end{aligned}$$

as $\Delta = -4b^2m^2 + 4b^2(c+d)m - 4b^2cd + a^2(d-c)^2$. The terms involving $\sqrt{\Delta}$ are

$$\begin{aligned}
& 2b^2((m-c)(a-d) + (m-d)(a-c))\sqrt{\Delta} - 2b^2(a-c+a-d)m\sqrt{\Delta} \\
& + 2b^2(ac+ad-2cd)\sqrt{\Delta} \\
& = 0.
\end{aligned}$$

So we have established $m_{ME} \times m_{MF} = -1$ and we are done. $\square$

Problem 17. (IMO Shortlist 2003 G1/Problem 4) $ABCD$ is cyclic. The feet of the perpendicular from D to the lines AB, BC, CA are P, Q, R respectively. Show that the angle bisectors of ABC and CDA meet on the line AC iff $RP = RQ$.



Solution. Let O be the intersection point of AD and BC . Let $O(0,0)$, $B(b,0)$, $C(c,0)$, $A(a,ma)$, $D(d,md)$ (so that the equation of AD is $y = mx$) where $b, c > 0$ and a, d are of the same sign. Then

$$bc = (m^2 + 1)ad \quad (1)$$

and

$$AB: \quad y = \frac{ma}{a-b}x - \frac{mab}{a-b}. \quad (2)$$

limiting case when m is undefined, and is usually handled by continuity),

$$\begin{aligned}
& RP = RQ \\
\Leftrightarrow & 2c((bc - ad)a + d(a - b)^2) = b((bc - ad)a + d(a - c)^2) \\
\Leftrightarrow & -ab^2c + 2abc^2 - 2abcd + 2b^2cd - bc^2d = 0 \\
\Leftrightarrow & -ab + 2ac - 2ad + 2bd - cd = 0.
\end{aligned} \tag{4}$$

On the other hand, the angle bisector of $\angle ABC$ cuts AC in the ratio $AB : BC$ by the angle bisector theorem. So the angle bisectors of $\angle ABC$ and $\angle CDA$ meet on AC if and only if $AB : BC = AD : DC$. Using (1), we have

$$\begin{aligned}
& \frac{AB}{BC} = \frac{AD}{DC} \\
\Leftrightarrow & \frac{(a - b)^2 + m^2a^2}{(c - b)^2} = \frac{(d - a)^2 + m^2(d - a)^2}{(d - c)^2 + m^2d^2} \\
\Leftrightarrow & \frac{d(a - b)^2 + (bc - ad)a}{d(c - b)^2} = \frac{bc(d - a)^2}{ad(d - c)^2 + (bc - ad)d^2} \\
\Leftrightarrow & (d(a - b)^2 + (bc - ad)a)(a(d - c)^2 + (bc - ad)d) = bc(d - a)^2(c - b)^2 \\
\Leftrightarrow & -a^2b^3c + 2a^2b^2c^2 - 4a^2bc^2d + 4a^2bcd^2 + 2ab^3cd \\
& -2ab^2c^2d - 4ab^2cd^2 + 2abc^3d + 2b^2c^2d^2 - bc^3d^2 = 0 \\
\Leftrightarrow & bc(-ab + 2ac - 2ad + 2bd - cd)(ab - 2ad + cd) = 0.
\end{aligned} \tag{5}$$

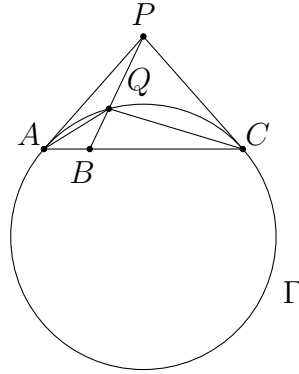
In order to show that (4) and (5) are equivalent, it remains to show that it is impossible to have $ab - 2ad + cd = 0$. Indeed, this yields $d = \frac{ab}{2a - c}$. Putting this in (1), we have

$$m^2 + 1 = \frac{bc(2a - c)}{a^2b} = \frac{c(2a - c)}{a^2}.$$

As $m^2 \geq 0$, this implies $0 \geq a^2 - 2ac + c^2$, i.e. $0 \geq (a - c)^2$. This cannot hold since $a \neq c$ (or otherwise $d = b$, which cannot happen as $ABCD$ is convex). So we have established the equivalence. $\square$

Problem 18. (IMO Shortlist 2003 G2) Three distinct points A, B , and C are fixed on a line in this order. Let Γ be a circle passing through A and C whose centre does not lie on the line AC . Denote by P the intersection of the tangents to Γ at A and C . Suppose Γ meets the segment PB at Q . Prove that the intersection of the bisector of $\angle AQC$ and the line AC does not depend on the choice of Γ .

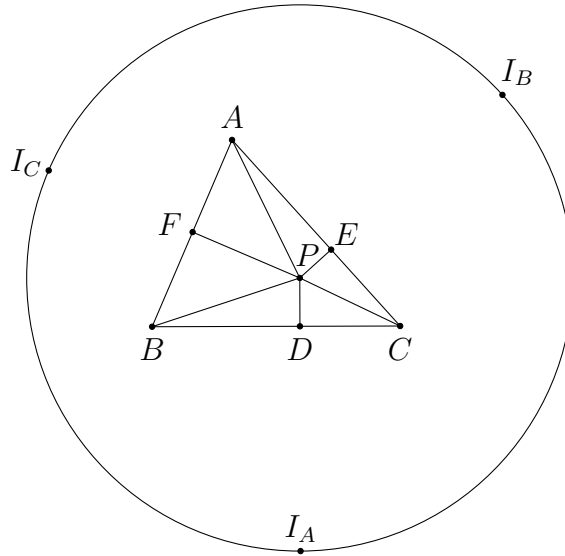
Solution. See [problem 4.2](#) of **Coordinate Geometry 5.0**. $\square$



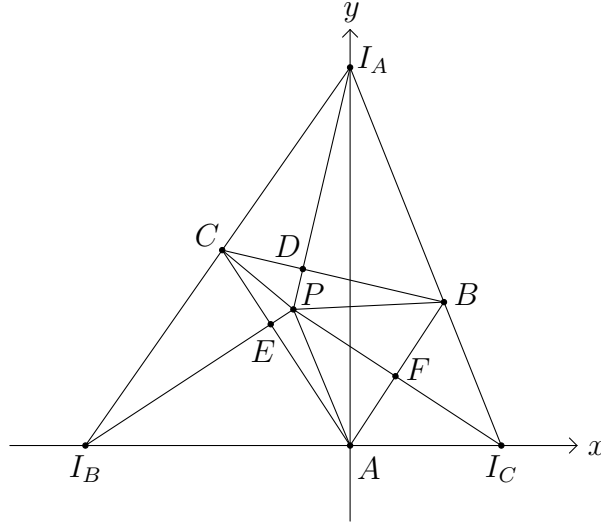
Problem 19. (IMO Shortlist 2003 G3) Let ABC be a triangle and let P be a point in its interior. Denote by D, E, F the feet of the perpendiculars from P to the lines BC, CA, AB , respectively. Suppose that

$$AP^2 + PD^2 = BP^2 + PE^2 = CP^2 + PF^2.$$

Denote by I_A, I_B, I_C the excentres of the triangle ABC . Prove that P is the circumcentre of the triangle $I_AI_BI_C$.



Solution. From $AP^2 + PD^2 = BP^2 + PE^2$, we get $AP^2 - PE^2 = BP^2 - PD^2$. This implies $AE^2 = BD^2$ by Pythagoras' theorem. Thus, we have $AE = BD$. Similarly, we can deduce $BF = CE$ and $CD = AF$. It follows that $BD + BF + AF$ is the semiperimeter s of $\triangle ABC$, so that $BD = s - AB$. Thus, D is the contact point of the A -excircle to the side BC . Therefore, P, D, I_A are collinear. Similarly, P also lies on EI_B and FI_C . (This part only involves Pythagoras' theorem and the most basic properties of the excentres, and we will show how to deal with the remaining simplified problem.)



Recall that A, B, C are the feet of the altitudes of $\triangle I_A I_B I_C$ from I_A, I_B, I_C respectively. Let $I_A(0, a)$, $I_B(b, 0)$, $I_C(c, 0)$. Then $A(0, 0)$, $B\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right)$, $C\left(\frac{b(a^2 + bc)}{a^2 + b^2}, \frac{ab(b - c)}{a^2 + b^2}\right)$. This yields

$$m_{AB} = \frac{a(c - b)}{a^2 + bc}$$

so that

$$I_C F : \quad y = -\frac{a^2 + bc}{a(c - b)}x + \frac{c(a^2 + bc)}{a(c - b)}.$$

By symmetry, we also have

$$I_B E : \quad y = -\frac{a^2 + bc}{a(b - c)}x + \frac{b(a^2 + bc)}{a(b - c)}.$$

Solving these, we get

$$x = \frac{c(a^2 + bc) + b(a^2 + bc)}{(a^2 + bc) + (a^2 + bc)} = \frac{b + c}{2}$$

and

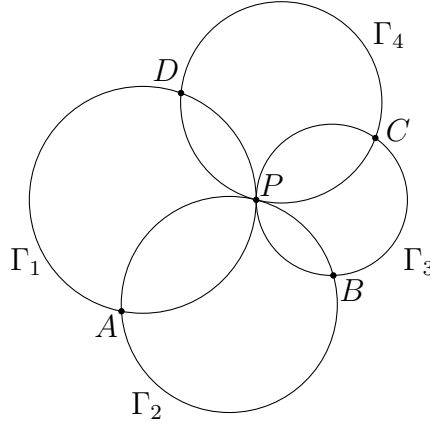
$$y = \frac{a^2 + bc}{a(c - b)} \cdot \left(c - \frac{b + c}{2}\right) = \frac{a^2 + bc}{2a}.$$

Thus, $P\left(\frac{b + c}{2}, \frac{a^2 + bc}{2a}\right)$. This shows P is the circumcentre of $\triangle I_A I_B I_C$. $\square$

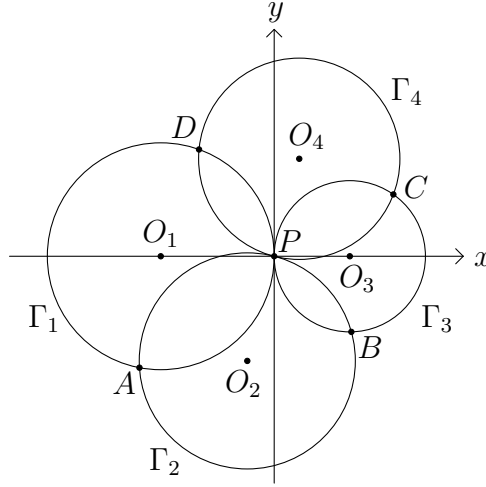
Problem 20. (IMO Shortlist 2003 G4) Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P , and Γ_2, Γ_4 are externally tangent at the same point

P . Suppose that Γ_1 and Γ_2 ; Γ_2 and Γ_3 ; Γ_3 and Γ_4 ; Γ_4 and Γ_1 meet at A, B, C, D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$



Solution.



For $j = 1, 2, 3, 4$, let O_j be the centre of Γ_j . Note that P is the intersection of O_1O_3 and O_2O_4 . Let $P(0,0)$, $O_1(a,0)$, $O_3(c,0)$, $O_2(b,mb)$, $O_4(d,md)$ (so that the equation of O_2O_4 is $y = mx$). As O_1O_2 is the perpendicular bisector of AP , we get $m_{AP} = -\frac{b-a}{mb}$. So

$$AP: \quad y = -\frac{b-a}{mb}x.$$

Solving with

$$O_1O_2: \quad y = \frac{mb}{b-a}x - \frac{mab}{b-a},$$

we easily get the coordinates $\left(\frac{m^2ab^2}{m^2b^2 + (b-a)^2}, -\frac{mab(b-a)}{m^2b^2 + (b-a)^2}\right)$ of the intersection of AP and O_1O_2 . This is the midpoint of AP . So $A\left(\frac{2m^2ab^2}{m^2b^2 + (b-a)^2}, -\frac{2mab(b-a)}{m^2b^2 + (b-a)^2}\right)$. By symmetry, we also have

$$\begin{aligned} B & \left(\frac{2m^2cb^2}{m^2b^2 + (b-c)^2}, -\frac{2mcb(b-c)}{m^2b^2 + (b-c)^2}\right), \\ C & \left(\frac{2m^2cd^2}{m^2d^2 + (d-c)^2}, -\frac{2mcd(d-c)}{m^2d^2 + (d-c)^2}\right), \\ D & \left(\frac{2m^2ad^2}{m^2d^2 + (d-a)^2}, -\frac{2mad(d-a)}{m^2d^2 + (d-a)^2}\right). \end{aligned}$$

It remains to compute all distances. Firstly, we have

$$\begin{aligned} AB^2 &= \left(\frac{2m^2ab^2}{m^2b^2 + (b-a)^2} - \frac{2m^2cb^2}{m^2b^2 + (b-c)^2}\right)^2 + \left(\frac{2mab(b-a)}{m^2b^2 + (b-a)^2} - \frac{2mcb(b-c)}{m^2b^2 + (b-c)^2}\right)^2 \\ &= \frac{4m^2b^2}{(m^2b^2 + (b-a)^2)^2(m^2b^2 + (b-c)^2)^2} \times \left(m^2b^2(m^2b^2(a-c) + a(b-c)^2 - c(b-a)^2)^2\right. \\ &\quad \left.+ (m^2b^2(a(b-a) - c(b-c)) + a(b-a)(b-c)^2 - c(b-c)(b-a)^2)^2\right) \\ &= \frac{4m^2b^4(a-c)^2}{(m^2b^2 + (b-a)^2)^2(m^2b^2 + (b-c)^2)^2} \\ &\quad \times \left(m^2(m^2b^2 + b^2 - ac)^2 + (m^2b(b-a-c) + (b-a)(b-c))^2\right) \\ &= \frac{4m^2b^4(a-c)^2}{(m^2b^2 + (b-a)^2)^2(m^2b^2 + (b-c)^2)^2} \times (m^6b^4 + m^4b^2(3b^2 - 2ab - 2cb + a^2 + c^2) \\ &\quad + m^2(3b^4 - 4ab^3 - 4cb^3 + 2a^2b^2 + 4acb^2 + 2c^2b^2 - 2a^2cb - 2ac^2b + a^2c^2) \\ &\quad + (b-a)^2(b-c)^2). \end{aligned}$$

The formula for CD^2 can be found by replacing b by d .

Secondly, we have

$$\begin{aligned} BC^2 &= \left(\frac{2m^2cb^2}{m^2b^2 + (b-c)^2} - \frac{2m^2cd^2}{m^2d^2 + (d-c)^2}\right)^2 + \left(\frac{2mcb(b-c)}{m^2b^2 + (b-c)^2} - \frac{2mcd(d-c)}{m^2d^2 + (d-c)^2}\right)^2 \\ &= \frac{4m^2c^2}{(m^2b^2 + (b-c)^2)^2(m^2d^2 + (d-c)^2)^2} \times \left(m^2(b^2(d-c)^2 - d^2(b-c)^2)^2\right. \\ &\quad \left.+ (m^2bd(d(b-c) - b(d-c)) + b(b-c)(d-c)^2 - d(d-c)(b-c)^2)^2\right) \end{aligned}$$

$$\begin{aligned}
&= \frac{4m^2c^4(b-d)^2}{(m^2b^2 + (b-c)^2)^2(m^2d^2 + (d-c)^2)^2} \\
&\quad \times \left(m^2(bc + dc - 2bd)^2 + (m^2bd - (b-c)(d-c))^2 \right) \\
&= \frac{4m^2c^4(b-d)^2}{(m^2b^2 + (b-c)^2)^2(m^2d^2 + (d-c)^2)^2} \\
&\quad \times (m^4b^2d^2 + m^2(b^2c^2 + d^2c^2 - 2b^2dc - 2bd^2c + 2b^2d^2) + (b-c)^2(d-c)^2).
\end{aligned}$$

The formula for AD^2 can be found by replacing c by a .

Thirdly, we have

$$PB^2 = \frac{4m^4c^2b^4}{(m^2b^2 + (b-c)^2)^2} + \frac{4m^2c^2b^2(b-c)^2}{(m^2b^2 + (b-c)^2)^2} = \frac{4m^2c^2b^2}{m^2b^2 + (b-c)^2}.$$

The formula for PD^2 can be found by replacing b by d and c by a .

Now, combining everything and cancelling some obvious factors, we need to prove

$$\begin{aligned}
&(m^2d^2 + (d-a)^2)^2 \\
&\quad \times (m^6b^4 + m^4b^2(3b^2 - 2ab - 2cb + a^2 + c^2) + m^2(3b^4 - 4ab^3 - 4cb^3 + 2a^2b^2 \\
&\quad + 4acb^2 + 2c^2b^2 - 2a^2cb - 2ac^2b + a^2c^2) + (b-a)^2(b-c)^2) \\
&\quad \times (m^4b^2d^2 + m^2(b^2c^2 + d^2c^2 - 2b^2dc - 2bd^2c + 2b^2d^2) + (b-c)^2(d-c)^2) \\
&= (m^2b^2 + (b-c)^2)^2 \\
&\quad \times (m^6d^4 + m^4d^2(3d^2 - 2ad - 2cd + a^2 + c^2) + m^2(3d^4 - 4ad^3 - 4cd^3 + 2a^2d^2 \\
&\quad + 4acd^2 + 2c^2d^2 - 2a^2cd - 2ac^2d + a^2c^2) + (d-a)^2(d-c)^2) \\
&\quad \times (m^4b^2d^2 + m^2(b^2a^2 + d^2a^2 - 2b^2da - 2bd^2a + 2b^2d^2) + (b-a)^2(d-a)^2).
\end{aligned}$$

In order that this is an identity, both sides must share the same factors. Therefore, it is natural to check whether $m^2b^2 + (b-c)^2$ is a factor of the terms on the left. We easily obtain the factorization

$$\begin{aligned}
&m^4b^2d^2 + m^2(b^2c^2 + d^2c^2 - 2b^2dc - 2bd^2c + 2b^2d^2) + (b-c)^2(d-c)^2 \\
&= (m^2b^2 + (b-c)^2)(m^2d^2 + (d-c)^2).
\end{aligned}$$

By symmetry, we also have

$$\begin{aligned}
&m^4b^2d^2 + m^2(b^2a^2 + d^2a^2 - 2b^2da - 2bd^2a + 2b^2d^2) + (b-a)^2(d-a)^2 \\
&= (m^2b^2 + (b-a)^2)(m^2d^2 + (d-a)^2).
\end{aligned}$$

In view of this, $m^2b^2 + (b-a)^2$ should also be a factor of the first term. Indeed,

$$\begin{aligned}
&m^6b^4 + m^4b^2(3b^2 - 2ab - 2cb + a^2 + c^2) + m^2(3b^4 - 4ab^3 - 4cb^3 + 2a^2b^2 \\
&\quad + 4acb^2 + 2c^2b^2 - 2a^2cb - 2ac^2b + a^2c^2) + (b-a)^2(b-c)^2
\end{aligned}$$

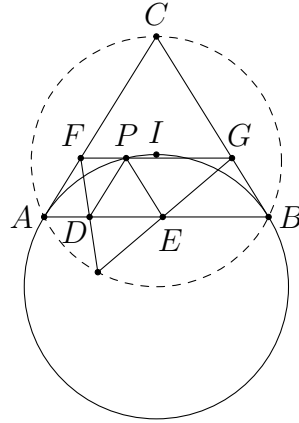
$$\begin{aligned}
&= (m^2b^2 + (b-a)^2)(m^4b^2 + m^2(2b^2 - 2bc + c^2) + (b-c)^2) \\
&= (m^2b^2 + (b-a)^2)(m^2b^2 + (b-c)^2)(m^2 + 1).
\end{aligned}$$

By symmetry, we have

$$\begin{aligned}
&m^6d^4 + m^4d^2(3d^2 - 2ad - 2cd + a^2 + c^2) + m^2(3d^4 - 4ad^3 - 4cd^3 + 2a^2d^2 \\
&\quad + 4acd^2 + 2c^2d^2 - 2a^2cd - 2ac^2d + a^2c^2) + (d-a)^2(d-c)^2 \\
&= (m^2d^2 + (d-a)^2)(m^2d^2 + (d-c)^2)(m^2 + 1).
\end{aligned}$$

Combining all factorizations, we get the desired result. $\square$

Problem 21. (IMO Shortlist 2003 G5) Let ABC be an isosceles triangle with $AC = BC$, whose incentre is I . Let P be a point on the circumcircle of the triangle AIB lying inside the triangle ABC . The lines through P parallel to CA and CB meet AB at D and E , respectively. The line through P parallel to AB meets CA and CB at F and G , respectively. Prove that the lines DF and EG intersect on the circumcircle of the triangle ABC .



Solution. Let $I(0, 1)$, $A(-a, 0)$, $B(a, 0)$. Then $C\left(0, \frac{2a^2}{a^2 - 1}\right)$,

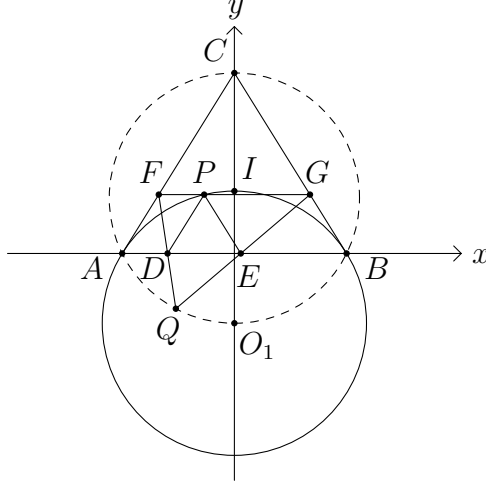
$$AC: \quad y = \frac{2a}{a^2 - 1}x + \frac{2a^2}{a^2 - 1} \quad (1)$$

and

$$BC: \quad y = -\frac{2a}{a^2 - 1}x + \frac{2a^2}{a^2 - 1}. \quad (2)$$

Let O_1 be the circumcentre of $\triangle AIB$. Its coordinates are $\left(0, \frac{1-a^2}{2}\right)$. Hence, the equation of (AIB) is $x^2 + y^2 + (a^2 - 1)y - a^2 = 0$. Thus, the point $P(b, c)$ satisfies

$$b^2 + c^2 + (a^2 - 1)c - a^2 = 0. \quad (3)$$



Solving FG (with equation $y = c$) and (1), we get $F\left(\frac{(a^2 - 1)c - 2a^2}{2a}, c\right)$. Similarly, using (2), we get $G\left(\frac{2a^2 - (a^2 - 1)c}{2a}, c\right)$. From the parallelograms,

$$D = A + P - F = \left(\frac{2ab - (a^2 - 1)c}{2a}, 0\right)$$

and

$$E = B + P - G = \left(\frac{2ab + (a^2 - 1)c}{2a}, 0\right).$$

Let $Q = DF \cap EG$. We need to solve

$$\begin{aligned} DF: \quad y &= \frac{ac}{-ab + (a^2 - 1)c - a^2}x - \frac{c(2ab - (a^2 - 1)c)}{2(-ab + (a^2 - 1)c - a^2)}, \\ EG: \quad y &= \frac{ac}{-ab - (a^2 - 1)c + a^2}x - \frac{c(2ab + (a^2 - 1)c)}{2(-ab - (a^2 - 1)c + a^2)}. \end{aligned}$$

This yields

$$\begin{aligned} x &= \frac{(2ab - (a^2 - 1)c)(-ab - (a^2 - 1)c + a^2) - (2ab + (a^2 - 1)c)(-ab + (a^2 - 1)c - a^2)}{2a(-ab - (a^2 - 1)c + a^2) - 2a(-ab + (a^2 - 1)c - a^2)} \\ &= \frac{-2a(a^2 - 1)bc + 4a^3b}{-4a((a^2 - 1)c - a^2)} = \frac{(a^2 - 1)bc - 2a^2b}{2((a^2 - 1)c - a^2)} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{c}{2(-ab - (a^2 - 1)c + a^2)} \cdot \left(2a \cdot \frac{(a^2 - 1)bc - 2a^2b}{2((a^2 - 1)c - a^2)} - 2ab - (a^2 - 1)c\right) \\ &= \frac{c}{2(-ab - (a^2 - 1)c + a^2)} \cdot \frac{-a(a^2 - 1)bc - (a^2 - 1)^2c^2 + a^2(a^2 - 1)c}{(a^2 - 1)c - a^2} \end{aligned}$$

$$\begin{aligned}
&= \frac{c}{2(-ab - (a^2 - 1)c + a^2)} \cdot \frac{(-ab - (a^2 - 1)c + a^2)(a^2 - 1)c}{(a^2 - 1)c - a^2} \\
&= \frac{(a^2 - 1)c^2}{2((a^2 - 1)c - a^2)}.
\end{aligned}$$

That means $Q \left(\frac{(a^2 - 1)bc - 2a^2b}{2((a^2 - 1)c - a^2)}, \frac{(a^2 - 1)c^2}{2((a^2 - 1)c - a^2)} \right)$.

To prove that A, B, C, Q are concyclic, we first note that O_1 lies on (ABC) . Indeed, it is the midpoint of $\widehat{AB}$ opposite to C . Therefore, CO_1 is a diameter of (ABC) . We just have to check $\angle O_1QC = 90^\circ$. Now,

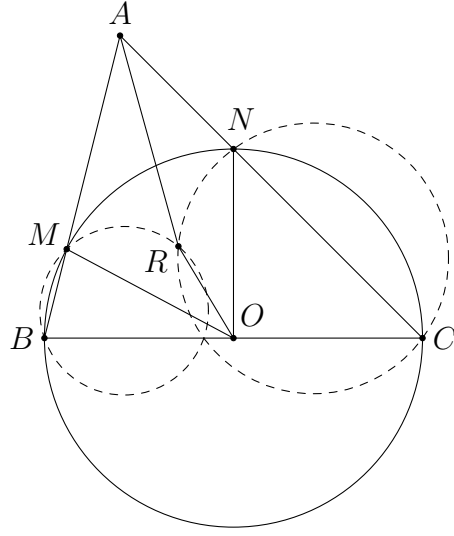
$$\begin{aligned}
m_{CQ} &= \frac{(a^2 - 1)^2c^2 - 4a^2((a^2 - 1)c - a^2)}{(a^2 - 1)((a^2 - 1)bc - 2a^2b)} \\
&= \frac{(a^2 - 1)^2c^2 - 4a^2(a^2 - 1)c + 4a^4}{(a^2 - 1)((a^2 - 1)bc - 2a^2b)} \\
&= \frac{((a^2 - 1)c - 2a^2)^2}{(a^2 - 1)b((a^2 - 1)c - 2a^2)} \\
&= \frac{(a^2 - 1)c - 2a^2}{(a^2 - 1)b}, \\
m_{O_1Q} &= \frac{(a^2 - 1)c^2 + (a^2 - 1)((a^2 - 1)c - a^2)}{(a^2 - 1)bc - 2a^2b}.
\end{aligned}$$

Multiplying these, we have

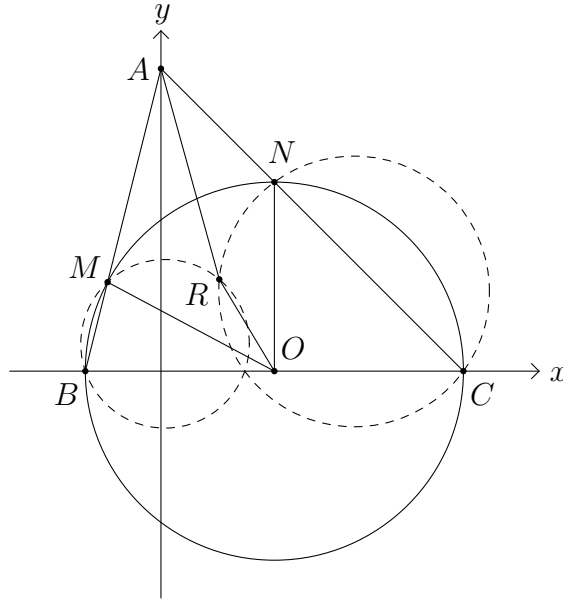
$$m_{CQ} \cdot m_{O_1Q} = \frac{(a^2 - 1)c - 2a^2}{(a^2 - 1)b} \cdot \frac{(a^2 - 1)(c^2 + (a^2 - 1)c - a^2)}{b((a^2 - 1)c - 2a^2)} = \frac{c^2 + (a^2 - 1)c - a^2}{b^2} = -1$$

by (3). The conclusion then follows. $\square$

Problem 22. (IMO Shortlist 2004 G1/Problem 1) Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC . The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$, $O\left(\frac{b+c}{2}, 0\right)$ where $b < 0 < c$ and $a > 0$. Note that M and N are just the feet of the altitudes from C and B respectively. So we have $M\left(\frac{b(a^2+bc)}{a^2+b^2}, \frac{ab(b-c)}{a^2+b^2}\right)$ and $N\left(\frac{c(a^2+bc)}{a^2+c^2}, \frac{ac(c-b)}{a^2+c^2}\right)$.

To find the internal angle bisector of $\angle BAC$, we notice that it cuts the side BC in the ratio $AB : AC = \sqrt{a^2+b^2} : \sqrt{a^2+c^2}$. So the foot of angle bisector from A has coordinates

$$\left(\frac{c\sqrt{a^2+b^2} + b\sqrt{a^2+c^2}}{\sqrt{a^2+b^2} + \sqrt{a^2+c^2}}, 0\right).$$

Note that

$$\begin{aligned}
\frac{c\sqrt{a^2+b^2}+b\sqrt{a^2+c^2}}{\sqrt{a^2+b^2}+\sqrt{a^2+c^2}} &= \frac{c\sqrt{a^2+b^2}+b\sqrt{a^2+c^2}}{\sqrt{a^2+b^2}+\sqrt{a^2+c^2}} \times \frac{\sqrt{a^2+b^2}-\sqrt{a^2+c^2}}{\sqrt{a^2+b^2}-\sqrt{a^2+c^2}} \\
&= \frac{c(a^2+b^2)-b(a^2+c^2)+(b-c)\sqrt{(a^2+b^2)(a^2+c^2)}}{b^2-c^2} \\
&= \frac{-a^2+bc+\sqrt{(a^2+b^2)(a^2+c^2)}}{b+c}.
\end{aligned}$$

Thus, we get

$$AR: \quad y = \frac{a(b+c)}{a^2-bc-\sqrt{(a^2+b^2)(a^2+c^2)}}x + a,$$

i.e.

$$AR: \quad y = \frac{-a^2+bc-\sqrt{(a^2+b^2)(a^2+c^2)}}{a(b+c)}x + a. \quad (1)$$

For the internal angle bisector of $\angle MON$, since $OM = ON$, it is simply the perpendicular bisector of MN . From

$$\begin{aligned}
m_{OR} &= \frac{-1}{m_{MN}} = \frac{b(a^2+bc)(a^2+c^2)-c(a^2+bc)(a^2+b^2)}{ac(c-b)(a^2+b^2)-ab(b-c)(a^2+c^2)} \\
&= \frac{-(a^2+bc)(c-b)(a^2-bc)}{a(c-b)(b+c)(a^2+bc)} = -\frac{a^2-bc}{a(b+c)},
\end{aligned}$$

we get

$$OR: \quad y = -\frac{a^2-bc}{a(b+c)}x + \frac{a^2-bc}{2a}. \quad (2)$$

We solve for x using (1) and (2). This yields

$$x = \frac{(b+c)((a^2-bc)-2a^2)}{2(a^2-bc-a^2+bc-\sqrt{(a^2+b^2)(a^2+c^2)})} = \frac{(b+c)(a^2+bc)}{2\sqrt{(a^2+b^2)(a^2+c^2)}}.$$

Therefore, we can write $R\left(t, \frac{(a^2-bc)(-2t+b+c)}{2a(b+c)}\right)$ where

$$t = \frac{(b+c)(a^2+bc)}{2\sqrt{(a^2+b^2)(a^2+c^2)}}. \quad (3)$$

Next, from $m_{MB} = m_{AB} = -\frac{a}{b}$, the slope of the perpendicular bisector of MB is $\frac{b}{a}$.

As this line passes through the midpoint $\left(\frac{b(2a^2+b^2+bc)}{2(a^2+b^2)}, \frac{ab(b-c)}{2(a^2+b^2)}\right)$ of BM (or one may notice that O also lies on this line), its equation is

$$y = \frac{b}{a}x - \frac{b(b+c)}{2a}. \quad (4)$$

Similarly, the slope of the perpendicular bisector of BR is $\frac{2a(b+c)(t-b)}{(a^2-bc)(2t-b-c)}$, and it passes through the midpoint $\left(\frac{b+t}{2}, \frac{(a^2-bc)(-2t+b+c)}{4a(b+c)}\right)$ of BR . So we compute

$$\begin{aligned} & \frac{(a^2-bc)(-2t+b+c)}{4a(b+c)} - \frac{2a(b+c)(t-b)}{(a^2-bc)(2t-b-c)} \cdot \frac{b+t}{2} \\ &= -\frac{4(a^2+b^2)(a^2+c^2)t^2 - 4(b+c)(a^2-bc)^2t + (b+c)^2(a^4 - 4a^2b^2 - 2a^2bc + b^2c^2)}{4a(b+c)(a^2-bc)(2t-b-c)}. \end{aligned}$$

Hence, the equation of this line is

$$\begin{aligned} y &= \frac{2a(b+c)(t-b)}{(a^2-bc)(2t-b-c)}x \\ &\quad - \frac{4(a^2+b^2)(a^2+c^2)t^2 - 4(b+c)(a^2-bc)^2t + (b+c)^2(a^4 - 4a^2b^2 - 2a^2bc + b^2c^2)}{4a(b+c)(a^2-bc)(2t-b-c)}. \end{aligned} \quad (5)$$

We now solve (4) and (5). We have

$$\begin{aligned} x &= \frac{2b(b+c)^2(a^2-bc)(2t-b-c) - 4(a^2+b^2)(a^2+c^2)t^2 + 4(b+c)(a^2-bc)^2t - (b+c)^2(a^4 - 4a^2b^2 - 2a^2bc + b^2c^2)}{4(b+c)(b(a^2-bc)(2t-b-c) - 2a^2(b+c)(t-b))} \\ &= \frac{-4(a^2+b^2)(a^2+c^2)t^2 + 4(b+c)(a^2-bc)(a^2+b^2)t - (b+c)^2(a^4 - 2a^2b^2 - 2b^3c - b^2c^2)}{4(b+c)(-2c(a^2+b^2)t + b(b+c)(a^2+bc))}. \end{aligned}$$

This is the x -coordinate of the circumcentre O' of $\triangle BMR$. Let D be the second intersection of (BMR) and BC . Since O' lies on the perpendicular bisector of BD , the x -coordinate of D is

$$\begin{aligned} & \frac{-4(a^2+b^2)(a^2+c^2)t^2 + 4(b+c)(a^2-bc)(a^2+b^2)t - (b+c)^2(a^4 - 2a^2b^2 - 2b^3c - b^2c^2)}{2(b+c)(-2c(a^2+b^2)t + b(b+c)(a^2+bc))} - b \\ &= \frac{-4(a^2+b^2)(a^2+c^2)t^2 + 4(b+c)(a^2-bc)(a^2+b^2)t - (b+c)^2(a^4 - 2a^2b^2 - 2b^3c - b^2c^2) + 4bc(b+c)(a^2+b^2)t - 2b^2(b+c)^2(a^2+bc)}{2(b+c)(-2c(a^2+b^2)t + b(b+c)(a^2+bc))} \\ &= \frac{-4(a^2+b^2)(a^2+c^2)t^2 + 4a^2(b+c)(a^2+b^2)t - (b+c)^2(a^4 - b^2c^2)}{2(b+c)(-2c(a^2+b^2)t + b(b+c)(a^2+bc))}. \end{aligned} \quad (6)$$

The x -coordinate of the second point of intersection of (CNR) and BC can be found by swapping b and c . We need to show that this is the same as (6) so that (BMR)

and (CNR) meet on the line BC . Thus, we need to show

$$\begin{aligned} & \left(-4(a^2 + b^2)(a^2 + c^2)t^2 + 4a^2(b + c)(a^2 + b^2)t - (b + c)^2(a^4 - b^2c^2) \right) \\ & \quad \times \left(-2b(a^2 + c^2)t + c(b + c)(a^2 + bc) \right) \\ & - \left(-4(a^2 + b^2)(a^2 + c^2)t^2 + 4a^2(b + c)(a^2 + c^2)t - (b + c)^2(a^4 - b^2c^2) \right) \\ & \quad \times \left(-2c(a^2 + b^2)t + b(b + c)(a^2 + bc) \right) = 0. \end{aligned}$$

After expansion, simplification and removing the obvious factor $(b - c)(a^2 - bc)$ (the factor $b - c$ is obvious, while the factor $a^2 - bc$ will appear after expansion), it suffices to prove

$$\begin{aligned} & 8(a^2 + b^2)(a^2 + c^2)t^3 - 4(a^2 + b^2)(a^2 + c^2)(b + c)t^2 \\ & - 2(b + c)^2(a^2 + bc)^2t + (b + c)^3(a^2 + bc)^2 = 0. \end{aligned}$$

This is the same as

$$(2t - b - c)(4(a^2 + b^2)(a^2 + c^2)t^2 - (b + c)^2(a^2 + bc)^2) = 0.$$

The second factor is 0 by (3).

According to the marking scheme, we also need to show that the intersection point does lie on the side BC (instead of the line BC only). Thus, we first show that the x -coordinate of D , which is given by (6), is larger than b . Therefore, we consider

$$\begin{aligned} & \frac{-4(a^2 + b^2)(a^2 + c^2)t^2 + 4a^2(b + c)(a^2 + b^2)t - (b + c)^2(a^4 - b^2c^2)}{2(b + c)(-2c(a^2 + b^2)t + b(b + c)(a^2 + bc))} - b \\ & = \frac{-4(a^2 + b^2)(a^2 + c^2)t^2 + 4(b + c)(a^2 + b^2)(a^2 + bc)t - (b + c)^2(a^2 + bc)(a^2 - bc + 2b^2)}{2(b + c)(-2c(a^2 + b^2)t + b(b + c)(a^2 + bc))}. \end{aligned}$$

By (3), we have

$$4(a^2 + b^2)(a^2 + c^2)t^2 = (b + c)^2(a^2 + bc)^2.$$

So this becomes

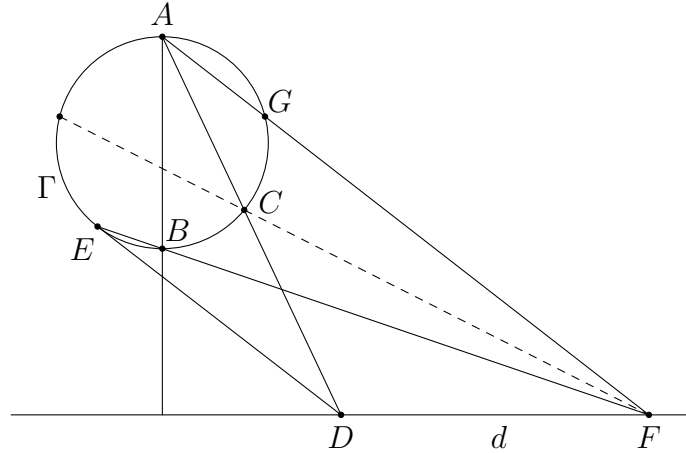
$$\begin{aligned} & \frac{4(b + c)(a^2 + b^2)(a^2 + bc)t - 2(b + c)^2(a^2 + bc)(a^2 + b^2)}{2(b + c)(-2c(a^2 + b^2)t + b(b + c)(a^2 + bc))} \\ & = \frac{(a^2 + b^2)(a^2 + bc)(2t - b - c)}{-2c(a^2 + b^2)t + b(b + c)(a^2 + bc)} \\ & = \frac{(a^2 + b^2)(a^2 + bc)(b + c)(a^2 + bc - \sqrt{(a^2 + b^2)(a^2 + c^2)})}{(b + c)(a^2 + bc)(-c(a^2 + b^2) + b\sqrt{(a^2 + b^2)(a^2 + c^2)})} \\ & = \frac{(a^2 + b^2)(a^2 + bc - \sqrt{(a^2 + b^2)(a^2 + c^2)})}{-c(a^2 + b^2) + b\sqrt{(a^2 + b^2)(a^2 + c^2)}} \\ & = \frac{(a^2 + bc - \sqrt{(a^2 + b^2)(a^2 + c^2)})(-c(a^2 + b^2) - b\sqrt{(a^2 + b^2)(a^2 + c^2)})}{a^2(c^2 - b^2)} \end{aligned}$$

$$= \frac{-(a^2 + b^2) + \sqrt{(a^2 + b^2)(a^2 + c^2)}}{b + c} = \frac{(a^2 + b^2)(c - b)}{a^2 + b^2 + \sqrt{(a^2 + b^2)(a^2 + c^2)}}.$$

This must be positive. Thus, D lies on the ray BC . By symmetry, D also lies on the ray CB . This shows D lies on the side BC . $\square$

Problem 23. (IMO Shortlist 2004 G2) Let Γ be a circle and let d be a line such that Γ and d have no common points. Further, let AB be a diameter of the circle Γ ; assume that this diameter AB is perpendicular to the line d , and the point B is nearer to the line d than the point A . Let C be an arbitrary point on the circle Γ , different from the points A and B . Let D be the point of intersection of the lines AC and d . One of the two tangents from the point D to the circle Γ touches this circle Γ at a point E ; hereby, we assume that the points B and E lie in the same halfplane with respect to the line AC . Denote by F the point of intersection of the lines BE and d . Let the line AF intersect the circle Γ at a point G , different from A .

Prove that the reflection of the point G in the line AB lies on the line CF .



Solution. Set the intersection point of d and AB as the origin. Let $A(0, 2a)$, $B(0, 2b)$, $D(c, 0)$ so that d is the x -axis. The centre O of Γ has coordinates $(0, a + b)$. So

$$\Gamma: x^2 + y^2 - 2(a + b)y + 4ab = 0. \quad (1)$$

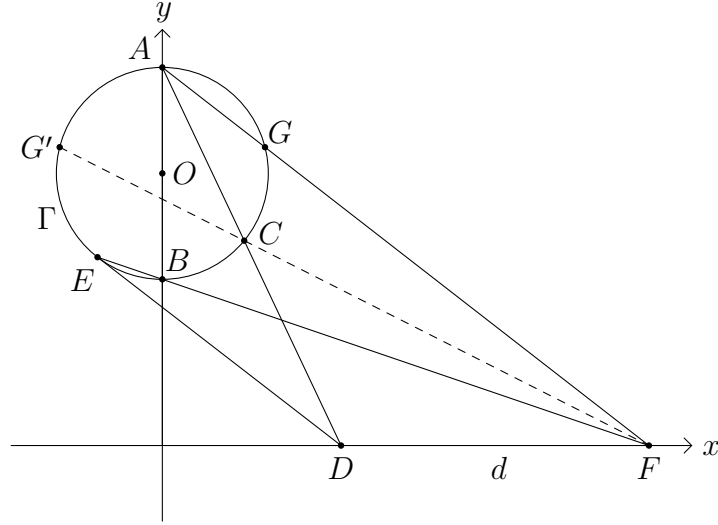
We have

$$AD: y = -\frac{2a}{c}x + 2a. \quad (2)$$

As $BC \perp AD$, we easily get

$$BC: y = \frac{c}{2a}x + 2b. \quad (3)$$

Solving (2) and (3), we get $C\left(\frac{4ac(a - b)}{4a^2 + c^2}, \frac{2a(4ab + c^2)}{4a^2 + c^2}\right)$.



Consider the point E . From $OE \perp DE$, we need $m_{OE} \times m_{DE} = -1$, i.e.

$$\frac{y - a - b}{x} \times \frac{y}{x - c} = -1.$$

After expansion, this becomes

$$x^2 + y^2 - cx - (a + b)y = 0. \quad (4)$$

We take the difference of (1) and (4) to get

$$cx - (a + b)y + 4ab = 0.$$

This means

$$y = \frac{c}{a + b}x + \frac{4ab}{a + b}.$$

Putting this in (1), we solve

$$\left(1 + \frac{c^2}{(a + b)^2}\right)x^2 + \left(\frac{8abc}{(a + b)^2} - 2c\right)x + \left(\frac{16a^2b^2}{(a + b)^2} - 8ab + 4ab\right) = 0.$$

This can be simplified to

$$((a + b)^2 + c^2)x^2 - 2c(a - b)^2x - 4ab(a - b)^2 = 0.$$

For simplicity, we write $E\left(t, \frac{ct + 4ab}{a + b}\right)$ where t satisfies

$$((a + b)^2 + c^2)t^2 - 2c(a - b)^2t - 4ab(a - b)^2 = 0. \quad (5)$$

Then we easily get

$$BE: \quad y = \frac{ct + 2b(a - b)}{(a + b)t}x + 2b.$$

This yields $F\left(-\frac{2b(a+b)t}{ct+2b(a-b)}, 0\right)$. Hence,

$$AF: \quad y = \frac{a(ct+2b(a-b))}{b(a+b)t}x + 2a.$$

Putting this in (1), we solve

$$\left(1 + \frac{a^2(ct+2b(a-b))^2}{b^2(a+b)^2t^2}\right)x^2 + \left(\frac{4a^2(ct+2b(a-b))}{b(a+b)t} - \frac{2a(ct+2b(a-b))}{bt}\right)x + (4a^2 - 4a(a+b) + 4ab) = 0.$$

This means

$$\left(b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2\right)x^2 + 2ab(a^2 - b^2)t(ct+2b(a-b))x = 0.$$

We reject the root $x = 0$ which corresponds to the point A . For the other root

$$x = -\frac{2ab(a^2 - b^2)t(ct+2b(a-b))}{b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2},$$

we compute

$$\begin{aligned} y &= \frac{a(ct+2b(a-b))}{b(a+b)t} \cdot \left(-\frac{2ab(a^2 - b^2)t(ct+2b(a-b))}{b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2}\right) + 2a \\ &= \frac{2ab(b(a+b)^2t^2 + a(ct+2b(a-b))^2)}{b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2}. \end{aligned}$$

This pair of x, y gives the coordinates of G . Then the reflection point G' of G in the y -axis is

$$G' \left(\frac{2ab(a^2 - b^2)t(ct+2b(a-b))}{b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2}, \frac{2ab(b(a+b)^2t^2 + a(ct+2b(a-b))^2)}{b^2(a+b)^2t^2 + a^2(ct+2b(a-b))^2} \right).$$

Now, from $C\left(\frac{4ac(a-b)}{4a^2+c^2}, \frac{2a(4ab+c^2)}{4a^2+c^2}\right)$ and $F\left(-\frac{2b(a+b)t}{ct+2b(a-b)}, 0\right)$, we compute

$$\begin{aligned} m_{CF} &= \frac{2a(4ab+c^2)(ct+2b(a-b))}{4ac(a-b)(ct+2b(a-b)) + 2b(a+b)(4a^2+c^2)t} \\ &= \frac{a(4ab+c^2)(ct+2b(a-b))}{(2a^2c^2 + 4a^3b + 4a^2b^2 - abc^2 + b^2c^2)t + 4abc(a-b)^2}. \end{aligned}$$

Also, we have

$$\begin{aligned}
& m_{FG'} \\
&= \frac{2ab(b(a+b)^2t^2 + a(ct + 2b(a-b))^2)(ct + 2b(a-b))}{2ab(a^2 - b^2)t(ct + 2b(a-b))^2 + 2b(a+b)t(b^2(a+b)^2t^2 + a^2(ct + 2b(a-b))^2)} \\
&= \frac{a((a^2b + 2ab^2 + b^3 + ac^2)t^2 + 4abc(a-b)t + 4ab^2(a-b)^2)(ct + 2b(a-b))}{(a+b)t((a(2a-b)c^2 + b^2(a+b)^2)t^2 + 4abc(a-b)(2a-b)t + 4ab^2(a-b)^2(2a-b))}.
\end{aligned}$$

We want to show that $m_{CF} = m_{FG'}$. Removing the common factor $a(ct + 2b(a-b))$ in the two terms, we expand everything and simplify. Then we need to prove

$$\begin{aligned}
& 2ab(a-b)(c^4 + (a^2 - b^2)c^2 - 2b(a+b)^3)t^3 \\
& + 4abc(a-b)^2((2b-a)c^2 + b(a+b)(3a-b))t^2 \\
& + 8ab^2(a-b)^3((b-2a)c^2 + 2ab(a+b))t - 16a^2b^3c(a-b)^4 = 0,
\end{aligned}$$

i.e.

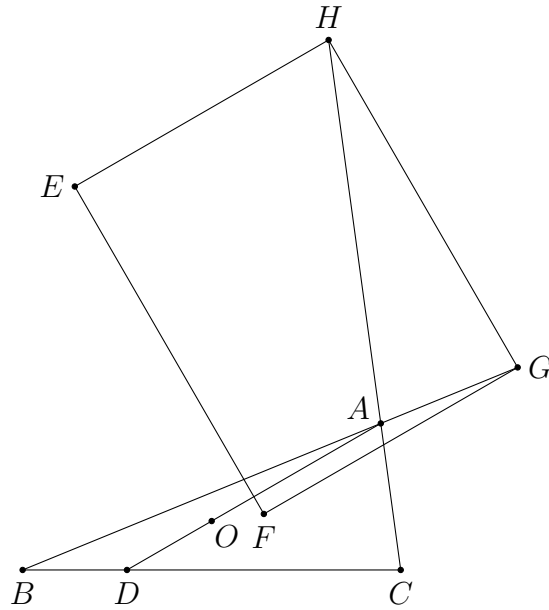
$$\begin{aligned}
& (c^4 + (a^2 - b^2)c^2 - 2b(a+b)^3)t^3 + 2c(a-b)((2b-a)c^2 + b(a+b)(3a-b))t^2 \\
& + 4b(a-b)^2((b-2a)c^2 + 2ab(a+b))t - 8ab^2c(a-b)^3 = 0.
\end{aligned}$$

We use the condition (5) repeatedly. Then

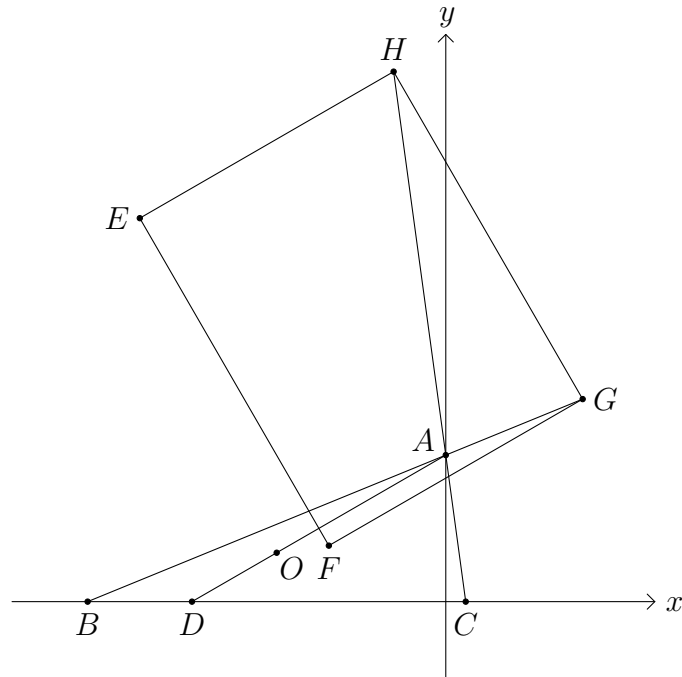
$$\begin{aligned}
& (c^4 + (a^2 - b^2)c^2 - 2b(a+b)^3)t^3 + 2c(a-b)((2b-a)c^2 + b(a+b)(3a-b))t^2 \\
& + 4b(a-b)^2((b-2a)c^2 + 2ab(a+b))t - 8ab^2c(a-b)^3 \\
&= (c^2 + (a+b)^2)(c^2 - 2b(a+b))t^3 + 2c(a-b)((2b-a)c^2 + b(a+b)(3a-b))t^2 \\
& + 4b(a-b)^2((b-2a)c^2 + 2ab(a+b))t - 8ab^2c(a-b)^3 \\
&= 2(a-b)^2(ct + 2ab)(c^2 - 2b(a+b))t + 2c(a-b)((2b-a)c^2 + b(a+b)(3a-b))t^2 \\
& + 4b(a-b)^2((b-2a)c^2 + 2ab(a+b))t - 8ab^2c(a-b)^3 \\
&= 2bc(a-b)(c^2 + (a+b)^2)t^2 - 4b(a-b)^3c^2t - 8ab^2c(a-b)^3 \\
&= 0.
\end{aligned}$$

This implies F, C, G' are collinear. □

Problem 24. (IMO Shortlist 2004 G3) Let O be the circumcentre of an acute-angled triangle ABC with $\angle B < \angle C$. The line AO meets the side BC at D . The circumcentres of the triangles ABD and ACD are E and F , respectively. Extend the sides BA and CA beyond A , and choose on the respective extensions points G and H such that $AG = AC$ and $AH = AB$. Prove that the quadrilateral $EFGH$ is a rectangle if and only if $\angle ACB - \angle ABC = 60^\circ$.



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$. From

$$m_{AO} = \frac{(a^2+bc) - 2a^2}{a(b+c)} = \frac{bc-a^2}{a(b+c)},$$

we easily get

$$AO: \quad y = \frac{bc - a^2}{a(b + c)}x + a.$$

It follows that $D\left(\frac{a^2(b + c)}{a^2 - bc}, 0\right)$. The x -coordinate of E is

$$\frac{1}{2}\left(b + \frac{a^2(b + c)}{a^2 - bc}\right) = \frac{2a^2b + a^2c - b^2c}{2(a^2 - bc)}.$$

The perpendicular bisector of AD has slope $\frac{-1}{m_{AD}} = \frac{a(b + c)}{a^2 - bc}$. It passes through the midpoint of AD , whose coordinates are $\left(\frac{a^2(b + c)}{2(a^2 - bc)}, \frac{a}{2}\right)$. So its equation is

$$y = \frac{a(b + c)}{a^2 - bc}x - \frac{a(b + c)}{a^2 - bc} \cdot \frac{a^2(b + c)}{2(a^2 - bc)} + \frac{a}{2},$$

i.e.

$$y = \frac{a(b + c)}{a^2 - bc}x - \frac{a^3(b + c)^2 - a(a^2 - bc)^2}{2(a^2 - bc)^2}.$$

Putting $x = \frac{2a^2b + a^2c - b^2c}{2(a^2 - bc)}$, we obtain

$$\begin{aligned} y &= \frac{a(b + c)}{a^2 - bc} \cdot \frac{2a^2b + a^2c - b^2c}{2(a^2 - bc)} - \frac{a^3(b + c)^2 - a(a^2 - bc)^2}{2(a^2 - bc)^2} \\ &= \frac{a(b + c)(a^2b - b^2c) + a(a^2 - bc)^2}{2(a^2 - bc)^2} = \frac{a(a^2 + b^2)}{2(a^2 - bc)}. \end{aligned}$$

So $E\left(\frac{2a^2b + a^2c - b^2c}{2(a^2 - bc)}, \frac{a(a^2 + b^2)}{2(a^2 - bc)}\right)$. By symmetry, $F\left(\frac{a^2b + 2a^2c - bc^2}{2(a^2 - bc)}, \frac{a(a^2 + c^2)}{2(a^2 - bc)}\right)$.

We proceed to find G and H . Firstly, as the equation of AB is $y = -\frac{a}{b}x + a$, we let

$G\left(g, \frac{a(b - g)}{b}\right)$. Then

$$\begin{aligned} &AG = AC \\ \Rightarrow &g^2 + \left(\frac{ag}{b}\right)^2 = a^2 + c^2 \\ \Rightarrow &g^2 = \frac{b^2(a^2 + c^2)}{a^2 + b^2}. \end{aligned}$$

From the configuration, we should choose the larger root. So $g = \sqrt{\frac{b^2(a^2 + c^2)}{a^2 + b^2}} = -b\Delta$ where

$$\Delta = \sqrt{\frac{a^2 + c^2}{a^2 + b^2}}.$$

Similarly, we can write $H\left(h, \frac{a(c-h)}{c}\right)$ where

$$h = -\sqrt{\frac{c^2(a^2 + b^2)}{a^2 + c^2}} = -\frac{c}{\Delta}.$$

We find the slopes of the sidelines of $EFGH$ one by one. (In fact, one may use the fact $EF \perp AO$ to find m_{EF} .)

$$\begin{aligned} m_{EF} &= \frac{a(a^2 + c^2) - a(a^2 + b^2)}{(a^2b + 2a^2c - bc^2) - (2a^2b + a^2c - b^2c)} \\ &= \frac{a(c^2 - b^2)}{(c - b)(a^2 - bc)} = \frac{a(b + c)}{a^2 - bc}, \\ m_{FG} &= \frac{2a(b - g)(a^2 - bc) - ab(a^2 + c^2)}{2b(a^2 - bc)g - b(a^2b + 2a^2c - bc^2)} \\ &= \frac{ab(a^2 - 2bc - c^2) - 2a(a^2 - bc)g}{-b(a^2b + 2a^2c - bc^2) + 2b(a^2 - bc)g} \\ &= \frac{a(a^2 - 2bc - c^2) + 2a(a^2 - bc)\Delta}{-(a^2b + 2a^2c - bc^2) - 2b(a^2 - bc)\Delta}, \\ m_{GH} &= \frac{a(b(c - h) - c(b - g))}{bc(h - g)} = \frac{a(cg - bh)}{bc(h - g)} = \frac{a(-bc\Delta^2 + bc)}{bc(-c + b\Delta^2)} \\ &= \frac{a(1 - \Delta^2)}{-c + b\Delta^2} = \frac{a((a^2 + b^2) - (a^2 + c^2))}{-c(a^2 + b^2) + b(a^2 + c^2)} = \frac{a(b + c)}{a^2 - bc}, \\ m_{HE} &= \frac{2a(c - h)(a^2 - bc) - ac(a^2 + b^2)}{2c(a^2 - bc)h - c(2a^2b + a^2c - b^2c)} \\ &= \frac{ac(a^2 - 2bc - b^2) - 2a(a^2 - bc)h}{-c(2a^2b + a^2c - b^2c) + 2c(a^2 - bc)h} \\ &= \frac{a(a^2 - 2bc - b^2)\Delta + 2a(a^2 - bc)}{-(2a^2b + a^2c - b^2c)\Delta - 2c(a^2 - bc)}. \end{aligned}$$

Notice that $m_{EF} = m_{GH}$. This means $EF \parallel GH$. Therefore, $EFGH$ is a rectangle iff $EF \perp FG$ and $EF \perp HE$. Firstly,

$$\begin{aligned} &m_{EF} \cdot m_{FG} = -1 \\ \Leftrightarrow &\frac{a(b + c)}{a^2 - bc} \cdot \frac{a(a^2 - 2bc - c^2) + 2a(a^2 - bc)\Delta}{-(a^2b + 2a^2c - bc^2) - 2b(a^2 - bc)\Delta} = -1 \\ \Leftrightarrow &a^2(b + c)(a^2 - 2bc - c^2) + 2a^2(b + c)(a^2 - bc)\Delta \\ &\quad - (a^2 - bc)(a^2b + 2a^2c - bc^2) - 2b(a^2 - bc)^2\Delta = 0 \\ \Leftrightarrow &-(a^2 + b^2)(a^2 + c^2)c + 2(a^2 - bc)(a^2 + b^2)c\Delta = 0 \\ \Leftrightarrow &-(a^2 + c^2) + 2(a^2 - bc)\Delta = 0 \\ \Leftrightarrow &2(a^2 - bc) = \sqrt{(a^2 + b^2)(a^2 + c^2)}. \end{aligned}$$

Secondly,

$$\begin{aligned}
& m_{EF} \cdot m_{HE} = -1 \\
\Leftrightarrow & \frac{a(b+c)}{a^2-bc} \cdot \frac{a(a^2-2bc-b^2)\Delta + 2a(a^2-bc)}{-(2a^2b+a^2c-b^2c)\Delta - 2c(a^2-bc)} = -1 \\
\Leftrightarrow & a^2(b+c)(a^2-2bc-b^2)\Delta + 2a^2(b+c)(a^2-bc) \\
& \quad - (a^2-bc)(2a^2b+a^2c-b^2c)\Delta - 2c(a^2-bc)^2 = 0 \\
\Leftrightarrow & -(a^2+b^2)(a^2+c^2)b\Delta + 2(a^2-bc)(a^2+c^2)b = 0 \\
\Leftrightarrow & -(a^2+b^2)\Delta + 2(a^2-bc) = 0 \\
\Leftrightarrow & 2(a^2-bc) = \sqrt{(a^2+b^2)(a^2+c^2)}.
\end{aligned}$$

The two expressions are the same. Therefore,

$$\begin{aligned}
& EFGH \text{ is a rectangle} \\
\Leftrightarrow & 2(a^2-bc) = \sqrt{(a^2+b^2)(a^2+c^2)} \\
\Leftrightarrow & 4(a^2-bc)^2 = (a^2+b^2)(a^2+c^2) \\
\Leftrightarrow & 3a^4 - a^2b^2 - 8a^2bc - a^2c^2 + 3b^2c^2 = 0. \tag{1}
\end{aligned}$$

Here, we have used the fact that $a^2 - bc > 0$ since we may assume $b < 0 < c$.

It remains to show that $\angle ACB - \angle ABC = 60^\circ$ is equivalent to this relation. Indeed,

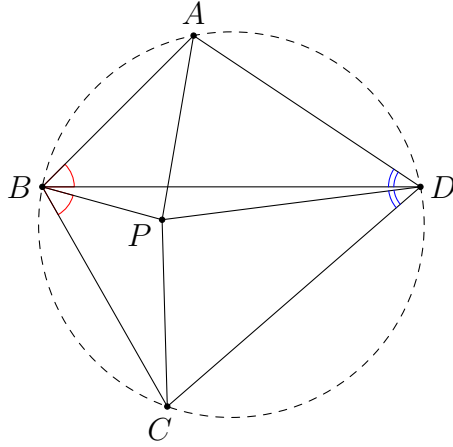
$$\begin{aligned}
& \angle ACB - \angle ABC = 60^\circ \\
\Leftrightarrow & \tan(\angle ACB - \angle ABC) = \sqrt{3} \\
& \frac{\frac{a}{c} + \frac{a}{b}}{1 - \frac{a^2}{bc}} = \sqrt{3} \\
\Leftrightarrow & \frac{a(b+c)}{bc-a^2} = \sqrt{3} \\
\Leftrightarrow & a^2(b+c)^2 = 3(bc-a^2)^2 \\
\Leftrightarrow & 3a^4 - a^2b^2 - 8a^2bc - a^2c^2 + 3b^2c^2 = 0.
\end{aligned}$$

This is the same as (1). So the equivalence is established. $\square$

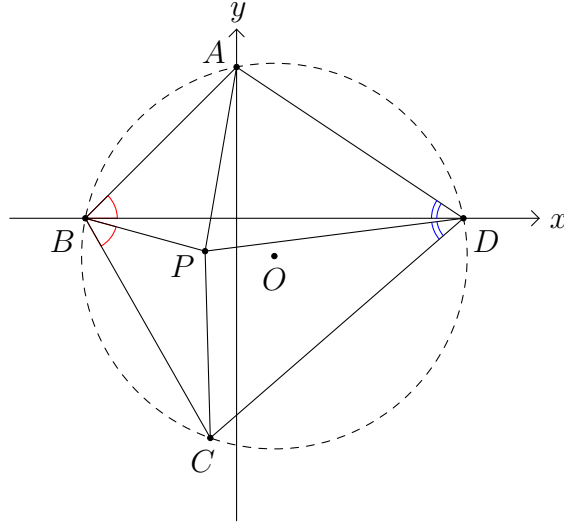
Problem 25. (IMO Shortlist 2004 G4/Problem 5) In a convex quadrilateral $ABCD$ the diagonal BD does not bisect the angles ABC and CDA . The point P lies inside $ABCD$ and satisfies

$$\angle PBC = \angle DBA \quad \text{and} \quad \angle PDC = \angle BDA.$$

Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.



Solution.



Let $A(0, a)$, $B(b, 0)$, $D(d, 0)$, $C(s, t)$. Then the circumcentre O of $\triangle ABD$ has coordinates $\left(\frac{b+d}{2}, \frac{a^2+bd}{2a}\right)$ and

$$(ABD): \quad x^2 + y^2 - (b+d)x - \frac{a^2+bd}{a}y + bd = 0.$$

Therefore, A, B, C, D are concyclic iff

$$s^2 + t^2 - (b+d)s - \frac{a^2+bd}{a}t + bd = 0. \quad (1)$$

WLOG assume P lies below the x -axis. Note that

$$m_{BP} = -\tan \angle DBP = -\tan (\angle DBC - \angle PBC) = -\tan (\angle DBC - \angle ABD)$$

$$= -\frac{(-m_{BC}) - m_{AB}}{1 + (-m_{BC})m_{AB}} = -\frac{\left(-\frac{t}{s-b}\right) - \left(-\frac{a}{b}\right)}{1 + \frac{at}{b(s-b)}} = \frac{bt - a(s-b)}{b(s-b) + at}.$$

This gives

$$BP: \quad y = \frac{bt - as + ab}{bs - b^2 + at}x - \frac{b(bt - as + ab)}{bs - b^2 + at}.$$

By symmetry, we have

$$DP: \quad y = \frac{dt - as + ad}{ds - d^2 + at}x - \frac{d(dt - as + ad)}{ds - d^2 + at}.$$

Solving these, we find that

$$\begin{aligned} x &= \frac{b(bt - as + ab)(ds - d^2 + at) - d(dt - as + ad)(bs - b^2 + at)}{(bt - as + ab)(ds - d^2 + at) - (dt - as + ad)(bs - b^2 + at)} \\ &= \frac{(b-d)(bdst + a(b+d)t^2 - a^2st + a^2(b+d)t)}{(b-d)(bdt + at^2 + as^2 - a(b+d)s + abd + a^2t)} \\ &= \frac{(bd - a^2)st + a(b+d)t^2 + a^2(b+d)t}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{bt - as + ab}{bs - b^2 + at} \cdot \left(\frac{(bd - a^2)st + a(b+d)t^2 + a^2(b+d)t}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} - b \right) \\ &= \frac{bt - as + ab}{bs - b^2 + at} \cdot \frac{-abs^2 + (bd - a^2)st + adt^2 + ab(b+d)s + d(a^2 - b^2)t - ab^2d}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} \\ &= \frac{bt - as + ab}{bs - b^2 + at} \cdot \frac{(bs - b^2 + at)(dt - as + ad)}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} \\ &= \frac{(bt - as + ab)(dt - as + ad)}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd}. \end{aligned}$$

This gives

$$P \left(\frac{(bd - a^2)st + a(b+d)t^2 + a^2(b+d)t}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd}, \frac{(bt - as + ab)(dt - as + ad)}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} \right).$$

The slope of the perpendicular bisector of AC is $\frac{-1}{m_{AC}} = -\frac{s}{t-a}$. So its equation is

$$y = -\frac{s}{t-a}x + \frac{s}{t-a} \cdot \frac{s}{2} + \frac{a+t}{2},$$

i.e.

$$y = -\frac{s}{t-a}x + \frac{s^2 + t^2 - a^2}{2(t-a)}.$$

Note that $AP = CP$ iff P lies on this perpendicular bisector. This means

$$\frac{(bt - as + ab)(dt - as + ad)}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} = -\frac{s}{t-a} \cdot \frac{(bd - a^2)st + a(b+d)t^2 + a^2(b+d)t}{as^2 + at^2 - a(b+d)s + (a^2 + bd)t + abd} + \frac{s^2 + t^2 - a^2}{2(t-a)}.$$

Clearing denominators and expanding everything, this is equivalent to

$$as^4 + 2as^2t^2 + at^4 - a(b+d)s^3 + (a^2 - bd)s^2t - a(b+d)st^2 + (a^2 - bd)t^3 + a(a^2 + bd)s^2 - 2a^2(b+d)st - a(a^2 + bd)t^2 - a^3(b+d)s - a^2(a^2 - bd)t + a^3bd = 0.$$

In view of (1), we factorize this as

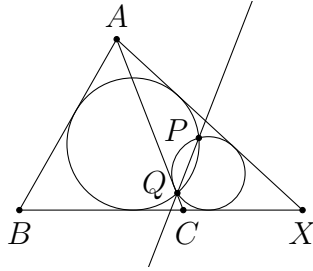
$$(as^2 + at^2 - a(b+d)s - (a^2 + bd)t + abd)(s^2 + t^2 + 2at + a^2) = 0.$$

The second factor is $s^2 + (t+a)^2 > 0$ (as $(s, t) \neq (0, -a)$ from the given condition). So $AP = CP$ iff

$$as^2 + at^2 - a(b+d)s - (a^2 + bd)t + abd = 0,$$

which is the same as (1). We have established the equivalence. $\square$

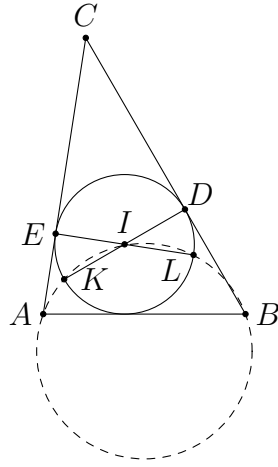
Problem 26. (IMO Shortlist 2004 G7) For a given triangle ABC , let X be a variable point on the line BC such that C lies between B and X and the incircles of the triangles ABX and ACX intersect at two distinct points P and Q . Prove that the line PQ passes through a point independent of X .



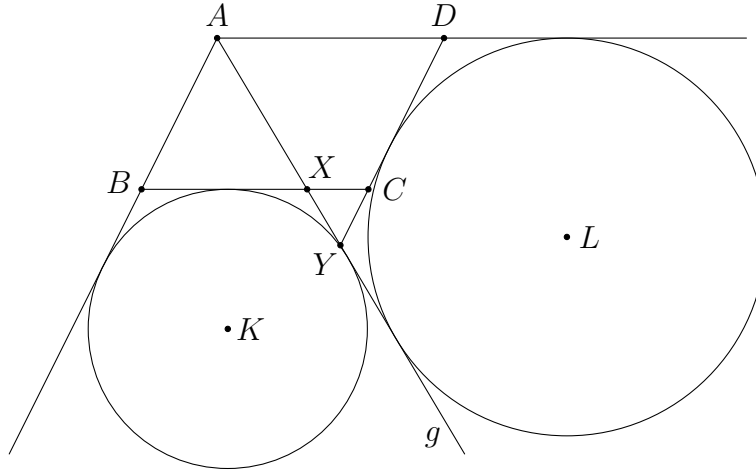
Solution. See [problem 5.6](#) of **Coordinate Geometry 5.0**. $\square$

Problem 27. (IMO Shortlist 2005 G1) Given a triangle ABC satisfying $AC + BC = 3AB$. The incircle of triangle ABC has centre I and touches the sides BC and CA at the points D and E , respectively. Let K and L be the reflections of the points D and E with respect to I . Prove that the points A, B, K, L lie on one circle.

Solution. See [problem 5.4](#) of **Coordinate Geometry 5.0**. $\square$



Problem 28. (IMO Shortlist 2005 G3) Let $ABCD$ be a parallelogram. A variable line g through the vertex A intersects the rays BC and DC at the points X and Y , respectively. Let K and L be the A -excentres of the triangles ABX and ADY . Show that the angle $\angle KCL$ is independent of the line g .

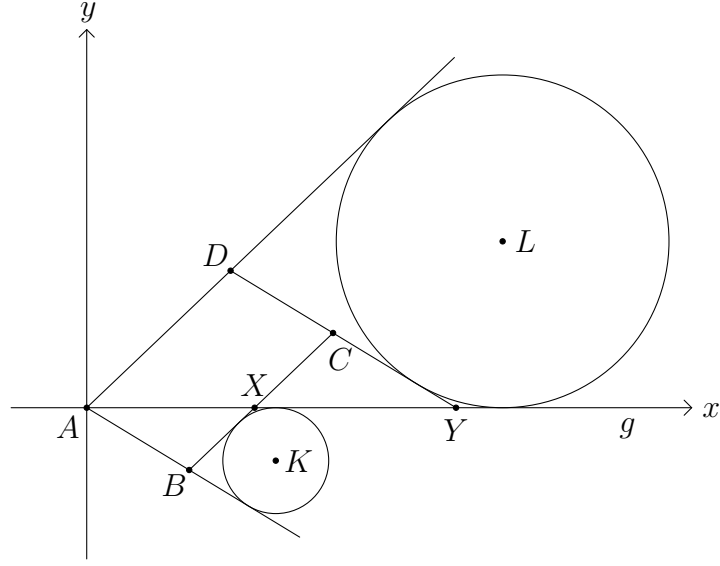


Solution. We set the origin at A and set g to be the x -axis. Let $K(a, b)$ and $L(c, d)$ where $a, c > 0$ and $b < 0 < d$. Then the radii of $\odot K$ and $\odot L$ are $-b$ and d respectively. This gives

$$\begin{aligned}\odot K : \quad x^2 + y^2 - 2ax - 2by + a^2 &= 0, \\ \odot L : \quad x^2 + y^2 - 2cx - 2dy + c^2 &= 0.\end{aligned}$$

Note that

$$m_{AD} = \tan \angle YAD = \tan 2\angle YAL = \frac{2m_{AL}}{1 - m_{AL}^2} = \frac{2cd}{c^2 - d^2}.$$



The line through K and parallel to AD has equation

$$2cdx - (c^2 - d^2)y = 2acd - (c^2 - d^2)b.$$

As the radius of $\odot K$ is $-b$, the line BC is obtained by translating the above line by $-b$ units in the parallel direction (towards the line AD). Since

$$\sqrt{(2cd)^2 + (d^2 - c^2)^2} = c^2 + d^2,$$

we have

$$BC : \quad 2cdx - (c^2 - d^2)y = 2acd - (c^2 - d^2)b - (-b)(c^2 + d^2),$$

i.e.

$$BC : \quad 2cdx - (c^2 - d^2)y = 2d(ac + bd). \quad (1)$$

By symmetry, we have

$$CD : \quad 2abx - (a^2 - b^2)y = 2b(ac + bd). \quad (2)$$

We now solve (1) and (2). Cancelling y , we obtain

$$2cd(a^2 - b^2)x - 2ab(c^2 - d^2)x = 2d(ac + bd)(a^2 - b^2) - 2b(ac + bd)(c^2 - d^2).$$

This implies

$$x = \frac{(ac + bd)(d(a^2 - b^2) - b(c^2 - d^2))}{cd(a^2 - b^2) - ab(c^2 - d^2)} = \frac{d(a^2 - b^2) - b(c^2 - d^2)}{ad - bc}.$$

Similarly, cancelling x , we obtain

$$cd(a^2 - b^2)y - ab(c^2 - d^2)y = 2abd(ac + bd) - 2bcd(ac + bd).$$

This implies

$$y = \frac{2bd(a-c)(ac+bd)}{cd(a^2-b^2) - ab(c^2-d^2)} = \frac{2bd(a-c)}{ad-bc}.$$

Thus, we have $C \left(\frac{d(a^2-b^2) - b(c^2-d^2)}{ad-bc}, \frac{2bd(a-c)}{ad-bc} \right)$. It is easy to compute

$$\begin{aligned} m_{CK} &= \frac{2bd(a-c) - b(ad-bc)}{d(a^2-b^2) - b(c^2-d^2) - a(ad-bc)} = \frac{ad+bc-2cd}{ac-bd-c^2+d^2}, \\ m_{CL} &= \frac{2bd(a-c) - d(ad-bc)}{d(a^2-b^2) - b(c^2-d^2) - c(ad-bc)} = \frac{ad+bc-2ab}{ac-bd-a^2+b^2}. \end{aligned}$$

It follows that

$$\begin{aligned} & \tan \angle KCL \\ &= \frac{m_{CL} - m_{CK}}{1 + m_{CL} \cdot m_{CK}} \\ &= \frac{(ad+bc-2ab)(ac-bd-c^2+d^2) - (ad+bc-2cd)(ac-bd-a^2+b^2)}{(ac-bd-a^2+b^2)(ac-bd-c^2+d^2) + (ad+bc-2ab)(ad+bc-2cd)} \\ &= \frac{-a^2bc + ab^2d + ac^2d - bc^3 + ad^3 - bcd^2 + 2abc^2 - 2abd^2 + a^3d - b^3c - 2a^2cd + 2b^2cd}{2a^2c^2 + 2b^2d^2 - ac^3 - bc^2d - acd^2 - bd^3 - a^3c - ab^2c - a^2bd - b^3d + 4abcd} \\ &= \frac{(ad-bc)(a^2+b^2+c^2+d^2-2ac-2bd)}{-(ac+bd)(a^2+b^2+c^2+d^2-2ac-2bd)} \\ &= -\frac{ad-bc}{ac+bd}. \end{aligned}$$

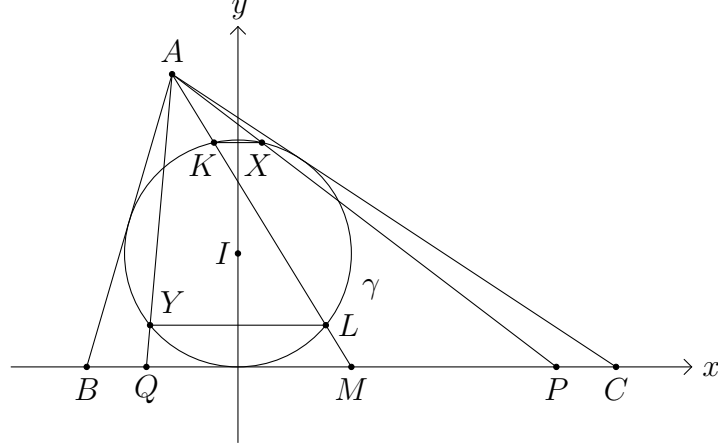
We wish to show that this can be expressed as some quantities that depend solely on $ABCD$. It is likely that this is related to the tangent of some angles. With some trial or with a good figure, one guess this is $-\tan \angle KAL = -\tan \frac{\angle BAD}{2}$. Indeed, it is easy to check that

$$\tan \angle KAL = \frac{m_{AL} - m_{AK}}{1 + m_{AL} \cdot m_{AK}} = \frac{\frac{d}{c} - \frac{b}{a}}{1 + \frac{bd}{ac}} = \frac{ad-bc}{ac+bd} = -\tan \angle KCL.$$

This proves $\angle KCL$ is independent of g . □

Problem 29. (IMO Shortlist 2005 G5) Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocentre of $\triangle ABC$, and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of $\triangle ABC$ and $\triangle ADE$.


$$\odot I: \quad x^2 + y^2 - 2y = 0. \quad (1)$$
$$AM: \quad y = \frac{4bc}{(b+c)(1-bc)}x - \frac{2bc}{1-bc}.$$
$$x^2 + \left(\frac{4bc}{(b+c)(1-bc)}x - \frac{2bc}{1-bc} \right)^2 - 2 \left(\frac{4bc}{(b+c)(1-bc)}x - \frac{2bc}{1-bc} \right) = 0.$$



This means

$$\left((b+c)^2(1-bc)^2 + 16b^2c^2\right)x^2 - 8bc(b+c)(1+bc)x + 4bc(b+c)^2 = 0. \quad (2)$$

As K and L lie on AM , we let

$$K\left(k, \frac{2bc(2k-b-c)}{(b+c)(1-bc)}\right), \quad L\left(\ell, \frac{2bc(2\ell-b-c)}{(b+c)(1-bc)}\right).$$

Then k and ℓ are the roots of (2). Thus, we have

$$k + \ell = \frac{8bc(b+c)(1+bc)}{(b+c)^2(1-bc)^2 + 16b^2c^2}, \quad (3)$$

$$k\ell = \frac{4bc(b+c)^2}{(b+c)^2(1-bc)^2 + 16b^2c^2}. \quad (4)$$

Since the perpendicular bisector of KX is the y -axis, we have $X\left(-k, \frac{2bc(2k-b-c)}{(b+c)(1-bc)}\right)$.

It follows from

$$\begin{aligned} m_{AX} &= \frac{2bc(b+c)(1-bc) - 2bc(bc+1)(2k-b-c)}{(b+c)(1-bc)((b+c) + (bc+1)k)} \\ &= \frac{4bc(-(bc+1)k + (b+c))}{(b+c)(1-bc)((bc+1)k + (b+c))} \end{aligned}$$

that

$$AX: \quad y = \frac{4bc(-(bc+1)k + (b+c))}{(b+c)(1-bc)((bc+1)k + (b+c))}x + \frac{2bc((3-bc)k - (b+c))}{(1-bc)((bc+1)k + (b+c))}.$$

Thus, we have $P\left(\frac{(b+c)((3-bc)k - (b+c))}{2((bc+1)k - (b+c))}, 0\right)$. The coordinates of Q can be found

by changing k to ℓ . Note that $BP = CQ$ iff the midpoint of PQ is M . Thus,

$$\begin{aligned}
& BP = CQ \\
\Leftrightarrow & \frac{(b+c)((3-bc)k - (b+c))}{2((bc+1)k - (b+c))} + \frac{(b+c)((3-bc)\ell - (b+c))}{2((bc+1)\ell - (b+c))} = b+c \\
\Leftrightarrow & \begin{aligned} & ((3-bc)k - (b+c))((bc+1)\ell - (b+c)) \\ & + ((3-bc)\ell - (b+c))((bc+1)k - (b+c)) \\ & - 2((bc+1)k - (b+c))((bc+1)\ell - (b+c)) = 0 \end{aligned} \\
\Leftrightarrow & 4(1-bc)(bc+1)k\ell - 2(b+c)(1-bc)(k+\ell) = 0.
\end{aligned}$$

Using (3) and (4), we have

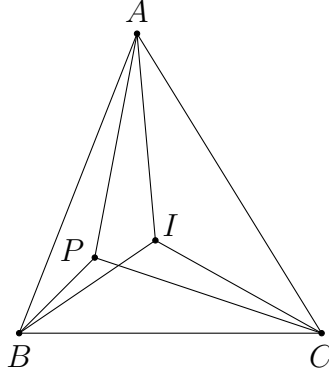
$$\begin{aligned}
& 4(1-bc)(bc+1)k\ell - 2(b+c)(1-bc)(k+\ell) \\
&= \frac{4(1-bc)(bc+1) \cdot 4bc(b+c)^2}{(b+c)^2(1-bc)^2 + 16b^2c^2} - \frac{2(b+c)(1-bc) \cdot 8bc(b+c)(1+bc)}{(b+c)^2(1-bc)^2 + 16b^2c^2} \\
&= 0.
\end{aligned}$$

So we are done. □

Problem 31. (IMO Shortlist 2006 G1/Problem 1) Let ABC be a triangle with incentre I . A point P in the interior of the triangle satisfies

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

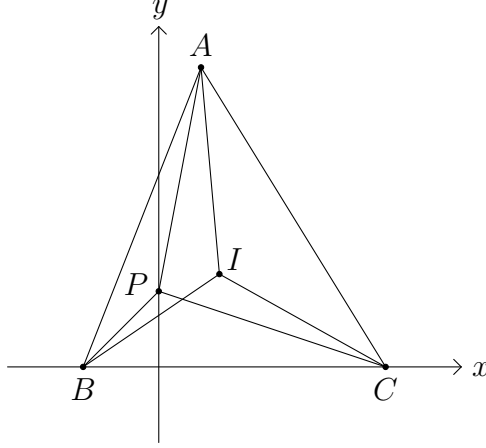
Show that $AP \geq AI$, and that equality holds if and only if $P = I$.



Solution. Let $B(b, 0)$, $C(c, 0)$, $P(0, p)$ where $p > 0$. Let $\angle PBC = \beta$ and $\angle PCB = \gamma$ so that

$$\tan \beta = -\frac{p}{b}, \quad \tan \gamma = \frac{p}{c}.$$

Since $\angle PBA + \angle PCA = \beta + \gamma$, we may let $\angle PBA = \beta + 2\theta$ and $\angle PCA = \gamma - 2\theta$ where θ can be negative. Then $\angle IBC = \frac{1}{2}\angle ABC = \beta + \theta$ and $\angle ICB = \frac{1}{2}\angle ACB = \gamma - \theta$.



Let $t = \tan \theta$. We get

$$m_{IB} = \tan(\beta + \theta) = \frac{\tan \beta + t}{1 - (\tan \beta)t} = \frac{bt - p}{pt + b}$$

and

$$m_{IC} = -\tan(\gamma - \theta) = -\frac{\tan \gamma - t}{1 + (\tan \gamma)t} = \frac{ct - p}{pt + c}.$$

These yield

$$\begin{aligned} IB: \quad y &= \frac{bt - p}{pt + b}x - \frac{b(bt - p)}{pt + b}, \\ IC: \quad y &= \frac{ct - p}{pt + c}x - \frac{c(ct - p)}{pt + c}. \end{aligned}$$

We solve these equations to find I . We have

$$\begin{aligned} x &= \frac{b(bt - p)(pt + c) - c(ct - p)(pt + b)}{(bt - p)(pt + c) - (ct - p)(pt + b)} \\ &= \frac{(b - c)t(bpt + cpt - p^2 + bc)}{(b - c)p(t^2 + 1)} = \frac{t(bpt + cpt - p^2 + bc)}{p(t^2 + 1)} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{bt - p}{pt + b} \cdot \left(\frac{t(bpt + cpt - p^2 + bc)}{p(t^2 + 1)} - b \right) \\ &= \frac{bt - p}{pt + b} \cdot \frac{cpt^2 - p^2t + bct - bp}{p(t^2 + 1)} = \frac{(bt - p)(ct - p)}{p(t^2 + 1)}. \end{aligned}$$

$$\text{So } I \left(\frac{t(bpt + cpt - p^2 + bc)}{p(t^2 + 1)}, \frac{(bt - p)(ct - p)}{p(t^2 + 1)} \right).$$

Next, we have

$$m_{AB} = \frac{2m_{IB}}{1 - m_{IB}^2} = \frac{2(bt - p)(pt + b)}{(pt + b)^2 - (bt - p)^2}.$$

This yields

$$AB: \quad y = \frac{2(bt-p)(pt+b)}{(pt+b)^2 - (bt-p)^2}x - \frac{2b(bt-p)(pt+b)}{(pt+b)^2 - (bt-p)^2}.$$

By symmetry, we have

$$AC: \quad y = \frac{2(ct-p)(pt+c)}{(pt+c)^2 - (ct-p)^2}x - \frac{2c(ct-p)(pt+c)}{(pt+c)^2 - (ct-p)^2}.$$

Let $A(x_A, y_A)$. Solving these equations, we get

$$\begin{aligned} x_A &= \frac{b(bt-p)(pt+b)((pt+c)^2 - (ct-p)^2) - c(ct-p)(pt+c)((pt+b)^2 - (bt-p)^2)}{(bt-p)(pt+b)((pt+c)^2 - (ct-p)^2) - (ct-p)(pt+c)((pt+b)^2 - (bt-p)^2)} \\ &= \frac{(pt+b)(pt+c)(b-c)t(bpt+cpt-p^2+bc) + (bt-p)(ct-p)(b-c)(p^2t-bct+bp+cp)}{(pt+b)(pt+c)(b-c)p(t^2+1) + (bt-p)(ct-p)(b-c)p(t^2+1)} \\ &\quad \frac{(b+c)p^3t^4 + (-p^4 + (b^2 + 4bc + c^2)p^2 - b^2c^2)t^3 + (-2(b+c)p^3 + 4bc(b+c)p)t^2 + (p^4 - (b^2 + 4bc + c^2)p^2 + b^2c^2)t + (b+c)p^3}{p(p^2+bc)(t^2+1)^2} \end{aligned}$$

and

$$\begin{aligned} y_A &= \frac{2(bt-p)(pt+b)}{(pt+b)^2 - (bt-p)^2} \cdot (x_A - b) \\ &= \frac{2(bt-p)(pt+b)}{(pt+b)^2 - (bt-p)^2} \cdot \frac{(cp^3 - b^2cp)t^4 + (-p^4 + (b^2 + 4bc + c^2)p^2 - b^2c^2)t^3 + (-2(2b+c)p^3 + 2bc(b+2c)p)t^2 + (p^4 - (b^2 + 4bc + c^2)p^2 + b^2c^2)t + (cp^3 - b^2cp)}{p(p^2+bc)(t^2+1)^2} \\ &= \frac{2(bt-p)(pt+b)(ct-p)(pt+c)((p^2-b^2)t^2 + 4bpt + (-p^2+b^2))}{p(p^2+bc)(t^2+1)^2((pt+b)^2 - (bt-p)^2)} \\ &= \frac{2(bt-p)(pt+b)(ct-p)(pt+c)}{p(p^2+bc)(t^2+1)^2}. \end{aligned}$$

The factorization is obvious since the expression should be symmetric in b and c . Now, $AP \geq AI$ iff $AP^2 \geq AI^2$. So we rewrite

$$\begin{aligned} &AP^2 - AI^2 \\ &= x_A^2 + (y_A - p)^2 - \left(x_A - \frac{t(bpt+cpt-p^2+bc)}{p(t^2+1)} \right)^2 - \left(y_A - \frac{(bt-p)(ct-p)}{p(t^2+1)} \right)^2 \\ &= \frac{2t(bpt+cpt-p^2+bc)}{p(t^2+1)}x_A + \left(-2p + \frac{2(bt-p)(ct-p)}{p(t^2+1)} \right)y_A \\ &\quad + p^2 - \left(\frac{t(bpt+cpt-p^2+bc)}{p(t^2+1)} \right)^2 - \left(\frac{(bt-p)(ct-p)}{p(t^2+1)} \right)^2 \end{aligned}$$

$$\begin{aligned}
&= \frac{2t(bpt + cpt - p^2 + bc)}{p(t^2 + 1)}x_A + \frac{2t((bc - p^2)t - (b + c)p)}{p(t^2 + 1)}y_A \\
&\quad + \frac{t((p^4 - (b + c)^2p^2 - b^2c^2)t + 2(b + c)p^3)}{p^2(t^2 + 1)} \\
&= \frac{t}{p^2(p^2 + bc)(t^2 + 1)^3} \cdot \left[2(bpt + cpt - p^2 + bc)((b + c)p^3t^4 \right. \\
&\quad + (-p^4 + (b^2 + 4bc + c^2)p^2 - b^2c^2)t^3 + (-2(b + c)p^3 + 4bc(b + c)p)t^2 \\
&\quad + (p^4 - (b^2 + 4bc + c^2)p^2 + b^2c^2)t + (b + c)p^3) \\
&\quad + 4((bc - p^2)t - (b + c)p)(bt - p)(pt + b)(ct - p)(pt + c) \\
&\quad \left. + ((p^4 - (b + c)^2p^2 - b^2c^2)t + 2(b + c)p^3)(p^2 + bc)(t^2 + 1)^2 \right] \\
&= \frac{t}{p^2(p^2 + bc)(t^2 + 1)^3} \cdot \left[(p^6 + (b^2 - bc + c^2)p^4 - bc(b^2 - bc + c^2)p^2 - b^3c^3)(t^5 - t) \right. \\
&\quad \left. + (2(b + c)p^5 + 2(b + c)(b^2 + c^2)p^3 + 2b^2c^2(b + c)p)(t^4 + t^2) \right] \\
&= \frac{t^2(p^2 + b^2)(p^2 + c^2)((p^2 - bc)(t^2 - 1) + 2(b + c)pt)}{p^2(p^2 + bc)(t^2 + 1)^2}.
\end{aligned}$$

Since $p^2 + b^2$, $p^2 + c^2$, $p^2(t^2 + 1)^2$ and $(p^2 + bc)^2$ are positive, it suffices to prove

$$t^2(p^2 + bc)((p^2 - bc)(t^2 - 1) + 2(b + c)pt) \geq 0. \quad (1)$$

By considering the y -coordinate of A , we have

$$\frac{2(bt - p)(pt + b)(ct - p)(pt + c)}{p(p^2 + bc)(t^2 + 1)^2} > 0.$$

We do not consider the cases $b = 0$ or $c = 0$, which can be done by continuity. As $p > 0$, this inequality implies

$$(p^2 + bc) \left(t - \frac{p}{b} \right) \left(\frac{p}{b}t + 1 \right) \left(t - \frac{p}{c} \right) \left(\frac{p}{c}t + 1 \right) > 0,$$

i.e.

$$(p^2 + bc)(\tan \theta + \tan \beta)(-\tan \beta \tan \theta + 1)(\tan \theta - \tan \gamma)(\tan \gamma \tan \theta + 1) > 0. \quad (2)$$

Note that $\angle ABC = 2(\beta + \theta)$ lies between 0° and 180° . This implies $\theta > -\beta$ and hence

$$\tan \theta + \tan \beta > 0 \quad (\text{as } \beta > 0^\circ).$$

From $2(\beta + \theta) < 180^\circ$, we get $\beta + \theta < 90^\circ$. This yields

$$-\tan \beta \tan \theta + 1 > 0.$$

Similarly, $\angle ACB = 2(\gamma - \theta)$ lies between 0° and 180° . This implies $\theta < \gamma$ and hence

$$\tan \theta - \tan \gamma < 0.$$

From $2(\gamma - \theta) < 180^\circ$, we get $\gamma - \theta < 90^\circ$. This yields

$$\tan \gamma \tan \theta + 1 > 0.$$

Combining these, we note from (2) that $p^2 + bc < 0$. To prove (1), it remains to prove

$$t^2((p^2 - bc)(t^2 - 1) + 2(b + c)pt) \leq 0. \quad (3)$$

With some careful observations, one finds that

$$\begin{aligned} & (p^2 - bc)(t^2 - 1) + 2(b + c)pt \\ &= 2(b + c)pt \left(\frac{p^2 - bc}{(b + c)p} \cdot \frac{t^2 - 1}{2t} + 1 \right) \\ &= 2(b + c)pt \left(\frac{1 + \frac{p}{c} \cdot \left(-\frac{p}{b}\right)}{\frac{p}{c} - \left(-\frac{p}{b}\right)} \cdot \frac{1 - t^2}{2t} + 1 \right) \\ &= 2(b + c)pt \left(\frac{1}{\tan(\gamma - \beta)} \cdot \frac{1}{\tan 2\theta} + 1 \right). \end{aligned} \quad (4)$$

Due to symmetry, we may assume $b + c > 0$ (again we ignore the limiting case $b + c = 0$). This implies the projection of P onto BC is closer to B . Thus, we have $\beta > \gamma$.

If $\theta > 0^\circ$, then $t > 0$ and $\tan 2\theta > 0$. Note that

$$(\beta - \gamma) + 2\theta = (\beta + \theta) - (\gamma - \theta) = \angle IBA - \angle ICA < 90^\circ.$$

So $\tan(\beta - \gamma) \tan 2\theta < 1$. This yields

$$\frac{1}{\tan(\gamma - \beta)} \cdot \frac{1}{\tan 2\theta} + 1 < 0.$$

From (4), we find that (3) holds.

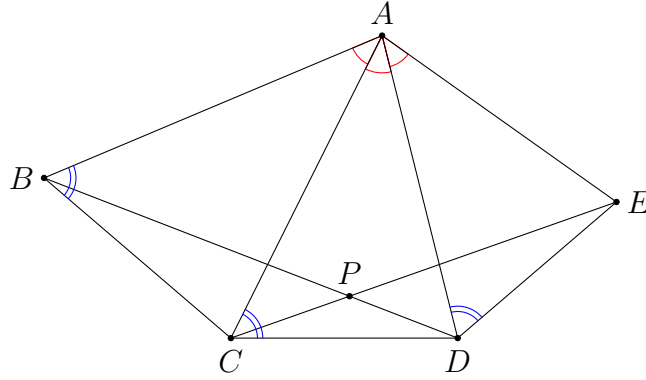
If $\theta < 0^\circ$, then $t < 0$ and $\tan 2\theta < 0$. So (4) is negative and (3) still holds.

In any case, we have proven the inequality. Equality of (3) holds when $t = 0$, i.e. when $\angle PBA = \angle PBC$ and $\angle PCA = \angle PCB$. This means $P = I$. $\square$

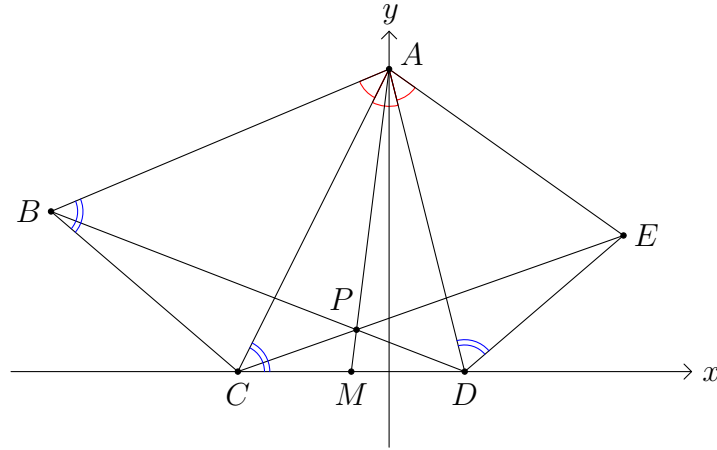
Problem 32. (IMO Shortlist 2006 G3) Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \quad \text{and} \quad \angle ABC = \angle ACD = \angle ADE.$$

The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .



Solution.



Let $A(0, a)$, $C(c, 0)$, $D(d, 0)$. Firstly, we have

$$\begin{aligned}
 & \angle BAC = \angle CAD \\
 \Rightarrow & \tan \angle BAC = \tan \angle CAD \\
 \Rightarrow & \frac{-\frac{a}{c} - m_{AB}}{1 - \frac{a}{c}m_{AB}} = \frac{-\frac{a}{d} + \frac{a}{c}}{1 + \frac{a^2}{cd}} \\
 \Rightarrow & \frac{a + cm_{AB}}{am_{AB} - c} = \frac{a(d - c)}{a^2 + cd} \\
 \Rightarrow & a(a^2 + cd) + c(a^2 + cd)m_{AB} = a^2(d - c)m_{AB} - ac(d - c) \\
 \Rightarrow & (a^2d - 2a^2c - c^2d)m_{AB} = a(a^2 + 2cd - c^2) \\
 \Rightarrow & m_{AB} = \frac{a(a^2 + 2cd - c^2)}{a^2d - 2a^2c - c^2d}.
 \end{aligned}$$

From this we get

$$AB: \quad y = \frac{a(a^2 + 2cd - c^2)}{a^2d - 2a^2c - c^2d}x + a. \quad (1)$$

Secondly, note that

$$\angle ACB = 180^\circ - \angle BAC - \angle ABC = 180^\circ - \angle CAD - \angle ACD = \angle ADC.$$

Also, we have

$$\begin{aligned}
& \angle ACB = \angle ADC \\
\Rightarrow & \tan \angle ACB = \tan \angle ADC \\
& \frac{m_{BC} + \frac{a}{c}}{1 - m_{BC}\frac{a}{c}} = \frac{a}{d} \\
\Rightarrow & \frac{cm_{BC} + a}{c - am_{BC}} = \frac{a}{d} \\
\Rightarrow & cdm_{BC} + ad = ac - a^2m_{BC} \\
\Rightarrow & (a^2 + cd)m_{BC} = a(c - d) \\
\Rightarrow & m_{BC} = \frac{a(c - d)}{a^2 + cd}.
\end{aligned}$$

This yields

$$BC: \quad y = \frac{a(c - d)}{a^2 + cd}x - \frac{ac(c - d)}{a^2 + cd}. \quad (2)$$

Solving (1) and (2), we get

$$x = \frac{(a^2d - 2a^2c - c^2d)(a(a^2 + cd) + ac(c - d))}{a(c - d)(a^2d - 2a^2c - c^2d) - a(a^2 + cd)(a^2 + 2cd - c^2)} = \frac{2a^2c + c^2d - a^2d}{a^2 + d^2}$$

and

$$y = \frac{a(a^2 + 2cd - c^2)}{a^2d - 2a^2c - c^2d} \cdot \frac{2a^2c + c^2d - a^2d}{a^2 + d^2} + a = \frac{a(c - d)^2}{a^2 + d^2}.$$

Thus, $B\left(\frac{2a^2c + c^2d - a^2d}{a^2 + d^2}, \frac{a(c - d)^2}{a^2 + d^2}\right)$. Then we have

$$m_{BD} = \frac{a(c - d)^2}{(2a^2c + c^2d - a^2d) - d(a^2 + d^2)} = \frac{a(c - d)}{2a^2 + cd + d^2}$$

so that

$$BD: \quad y = \frac{a(c - d)}{2a^2 + cd + d^2}x - \frac{ad(c - d)}{2a^2 + cd + d^2}.$$

By symmetry, we have

$$CE: \quad y = \frac{a(d - c)}{2a^2 + cd + c^2}x - \frac{ac(d - c)}{2a^2 + cd + c^2}.$$

Solving these, we get

$$\begin{aligned}
x &= \frac{ac(d - c)(2a^2 + cd + d^2) - ad(c - d)(2a^2 + cd + c^2)}{a(d - c)(2a^2 + cd + d^2) - a(c - d)(2a^2 + cd + c^2)} \\
&= \frac{2a(d - c)(c + d)(a^2 + cd)}{a(d - c)(4a^2 + c^2 + 2cd + d^2)} = \frac{2(c + d)(a^2 + cd)}{4a^2 + c^2 + 2cd + d^2}
\end{aligned}$$

and

$$\begin{aligned}
 y &= \frac{a(c-d)}{2a^2+cd+d^2} \cdot \left(\frac{2(c+d)(a^2+cd)}{4a^2+c^2+2cd+d^2} - d \right) \\
 &= \frac{a(c-d)}{2a^2+cd+d^2} \cdot \frac{(c-d)(2a^2+cd+d^2)}{4a^2+c^2+2cd+d^2} \\
 &= \frac{a(c-d)^2}{4a^2+c^2+2cd+d^2}.
 \end{aligned}$$

This shows $P \left(\frac{2(c+d)(a^2+cd)}{4a^2+c^2+2cd+d^2}, \frac{a(c-d)^2}{4a^2+c^2+2cd+d^2} \right)$. It follows that

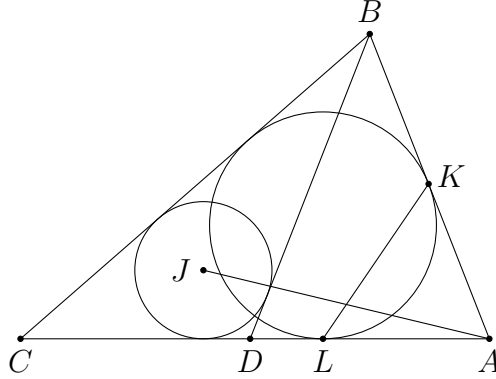
$$m_{AP} = \frac{a(c-d)^2 - a(4a^2+c^2+2cd+d^2)}{2(c+d)(a^2+cd)} = -\frac{2a}{c+d}.$$

Hence, the equation of AP is

$$y = -\frac{2a}{c+d}x + a.$$

Clearly, it passes through the midpoint $\left(\frac{c+d}{2}, 0 \right)$ of CD , as desired. $\square$

Problem 33. (IMO Shortlist 2006 G4) A point D is chosen on the side AC of a triangle ABC with $\angle C < \angle A < 90^\circ$ in such a way that $BD = BA$. The incircle of ABC is tangent to AB and AC at points K and L , respectively. Let J be the incentre of triangle BCD . Prove that the line KL intersects the line segment AJ at its midpoint.

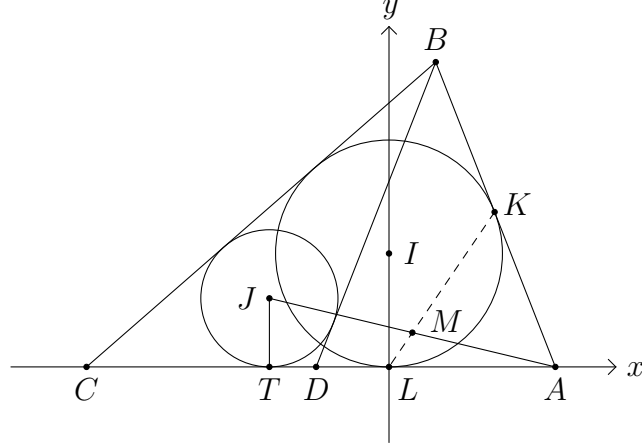


Solution. Let I be the incentre of $\triangle ABC$. Let $I(0,1)$, $A(a,0)$, $C(c,0)$, $L(0,0)$ where $a > 0 > c$. Then $B \left(\frac{a+c}{ac+1}, \frac{2ac}{ac+1} \right)$ and

$$KL: \quad y = ax. \tag{1}$$

Note that the projection of B on AC is the midpoint of AD . So the x -coordinate of D is

$$\frac{2(a+c)}{ac+1} - a = \frac{-a^2c + a + 2c}{ac+1}.$$



This gives $D\left(\frac{-a^2c + a + 2c}{ac + 1}, 0\right)$.

Let T be the contact point of the incircle of $\triangle BCD$ and the side CD . Then we get $CT = \frac{BC + CD - BD}{2}$ and $TJ = \frac{2[BCD]}{BC + CD + BD}$. Recall that

$$BC = -\frac{a(1 + c^2)}{ac + 1}.$$

Also, we have

$$CD = \frac{-a^2c + a + 2c}{ac + 1} - c = \frac{(a + c)(1 - ac)}{ac + 1}$$

and

$$BD = BA = \frac{c(1 + a^2)}{ac + 1}.$$

Using CD as the base, we have

$$[BCD] = \frac{1}{2} \cdot \frac{(a + c)(1 - ac)}{ac + 1} \cdot \frac{2ac}{ac + 1} = \frac{ac(a + c)(1 - ac)}{(ac + 1)^2}.$$

Combining all these, we obtain

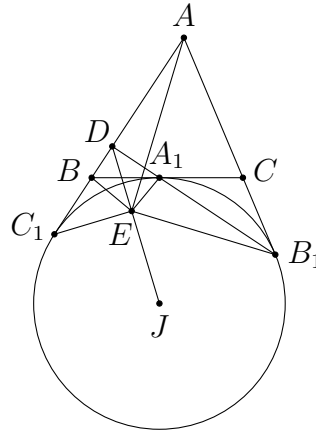
$$CT = \frac{1}{2} \left(-\frac{a(1 + c^2)}{ac + 1} + \frac{(a + c)(1 - ac)}{ac + 1} - \frac{c(1 + a^2)}{ac + 1} \right) = -\frac{ac(a + c)}{ac + 1}$$

and

$$TJ = \frac{2ac(a + c)(1 - ac)}{(ac + 1)^2} \div \left(-\frac{a(1 + c^2)}{ac + 1} + \frac{(a + c)(1 - ac)}{ac + 1} + \frac{c(1 + a^2)}{ac + 1} \right) = \frac{a(a + c)}{ac + 1}.$$

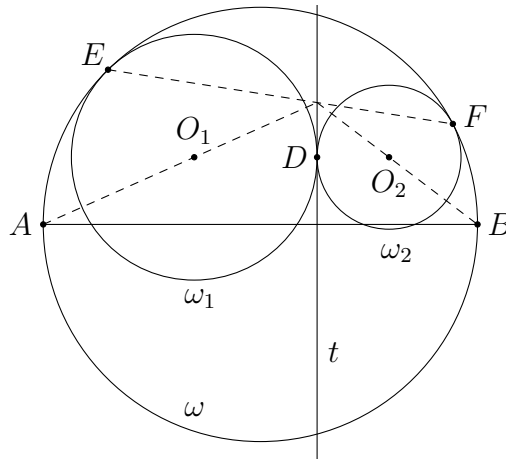
Therefore, we have $J = \left(c - \frac{ac(a + c)}{ac + 1}, \frac{a(a + c)}{ac + 1} \right) = \left(\frac{c(1 - a^2)}{ac + 1}, \frac{a(a + c)}{ac + 1} \right)$. Thus, the midpoint M of AJ has coordinates $\left(\frac{a + c}{2(ac + 1)}, \frac{a(a + c)}{2(ac + 1)} \right)$. We need to prove M lies on KL . This is obvious in view of (1). $\square$

Problem 34. (IMO Shortlist 2006 G5) In triangle ABC , let J be the centre of the excircle tangent to side BC at A_1 and to the extensions of sides AC and AB at B_1 and C_1 , respectively. Suppose that the lines A_1B_1 and AB are perpendicular and intersect at D . Let E be the foot of the perpendicular from C_1 to line DJ . Determine the angles $\angle BEA_1$ and $\angle AEB_1$.

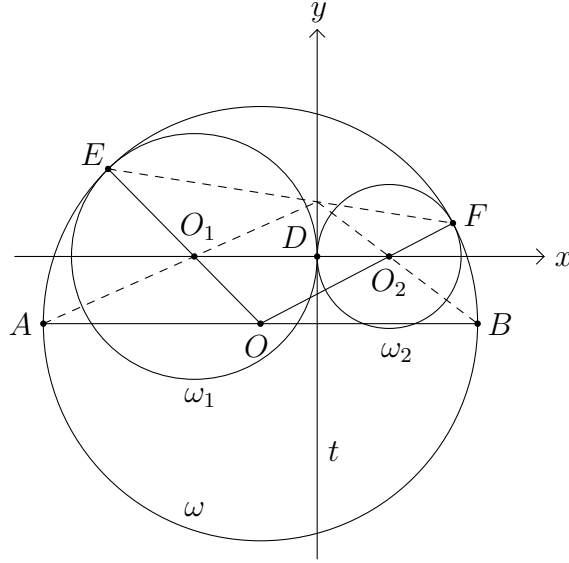


Solution. See [problem 5.5](#) of **Coordinate Geometry 5.0**. □

Problem 35. (IMO Shortlist 2006 G6) Circles ω_1 and ω_2 with centres O_1 and O_2 are externally tangent at point D and internally tangent to a circle ω at points E and F , respectively. Line t is the common tangent of ω_1 and ω_2 at D . Let AB be the diameter of ω perpendicular to t , so that A, E and O_1 are on the same side of t . Prove that lines AO_1, BO_2, EF and t are concurrent.



Solution. Let O be the centre of ω . Note that E, O_1, O are collinear, F, O_2, O are collinear. Let $D(0,0)$, $O_1(a,0)$, $O_2(b,0)$ (with $a < 0 < b$), $O(c,d)$. Let R be the radius of ω . Then $A(c-R, d)$. The radii of ω_1 and ω_2 are $-a$ and b respectively.



Using directed lengths, since $O_1E : EO = a : R$, we have $E\left(\frac{ac + Ra}{R + a}, \frac{ad}{R + a}\right)$. Similarly, $F\left(\frac{-bc + Rb}{R - b}, \frac{-bd}{R - b}\right)$. From these, we get

$$m_{EF} = \frac{ad(R - b) + bd(R + a)}{(ac + Ra)(R - b) - (-bc + Rb)(R + a)} = \frac{d(a + b)}{(a - b)R + ac + bc - 2ab}$$

and hence

$$EF: \quad y = \frac{d(a + b)}{(a - b)R + ac + bc - 2ab}x - \frac{2abd}{(a - b)R + ac + bc - 2ab}.$$

Also, we have

$$AO_1: \quad y = \frac{d}{c - a - R}x - \frac{ad}{c - a - R}.$$

To show that EF, AO_1, t are concurrent, we need to show EF and AO_1 have the same y -intercept, i.e.

$$-\frac{2abd}{(a - b)R + ac + bc - 2ab} = -\frac{ad}{c - a - R}.$$

This means we need to show

$$(a + b)R + ac - bc = 0. \quad (1)$$

Since E, O_1, O are collinear, we have $O_1O = EO - EO_1$. This yields

$$\sqrt{(c - a)^2 + d^2} = R + a. \quad (2)$$

Similarly, as F, O_2, O are collinear, we have

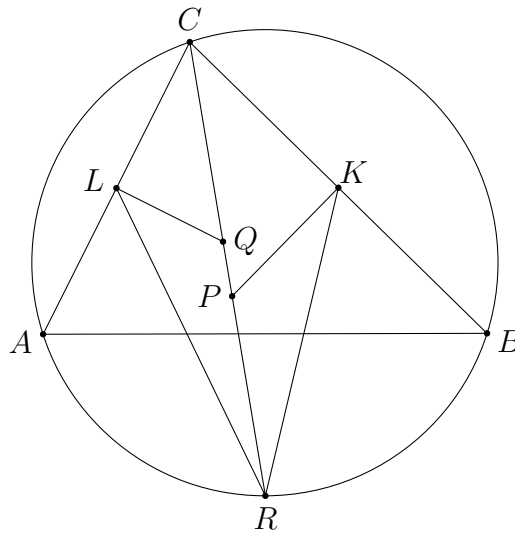
$$\sqrt{(c - b)^2 + d^2} = R - b. \quad (3)$$

Taking the difference of the squares of (2) and (3), we find that

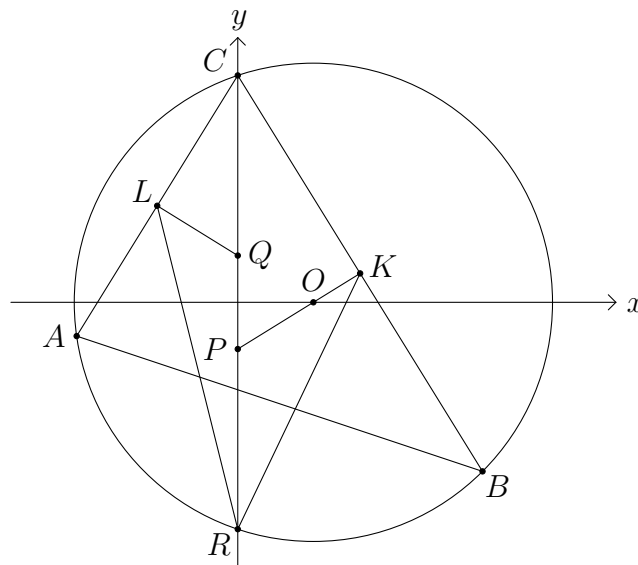
$$(c-a)^2 - (c-b)^2 = (R+a)^2 - (R-b)^2,$$

i.e. $-ac + bc = aR + bR$. This is our goal (1). By symmetry, EF, BO_2, t are also concurrent. So the four lines meet at a point. $\square$

Problem 36. (IMO Shortlist 2007 G1/Problem 4) In triangle ABC the bisector of angle BCA intersects the circumcircle again at R , the perpendicular bisector of BC at P , and the perpendicular bisector of AC at Q . The midpoint of BC is K and the midpoint of AC is L . Prove that the triangles RPK and RQL have the same area.



Solution.



Let $C(0,1)$, $R(0,-1)$. The circumcentre O of $\triangle ABC$ lies on the perpendicular bisector of CR . Let $O(a,0)$. As CR bisects $\angle ACB$, the slopes of AC and BC are negatives of each other. Let

$$AC: \quad y = mx + 1$$

and

$$BC: \quad y = -mx + 1$$

so that $m > 0$. Since POK is the perpendicular bisector of BC , its slope is $\frac{1}{m}$. As it passes through O , its equation is

$$y = \frac{1}{m}x - \frac{a}{m}.$$

This yields $P\left(0, -\frac{a}{m}\right)$ and $K\left(\frac{a+m}{m^2+1}, \frac{1-ma}{m^2+1}\right)$. Similarly, the perpendicular bisector of AC has equation $y = -\frac{1}{m}x + \frac{a}{m}$. This yields $Q\left(0, \frac{a}{m}\right)$ and $L\left(\frac{a-m}{m^2+1}, \frac{1+ma}{m^2+1}\right)$.

To compute $[RPK]$, we use PR as the base. The height is the x -coordinate of K . So

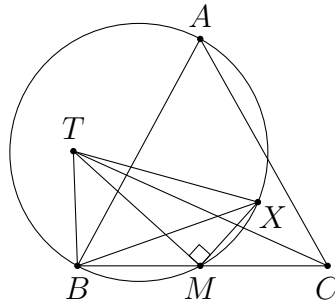
$$[RPK] = \frac{1}{2} \cdot \frac{m-a}{m} \cdot \frac{a+m}{m^2+1} = \frac{(m-a)(a+m)}{2m(m^2+1)}.$$

Similarly, we have

$$[RQL] = \frac{1}{2} \cdot \frac{a+m}{m} \cdot \left(-\frac{a-m}{m^2+1}\right) = \frac{(m-a)(a+m)}{2m(m^2+1)} = [RPK].$$

□

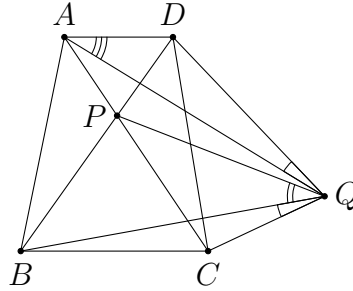
Problem 37. (IMO Shortlist 2007 G2) Given an isosceles triangle ABC with $AB = AC$. The midpoint of side BC is denoted by M . Let X be a variable point on the shorter arc MA of the circumcircle of triangle ABM . Let T be the point in the angle domain BMA , for which $\angle TMX = 90^\circ$ and $TX = BX$. Prove that $\angle MTB - \angle CTM$ does not depend on X .



Solution. See [problem 3.5](#) of **Coordinate Geometry 5.0**.

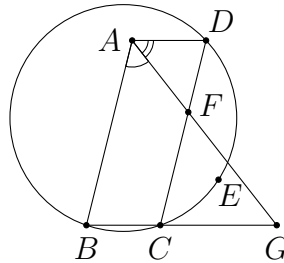
□

Problem 38. (IMO Shortlist 2007 G3) The diagonals of a trapezoid $ABCD$ intersect at point P . Point Q lies between the parallel lines BC and AD such that $\angle AQD = \angle CQB$, and line CD separates points P and Q . Prove that $\angle BQP = \angle DAQ$.



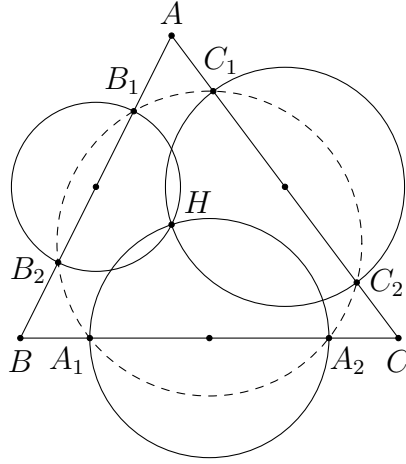
Solution. See [problem 3.6](#) of **Coordinate Geometry 5.0**. □

Problem 39. (IMO Shortlist 2007 G4/Problem 2) Consider five points A, B, C, D and E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A . Suppose that ℓ intersects the interior of the segment DC at F and intersects line BC at G . Suppose also that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB .

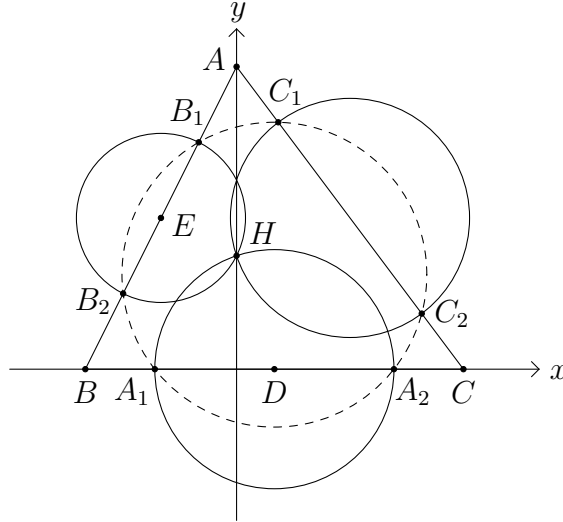


Solution. See [problem 3.4](#) of **Coordinate Geometry 5.0**. □

Problem 40. (IMO Shortlist 2008 G1/Problem 1) An acute-angled triangle ABC has orthocentre H . The circle passing through H with centre the midpoint of BC intersects the line BC at A_1 and A_2 . Similarly, the circle passing through H with centre the midpoint of CA intersects the line CA at B_1 and B_2 , and the circle passing through H with centre the midpoint of AB intersects the line AB at C_1 and C_2 . Show that $A_1, A_2, B_1, B_2, C_1, C_2$ lie on a circle.



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $H\left(0, -\frac{bc}{a}\right)$. The midpoint D of BC has coordinates $\left(\frac{b+c}{2}, 0\right)$. The equation of the circle with centre D passing through H is

$$x^2 + y^2 - (b+c)x - \frac{b^2c^2}{a^2} = 0.$$

Putting $y = 0$, the roots are

$$x = \frac{(b+c) \pm \sqrt{(b+c)^2 + \frac{4b^2c^2}{a^2}}}{2} = \frac{a(b+c) \pm \sqrt{a^2(b+c)^2 + 4b^2c^2}}{2a}.$$

(We have assumed $a > 0$.) So we have $A_1\left(\frac{a(b+c) - \sqrt{a^2(b+c)^2 + 4b^2c^2}}{2a}, 0\right)$.

Similarly, the midpoint E of AB has coordinates $\left(\frac{b}{2}, \frac{a}{2}\right)$. The equation of the circle with centre E passing through H is

$$x^2 + y^2 - bx - ay - \left(\frac{b^2c^2}{a^2} + bc\right) = 0.$$

Putting $y = -\frac{a}{b}x + a$, we solve

$$\left(1 + \frac{a^2}{b^2}\right)x^2 - \left(b + \frac{a^2}{b}\right)x - \left(\frac{b^2c^2}{a^2} + bc\right) = 0$$

to get

$$x = \frac{ab(a^2 + b^2) \pm b\sqrt{(a^2 + b^2)(a^2(a^2 + b^2) + 4bc(a^2 + bc))}}{2a(a^2 + b^2)}.$$

This yields

$$B_2 \left(\frac{ab(a^2 + b^2) - b\sqrt{\Delta}}{2a(a^2 + b^2)}, \frac{a(a^2 + b^2) + \sqrt{\Delta}}{2(a^2 + b^2)} \right)$$

where $\Delta = (a^2 + b^2)(a^2(a^2 + b^2) + 4bc(a^2 + bc))$.

Now, the perpendicular bisector of A_1A_2 is the same as that of BC . Similarly, the perpendicular bisector of B_1B_2 is the same as that of AB . Thus, the two perpendicular bisectors meet at the circumcentre $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$ of $\triangle ABC$. We now check that $OA_1 = OB_2$. Indeed, we have

$$\begin{aligned} OA_1^2 &= \left(\frac{b+c}{2} - \frac{a(b+c) - \sqrt{a^2(b+c)^2 + 4b^2c^2}}{2a} \right)^2 + \left(\frac{a^2+bc}{2a} \right)^2 \\ &= \frac{a^2(b+c)^2 + 4b^2c^2}{4a^2} + \frac{(a^2+bc)^2}{4a^2} \end{aligned} \quad (1)$$

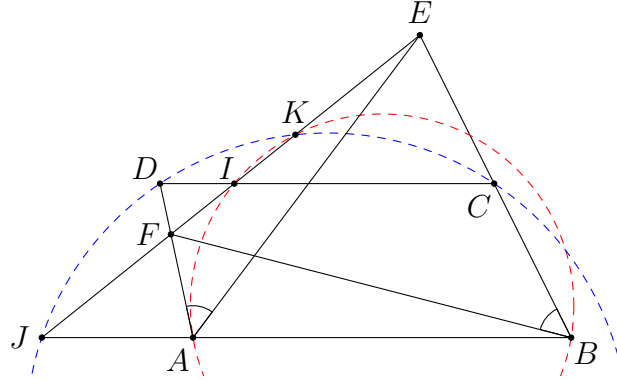
and

$$\begin{aligned} OB_2^2 &= \left(\frac{b+c}{2} - \frac{ab(a^2+b^2) - b\sqrt{\Delta}}{2a(a^2+b^2)} \right)^2 + \left(\frac{a^2+bc}{2a} - \frac{a(a^2+b^2) + \sqrt{\Delta}}{2(a^2+b^2)} \right)^2 \\ &= \left(\frac{c}{2} + \frac{b\sqrt{\Delta}}{2a(a^2+b^2)} \right)^2 + \left(\frac{bc}{2a} - \frac{\sqrt{\Delta}}{2(a^2+b^2)} \right)^2 \\ &= \frac{c^2}{4} + \frac{b^2\Delta}{4a^2(a^2+b^2)^2} + \frac{b^2c^2}{4a^2} + \frac{\Delta}{4(a^2+b^2)^2} \\ &= \frac{c^2}{4} + \frac{b^2c^2}{4a^2} + \frac{a^2(a^2+b^2) + 4bc(a^2+bc)}{4a^2}. \end{aligned} \quad (2)$$

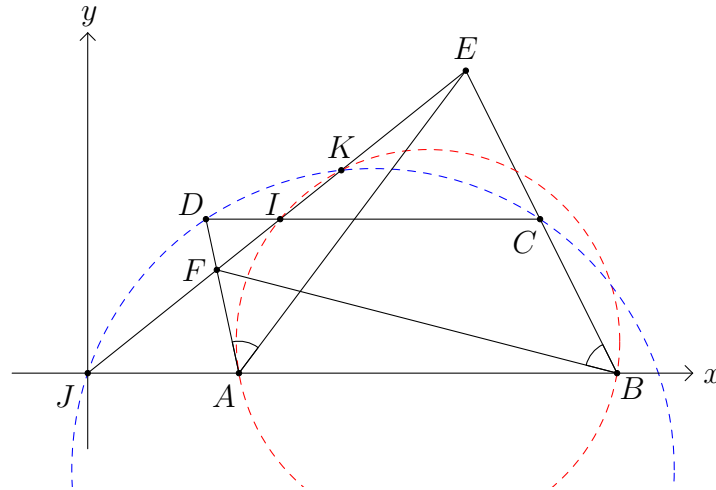
One readily checks that (1) and (2) are equal. It follows that A_1, A_2, B_1, B_2 lie on the same circle with centre O . By symmetry, A_1, A_2, C_1, C_2 also lie on the same circle with centre O . The two circles are clearly the same. So all the six points are concyclic. $\square$

Problem 41. (IMO Shortlist 2008 G2) Given trapezoid $ABCD$ with parallel sides AB and CD , assume that there exist points E on line BC outside segment BC , and F inside segment AD , such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF , and by J the point of intersection of AB and EF . Let K be the midpoint of segment EF ; assume it does not lie on line AB .

Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ .



Solution.



Let $J(0,0)$, $A(a,0)$, $B(b,0)$, $E(e,me)$, $F(f,mf)$ (so that the equation of EF is $y = mx$). Then $K\left(\frac{e+f}{2}, \frac{m(e+f)}{2}\right)$. From $\angle DAE = \angle CBF$, the points A, B, E, F are concyclic. This implies

$$ab = (m^2 + 1)ef. \quad (1)$$

Let $I(c, mc)$. As

$$BE: \quad y = \frac{me}{e-b}x - \frac{mbe}{e-b},$$

$$CD: \quad y = mc,$$

we obtain $C\left(\frac{ce - bc + be}{e}, mc\right)$. Similarly, we have $D\left(\frac{cf - ac + af}{f}, mc\right)$.

Now, A, B, K, I are concyclic iff $JA \times JB = JI \times JK$. This is the same as

$$ab = (m^2 + 1)c\left(\frac{e + f}{2}\right).$$

Using (1) to remove the term $m^2 + 1$, this is equivalent to $2ef = c(e + f)$, i.e.

$$ce + cf - 2ef = 0. \quad (2)$$

Similarly, C, K, D, J are concyclic iff $IC \times ID = IJ \times IK$. This is the same as

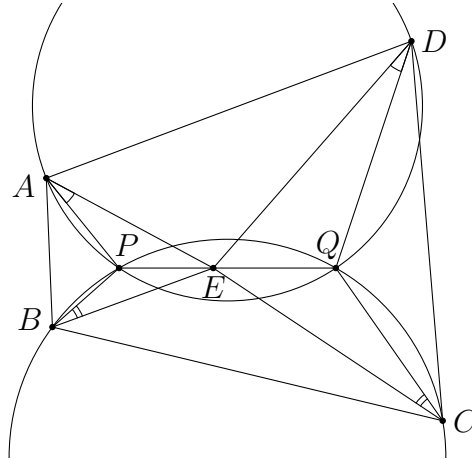
$$\left(\frac{ce - bc + be}{e} - c\right)\left(c - \frac{cf - ac + af}{f}\right) = (m^2 + 1)c\left(\frac{e + f}{2} - c\right).$$

Using (1), this is equivalent to

$$\frac{ab(e - c)(c - f)}{ef} = \frac{abc(e + f - 2c)}{2ef},$$

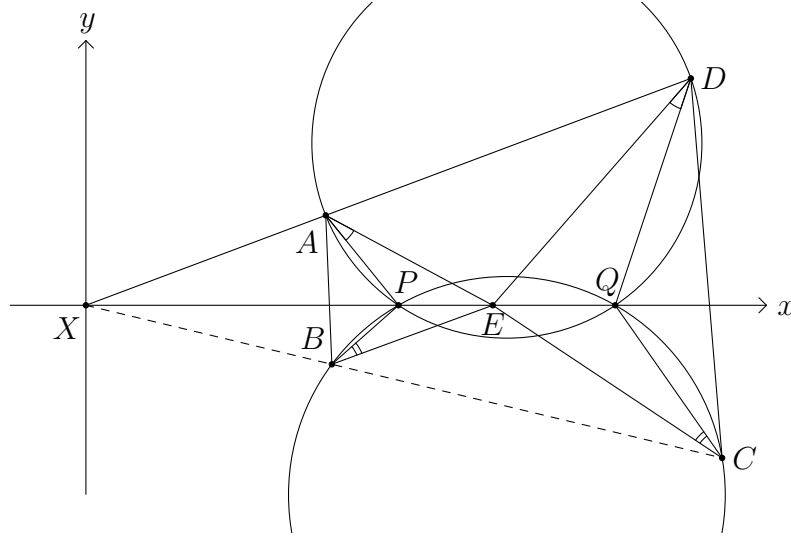
i.e. $2(e - c)(c - f) = c(e + f - 2c)$. After expansion, we find that it is exactly the same as (2). So the equivalence is established. $\square$

Problem 42. (IMO Shortlist 2008 G3) Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.



Solution. Let AD meet PQ at X . Let $X(0,0)$, $A(a, ma)$, $D(d, md)$, $P(p, 0)$, $Q(q, 0)$, $E(e, 0)$ so that the equation of AD is $y = mx$. Since A, P, Q, D are concyclic, we have

$$ad(m^2 + 1) = pq. \quad (1)$$



Note that

$$\begin{aligned}
 & \angle PAE = \angle QDE \\
 \Rightarrow & \tan \angle PAE = \tan \angle QDE \\
 \Rightarrow & \frac{m_{AE} - m_{AP}}{1 + m_{AE} \cdot m_{AP}} = \frac{m_{DQ} - m_{DE}}{1 + m_{DQ} \cdot m_{DE}} \\
 \Rightarrow & \frac{\frac{ma}{a-e} - \frac{ma}{a-p}}{1 + \frac{ma}{a-e} \cdot \frac{ma}{a-p}} = \frac{\frac{md}{d-q} - \frac{md}{d-e}}{1 + \frac{md}{d-q} \cdot \frac{md}{d-e}} \\
 \Rightarrow & \frac{a(e-p)}{(a-e)(a-p) + m^2 a^2} = \frac{d(q-e)}{(d-e)(d-q) + m^2 d^2} \\
 \Rightarrow & a(e-p)(d-e)(d-q) + m^2 a d^2 (e-p) = d(q-e)(a-e)(a-p) + m^2 a^2 d(q-e).
 \end{aligned}$$

Using (1) to write $m^2 ad = pq - ad$ and expanding everything, this implies

$$(aq + dp - 2ad)e^2 - pq(aq + dp - 2ad) = 0,$$

i.e. $(aq + dp - 2ad)(e^2 - pq) = 0$. This yields $aq + dp - 2ad = 0$ or $e^2 = pq$. For the former case $q = \frac{d(2a-p)}{a}$, by (1), we have

$$m^2 + 1 = \frac{pq}{ad} = \frac{p(2a-p)}{a^2} \leq 1,$$

since the last inequality is the same as $(a-p)^2 \geq 0$. This is impossible since $m \neq 0$. Thus, we must have $e^2 = pq$. Since E lies on the segment PQ , this yields $e = \sqrt{pq}$. In other words, we have

$$XE = \sqrt{XP \cdot XQ}.$$

By symmetry, if we let BC meet PQ at Y , then we also have

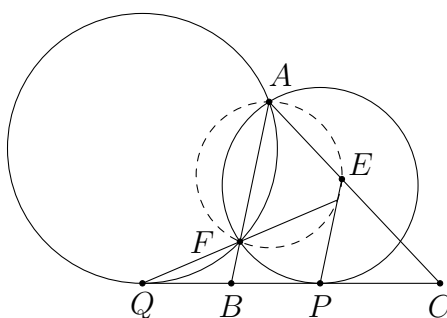
$$YE = \sqrt{YP \cdot YQ}.$$

Note that there is only one point on the line PQ outside the segment PQ satisfying this relation. Thus, we have $X = Y$. It follows that

$$XA \cdot XD = XP \cdot XQ = XB \cdot XC.$$

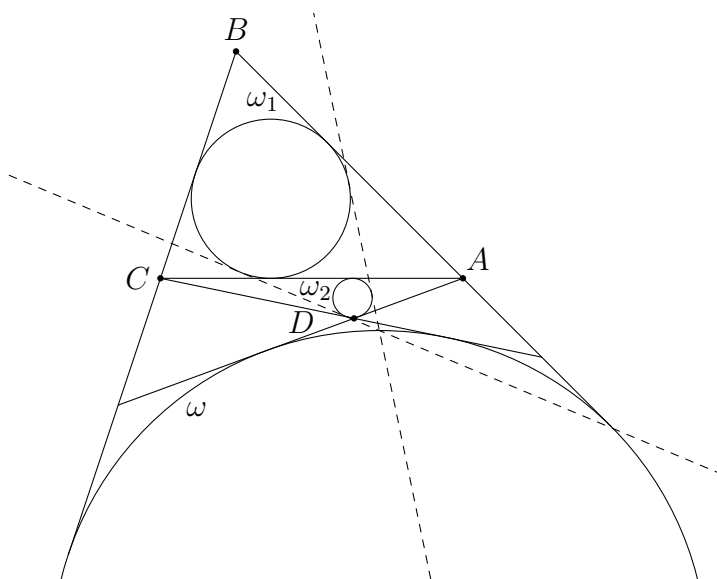
So A, B, C, D are concyclic as desired. □

Problem 43. (IMO Shortlist 2008 G4) In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the points A and F are tangent to the line BC at the points P and Q so that B lies between C and Q . Prove that the lines PE and QF intersect on the circumcircle of triangle AEF .

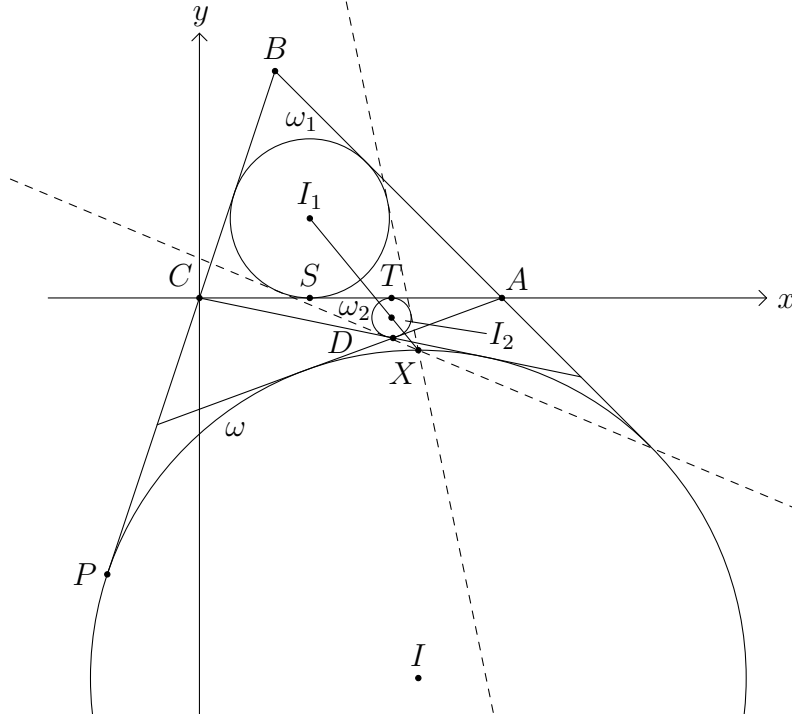


Solution. See [problem 4.3](#) of **Coordinate Geometry 5.0**. □

Problem 44. (IMO Shortlist 2008 G7/Problem 6) Let $ABCD$ be a convex quadrilateral with $|BA| \neq |BC|$. Denote the incircles of triangles ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle ω tangent to the ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents of ω_1 and ω_2 intersect on ω .



Solution.



Let I_1, I_2, I be the centres of $\omega_1, \omega_2, \omega$ respectively. Let ω_1 and ω_2 touch AC at S and T respectively. Let $C(0, 0)$, $A(a, 0)$, $S(s, 0)$, $T(t, 0)$, $I_1(s, b)$, $I_2(t, c)$ where $c < 0 < s, t, a, b$. Since ω is tangent to AB, BC, CD, DA , the centre I lies on the angle bisectors of $\angle BAD$, $\angle CBA$, $\angle DCB$, $\angle ADC$. From the configuration, the angle bisectors at A and C are external, while the angle bisectors at B and D are internal. We find the equations of CI and AI as follows.

Note that

$$\begin{aligned}\angle ICA &= \angle ICD + \angle DCA = \frac{180^\circ - \angle DCB}{2} + \angle DCA \\ &= 90^\circ + \frac{\angle DCA - \angle ACB}{2} = 90^\circ + \angle I_2CA - \angle ACI_1.\end{aligned}$$

Since $\tan \angle I_2CA = \frac{-c}{t}$ and $\tan \angle ACI_1 = \frac{b}{s}$, we have

$$m_{IC} = -\tan \angle ICA = \frac{1}{\tan(\angle I_2CA - \angle ACI_1)} = \frac{bc - st}{cs + bt}.$$

Thus, we get

$$IC: \quad y = \frac{bc - st}{cs + bt}x. \quad (1)$$

Similarly, we have

$$\angle CAI = 90^\circ + \angle CAI_2 - \angle I_1AC.$$

Since $\tan \angle CAI_2 = \frac{-c}{a-t}$ and $\tan \angle I_1AC = \frac{b}{a-s}$, this yields

$$m_{IA} = \tan \angle CAI = -\frac{1}{\tan(\angle CAI_2 - \angle I_1AC)} = \frac{(a-s)(a-t) - bc}{b(a-t) + c(a-s)}.$$

Hence,

$$IA: \quad y = \frac{(a-s)(a-t) - bc}{b(a-t) + c(a-s)}x - \frac{a((a-s)(a-t) - bc)}{b(a-t) + c(a-s)}. \quad (2)$$

Solving (1) and (2), we get

$$\begin{aligned} x &= \frac{a((a-s)(a-t) - bc)(cs + bt)}{((a-s)(a-t) - bc)(cs + bt) - (b(a-t) + c(a-s))(bc - st)} \\ &= \frac{a((a-s)(a-t) - bc)(cs + bt)}{a^2bt + a^2cs - ab^2c - abc^2 - abt^2 - acs^2} \\ &= \frac{((a-s)(a-t) - bc)(cs + bt)}{abt + acs - b^2c - bc^2 - bt^2 - cs^2} \end{aligned}$$

and

$$y = \frac{bc - st}{cs + bt} \cdot \frac{((a-s)(a-t) - bc)(cs + bt)}{abt + acs - b^2c - bc^2 - bt^2 - cs^2} = \frac{((a-s)(a-t) - bc)(bc - st)}{abt + acs - b^2c - bc^2 - bt^2 - cs^2}.$$

$$\text{So } I \left(\frac{((a-s)(a-t) - bc)(cs + bt)}{abt + acs - b^2c - bc^2 - bt^2 - cs^2}, \frac{((a-s)(a-t) - bc)(bc - st)}{abt + acs - b^2c - bc^2 - bt^2 - cs^2} \right).$$

Next, we have $m_{BC} = \frac{2m_{I_1C}}{1 - m_{I_1C}^2} = \frac{2bs}{s^2 - b^2}$. This yields

$$BC: \quad y = \frac{2bs}{s^2 - b^2}x. \quad (3)$$

Also, we have $m_{AB} = \frac{2m_{I_1A}}{1 - m_{I_1A}^2} = \frac{2b(s-a)}{(s-a)^2 - b^2}$. This yields

$$AB: \quad y = \frac{2b(s-a)}{(s-a)^2 - b^2}x - \frac{2ab(s-a)}{(s-a)^2 - b^2}. \quad (4)$$

Solving (3) and (4), we find that

$$x = \frac{a(s-a)(s^2 - b^2)}{(s-a)(s^2 - b^2) - s((s-a)^2 - b^2)} = \frac{(s-a)(s^2 - b^2)}{s^2 - as + b^2}$$

and

$$y = \frac{2bs}{s^2 - b^2} \cdot \frac{(s-a)(s^2 - b^2)}{s^2 - as + b^2} = \frac{2bs(s-a)}{s^2 - as + b^2}.$$

Thus, we have $B\left(\frac{(s-a)(s^2-b^2)}{s^2-as+b^2}, \frac{2bs(s-a)}{s^2-as+b^2}\right)$. Since I_1 lies on the internal angle bisector of $\angle ABC$, B, I_1, I should be collinear. From

$$m_{BI_1} = \frac{2bs(s-a) - b(s^2-as+b^2)}{(s-a)(s^2-b^2) - s(s^2-as+b^2)} = \frac{b(s^2-as-b^2)}{-2b^2s+ab^2} = \frac{s^2-as-b^2}{b(a-2s)},$$

the equation of BI_1 is

$$y = \frac{s^2-as-b^2}{b(a-2s)}x - \frac{s^2-as-b^2}{b(a-2s)} \cdot s + b,$$

i.e.

$$BI_1: \quad y = \frac{s^2-as-b^2}{b(a-2s)}x - \frac{(s-a)(s^2+b^2)}{b(a-2s)}.$$

Since I lies on this straight line, we have

$$\begin{aligned} \frac{((a-s)(a-t)-bc)(bc-st)}{abt+acs-b^2c-bc^2-bt^2-cs^2} &= \frac{s^2-as-b^2}{b(a-2s)} \cdot \frac{((a-s)(a-t)-bc)(cs+bt)}{abt+acs-b^2c-bc^2-bt^2-cs^2} \\ &\quad - \frac{(s-a)(s^2+b^2)}{b(a-2s)}. \end{aligned}$$

Clearing denominators, we get

$$\begin{aligned} b(a-2s)((a-s)(a-t)-bc)(bc-st) &- (s^2-as-b^2)((a-s)(a-t)-bc)(cs+bt) \\ &+ (s-a)(s^2+b^2)(abt+acs-b^2c-bc^2-bt^2-cs^2) = 0. \end{aligned}$$

Combining the first two terms and simplifying, this becomes

$$\begin{aligned} &((a-s)(a-t)-bc)(s^2+b^2)(ac+bt-cs) \\ &+ (s-a)(s^2+b^2)(abt+acs-b^2c-bc^2-bt^2-cs^2) = 0. \end{aligned}$$

We can remove the factor s^2+b^2 and simplify further. This yields

$$c(s^3+s^2t-3as^2+3a^2s-2ast+b^2s+b^2t+a^2t-ab^2-a^3) = 0,$$

which means

$$c(s+t-a)(b^2+s^2-2as+a^2) = 0.$$

Since $c < 0$ and $b^2+s^2-2as+a^2 = b^2+(s-a)^2 > 0$, this implies $a = s+t$. Thus, we can simplify

$$\begin{aligned} I &= \left(\frac{((a-s)(a-t)-bc)(cs+bt)}{abt+acs-b^2c-bc^2-bt^2-cs^2}, \frac{((a-s)(a-t)-bc)(bc-st)}{abt+acs-b^2c-bc^2-bt^2-cs^2} \right) \\ &= \left(\frac{(st-bc)(cs+bt)}{bst+cst-b^2c-bc^2}, \frac{(st-bc)(bc-st)}{bst+cst-b^2c-bc^2} \right) \\ &= \left(\frac{cs+bt}{b+c}, \frac{bc-st}{b+c} \right). \end{aligned}$$

Now, recall from (3) that the equation of BC is $y = \frac{2bs}{s^2 - b^2}x$. Let P be the projection of I onto BC . Then the equation of IP is

$$y = -\frac{s^2 - b^2}{2bs}x + \frac{s^2 - b^2}{2bs} \cdot \frac{cs + bt}{b + c} + \frac{bc - st}{b + c},$$

i.e.

$$y = -\frac{s^2 - b^2}{2bs}x + \frac{(s^2 + b^2)(cs - bt)}{2bs(b + c)}.$$

Solving this with the equation of BC , we get

$$x = \frac{(s^2 + b^2)(cs - bt)(s^2 - b^2)}{(b + c)((2bs)^2 + (s^2 - b^2)^2)} = \frac{(s^2 - b^2)(cs - bt)}{(b + c)(s^2 + b^2)}$$

and

$$y = \frac{2bs}{s^2 - b^2} \cdot \frac{(s^2 - b^2)(cs - bt)}{(b + c)(s^2 + b^2)} = \frac{2bs(cs - bt)}{(b + c)(s^2 + b^2)}.$$

Thus, $P \left(\frac{(s^2 - b^2)(cs - bt)}{(b + c)(s^2 + b^2)}, \frac{2bs(cs - bt)}{(b + c)(s^2 + b^2)} \right)$. Note that P lies on ω . So the equation of ω is

$$\begin{aligned} x^2 + y^2 - \frac{2(cs + bt)}{b + c}x - \frac{2(bc - st)}{b + c}y - \left(\frac{(s^2 - b^2)(cs - bt)}{(b + c)(s^2 + b^2)} \right)^2 - \left(\frac{2bs(cs - bt)}{(b + c)(s^2 + b^2)} \right)^2 \\ + \frac{2(cs + bt)}{b + c} \cdot \frac{(s^2 - b^2)(cs - bt)}{(b + c)(s^2 + b^2)} + \frac{2(bc - st)}{b + c} \cdot \frac{2bs(cs - bt)}{(b + c)(s^2 + b^2)} = 0. \end{aligned}$$

This can be simplified to

$$x^2 + y^2 - \frac{2(cs + bt)}{b + c}x - \frac{2(bc - st)}{b + c}y + \frac{(cs - bt)^2}{(b + c)^2} = 0.$$

Let X be the intersection of the external common tangents of ω_1 and ω_2 . Then X is the external homothetic centre between ω_1 and ω_2 . Hence, X lies on the line I_1I_2 with ratio $I_1X : I_2X = I_1S : I_2T = b : (-c)$. Using directed lengths, this means $I_1X : XI_2 = b : c$. Therefore, using the sectional formula, we get

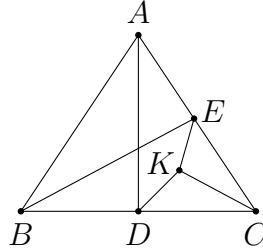
$$X = \left(\frac{bt + cs}{b + c}, \frac{bc + cb}{b + c} \right) = \left(\frac{bt + cs}{b + c}, \frac{2bc}{b + c} \right).$$

We now check that it lies on ω . Indeed,

$$\begin{aligned} & \left(\frac{bt + cs}{b + c} \right)^2 + \left(\frac{2bc}{b + c} \right)^2 - \frac{2(cs + bt)}{b + c} \cdot \frac{bt + cs}{b + c} - \frac{2(bc - st)}{b + c} \cdot \frac{2bc}{b + c} + \frac{(cs - bt)^2}{(b + c)^2} \\ &= -\frac{(bt + cs)^2}{(b + c)^2} + \frac{4bcst}{(b + c)^2} + \frac{(cs - bt)^2}{(b + c)^2} \\ &= 0. \end{aligned}$$

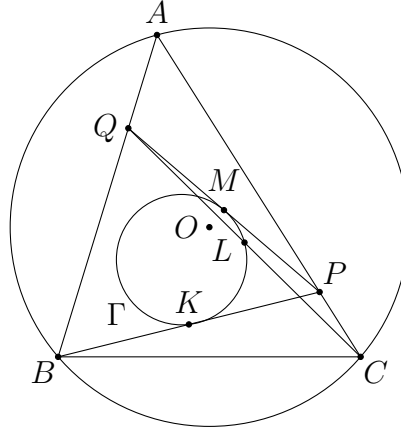
So we are done. □

Problem 45. (IMO Shortlist 2009 G1/Problem 4) Let ABC be a triangle with $AB = AC$. The angle bisectors of $\angle CAB$ and $\angle ABC$ meet the sides BC and CA at D and E , respectively. Let K be the incentre of triangle ADC . Suppose that $\angle BEK = 45^\circ$. Find all possible values of $\angle CAB$.



Solution. See [problem 5.1](#) of **Coordinate Geometry 5.0**. □

Problem 46. (IMO Shortlist 2009 G2/Problem 2) Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB , respectively. Let K, L and M be the midpoints of the segments BP, CQ and PQ , respectively, and let Γ be the circle passing through K, L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

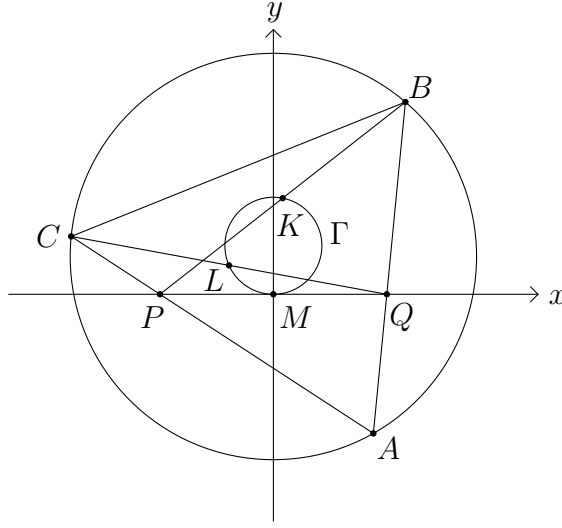


Solution. Let $M(0,0)$, $K(a,b)$, $L(c,d)$ and let the equation of Γ be $x^2 + y^2 - 2y = 0$. Then

$$a^2 + b^2 - 2b = 0 \quad \text{and} \quad c^2 + d^2 - 2d = 0.$$

Let $P(-s,0)$ and $Q(s,0)$. Since K is the midpoint of BP , we have $B(2a+s, 2b)$. Similarly, we have $C(2c-s, 2d)$. Solving

$$\begin{aligned} BQ: \quad y &= \frac{b}{a}x - \frac{bs}{a}, \\ CP: \quad y &= \frac{d}{c}x + \frac{ds}{c}, \end{aligned}$$



we find $A\left(\frac{(bc+ad)s}{bc-ad}, \frac{2bds}{bc-ad}\right)$. The perpendicular bisector of AB has slope $-\frac{a}{b}$. It passes through the midpoint $\left(\frac{bcs+a(bc-ad)}{bc-ad}, \frac{bds+b(bc-ad)}{bc-ad}\right)$ of AB . Therefore, its equation is

$$y = -\frac{a}{b}x + \frac{(ac+bd)bs + (a^2+b^2)(bc-ad)}{b(bc-ad)}.$$

Using the relation $a^2 + b^2 = 2b$, this can be simplified to

$$y = -\frac{a}{b}x + \frac{(ac+bd)s + 2(bc-ad)}{bc-ad}.$$

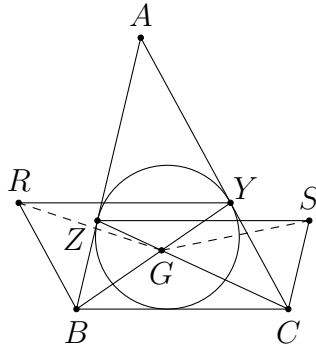
Swapping a and c , swapping b and d , and changing s to $-s$, the perpendicular bisector of AC has equation

$$y = -\frac{c}{d}x + \frac{(ac+bd)s + 2(bc-ad)}{bc-ad}.$$

It is obvious that the perpendicular bisectors have the same y -intercept. So the coordinates of O are $\left(0, \frac{(ac+bd)s + 2(bc-ad)}{bc-ad}\right)$. So we must have $OP = OQ$ as desired. $\square$

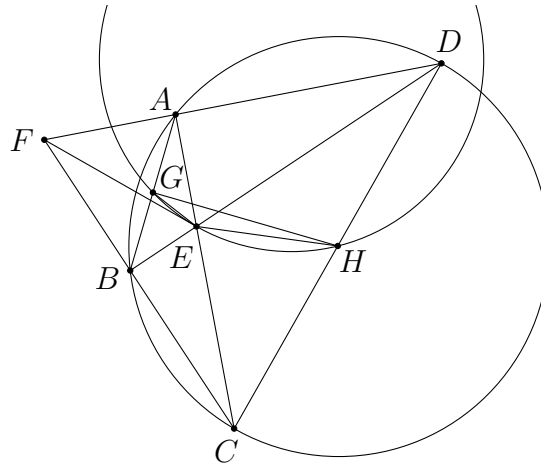
Problem 47. (IMO Shortlist 2009 G3) Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y , respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms.

Prove that $GR = GS$.



Solution. See [problem 5.3](#) of **Coordinate Geometry 5.0**. □

Problem 48. (IMO Shortlist 2009 G4) Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F . The midpoints of AB and CD are G and H , respectively. Show that EF is tangent at E to the circle through the points E, G , and H .



Solution. Let $E(0,0)$, $A(a,0)$, $C(c,0)$, $B(b,mb)$, $D(d,md)$ (so that the equation of BD is $y = mx$). Then

$$bd(m^2 + 1) = ac$$

and $F\left(\frac{abc + abd - acd - bcd}{ab - cd}, \frac{mbd(a - c)}{ab - cd}\right)$. Also, $G\left(\frac{a + b}{2}, \frac{mb}{2}\right)$ and $H\left(\frac{c + d}{2}, \frac{md}{2}\right)$.

We use angles to prove the assertion. Indeed,

$$\begin{aligned} EF \text{ is tangential to } (EGH) & \\ \Leftrightarrow \angle GEF = \angle GHE & \\ \Leftrightarrow \tan \angle GEF = \tan \angle GHE & \\ \Leftrightarrow \frac{m_{FE} - m_{GE}}{1 + m_{FE} \cdot m_{GE}} = \frac{m_{EH} - m_{GH}}{1 + m_{EH} \cdot m_{GH}}. & \end{aligned}$$

we find that

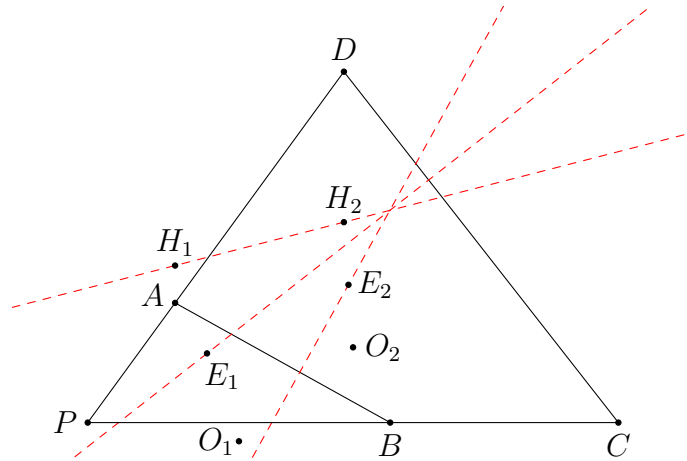
$$\begin{aligned}
& ab(c+d)(a+b-c-d) + m^2 abd(b-d) \\
&= (abc + abd - acd - bcd)(a+b) + m^2 b^2 d(a-c) \\
\Leftrightarrow & a^2 bc + a^2 bd + ab^2 c + ab^2 d - abc^2 - abcd - abcd - abd^2 + m^2 bd(ab - ad) \\
&= a^2 bc + a^2 bd - a^2 cd - abcd + ab^2 c + ab^2 d - abcd - b^2 cd + m^2 bd(ab - bc).
\end{aligned}$$

Cancelling like terms, we need to show $a^2 cd + b^2 cd - abc^2 - abd^2 + m^2 bd(bc - ad) = 0$.
Note that

$$\begin{aligned}
& a^2 cd + b^2 cd - abc^2 - abd^2 + m^2 bd(bc - ad) = 0 \\
\Leftrightarrow & (bc - ad)(bd - ac) + m^2 bd(bc - ad) = 0 \\
\Leftrightarrow & (bc - ad)(bd - ac + m^2 bd) = 0.
\end{aligned}$$

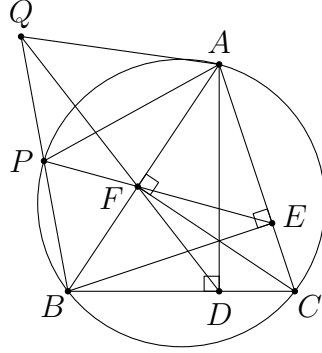
This is true since $m^2 = \frac{ac - bd}{bd}$. The proof is thus complete. $\square$

Problem 49. (IMO Shortlist 2009 G6) Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P . Points O_1 and O_2 are the circumcentres and points H_1 and H_2 are the orthocentres of triangles ABP and DCP , respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2 , respectively. Prove that the perpendicular from E_1 on CD , the perpendicular from E_2 on AB and the line H_1H_2 are concurrent.

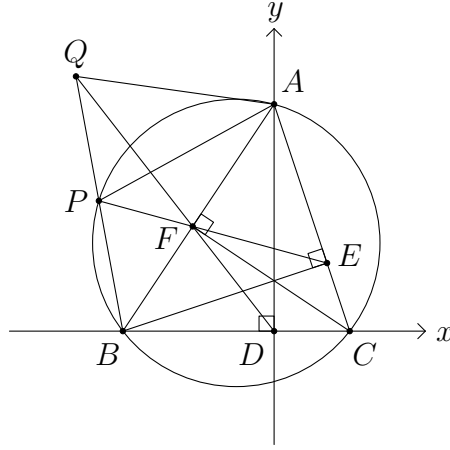


Solution. See [problem 2.3](#) of **Coordinate Geometry 5.0**. $\square$

Problem 50. (IMO Shortlist 2010 G1) Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$.



Solution.



Let $D(0,0)$, $A(0,a)$, $B(b,0)$ and $C(c,0)$. Then we have $F\left(\frac{b(a^2+bc)}{a^2+b^2}, \frac{ab(b-c)}{a^2+b^2}\right)$ and $E\left(\frac{c(a^2+bc)}{a^2+c^2}, \frac{ac(c-b)}{a^2+c^2}\right)$. The equation of EF is

$$y = \frac{a(b+c)}{a^2-bc}x - \frac{2abc}{a^2-bc}.$$

Also, we have $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$ and

$$(ABC): \quad x^2 + y^2 - (b+c)x - \frac{a^2+bc}{a}y + bc = 0. \quad (1)$$

Let $P\left(p, \frac{a(b+c)p-2abc}{a^2-bc}\right)$. We find that

$$DF: \quad y = \frac{a(b-c)}{a^2+bc}x$$

and

$$BP: \quad y = \frac{a(b+c)p-2abc}{(a^2-bc)(p-b)}x - \frac{ab(b+c)p-2ab^2c}{(a^2-bc)(p-b)}. \quad (2)$$

Solving these equations, we get

$$\begin{aligned} x &= \frac{(a^2 + bc)(ab(b + c)p - 2ab^2c)}{(a^2 + bc)(a(b + c)p - 2abc) - a(b - c)(a^2 - bc)(p - b)} \\ &= \frac{b(a^2 + bc)((b + c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{a(b - c)}{a^2 + bc} \cdot \frac{b(a^2 + bc)((b + c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)} \\ &= \frac{ab(b - c)((b + c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)}. \end{aligned}$$

Therefore,

$$Q \left(\frac{b(a^2 + bc)((b + c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)}, \frac{ab(b - c)((b + c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)} \right).$$

Recall that $m_{BP} = \frac{a(b + c)p - 2abc}{(a^2 - bc)(p - b)}$. So the equation of the perpendicular line from A to BP is

$$y = -\frac{(a^2 - bc)(p - b)}{a(b + c)p - 2abc}x + a. \quad (3)$$

Solving (2) and (3), we get

$$\begin{aligned} x &= \frac{a + \frac{ab(b + c)p - 2ab^2c}{(a^2 - bc)(p - b)}}{\frac{a(b + c)p - 2abc}{(a^2 - bc)(p - b)} + \frac{(a^2 - bc)(p - b)}{a(b + c)p - 2abc}} \\ &= \frac{a^2((b + c)p - 2bc)((a^2 + b^2)p - b(a^2 + bc))}{(a^2 + b^2)(a^2 + c^2)p^2 - 2b(a^4 + 2a^2c^2 + b^2c^2)p + b^2(a^4 - 2a^2bc + 4a^2c^2 + b^2c^2)}. \end{aligned} \quad (4)$$

As $P \left(p, \frac{a(b + c)p - 2abc}{a^2 - bc} \right)$ lies on (ABC) , we put the coordinates in (1). After expansion and simplification, we get

$$(a^2 + b^2)(a^2 + c^2)p^2 - 2a^2(b + c)(a^2 + bc)p + bc(3a^4 + 2a^2bc - b^2c^2) = 0. \quad (5)$$

We now use (5) to simplify the denominator of (4). In fact, the denominator becomes

$$\begin{aligned} &2a^2(b + c)(a^2 + bc)p - 2b(a^4 + 2a^2c^2 + b^2c^2)p \\ &\quad - bc(3a^4 + 2a^2bc - b^2c^2) + b^2(a^4 - 2a^2bc + 4a^2c^2 + b^2c^2) \\ &= (a^2 - bc)(2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)). \end{aligned}$$

Thus,

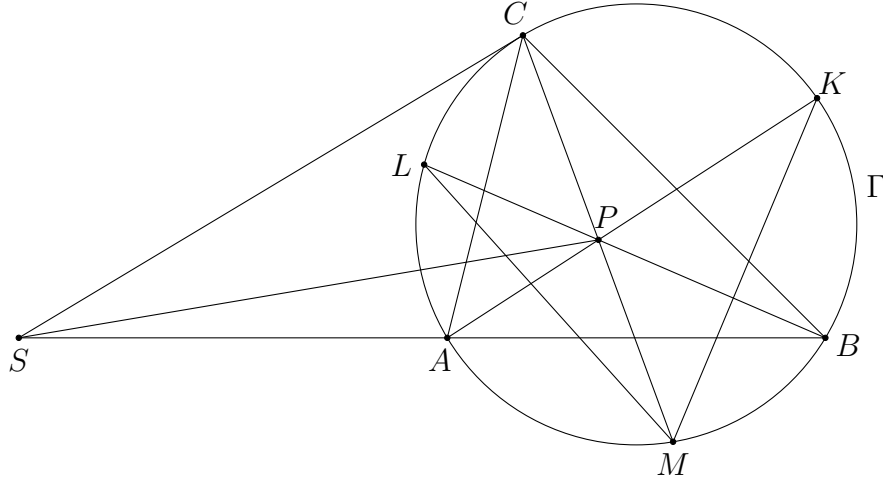
$$\frac{a^2((b+c)p - 2bc)((a^2 + b^2)p - b(a^2 + bc))}{(a^2 - bc)(2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2))}$$

is the x -coordinate of the foot of perpendicular from A to PB . Now, $AP = AQ$ iff this foot of perpendicular is the midpoint of PQ . So it suffices to check the above expression is the average of the x -coordinates of P and Q . Indeed,

$$\begin{aligned} & \frac{a^2((b+c)p - 2bc)((a^2 + b^2)p - b(a^2 + bc))}{(a^2 - bc)(2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2))} \\ &= \frac{1}{2} \left(p + \frac{b(a^2 + bc)((b+c)p - 2bc)}{2c(a^2 + b^2)p + b(a^2b - 3a^2c - b^2c - bc^2)} \right) \\ \Leftrightarrow & \quad a^2((b+c)p - 2bc)((a^2 + b^2)p - b(a^2 + bc)) \\ &= (a^2 - bc)(c(a^2 + b^2)p^2 + a^2b(b-c)p - b^2c(a^2 + bc)) \\ \Leftrightarrow & \quad (a^2 + b^2)(a^2 + c^2)p^2 - 2a^2(b+c)(a^2 + bc)p + bc(3a^4 + 2a^2bc - b^2c^2) = 0. \end{aligned}$$

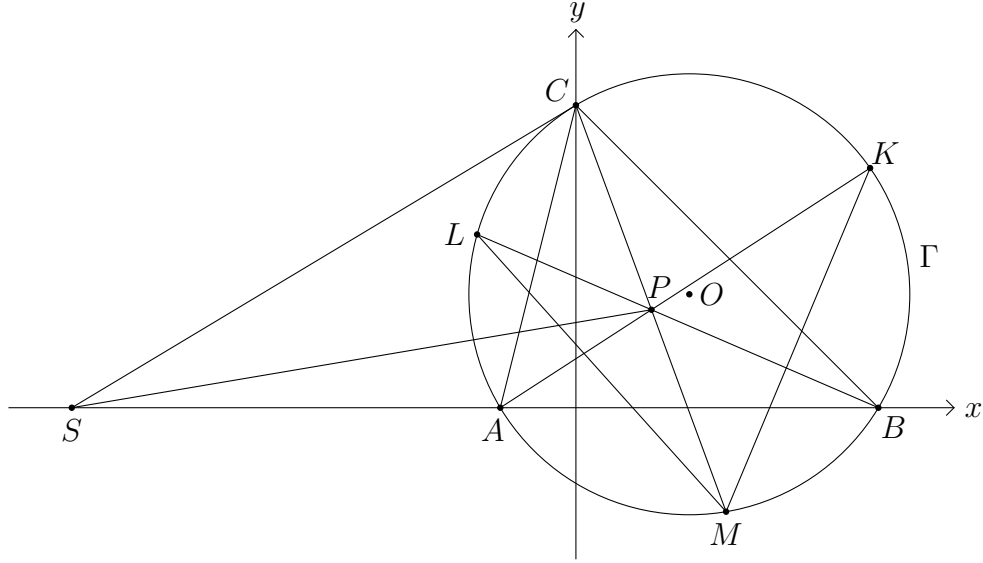
This is exactly the condition (5). So we are done. $\square$

Problem 51. (IMO Shortlist 2010 G2/Problem 4) Let P be a point inside the triangle ABC . The lines AP, BP and CP intersect the circumcircle Γ of triangle ABC again at the points K, L and M respectively. The tangent to Γ at C intersects the line AB at S . Suppose that $SC = SP$. Prove that $MK = ML$.



Solution. Let $A(a, 0)$, $B(b, 0)$, $C(0, c)$, $P(s, t)$. The circumcentre O of $\triangle ABC$ has coordinates $\left(\frac{a+b}{2}, \frac{c^2 + ab}{2c}\right)$ and

$$\Gamma: \quad x^2 + y^2 - (a+b)x - \frac{c^2 + ab}{c}y + ab = 0. \quad (1)$$



From $m_{CS} = \frac{-1}{m_{CO}} = -\frac{c(a+b)}{ab-c^2}$, we obtain

$$CS: \quad y = -\frac{c(a+b)}{ab-c^2}x + c.$$

This yields $S\left(\frac{ab-c^2}{a+b}, 0\right)$. We have

$$\begin{aligned} SC &= SP \\ \Leftrightarrow \quad \left(\frac{ab-c^2}{a+b}\right)^2 + c^2 &= \left(s - \frac{ab-c^2}{a+b}\right)^2 + t^2 \\ \Leftrightarrow \quad s^2 + t^2 - \frac{2(ab-c^2)}{a+b}s - c^2 &= 0. \end{aligned} \tag{2}$$

Next,

$$AP: \quad y = \frac{t}{s-a}x - \frac{at}{s-a}.$$

Putting this in (1), we solve

$$\left(1 + \frac{t^2}{(s-a)^2}\right)x^2 + (\dots)x + \left(\frac{a^2t^2}{(s-a)^2} + \frac{at(c^2+ab)}{c(s-a)} + ab\right) = 0.$$

This means

$$\frac{(s-a)^2 + t^2}{(s-a)^2}x^2 + (\dots)x + \frac{a^2ct^2 + a(c^2+ab)(s-a)t + abc(s-a)^2}{c(s-a)^2} = 0.$$

From the product of roots, since one of the roots is a , the other root is

$$\begin{aligned} x &= \frac{a^2 ct^2 + a(c^2 + ab)(s-a)t + abc(s-a)^2}{c(s-a)^2} \div \frac{(s-a)^2 + t^2}{(s-a)^2} \div a \\ &= \frac{act^2 + (c^2 + ab)(s-a)t + bc(s-a)^2}{c((s-a)^2 + t^2)}. \end{aligned}$$

Then

$$\begin{aligned} y &= \frac{t}{s-a} \cdot \left(\frac{act^2 + (c^2 + ab)(s-a)t + bc(s-a)^2}{c((s-a)^2 + t^2)} - a \right) \\ &= \frac{t}{s-a} \cdot \frac{(c^2 + ab)(s-a)t + c(b-a)(s-a)^2}{c((s-a)^2 + t^2)} \\ &= \frac{(c^2 + ab)t^2 + c(b-a)(s-a)t}{c((s-a)^2 + t^2)}. \end{aligned}$$

This gives $K \left(\frac{act^2 + (c^2 + ab)(s-a)t + bc(s-a)^2}{c((s-a)^2 + t^2)}, \frac{(c^2 + ab)t^2 + c(b-a)(s-a)t}{c((s-a)^2 + t^2)} \right)$. We have $L \left(\frac{bct^2 + (c^2 + ab)(s-b)t + ac(s-b)^2}{c((s-b)^2 + t^2)}, \frac{(c^2 + ab)t^2 + c(a-b)(s-b)t}{c((s-b)^2 + t^2)} \right)$ by symmetry.

Similarly,

$$CP: \quad y = \frac{t-c}{s}x + c.$$

Putting this in (1), we solve

$$\left(1 + \frac{(t-c)^2}{s^2} \right) x^2 + \left(\frac{2c(t-c)}{s} - (a+b) - \frac{(c^2 + ab)(t-c)}{cs} \right) x = 0,$$

i.e.

$$\frac{s^2 + (t-c)^2}{s^2} x^2 + \frac{(c^2 - ab)(t-c) - c(a+b)s}{cs} x = 0.$$

The nonzero root is

$$x = \frac{c(a+b)s^2 - (c^2 - ab)s(t-c)}{c(s^2 + (t-c)^2)}.$$

Then

$$\begin{aligned} y &= \frac{t-c}{s} \cdot \frac{c(a+b)s^2 - (c^2 - ab)s(t-c)}{c(s^2 + (t-c)^2)} + c \\ &= \frac{c(a+b)s(t-c) + ab(t-c)^2 + c^2 s^2}{c(s^2 + (t-c)^2)}. \end{aligned}$$

This means $M \left(\frac{c(a+b)s^2 - (c^2 - ab)s(t-c)}{c(s^2 + (t-c)^2)}, \frac{c(a+b)s(t-c) + ab(t-c)^2 + c^2 s^2}{c(s^2 + (t-c)^2)} \right)$.

Note that $MK = ML$ iff $KL \perp MO$. We find that

$$\begin{aligned}
m_{KL} &= \frac{((c^2 + ab)t^2 + c(b-a)(s-a)t)((s-b)^2 + t^2)}{-((c^2 + ab)t^2 + c(a-b)(s-b)t)((s-a)^2 + t^2)} \\
&= \frac{(act^2 + (c^2 + ab)(s-a)t + bc(s-a)^2)((s-b)^2 + t^2)}{- (bct^2 + (c^2 + ab)(s-b)t + ac(s-b)^2)((s-a)^2 + t^2)} \\
&= \frac{(a-b)(2s-a-b)(-ct^2 + (c^2 + ab)t - c(s-a)(s-b))t}{(a-b)(-ct^2 + (c^2 + ab)t - c(s-a)(s-b))(-t^2 + (s-a)(s-b))} \\
&= \frac{(2s-a-b)t}{-t^2 + (s-a)(s-b)}
\end{aligned}$$

and

$$\begin{aligned}
m_{MO} &= \frac{2(c(a+b)s(t-c) + ab(t-c)^2 + c^2s^2) - (c^2 + ab)(s^2 + (t-c)^2)}{2(c(a+b)s^2 - (c^2 - ab)s(t-c)) - (a+b)c(s^2 + (t-c)^2)} \\
&= \frac{(c^2 - ab)s^2 + 2c(a+b)s(t-c) + (ab - c^2)(t-c)^2}{c(a+b)s^2 - 2(c^2 - ab)s(t-c) - c(a+b)(t-c)^2}.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\Leftrightarrow \frac{KL \perp MO}{m_{KL} \cdot m_{MO} = -1} \\
&\Leftrightarrow \frac{(2s-a-b)t}{-t^2 + (s-a)(s-b)} \cdot \frac{(c^2 - ab)s^2 + 2c(a+b)s(t-c) + (ab - c^2)(t-c)^2}{c(a+b)s^2 - 2(c^2 - ab)s(t-c) - c(a+b)(t-c)^2} = -1.
\end{aligned}$$

Expanding everything, this is equivalent to

$$\begin{aligned}
&c(a+b)s^4 + 2c(a+b)s^2t^2 + c(a+b)t^4 + c(2(c^2 - ab) - (a+b)^2)s^3 \\
&- (a+b)(c^2 + ab)s^2t + c(2(c^2 - ab) - (a+b)^2)st^2 - (a+b)(c^2 + ab)t^3 \\
&- 3c(a+b)(c^2 - ab)s^2 - 2(c^2 - ab)(ab + c^2)st + c(a+b)(ab - c^2)t^2 \\
&+ c(2ab(c^2 - ab) + c^2(a+b)^2)s + c^2(a+b)(ab + c^2)t - abc^3(a+b) = 0.
\end{aligned}$$

In view of (2), we factorize this as

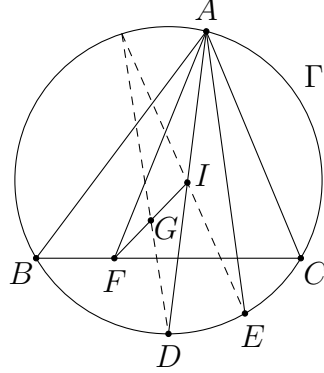
$$((a+b)s^2 + (a+b)t^2 - 2(ab - c^2)s - (a+b)c^2)(cs^2 + ct^2 - c(a+b)s - (c^2 + ab)t + abc) = 0.$$

The second bracket is exactly the same as (1). This means P lies on Γ . However, it is known that P lies inside $\triangle ABC$. So the second bracket is nonzero. The first bracket is the same as (2). So $MK = ML$ iff $SC = SP$ as desired. $\square$

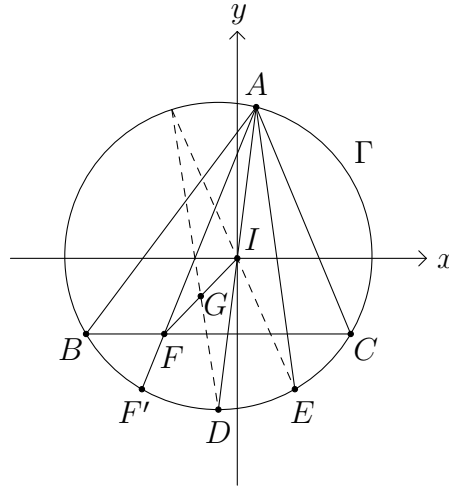
Problem 52. (IMO Shortlist 2010 G4/Problem 2) Let I be the incentre of triangle ABC and let Γ be its circumcircle. Let the line AI intersect Γ again at D . Let E be a point on the arc $\widehat{BDC}$ and F a point on the side BC such that

$$\angle BAF = \angle CAE < \frac{1}{2} \angle BAC.$$

Finally, let G be the midpoint of the segment IF . Prove that the lines DG and EI intersect on Γ .



Solution.



Let $I(0,0)$, $B(b, -1)$, $C(c, -1)$. Then $A\left(\frac{b+c}{bc+1}, \frac{bc-1}{bc+1}\right)$ and $D\left(\frac{b+c}{2}, \frac{bc-1}{2}\right)$ analogous to the usual setting.

Let O be the centre of Γ . Then $O\left(\frac{b+c}{2}, \frac{b^2c^2 - b^2 - c^2 - 3}{4(bc+1)}\right)$ and

$$\Gamma: x^2 + y^2 - (b+c)x - \frac{b^2c^2 - b^2 - c^2 - 3}{2(bc+1)}y + \frac{b^2c^2 + b^2 + c^2 + 1}{2(bc+1)} = 0.$$

Let AF meet Γ again at F' . Then the angle conditions imply E and F' have the same y -coordinate. Let $E(\alpha, \gamma)$ and $F'(\beta, \gamma)$. Then these coordinates satisfy the equation of Γ . This means if we put $y = \gamma$ in the equation, then the roots are $x = \alpha$ and $x = \beta$. So

$$\alpha + \beta = b + c, \tag{1}$$

$$\alpha\beta = \gamma^2 - \frac{b^2c^2 - b^2 - c^2 - 3}{2(bc+1)}\gamma + \frac{b^2c^2 + b^2 + c^2 + 1}{2(bc+1)}. \tag{2}$$

Now, we find that

$$AF' : y = \frac{(bc+1)\gamma - (bc-1)}{(bc+1)\beta - (b+c)}x + \frac{(bc-1)\beta - (b+c)\gamma}{(bc+1)\beta - (b+c)}.$$

Solving with

$$BC : y = -1,$$

we get $F\left(\frac{(b+c)\gamma - 2bc\beta + (b+c)}{(bc+1)\gamma - (bc-1)}, -1\right)$ and hence $G\left(\frac{(b+c)\gamma - 2bc\beta + (b+c)}{2((bc+1)\gamma - (bc-1))}, -\frac{1}{2}\right)$.

To show that the intersection of DG and EI lies on Γ , it is the same as showing $\angle GDA = \angle IEA$, i.e. $\frac{m_{GD} - m_{AD}}{1 + m_{GD} \cdot m_{AD}} = \frac{m_{IE} - m_{AE}}{1 + m_{IE} \cdot m_{AE}}$.

We find that

$$\begin{aligned} \frac{m_{GD} - m_{AD}}{1 + m_{GD} \cdot m_{AD}} &= \frac{\frac{(bc+1)\gamma - (bc-1)}{(b+c)\gamma + 2\beta - (b+c)} - \frac{bc-1}{b+c}}{1 + \frac{(bc+1)\gamma - (bc-1)}{(b+c)\gamma + 2\beta - (b+c)} \cdot \frac{bc-1}{b+c}} \\ &= \frac{2(b+c)\gamma - 2(bc-1)\beta}{(b^2c^2 + b^2 + 2bc + c^2 - 1)\gamma + 2(b+c)\beta - (b^2c^2 + b^2 + c^2 + 1)} \end{aligned} \quad (3)$$

and

$$\begin{aligned} \frac{m_{IE} - m_{AE}}{1 + m_{IE} \cdot m_{AE}} &= \frac{\frac{\gamma}{\alpha} - \frac{(bc+1)\gamma - (bc-1)}{(bc+1)\alpha - (b+c)}}{1 + \frac{\gamma}{\alpha} \cdot \frac{(bc+1)\gamma - (bc-1)}{(bc+1)\alpha - (b+c)}} \\ &= \frac{-(b+c)\gamma + (bc-1)\alpha}{(bc+1)(\alpha^2 + \gamma^2) - (bc-1)\gamma - (b+c)\alpha}. \end{aligned}$$

To further simplify the expression, we use the fact that

$$\alpha^2 + \gamma^2 - (b+c)\alpha - \frac{b^2c^2 - b^2 - c^2 - 3}{2(bc+1)}\gamma + \frac{b^2c^2 + b^2 + c^2 + 1}{2(bc+1)} = 0$$

from the equation of Γ . We use this to convert the term $\alpha^2 + \gamma^2$ to degree 1. Thus,

$$\frac{m_{IE} - m_{AE}}{1 + m_{IE} \cdot m_{AE}} = \frac{-2(b+c)\gamma + 2(bc-1)\alpha}{(b^2c^2 - b^2 - 2bc - c^2 - 1)\gamma + 2bc(b+c)\alpha - (b^2c^2 + b^2 + c^2 + 1)}. \quad (4)$$

Combining (3) and (4), we need to prove

$$\begin{aligned} &((b+c)\gamma - (bc-1)\beta)((b^2c^2 - b^2 - 2bc - c^2 - 1)\gamma + 2bc(b+c)\alpha - (b^2c^2 + b^2 + c^2 + 1)) \\ &= (-(b+c)\gamma + (bc-1)\alpha)((b^2c^2 + b^2 + 2bc + c^2 - 1)\gamma + 2(b+c)\beta - (b^2c^2 + b^2 + c^2 + 1)). \end{aligned}$$

After simplification, this is the same as

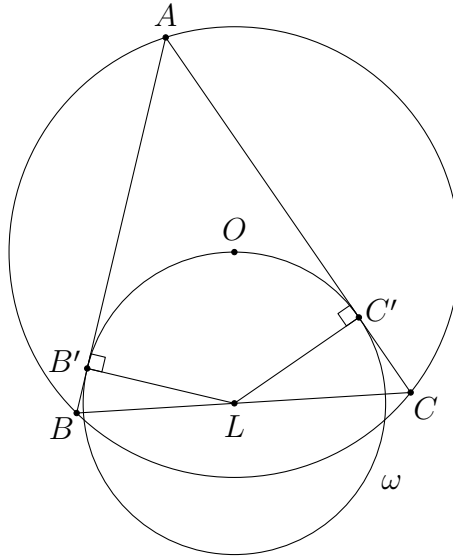
$$\begin{aligned} &2(b+c)(1 - b^2c^2)\gamma^2 + (b^3c^3 - b^3c - 3b^2c^2 - bc^3 - b^2 - 3bc - c^2 + 1)(\alpha + \beta)\gamma \\ &+ 2(b+c)(b^2c^2 + b^2 + c^2 + 1)\gamma + 2(b+c)(b^2c^2 - 1)\alpha\beta \\ &- (bc-1)(b^2c^2 + b^2 + c^2 + 1)(\alpha + \beta) = 0. \end{aligned}$$

We now make use of the relations (1) and (2). Then the left-hand side is equal to

$$\begin{aligned}
& 2(b+c)(1-b^2c^2)\gamma^2 + (b^3c^3 - b^3c - 3b^2c^2 - bc^3 - b^2 - 3bc - c^2 + 1)(b+c)\gamma \\
& + 2(b+c)(b^2c^2 + b^2 + c^2 + 1)\gamma - (bc-1)(b^2c^2 + b^2 + c^2 + 1)(b+c) \\
& + 2(b+c)(b^2c^2 - 1)\left(\gamma^2 - \frac{b^2c^2 - b^2 - c^2 - 3}{2(bc+1)}\gamma + \frac{b^2c^2 + b^2 + c^2 + 1}{2(bc+1)}\right) \\
& = (2(b+c)(1-b^2c^2) + 2(b+c)(b^2c^2 - 1))\gamma^2 \\
& + ((b^3c^3 - b^3c - 3b^2c^2 - bc^3 - b^2 - 3bc - c^2 + 1)(b+c) + 2(b+c)(b^2c^2 + b^2 + c^2 + 1) \\
& - (b+c)(bc-1)(b^2c^2 - b^2 - c^2 - 3))\gamma \\
& - (bc-1)(b^2c^2 + b^2 + c^2 + 1)(b+c) + (b+c)(bc-1)(b^2c^2 + b^2 + c^2 + 1) = 0
\end{aligned}$$

as desired. $\square$

Problem 53. (IMO Shortlist 2011 G1) Let ABC be an acute triangle. Let ω be a circle whose centre L lies on the side BC . Suppose that ω is tangent to AB at B' and to AC at C' . Suppose also that the circumcentre O of the triangle ABC lies on the shorter arc $B'C'$ of ω . Prove that the circumcircle of ABC and ω meet at two points.

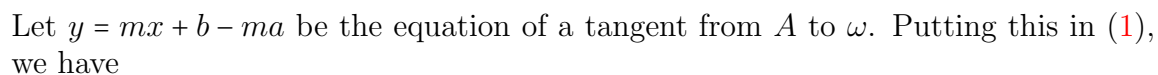


Solution. Let $O(0,0)$, $L(1,0)$ and $A(a,b)$. Let $x^2 + y^2 = R^2$ be the equation of (ABC) . This means

$$a^2 + b^2 = R^2.$$

As L is the centre of ω and O lies on ω , we have

$$\omega : x^2 + y^2 - 2x = 0. \quad (1)$$



As there is a unique solution, we need

This implies

The two roots of this quadratic equation correspond to the slopes of AB and AC . For simplicity, we write $m = m_{AB}$ and $n = m_{AC}$. Then we have

$$mn = \frac{1 - b^2}{a(2 - a)}. \quad (3)$$

Putting $y = mx + b - ma$ in $x^2 + y^2 = R^2$, we solve

As a is one of the roots, the other root is

This is the x -coordinate of B , and hence $B\left(\frac{(m^2-1)a-2mb}{m^2+1}, \frac{-2ma-(m^2-1)b}{m^2+1}\right)$.

By symmetry, we have $C\left(\frac{(n^2-1)a-2nb}{n^2+1}, \frac{-2na-(n^2-1)b}{n^2+1}\right)$. It follows that

$$\begin{aligned} m_{BC} &= \frac{(-2na-(n^2-1)b)(m^2+1) - (-2ma-(m^2-1)b)(n^2+1)}{((n^2-1)a-2nb)(m^2+1) - ((m^2-1)a-2mb)(n^2+1)} \\ &= \frac{2(m-n)(-mna+(m+n)b+a)}{2(m-n)(-(m+n)a-mnb+b)} \\ &= \frac{mna-(m+n)b-a}{(m+n)a+mnb-b}. \end{aligned}$$

This easily implies

$$BC: \quad y = \frac{mna-(m+n)b-a}{(m+n)a+mnb-b}x - \frac{(mn+1)(a^2+b^2)}{(m+n)a+mnb-b}.$$

Since L lies on BC , we must have

$$0 = \frac{mna-(m+n)b-a}{(m+n)a+mnb-b} - \frac{(mn+1)(a^2+b^2)}{(m+n)a+mnb-b}.$$

Using (2) and (3), this implies

$$\frac{1-b^2}{2-a} - \frac{2b^2(1-a)}{a(2-a)} - a - \frac{1-b^2+2a-a^2}{a(2-a)} \cdot (a^2+b^2) = 0.$$

After expansion and simplification, this becomes

$$a^4 + 2a^2b^2 + b^4 - a^3 - ab^2 - 3a^2 - 3b^2 + a = 0.$$

Using $a^2 + b^2 = R^2$, this yields $R^4 - aR^2 - 3R^2 + a = 0$, and hence

$$a = \frac{R^4 - 3R^2}{R^2 - 1}.$$

As O lies on the minor arc $B'C'$, we must have $a < 0$. Also, we must have $R > 1$ since $OL = 1$ and L lies inside (ABC) . This implies $R < \sqrt{3}$. This shows ω and (ABC) meet at two distinct points as the diameter of ω is 2, which is larger than R . This proves the assertion. $\square$

Problem 54. (IMO Shortlist 2011 G2) Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcentre and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

Solution. Let $A_1(a, b)$, $A_2(c, d)$, $A_3(e, f)$, $A_4(g, h)$. Since $m_{A_2A_3} = \frac{f-d}{e-c}$, the perpendicular bisector of A_2A_3 has slope $-\frac{e-c}{f-d}$, and hence its equation is

$$y = -\frac{e-c}{f-d}x + \frac{e^2 - c^2 + f^2 - d^2}{2(f-d)}.$$

By symmetry, the perpendicular bisector of A_2A_4 has equation

$$y = -\frac{g-c}{h-d}x + \frac{g^2 - c^2 + h^2 - d^2}{2(h-d)}.$$

Solving these equations, we get

$$\begin{aligned} x &= \frac{(g^2 - c^2 + h^2 - d^2)(f-d) - (e^2 - c^2 + f^2 - d^2)(h-d)}{2((g-c)(f-d) - (e-c)(h-d))} \\ &= \frac{(h-f)(c^2 + d^2) + (d-h)(e^2 + f^2) + (f-d)(g^2 + h^2)}{2(ch + ed + gf - cf - eh - gd)} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{-(e-c)((h-f)(c^2 + d^2) + (d-h)(e^2 + f^2) + (f-d)(g^2 + h^2)) + (e^2 - c^2 + f^2 - d^2)(ch + ed + gf - cf - eh - gd)}{2(f-d)(ch + ed + gf - cf - eh - gd)} \\ &= \frac{(e-g)(c^2 + d^2) + (g-c)(e^2 + f^2) + (c-e)(g^2 + h^2)}{2(ch + ed + gf - cf - eh - gd)}. \end{aligned}$$

So

$$O_1 \left(\frac{(h-f)(c^2 + d^2) + (d-h)(e^2 + f^2) + (f-d)(g^2 + h^2)}{2(ch + ed + gf - cf - eh - gd)}, \frac{(e-g)(c^2 + d^2) + (g-c)(e^2 + f^2) + (c-e)(g^2 + h^2)}{2(ch + ed + gf - cf - eh - gd)} \right).$$

It follows that

$$\begin{aligned} &\frac{1}{O_1A_1^2 - r_1^2} \\ &= \frac{1}{(x-a)^2 + (y-b)^2 - (x-c)^2 - (y-d)^2} \\ &= \frac{1}{(c-a)(2x-a-c) + (d-b)(2y-b-d)} \\ &= \frac{ch + ed + gf - cf - eh - gd}{(c-a)((h-f)(c^2 + d^2) + (d-h)(e^2 + f^2) + (f-d)(g^2 + h^2) - (a+c)(ch + ed + gf - cf - eh - gd)) + (d-b)((e-g)(c^2 + d^2) + (g-c)(e^2 + f^2) + (c-e)(g^2 + h^2) - (b+d)(ch + ed + gf - cf - eh - gd))} \end{aligned}$$

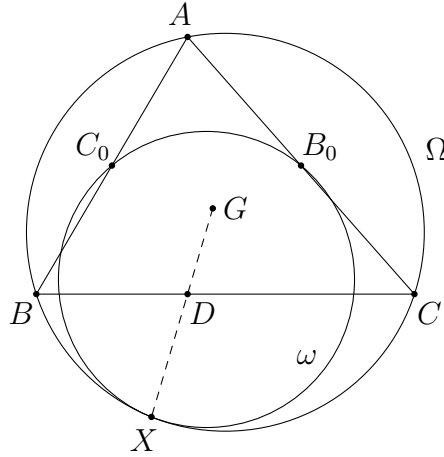
$$= \frac{ch + ed + gf - cf - eh - gd}{(ch + ed + gf - cf - eh - gd)(a^2 + b^2) + (af + eh + gb - ah - eb - gf)(c^2 + d^2) + (ah + cb + gd - ad - ch - gb)(e^2 + f^2) + (ad + cf + eb - af - cb - ed)(g^2 + h^2)}. \quad (1)$$

The second term $\frac{1}{O_2A_2^2 - r_2^2}$ can be found by swapping a and c , and swapping b and d . The numerator of this term is $ah + eb + gf - af - eh - gb$, while the denominator is -1 times that of (1). Similarly, the numerators of $\frac{1}{O_3A_3^2 - r_3^2}$ and $\frac{1}{O_4A_4^2 - r_4^2}$ after simplification are $ch + ad + gb - cb - ah - gd$ and $cb + ed + af - cf - eb - ad$ respectively, and both denominators are -1 times that of (1). Thus, by noting

$$\begin{aligned} & (ch + ed + gf - cf - eh - gd) - (ah + eb + gf - af - eh - gb) \\ & - (ch + ad + gb - cb - ah - gd) - (cb + ed + af - cf - eb - ad) = 0, \end{aligned}$$

we get the desired identity. $\square$

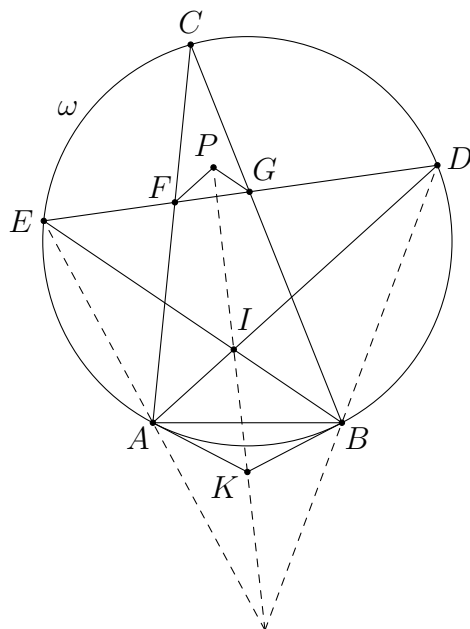
Problem 55. (IMO Shortlist 2011 G4) Let ABC be an acute triangle with circumcircle Ω . Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB . Let D be the foot of the altitude from A , and let G be the centroid of the triangle ABC . Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G , and X are collinear.



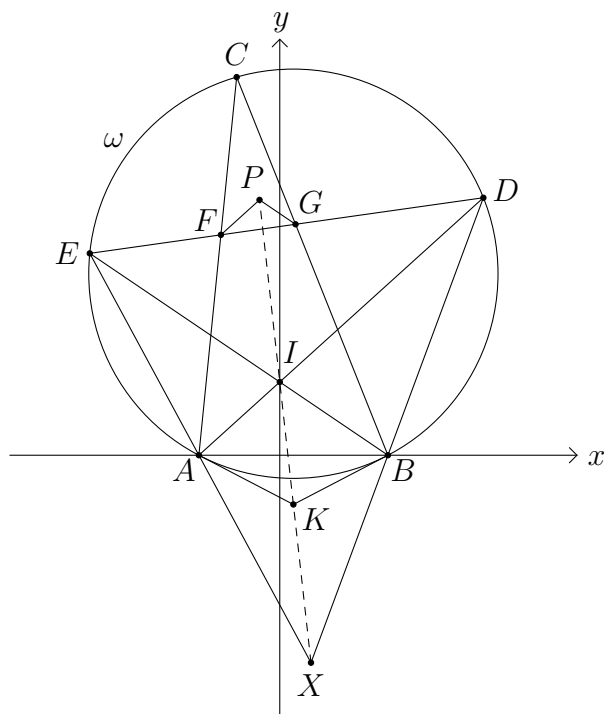
Solution. See [problem 4.4](#) of **Coordinate Geometry 5.0**. $\square$

Problem 56. (IMO Shortlist 2011 G5) Let ABC be a triangle with incentre I and circumcircle ω . Let D and E be the second intersection points of ω with the lines AI and BI , respectively. The chord DE meets AC at a point F , and BC at a point G . Let P be the intersection point of the line through F parallel to AD and the line through G parallel to BE . Suppose that the tangents to ω at A and at B meet

at a point K . Prove that the three lines AE, BD , and KP are either parallel or concurrent.



Solution.



Let $I(0,1)$, $A(a,0)$, $B(b,0)$. Then $C\left(\frac{a+b}{ab+1}, \frac{2ab}{ab+1}\right)$,

$$AC: \quad y = -\frac{2a}{a^2-1}x + \frac{2a^2}{a^2-1} \quad (1)$$

and

$$\omega: \quad x^2 + y^2 - (a+b)x - \frac{a^2b^2 - a^2 + 4ab - b^2 + 1}{2(ab+1)}y + ab = 0. \quad (2)$$

The equation of AI is

$$AI: \quad y = -\frac{1}{a}x + 1.$$

Putting this in (2), we get

$$\left(1 + \frac{1}{a^2}\right)x^2 + (\dots)x + \left(1 - \frac{a^2b^2 - a^2 + 4ab - b^2 + 1}{2(ab+1)} + ab\right) = 0.$$

This is the same as

$$\frac{a^2+1}{a^2}x^2 + (\dots)x + \frac{(a^2+1)(b^2+1)}{2(ab+1)} = 0.$$

From the product of roots, since one of the roots is a , the other root is

$$x = \frac{(a^2+1)(b^2+1)}{2(ab+1)} \div \frac{a^2+1}{a^2} \div a = \frac{a(b^2+1)}{2(ab+1)}.$$

Then

$$y = -\frac{1}{a} \cdot \frac{a(b^2+1)}{2(ab+1)} + 1 = \frac{2ab - b^2 + 1}{2(ab+1)}.$$

This gives $D\left(\frac{a(b^2+1)}{2(ab+1)}, \frac{2ab - b^2 + 1}{2(ab+1)}\right)$. So $E\left(\frac{b(a^2+1)}{2(ab+1)}, \frac{2ab - a^2 + 1}{2(ab+1)}\right)$ by symmetry.

Then we find that

$$m_{DE} = \frac{(2ab - b^2 + 1) - (2ab - a^2 + 1)}{a(b^2+1) - b(a^2+1)} = \frac{(a-b)(a+b)}{(a-b)(1-ab)} = \frac{a+b}{1-ab}.$$

(Indeed, it is well-known that $DE \perp CI$.) Hence, the equation of DE is

$$y = \frac{a+b}{1-ab}x - \frac{a+b}{1-ab} \cdot \frac{a(b^2+1)}{2(ab+1)} + \frac{2ab - b^2 + 1}{2(ab+1)},$$

i.e.

$$DE: \quad y = \frac{a+b}{1-ab}x - \frac{3a^2b^2 + a^2 + b^2 - 1}{2(ab+1)(1-ab)}.$$

Solving this with (1), we have

$$x = \frac{4a^2(ab+1)(1-ab) + (a^2-1)(3a^2b^2 + a^2 + b^2 - 1)}{2(ab+1)((a+b)(a^2-1) + 2a(1-ab))}$$

$$\begin{aligned}
&= \frac{-a^4b^2 + a^4 - 2a^2b^2 + 2a^2 - b^2 + 1}{2(ab+1)(a^3 - a^2b + a - b)} \\
&= \frac{(a^2+1)^2(1-b^2)}{2(ab+1)(a-b)(a^2+1)} \\
&= \frac{(a^2+1)(1-b^2)}{2(ab+1)(a-b)}
\end{aligned}$$

and

$$\begin{aligned}
y &= \frac{2a}{a^2-1} \cdot \left(a - \frac{(a^2+1)(1-b^2)}{2(ab+1)(a-b)} \right) \\
&= \frac{2a}{a^2-1} \cdot \frac{2a^3b - a^2b^2 + a^2 - 2ab + b^2 - 1}{2(ab+1)(a-b)} \\
&= \frac{2a}{a^2-1} \cdot \frac{(a^2-1)(2ab-b^2+1)}{2(ab+1)(a-b)} \\
&= \frac{a(2ab-b^2+1)}{(ab+1)(a-b)}.
\end{aligned}$$

This yields $F\left(\frac{(a^2+1)(1-b^2)}{2(ab+1)(a-b)}, \frac{a(2ab-b^2+1)}{(ab+1)(a-b)}\right)$. Since $m_{FP} = m_{AI} = -\frac{1}{a}$, we easily get

$$FP: \quad y = -\frac{1}{a}x + \frac{4a^3b - 3a^2b^2 + 3a^2 - b^2 + 1}{2a(ab+1)(a-b)}.$$

By symmetry, we have

$$GP: \quad y = -\frac{1}{b}x + \frac{4ab^3 - 3a^2b^2 + 3b^2 - a^2 + 1}{2b(ab+1)(b-a)}.$$

Solving these, we obtain

$$\begin{aligned}
x &= -\frac{a(4ab^3 - 3a^2b^2 + 3b^2 - a^2 + 1) + b(4a^3b - 3a^2b^2 + 3a^2 - b^2 + 1)}{2(ab+1)(b-a)^2} \\
&= -\frac{(a+b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{2(ab+1)(b-a)^2}
\end{aligned}$$

and

$$\begin{aligned}
y &= \frac{1}{a} \cdot \frac{(a+b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{2(ab+1)(b-a)^2} + \frac{4a^3b - 3a^2b^2 + 3a^2 - b^2 + 1}{2a(ab+1)(a-b)} \\
&= \frac{(a+b)(a^2b^2 - a^2 + 4ab - b^2 + 1) + (a-b)(4a^3b - 3a^2b^2 + 3a^2 - b^2 + 1)}{2a(ab+1)(b-a)^2} \\
&= \frac{4a^4b - 6a^3b^2 + 4a^2b^3 + 2a^3 + 2ab^2 + 2a}{2a(ab+1)(b-a)^2} \\
&= \frac{2a^3b - 3a^2b^2 + 2ab^3 + a^2 + b^2 + 1}{(ab+1)(b-a)^2}.
\end{aligned}$$

Thus, $P \left(-\frac{(a+b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{2(ab+1)(b-a)^2}, \frac{2a^3b - 3a^2b^2 + 2ab^3 + a^2 + b^2 + 1}{(ab+1)(b-a)^2} \right)$.

Next, the circumcentre O of $\triangle ABC$ has coordinates $\left(\frac{a+b}{2}, \frac{a^2b^2 - a^2 + 4ab - b^2 + 1}{4(ab+1)} \right)$.

Thus,

$$m_{AK} = \frac{-1}{m_{OA}} = -\frac{2(b-a)(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1}.$$

This yields

$$AK: \quad y = -\frac{2(b-a)(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1}x + \frac{2a(b-a)(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1}.$$

Note that K lies on the perpendicular bisector of AB . Therefore, its x -coordinate is $\frac{a+b}{2}$. From

$$\frac{2(b-a)(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1} \cdot \left(a - \frac{a+b}{2} \right) = -\frac{(b-a)^2(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1},$$

we get $K \left(\frac{a+b}{2}, -\frac{(b-a)^2(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1} \right)$. From the coordinates of P and K , we compute

$$\begin{aligned} m_{KP} &= -\frac{2((2a^3b - 3a^2b^2 + 2ab^3 + a^2 + b^2 + 1)(a^2b^2 - a^2 + 4ab - b^2 + 1) + (b-a)^4(ab+1)^2)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)((ab+1)(b-a)^2 + (a^2b^2 - a^2 + 4ab - b^2 + 1))} \\ &= -\frac{2(a^6b^2 - 2a^5b^3 + 3a^4b^4 - 2a^3b^5 + a^2b^6 + 4a^4b^2 - 4a^3b^3 + 4a^2b^4 + 2a^3b + 2a^2b^2 + 2ab^3 + 4ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)(a^3b - a^2b^2 + ab^3 + 2ab + 1)} \\ &= -\frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)}. \end{aligned}$$

The equation of KP is

$$\begin{aligned} y &= -\frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)}x + \frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)} \cdot \frac{a+b}{2} \\ &\quad - \frac{(b-a)^2(ab+1)}{a^2b^2 - a^2 + 4ab - b^2 + 1}, \end{aligned}$$

i.e.

$$KP: \quad y = -\frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a+b)}x + 1. \quad (3)$$

(One may note the interesting result that $I(0,1)$ lies on KP .)

Lastly, from $A(a,0)$ and $E \left(\frac{b(a^2+1)}{2(ab+1)}, \frac{2ab - a^2 + 1}{2(ab+1)} \right)$, we get

$$m_{AE} = \frac{2ab - a^2 + 1}{b(a^2+1) - 2a(ab+1)} = \frac{2ab - a^2 + 1}{-a^2b - 2a + b}$$

so that

$$AE: \quad y = \frac{2ab - a^2 + 1}{-a^2b - 2a + b}x - \frac{a(2ab - a^2 + 1)}{-a^2b - 2a + b}.$$

By symmetry, we have

$$BD: \quad y = \frac{2ab - b^2 + 1}{-ab^2 - 2b + a}x - \frac{b(2ab - b^2 + 1)}{-ab^2 - 2b + a}.$$

The case $AE \parallel BD$ is easy to handle, or we regard this as the limiting case. In general, let X be the intersection point of AE and BD . By solving the above equations, we get

$$\begin{aligned} x &= \frac{a(2ab - a^2 + 1)(-ab^2 - 2b + a) - b(2ab - b^2 + 1)(-a^2b - 2a + b)}{(2ab - a^2 + 1)(-ab^2 - 2b + a) - (2ab - b^2 + 1)(-a^2b - 2a + b)} \\ &= \frac{(a - b)(a + b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{(a - b)(3a^2b^2 - a^2 + 8ab - b^2 + 3)} \\ &= \frac{(a + b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{2ab - a^2 + 1}{-a^2b - 2a + b} \cdot \left(\frac{(a + b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3} - a \right) \\ &= \frac{2ab - a^2 + 1}{-a^2b - 2a + b} \cdot \frac{-2a^3b^2 + a^2b^3 - 5a^2b + 4ab^2 - b^3 - 2a + b}{3a^2b^2 - a^2 + 8ab - b^2 + 3} \\ &= \frac{(2ab - a^2 + 1)(2ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3}. \end{aligned}$$

Therefore, we have $X \left(\frac{(a + b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3}, \frac{(2ab - a^2 + 1)(2ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3} \right)$.

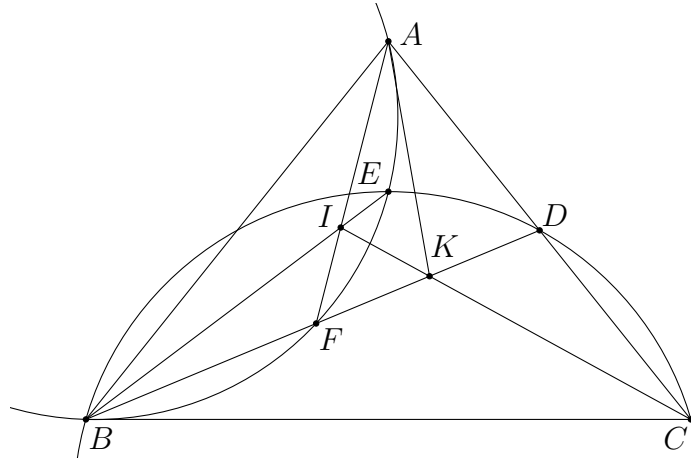
It remains to check that the coordinates satisfy (3). Indeed, we have

$$\begin{aligned} & - \frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{(a^2b^2 - a^2 + 4ab - b^2 + 1)(a + b)} \cdot \frac{(a + b)(a^2b^2 - a^2 + 4ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3} + 1 \\ &= - \frac{2(a^3b - a^2b^2 + ab^3 + 2ab + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3} + 1 \\ &= \frac{-2a^3b + 5a^2b^2 - 2ab^3 - a^2 + 4ab - b^2 + 1}{3a^2b^2 - a^2 + 8ab - b^2 + 3} \\ &= \frac{(2ab - a^2 + 1)(2ab - b^2 + 1)}{3a^2b^2 - a^2 + 8ab - b^2 + 3}. \end{aligned}$$

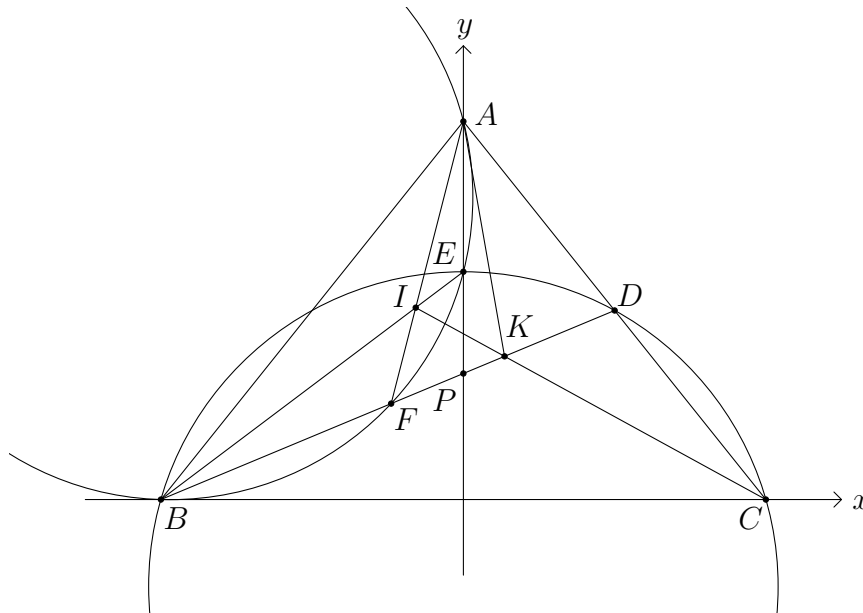
This shows X lies on KP . Thus, AE, BD, KP are concurrent (or parallel). $\square$

Problem 57. (IMO Shortlist 2011 G6) Let ABC be a triangle with $AB = AC$, and let D be the midpoint of AC . The angle bisector of $\angle BAC$ intersects the circle through D, B , and C in a point E inside the triangle ABC . The line BD intersects

the circle through A, E , and B in two points B and F . The lines AF and BE meet at a point I , and the lines CI and BD meet at a point K . Show that I is the incentre of triangle KAB .



Solution.



Let $A(0, a)$, $B(-1, 0)$, $C(1, 0)$, $D\left(\frac{1}{2}, \frac{a}{2}\right)$. Note that E lies on the y -axis. Let $E(0, b)$. The circumcentre O of $\triangle BCD$ also lies on the y -axis. Also, from

$$\begin{aligned}
 & OB = OD \\
 \Rightarrow & 1^2 + y^2 = \left(\frac{1}{2}\right)^2 + \left(y - \frac{a}{2}\right)^2 \\
 \Rightarrow & 1 = \frac{1}{4} + \frac{a^2}{4} - ay \\
 \Rightarrow & y = \frac{a^2 - 3}{4a},
 \end{aligned}$$

we find that $O\left(0, \frac{a^2 - 3}{4a}\right)$. So

$$(BCD): \quad x^2 + y^2 - \frac{a^2 - 3}{2a}y - 1 = 0.$$

As E lies on this circle, we have $b^2 - \frac{a^2 - 3}{2a}b - 1 = 0$, i.e.

$$a^2b - 2ab^2 + 2a - 3b = 0. \quad (1)$$

Let P be the intersection of AE and BD . Since

$$BD: \quad y = \frac{a}{3}x + \frac{a}{3}, \quad (2)$$

we have $P\left(0, \frac{a}{3}\right)$. As A, B, F, E are concyclic, we have $PA \times PE = PB \times PF$. Noting that the slope of BFP is $\frac{a}{3}$, this implies

$$\left(a - \frac{a}{3}\right)\left(b - \frac{a}{3}\right) = (0 - (-1))(0 - x)\left(1 + \left(\frac{a}{3}\right)^2\right),$$

where x is the x -coordinate of F . This is the same as

$$\frac{2a(3b - a)}{9} = -\frac{a^2 + 9}{9}x.$$

Thus, we have

$$x = \frac{2a(a - 3b)}{a^2 + 9}$$

and

$$y = \frac{a}{3} \cdot \left(\frac{2a(a - 3b)}{a^2 + 9} + 1\right) = \frac{a(a^2 - 2ab + 3)}{a^2 + 9}.$$

It follows that $F\left(\frac{2a(a - 3b)}{a^2 + 9}, \frac{a(a^2 - 2ab + 3)}{a^2 + 9}\right)$. It is easy to get

$$AF: \quad y = -\frac{ab + 3}{a - 3b}x + a$$

and

$$BE: \quad y = bx + b.$$

Solving these, we obtain

$$x = \frac{(a - b)(a - 3b)}{b(a - 3b) + (ab + 3)} = \frac{(a - b)(a - 3b)}{2ab - 3b^2 + 3}$$

and

$$y = b \cdot \left(\frac{(a - b)(a - 3b)}{2ab - 3b^2 + 3} + 1\right) = \frac{b(a^2 - 2ab + 3)}{2ab - 3b^2 + 3}.$$

Therefore, we have $I\left(\frac{(a-b)(a-3b)}{2ab-3b^2+3}, \frac{b(a^2-2ab+3)}{2ab-3b^2+3}\right)$. The equation of CI is

$$y = \frac{b(a^2-2ab+3)}{a^2-6ab+6b^2-3}x - \frac{b(a^2-2ab+3)}{a^2-6ab+6b^2-3}. \quad (3)$$

Solving (2) and (3), we find that

$$x = \frac{a(a^2-6ab+6b^2-3) + 3b(a^2-2ab+3)}{3b(a^2-2ab+3) - a(a^2-6ab+6b^2-3)} = -\frac{a^3-3a^2b-3a+9b}{a^3-9a^2b+12ab^2-3a-9b}$$

and

$$y = \frac{a}{3} \cdot \left(-\frac{a^3-3a^2b-3a+9b}{a^3-9a^2b+12ab^2-3a-9b} + 1 \right) = -\frac{2ab(a^2-2ab+3)}{a^3-9a^2b+12ab^2-3a-9b}.$$

That means $K\left(-\frac{a^3-3a^2b-3a+9b}{a^3-9a^2b+12ab^2-3a-9b}, -\frac{2ab(a^2-2ab+3)}{a^3-9a^2b+12ab^2-3a-9b}\right)$.

We first show that BI bisects $\angle KBA$. From $m_{AB} = a$, $m_{BI} = b$ and $m_{BK} = \frac{a}{3}$, we find that

$$\begin{aligned} & \angle KBI = \angle IBA \\ \Leftrightarrow & \tan \angle KBI = \tan \angle IBA \\ \Leftrightarrow & \frac{m_{BI} - m_{BK}}{1 + m_{BI} \cdot m_{BK}} = \frac{m_{AB} - m_{BI}}{1 + m_{AB} \cdot m_{BI}} \\ \Leftrightarrow & \frac{3b-a}{3+ab} = \frac{a-b}{1+ab} \\ \Leftrightarrow & (3b-a)(1+ab) = (a-b)(3+ab) \\ \Leftrightarrow & -2a^2b + 4ab^2 - 4a + 6b = 0. \end{aligned}$$

This is true by (1).

Next, we show that KI bisects $\angle AKB$. From $m_{BK} = \frac{a}{3}$, $m_{IK} = \frac{b(a^2-2ab+3)}{a^2-6ab+6b^2-3}$

and $m_{AK} = \frac{a(a^3-7a^2b+8ab^2-3a-3b)}{a^3-3a^2b-3a+9b}$, we find that

$$\begin{aligned} \tan \angle IKB &= \frac{m_{BK} - m_{IK}}{1 + m_{BK} \cdot m_{IK}} \\ &= \frac{a(a^2-6ab+6b^2-3) - 3b(a^2-2ab+3)}{3(a^2-6ab+6b^2-3) + ab(a^2-2ab+3)} \\ &= \frac{a^3-9a^2b+12ab^2-3a-9b}{a^3b-2a^2b^2+3a^2-15ab+18b^2-9} \end{aligned} \quad (4)$$

and

$$\begin{aligned}
& \tan \angle AKI \\
&= \frac{m_{IK} - m_{AK}}{1 + m_{IK} \cdot m_{AK}} \\
&= \frac{b(a^2 - 2ab + 3)(a^3 - 3a^2b - 3a + 9b) - a(a^3 - 7a^2b + 8ab^2 - 3a - 3b)(a^2 - 6ab + 6b^2 - 3)}{(a^2 - 6ab + 6b^2 - 3)(a^3 - 3a^2b - 3a + 9b) + ab(a^2 - 2ab + 3)(a^3 - 7a^2b + 8ab^2 - 3a - 3b)} \\
&= \frac{-a^6 + 14a^5b - 61a^4b^2 + 96a^3b^3 - 48a^2b^4 + 6a^4 - 36a^3b + 30a^2b^2 - 9a^2 - 18ab + 27b^2}{a^6b - 9a^5b^2 + 22a^4b^3 - 16a^3b^4 + a^5 - 9a^4b + 6a^3b^2 + 12a^2b^3 - 6a^3 + 27a^2b - 81ab^2 + 54b^3 + 9a - 27b}.
\end{aligned}$$

In view of (4), it is natural to see if we can factorize the numerator or the denominator. Indeed, the numerator is

$$\begin{aligned}
& -a^6 + 14a^5b - 61a^4b^2 + 96a^3b^3 - 48a^2b^4 + 6a^4 - 36a^3b + 30a^2b^2 - 9a^2 - 18ab + 27b^2 \\
&= -(a^3 - 9a^2b + 12ab^2 - 3a - 9b)(a^3 - 5a^2b + 4ab^2 - 3a + 3b).
\end{aligned}$$

Comparing this with (4), it suffices to show

$$\begin{aligned}
& a^6b - 9a^5b^2 + 22a^4b^3 - 16a^3b^4 + a^5 - 9a^4b + 6a^3b^2 + 12a^2b^3 \\
& - 6a^3 + 27a^2b - 81ab^2 + 54b^3 + 9a - 27b \\
&= -(a^3 - 5a^2b + 4ab^2 - 3a + 3b)(a^3b - 2a^2b^2 + 3a^2 - 15ab + 18b^2 - 9).
\end{aligned}$$

After expansion and simplification, this is the same as

$$\begin{aligned}
& 2a^6b - 16a^5b^2 + 36a^4b^3 - 24a^3b^4 + 4a^5 - 42a^4b + 120a^3b^2 - 144a^2b^3 + 72ab^4 \\
& - 24a^3 + 126a^2b - 216ab^2 + 108b^3 + 36a - 54b = 0,
\end{aligned}$$

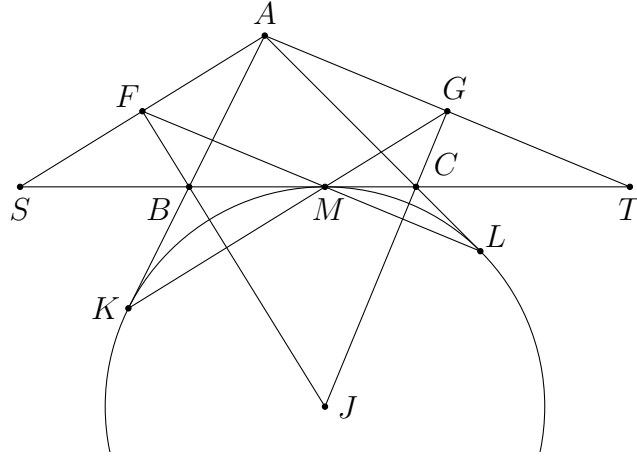
i.e.

$$(a^2b - 2ab^2 + 2a - 3b)(2a^4 - 12a^3b + 12a^2b^2 - 12a^2 + 36ab - 36b^2 + 18) = 0.$$

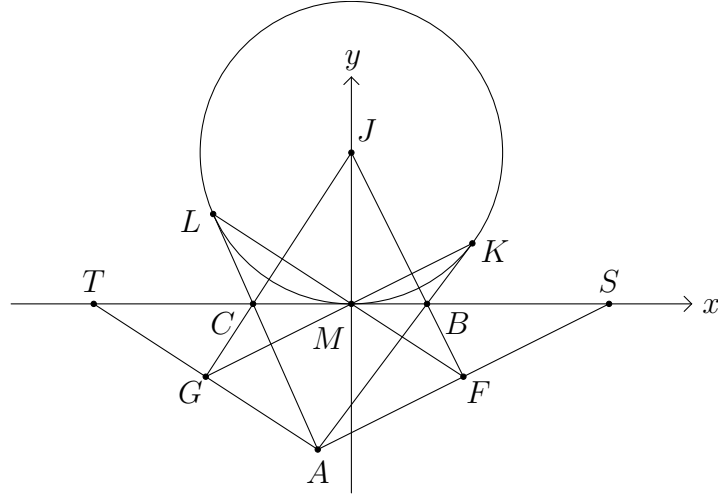
This is true by (1) (which also explains the factorization). So KI bisects $\angle AKB$. This implies I is the incentre of $\triangle ABK$ as desired. $\square$

Problem 58. (IMO Shortlist 2012 G1/Problem 1) Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC .

Prove that M is the midpoint of ST .



Solution.



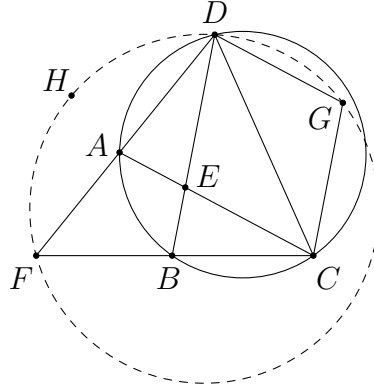
Let $M(0,0)$, $J(0,1)$, $B(b,0)$, $C(c,0)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$ and

$$CJ: \quad y = -\frac{1}{c}x + 1.$$

As $m_{MK} = \frac{-1}{m_{BJ}} = b$, the equation of MK is $y = bx$. Solving these, we easily obtain $G\left(\frac{c}{bc+1}, \frac{bc}{bc+1}\right)$. The equation of AG is $y = cx + \frac{(b-c)c}{bc+1}$. It follows that $T\left(-\frac{b-c}{bc+1}, 0\right)$. By symmetry, we have $S\left(-\frac{c-b}{bc+1}, 0\right)$. It is obvious that M is the midpoint of ST . $\square$

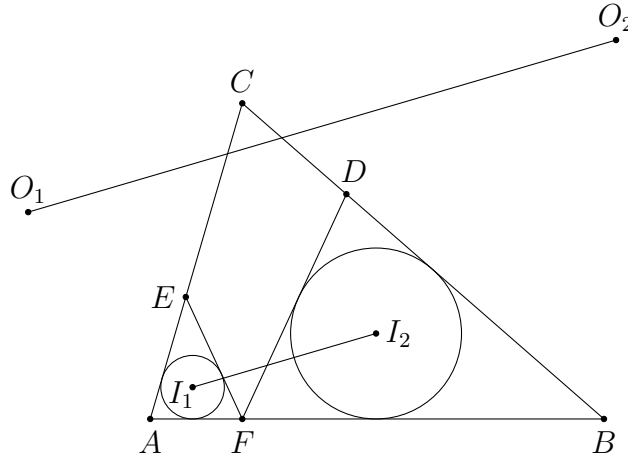
Problem 59. (IMO Shortlist 2012 G2) Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond

A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.



Solution. See [problem 4.7](#) of **Coordinate Geometry 5.0**.

Problem 60. (IMO Shortlist 2012 G3) In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incentres of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcentres of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

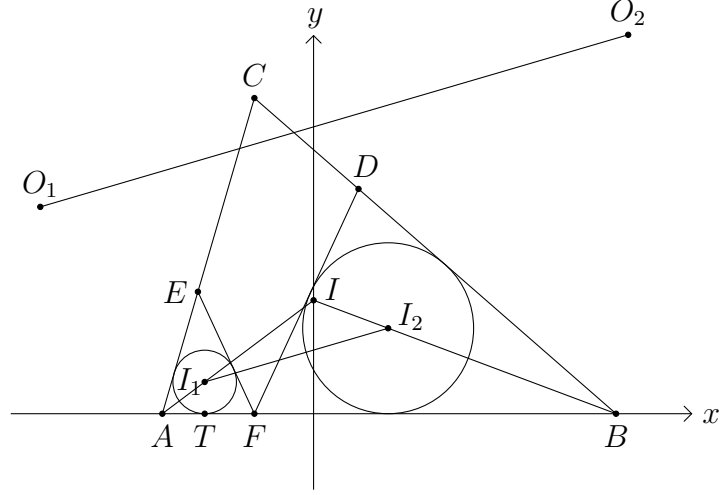


Solution. Let $I(0, 1)$ be the incentre of $\triangle ABC$. We set the origin to be the contact point of the incircle of $\triangle ABC$ and the side AB . Also, let $A(a, 0)$ and $B(b, 0)$ where $a < 0 < b$. Then $C\left(\frac{a+b}{ab+1}, \frac{2ab}{ab+1}\right)$. This yields $F\left(\frac{a+b}{ab+1}, 0\right)$. We have

$$AC: \quad y = -\frac{2a}{a^2-1}x + \frac{2a^2}{a^2-1} \quad (1)$$

and

$$BE: \quad y = \frac{a^2-1}{2a}x - \frac{b(a^2-1)}{2a}. \quad (2)$$



We solve (1) and (2) to get

$$x = \frac{4a^3 + b(a^2 - 1)^2}{(a^2 - 1)^2 + (2a)^2} = \frac{4a^3 + b(a^2 - 1)^2}{(a^2 + 1)^2}$$

and

$$y = \frac{a^2 - 1}{2a} \cdot \left(\frac{4a^3 + b(a^2 - 1)^2}{(a^2 + 1)^2} - b \right) = \frac{a^2 - 1}{2a} \cdot \frac{4a^3 - 4a^2b}{(a^2 + 1)^2} = \frac{2a(a - b)(a^2 - 1)}{(a^2 + 1)^2}.$$

This yields $E \left(\frac{4a^3 + b(a^2 - 1)^2}{(a^2 + 1)^2}, \frac{2a(a - b)(a^2 - 1)}{(a^2 + 1)^2} \right)$.

Let T be the contact point of the incircle of $\triangle AEF$ and the side AF . To find the length AT , one way is to use the formula

$$AT = \frac{AE + AF - EF}{2}$$

and compute the lengths of AE , AF , EF one by one like how we compute the lengths of AB , BC , CA . This requires more algebraic work but is not hard to do. Alternatively, we make use of the geometric fact that $\triangle AEF \sim \triangle ABC$. This provides a simpler way to find the length if one knows this similarity. As the ratio of similarity is $\frac{AF}{AC}$, we know that AT is $\frac{AF}{AC}$ times the length of the tangent from A to the incircle of $\triangle ABC$, which is $-a$. It follows that

$$AT = \frac{\frac{a+b}{ab+1} - a}{\frac{b(1+a^2)}{ab+1}} \cdot (-a) = -\frac{a(a^2 - 1)}{a^2 + 1}.$$

Therefore, $T \left(\frac{2a}{a^2 + 1}, 0 \right)$, and so the x -coordinate of I_1 is $\frac{2a}{a^2 + 1}$.

Next, note that I_1 lies on AI since AI_1I is the angle bisector of $\angle BAC$. As

$$AI: \quad y = -\frac{1}{a}x + 1,$$

we obtain $I_1 \left(\frac{2a}{a^2+1}, \frac{a^2-1}{a^2+1} \right)$. (In fact, we can find I_1 directly by using the similar triangles.) By symmetry, $I_2 \left(\frac{2b}{b^2+1}, \frac{b^2-1}{b^2+1} \right)$. Therefore, we have

$$m_{I_1I_2} = \frac{(a^2-1)(b^2+1) - (b^2-1)(a^2+1)}{2a(b^2+1) - 2b(a^2+1)} = \frac{a+b}{1-ab}. \quad (3)$$

The perpendicular bisector of AC has slope $\frac{-1}{m_{AC}} = \frac{a^2-1}{2a}$ and it passes through the midpoint $\left(\frac{a^2b+2a+b}{2(ab+1)}, \frac{ab}{ab+1} \right)$ of AC . So its equation is

$$y = \frac{a^2-1}{2a}x - \frac{a^2-1}{2a} \cdot \frac{a^2b+2a+b}{2(ab+1)} + \frac{ab}{ab+1},$$

i.e.

$$y = \frac{a^2-1}{2a}x - \frac{a^4b+2a^3-4a^2b-2a-b}{4a(ab+1)}. \quad (4)$$

Similarly, the perpendicular bisector of AI_1 has slope $\frac{-1}{m_{AI}} = a$ and it passes through the midpoint $\left(\frac{a(a^2+3)}{2(a^2+1)}, \frac{a^2-1}{2(a^2+1)} \right)$ of AI_1 . So its equation is

$$y = ax - \frac{a^2+1}{2}. \quad (5)$$

We now solve (4) and (5) to get

$$x = \frac{2a(a^2+1)(ab+1) - (a^4b+2a^3-4a^2b-2a-b)}{2(ab+1)(2a^2-(a^2-1))} = \frac{a^4b+6a^2b+4a+b}{2(ab+1)(a^2+1)}$$

and

$$y = a \cdot \frac{a^4b+6a^2b+4a+b}{2(ab+1)(a^2+1)} - \frac{a^2+1}{2} = \frac{4a^3b-a^4+2a^2-1}{2(ab+1)(a^2+1)}.$$

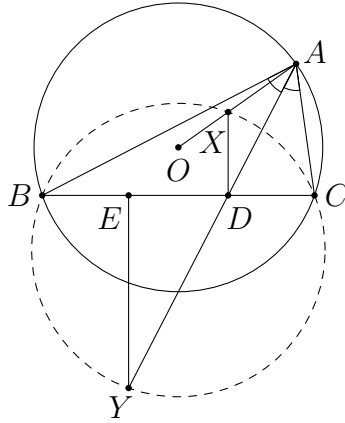
Therefore, we have $O_1 \left(\frac{a^4b+6a^2b+4a+b}{2(ab+1)(a^2+1)}, \frac{4a^3b-a^4+2a^2-1}{2(ab+1)(a^2+1)} \right)$. By symmetry, we also have $O_2 \left(\frac{ab^4+6ab^2+4b+a}{2(ab+1)(b^2+1)}, \frac{4ab^3-b^4+2b^2-1}{2(ab+1)(b^2+1)} \right)$. It follows that

$$m_{O_1O_2} = \frac{(4a^3b-a^4+2a^2-1)(b^2+1) - (4ab^3-b^4+2b^2-1)(a^2+1)}{(a^4b+6a^2b+4a+b)(b^2+1) - (ab^4+6ab^2+4b+a)(a^2+1)}$$

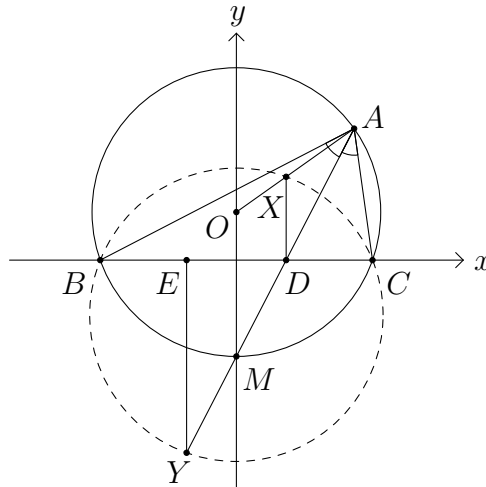
$$\begin{aligned}
&= \frac{(a-b)(-a^3b^2 - a^2b^3 - a^3 + 3a^2b + 3ab^2 - b^3 + 3a + 3b)}{(a-b)(a^3b^3 + a^3b - 5a^2b^2 + ab^3 - a^2 + ab - b^2 + 3)} \\
&= \frac{(a-b)(a+b)(-a^2b^2 - a^2 + 4ab - b^2 + 3)}{(a-b)(1-ab)(-a^2b^2 - a^2 + 4ab - b^2 + 3)} \\
&= \frac{a+b}{1-ab}.
\end{aligned}$$

This is equal to (3). Thus, $I_1I_2 \parallel O_1O_2$. □

Problem 61. (IMO Shortlist 2012 G4) Let ABC be a triangle with $AB \neq AC$ and circumcentre O . The bisector of $\angle BAC$ intersects BC at D . Let E be the reflection of D with respect to the midpoint of BC . The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.



Solution.



Let $B(-1, 0)$, $C(1, 0)$, $A(b, c)$ and $M(0, a)$ where M is the second intersection of AD and the circumcircle of $\triangle ABC$ (recall that M lies on the perpendicular bisector of BC). From

$$\begin{aligned} OB &= OM \\ \Rightarrow y^2 + 1 &= (y - a)^2 \\ \Rightarrow y &= \frac{a^2 - 1}{2a}, \end{aligned}$$

we get $O\left(0, \frac{a^2 - 1}{2a}\right)$. The equation of $\odot O$ is $x^2 + y^2 - \frac{a^2 - 1}{a}y - 1 = 0$. In particular, we have

$$b^2 + c^2 - \frac{a^2 - 1}{a}c - 1 = 0. \quad (1)$$

Next, we find

$$AM: \quad y = \frac{c - a}{b}x + a.$$

Then $D\left(\frac{ab}{a - c}, 0\right)$ and $E\left(-\frac{ab}{a - c}, 0\right)$. Hence, we get $Y\left(-\frac{ab}{a - c}, 2a\right)$. For the point X , we first obtain

$$AO: \quad y = \frac{2ac - a^2 + 1}{2ab}x + \frac{a^2 - 1}{2a}.$$

Then $X\left(\frac{ab}{a - c}, \frac{(a^2 + 1)c}{2a(a - c)}\right)$. Let O' be the circumcentre of $\triangle BCY$. Obviously, O' lies on the y -axis. From

$$\begin{aligned} O'B &= O'Y \\ \Rightarrow y^2 + 1 &= \left(\frac{ab}{a - c}\right)^2 + (y - 2a)^2 \\ \Rightarrow 1 &= \frac{a^2b^2}{(a - c)^2} + 4a^2 - 4ay \\ \Rightarrow y &= \frac{a^2b^2 + 4a^2(a - c)^2 - (a - c)^2}{4a(a - c)^2}, \end{aligned}$$

we have $O'\left(0, \frac{a^2b^2 + 4a^2(a - c)^2 - (a - c)^2}{4a(a - c)^2}\right)$. Lastly,

$$\begin{aligned} O'Y &= O'X \\ \Leftrightarrow \left(\frac{ab}{a - c}\right)^2 + (y - 2a)^2 &= \left(\frac{ab}{a - c}\right)^2 + \left(y - \frac{(a^2 + 1)c}{2a(a - c)}\right)^2 \\ \Leftrightarrow (y - 2a)^2 &= \left(y - \frac{(a^2 + 1)c}{2a(a - c)}\right)^2. \end{aligned}$$

Therefore it suffices to show

$$y - 2a = -y + \frac{(a^2 + 1)c}{2a(a - c)}.$$

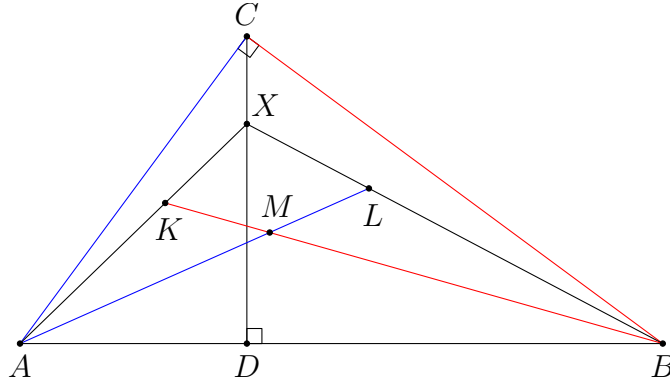
Note that

$$\begin{aligned} y - 2a &= -y + \frac{(a^2 + 1)c}{2a(a - c)} \\ \Leftrightarrow \frac{a^2b^2 + 4a^2(a - c)^2 - (a - c)^2}{2a(a - c)^2} &= 2a + \frac{(a^2 + 1)c}{2a(a - c)} \\ \Leftrightarrow ab^2 &= (ac + 1)(a - c) \\ \Leftrightarrow b^2 + c^2 - \frac{a^2 - 1}{a}c - 1 &= 0, \end{aligned}$$

which is the same as the condition (1). The proof is complete. $\square$

Problem 62. (IMO Shortlist 2012 G5/Problem 5) Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK .

Show that $MK = ML$.



Solution. Let $D(0,0)$, $A(a,0)$, $B(b,0)$, $C(0,c)$, $X(0,d)$ where $a < 0 < b$. Since $AC \perp BC$, we have $\left(-\frac{c}{a}\right)\left(-\frac{c}{b}\right) = -1$, i.e.

$$c^2 = -ab.$$

The equation of AX is

$$y = -\frac{d}{a}x + d.$$



After expansion and using the fact that $c^2 = -ab$, this becomes

From the configuration, k should be the smaller root of this quadratic equation. So

Similarly, we can write $L\left(\ell, \frac{d(b-\ell)}{b}\right)$, where ℓ is the larger root of the equation

Indeed, we have

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Now, using, $K\left(k, \frac{d(a-k)}{a}\right)$, we get

$$BK: \quad y = \frac{d(a-k)}{a(k-b)}x - \frac{bd(a-k)}{a(k-b)}.$$

By symmetry,

$$AL: \quad y = \frac{d(b-\ell)}{b(\ell-a)} - \frac{ad(b-\ell)}{b(\ell-a)}.$$

Solving these, we get

$$\begin{aligned} x &= \frac{a^2d(b-\ell)(k-b) - b^2d(a-k)(\ell-a)}{ad(b-\ell)(k-b) - bd(a-k)(\ell-a)} \\ &= \frac{(a-b)(abk + abl - ak\ell - bk\ell)}{(a-b)(ab - k\ell)} = \frac{abk + abl - ak\ell - bk\ell}{ab - k\ell} \end{aligned}$$

and

$$\begin{aligned} y &= \frac{d(a-k)}{a(k-b)} \cdot \left(\frac{abk + abl - ak\ell - bk\ell}{ab - k\ell} - b \right) \\ &= \frac{d(a-k)}{a(k-b)} \cdot \frac{a(k-b)(b-\ell)}{ab - k\ell} = \frac{d(a-k)(b-\ell)}{ab - k\ell}. \end{aligned}$$

This means $M\left(\frac{abk + abl - ak\ell - bk\ell}{ab - k\ell}, \frac{d(a-k)(b-\ell)}{ab - k\ell}\right)$. From this, we compute

$$\begin{aligned} MK^2 &= \left(\frac{abk + abl - ak\ell - bk\ell}{ab - k\ell} - k \right)^2 + \left(\frac{d(a-k)(b-\ell)}{ab - k\ell} - \frac{d(a-k)}{a} \right)^2 \\ &= \left(\frac{\ell(k-a)(k-b)}{ab - k\ell} \right)^2 + \left(-\frac{d(a-k)^2\ell}{a(ab - k\ell)} \right)^2 \\ &= \frac{\ell^2(k-a)^2}{a^2(ab - k\ell)^2} \cdot (a^2(k-b)^2 + d^2(a-k)^2) \\ &= \frac{\ell^2(k-a)^2}{a^2(ab - k\ell)^2} \cdot ((a^2 + d^2)k^2 - 2a(ab + d^2)k + a^2(b^2 + d^2)). \end{aligned}$$

Using (1), the last bracket becomes $a^2b(b-a)$. So

$$MK^2 = \frac{b(b-a)\ell^2(k-a)^2}{(ab - k\ell)^2}. \quad (4)$$

We now use the explicit formulae for k and ℓ in (2) and (3). These yield

$$\begin{aligned} &\ell^2(k-a)^2 \\ &= \left(\frac{b(ab + d^2) + b\sqrt{b(a-b)(ab + d^2)}}{b^2 + d^2} \right)^2 \left(\frac{a(ab + d^2) + a\sqrt{a(b-a)(ab + d^2)}}{a^2 + d^2} - a \right)^2 \end{aligned}$$

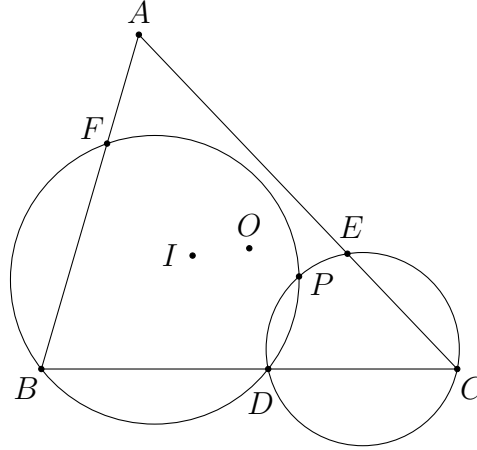
$$\begin{aligned}
&= \left(\frac{b(ab + d^2) + b\sqrt{b(a-b)(ab + d^2)}}{b^2 + d^2} \right)^2 \left(\frac{a^2(b-a) + a\sqrt{a(b-a)(ab + d^2)}}{a^2 + d^2} \right)^2 \\
&= \frac{(-a^3)b^2(b-a)(-(ab + d^2))}{(a^2 + d^2)^2(b^2 + d^2)^2} \\
&\quad \times \left(-\sqrt{-(ab + d^2)} + \sqrt{b(b-a)} \right)^2 \left(\sqrt{(-a)(b-a)} - \sqrt{-(ab + d^2)} \right)^2.
\end{aligned}$$

Putting this in (4), we find that

$$\begin{aligned}
MK^2 &= \frac{a^3b^3(b-a)^2(ab + d^2)}{(ab - k\ell)^2(a^2 + d^2)^2(b^2 + d^2)^2} \\
&\quad \times \left(-\sqrt{-(ab + d^2)} + \sqrt{b(b-a)} \right)^2 \left(\sqrt{-a(b-a)} - \sqrt{-(ab + d^2)} \right)^2
\end{aligned}$$

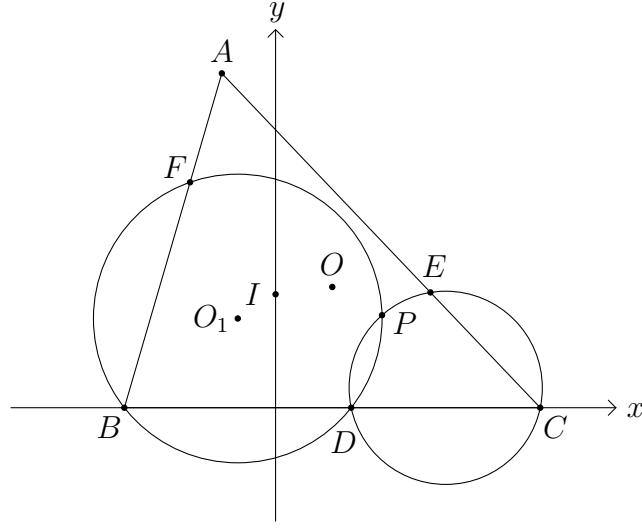
Swapping a and b , and swapping k and ℓ , we get the same expression. This implies $MK^2 = ML^2$, and thus $MK = ML$. $\square$

Problem 63. (IMO Shortlist 2012 G6) Let ABC be a triangle with circumcentre O and incentre I . The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.



Solution. Let $I(0, 1)$, $B(b, 0)$, $C(c, 0)$ and $D(d, 0)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$ and $O\left(\frac{b+c}{2}, \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{4(bc+1)}\right)$. By the given condition, we have

$$BF = CA - BD = \frac{b(1+c^2)}{bc+1} - (d-b) = \frac{b^2c + bc^2 + 2b - (bc+1)d}{bc+1}.$$



Note that

$$\cos B = \frac{x_A - x_B}{AB} = \frac{\frac{b+c}{bc+1} - b}{-\frac{c(1+b^2)}{bc+1}} = \frac{c(1-b^2)}{-c(1+b^2)} = \frac{b^2-1}{b^2+1}$$

and

$$\sin B = \frac{y_A}{AB} = \frac{\frac{2bc}{bc+1}}{-\frac{c(1+b^2)}{bc+1}} = -\frac{2b}{b^2+1}.$$

It follows that

$$\begin{aligned} F &= (b + BF \cos B, BF \sin B) \\ &= \left(b + \frac{(b^2c + bc^2 + 2b - (bc+1)d)(b^2-1)}{(bc+1)(b^2+1)}, -\frac{2b(b^2c + bc^2 + 2b - (bc+1)d)}{(bc+1)(b^2+1)} \right) \\ &= \left(\frac{b(2b^3c + b^2c^2 + 3b^2 - c^2 - 1) - (bc+1)(b^2-1)d}{(bc+1)(b^2+1)}, -\frac{2b(b^2c + bc^2 + 2b - (bc+1)d)}{(bc+1)(b^2+1)} \right). \end{aligned}$$

The x -coordinate of the circumcentre O_1 of $\triangle BDF$ is $\frac{b+d}{2}$. We have

$$\begin{aligned} &O_1B = O_1F \\ \Rightarrow &\left(\frac{b-d}{2}\right)^2 + y^2 = \left(\frac{b+d}{2} - x_F\right)^2 + (y - y_F)^2 \\ \Rightarrow &-bd = x_F^2 - (b+d)x_F + y_F^2 - 2y_Fy \\ \Rightarrow &y = \frac{(x_F - b)(x_F - d) + y_F^2}{2y_F}. \end{aligned}$$

Firstly, note that

$$x_F - b = \frac{(b^2c + bc^2 + 2b - (bc + 1)d)(b^2 - 1)}{(bc + 1)(b^2 + 1)}$$

and

$$\begin{aligned} x_F - d &= \frac{b(2b^3c + b^2c^2 + 3b^2 - c^2 - 1) - (bc + 1)(b^2 - 1)d}{(bc + 1)(b^2 + 1)} - d \\ &= \frac{b(2b^3c + b^2c^2 + 3b^2 - c^2 - 1) - 2b^2(bc + 1)d}{(bc + 1)(b^2 + 1)}. \end{aligned}$$

This gives

$$\begin{aligned} y &= \frac{(b^2 - 1)(b(2b^3c + b^2c^2 + 3b^2 - c^2 - 1) - 2b^2(bc + 1)d) + 4b^2(b^2c + bc^2 + 2b - (bc + 1)d)}{-4b(bc + 1)(b^2 + 1)} \\ &= \frac{b(2b^5c + b^4c^2 + 3b^4 + 2b^3c + 2b^2c^2 + 4b^2 + c^2 + 1) - 2b^2(bc + 1)(b^2 + 1)d}{-4b(bc + 1)(b^2 + 1)} \\ &= -\frac{(2b^3c + b^2c^2 + 3b^2 + c^2 + 1) - 2b(bc + 1)d}{4(bc + 1)}. \end{aligned}$$

Therefore,

$$(BDF): \quad x^2 + y^2 - (b + d)x + \frac{(2b^3c + b^2c^2 + 3b^2 + c^2 + 1) - 2b(bc + 1)d}{2(bc + 1)}y + bd = 0. \quad (1)$$

By symmetry,

$$(CDE): \quad x^2 + y^2 - (c + d)x + \frac{(2bc^3 + b^2c^2 + 3c^2 + b^2 + 1) - 2c(bc + 1)d}{2(bc + 1)}y + cd = 0. \quad (2)$$

Taking the difference of (1) and (2), we get

$$(c - b)x + \frac{(2b^3c - 2bc^3 + 2b^2 - 2c^2) + 2(c - b)(bc + 1)d}{2(bc + 1)}y + (b - c)d = 0.$$

This means $x - (b + c - d)y - d = 0$, or

$$x = (b + c - d)y + d. \quad (3)$$

This is the radical axis of (BDF) and (CDE) . Thus, P and D are the intersection points of this radical axis and one of these circles. Putting (3) in (1), we solve

$$\begin{aligned} & (1 + (b + c - d)^2)y^2 \\ & + \left(2d(b + c - d) - (b + d)(b + c - d) + \frac{(2b^3c + b^2c^2 + 3b^2 + c^2 + 1) - 2b(bc + 1)d}{2(bc + 1)} \right) y = 0. \end{aligned}$$

The root $y = 0$ corresponds to the point D . Therefore, the y -coordinate of P is the other root

$$\begin{aligned} y_P &= -\frac{2d(b+c-d) - (b+d)(b+c-d) + \frac{(2b^3c + b^2c^2 + 3b^2 + c^2 + 1) - 2b(bc+1)d}{2(bc+1)}}{1 + (b+c-d)^2} \\ &= \frac{2(bc+1)d^2 - 2(b+c)(bc+1)d + (b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc+1)((b+c-d)^2 + 1)}. \end{aligned}$$

The x -coordinate of P is

$$\begin{aligned} x_P &= (b+c-d) \cdot \frac{2(bc+1)d^2 - 2(b+c)(bc+1)d + (b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc+1)((b+c-d)^2 + 1)} + d \\ &= \frac{(b+c-d)(b^2c^2 - b^2 + 2bc - c^2 - 1) + 2(bc+1)d}{2(bc+1)((b+c-d)^2 + 1)} \\ &= \frac{(-b^2c^2 + b^2 + c^2 + 3)d + (b+c)(b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc+1)((b+c-d)^2 + 1)}. \end{aligned}$$

To show that $OP = OI$, we show that P lies on the circle with centre O and passing through I . It is easy to deduce the equation of this circle, which is

$$x^2 + y^2 - (b+c)x - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc+1)}y + \frac{b^2c^2 - b^2 + 2bc - c^2 - 1}{2(bc+1)} = 0.$$

We can rewrite this equation as

$$x(x-b-c) + (y-1)\left(y - \frac{b^2c^2 - b^2 + 2bc - c^2 - 1}{2(bc+1)}\right) = 0.$$

In view of this, we compute the following terms.

$$\begin{aligned} x_P - b - c &= \frac{(-b^2c^2 + b^2 + c^2 + 3)d + (b+c)(b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc+1)((b+c-d)^2 + 1)} - (b+c) \\ &= \frac{-2(b+c)(bc+1)d^2 + (4b^3c + 7b^2c^2 + 4bc^3 + 5b^2 + 8bc + 5c^2 + 3)d - (b+c)(2b^3c + 3b^2c^2 + 2bc^3 + 3b^2 + 4bc + 3c^2 + 3)}{2(bc+1)((b+c-d)^2 + 1)} \\ &= \frac{(b+c-d)(2(b+c)(bc+1)d - (2b^3c + 3b^2c^2 + 2bc^3 + 3b^2 + 4bc + 3c^2 + 3))}{2(bc+1)((b+c-d)^2 + 1)}, \\ y_P - 1 &= \frac{2(bc+1)d^2 - 2(b+c)(bc+1)d + (b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc+1)((b+c-d)^2 + 1)} - 1 \\ &= \frac{2(b+c)(bc+1)d - (2b^3c + 3b^2c^2 + 2bc^3 + 3b^2 + 4bc + 3c^2 + 3)}{2(bc+1)((b+c-d)^2 + 1)}, \end{aligned}$$

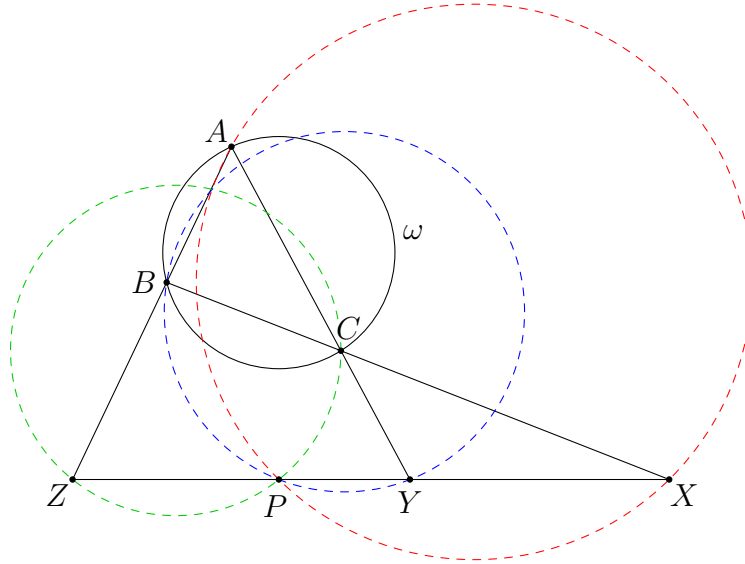
$$\begin{aligned}
& y_P - \frac{b^2c^2 - b^2 + 2bc - c^2 - 1}{2(bc + 1)} \\
&= \frac{2(bc + 1)d^2 - 2(b + c)(bc + 1)d + (b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc + 1)((b + c - d)^2 + 1)} - \frac{b^2c^2 - b^2 + 2bc - c^2 - 1}{2(bc + 1)} \\
&= \frac{(-b^2c^2 + b^2 + c^2 + 3)d^2 + 2(b + c)(b^2c^2 - b^2 + bc - c^2 - 2)d - (b + c)^2(b^2c^2 - b^2 + 2bc - c^2 - 1)}{2(bc + 1)((b + c - d)^2 + 1)} \\
&= \frac{-(b + c - d)((-b^2c^2 + b^2 + c^2 + 3)d + (b + c)(b^2c^2 - b^2 + 2bc - c^2 - 1))}{2(bc + 1)((b + c - d)^2 + 1)}
\end{aligned}$$

Combining the above expressions, it is now clear that

$$x_P(x_P - b - c) + (y_P - 1) \left(y_P - \frac{b^2c^2 - b^2 + 2bc - c^2 - 1}{2(bc + 1)} \right) = 0.$$

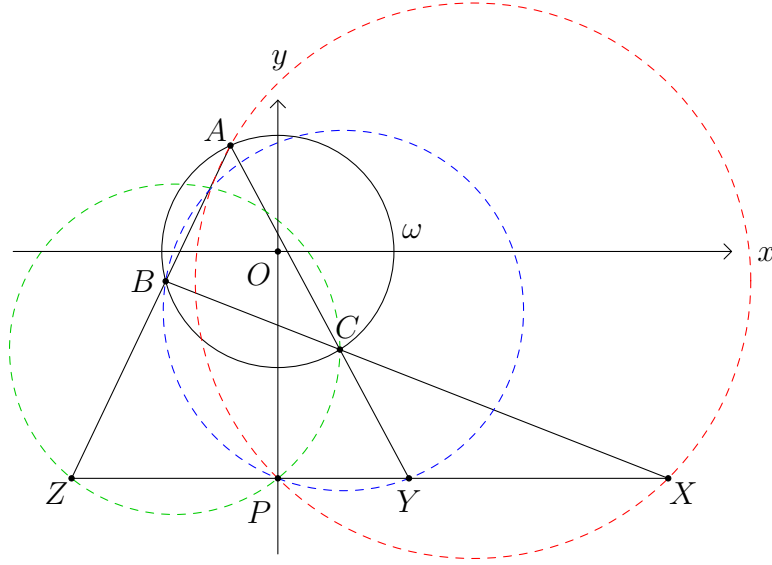
This proves $OP = OI$ as desired. $\square$

Problem 64. (IMO Shortlist 2012 G8) Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω . Denote by P the foot of the perpendicular from the centre of ω to ℓ . The sidelines BC, CA, AB intersect ℓ at the points X, Y, Z different from P . Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P .



Solution. Let $O(0,0)$ be the centre of ω , $A(a,b)$, $B(c,d)$, $C(e,f)$, $P(0,p)$ with

$$a^2 + b^2 = c^2 + d^2 = e^2 + f^2 = 1.$$



(This means ω has equation $x^2 + y^2 = 1$.) We easily get

$$BC: \quad y = \frac{f-d}{e-c}x + \frac{de-cf}{e-c}.$$

Thus, $X\left(\frac{(e-c)p - (de-cf)}{f-d}, p\right)$. Let O' be the circumcentre of $\triangle AXP$. Then $\frac{(e-c)p - (de-cf)}{2(f-d)}$ is the x -coordinate of O' . Since $m_{AP} = \frac{b-p}{a}$, the slope of the perpendicular bisector of AP is $-\frac{a}{b-p}$. Since this perpendicular bisector passes through the midpoint $\left(\frac{a}{2}, \frac{b+p}{2}\right)$ of AP , its equation is

$$y = -\frac{a}{b-p}x + \frac{a^2 + b^2 - p^2}{2(b-p)}.$$

It follows that

$$O' \left(\frac{(e-c)p - (de-cf)}{2(f-d)}, \frac{-a(e-c)p + a(de-cf) + (f-d)(a^2 + b^2 - p^2)}{2(b-p)(f-d)} \right).$$

Using $a^2 + b^2 = 1$, this yields

$$O' \left(\frac{(e-c)p - (de-cf)}{2(f-d)}, \frac{-(f-d)p^2 - a(e-c)p + (ade - acf + f-d)}{2(b-p)(f-d)} \right).$$

Hence, the equation of (AXP) is

$$\begin{aligned} x^2 + y^2 - \frac{(e-c)p - (de-cf)}{f-d}x - \frac{-(f-d)p^2 - a(e-c)p + (ade - acf + f-d)}{(b-p)(f-d)}y \\ + \frac{-(ae - ac + bf - bd)p + (ade - acf + f-d)}{(b-p)(f-d)}p = 0. \end{aligned}$$

By symmetry, the equation of (BYP) is

$$x^2 + y^2 - \frac{(a-e)p - (fa-eb)}{b-f}x - \frac{-(b-f)p^2 - c(a-e)p + (cfa-ceb+b-f)}{(d-p)(b-f)}y + \frac{-(ca-ce+db-df)p + (cfa-ceb+b-f)}{(d-p)(b-f)}p = 0.$$

Taking the difference of these equations, we get the equation of the radical axis of (AXP) and (BYP) , i.e.

$$\begin{aligned} & \left(\frac{(a-e)p - (fa-eb)}{b-f} - \frac{(e-c)p - (de-cf)}{f-d} \right) x \\ & + \left(\frac{-(b-f)p^2 - c(a-e)p + (cfa-ceb+b-f)}{(d-p)(b-f)} \right. \\ & \left. - \frac{-(f-d)p^2 - a(e-c)p + (ade-acf+f-d)}{(b-p)(f-d)} \right) y \\ & + \left(\frac{-(ae-ac+bf-bd)p + (ade-acf+f-d)}{(b-p)(f-d)} \right. \\ & \left. - \frac{-(ca-ce+db-df)p + (cfa-ceb+b-f)}{(d-p)(b-f)} \right) p = 0. \end{aligned}$$

Multiplying the whole equation by $(b-f)(f-d)(b-p)(d-p)$ and simplifying, the coefficient of x is

$$\begin{aligned} & (b-p)(d-p) \left(((a-e)(f-d) - (e-c)(b-f))p \right. \\ & \left. - ((fa-eb)(f-d) - (de-cf)(b-f)) \right) \\ & = (b-p)(d-p) \left((bc+de+fa-ad-be-cf)p + f(ad+be+cf-bc-de-fa) \right) \\ & = (b-p)(d-p)(f-p)(ad+be+cf-bc-de-fa), \end{aligned}$$

the coefficient of y is

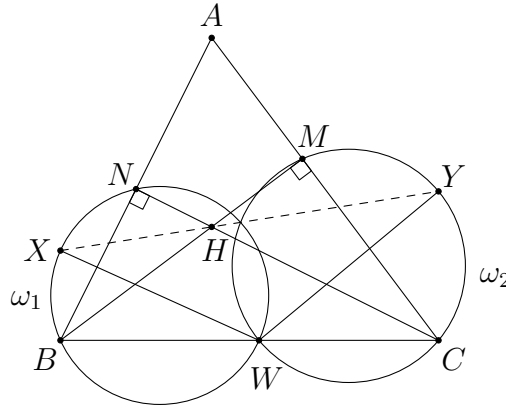
$$\begin{aligned} & \left(-(b-p)(f-d)(b-f) + (d-p)(b-f)(f-d) \right) p^2 \\ & - \left(c(b-p)(f-d)(a-e) - a(d-p)(b-f)(e-c) \right) p \\ & + \left((b-p)(f-d)(cfa-ceb+b-f) - (d-p)(b-f)(ade-acf+f-d) \right) \\ & = (d-b)(f-d)(b-f)p^2 + (abc+cde+efa-abe-cda-efc)p^2 \\ & + 2(abde+cdfa+efbc-abc f-cdeb-efad)p \\ & + (b^2cde+d^2efa+f^2abc-b^2cef-d^2eab-f^2acd) \\ & + (b^2f+d^2b+f^2d-b^2d-d^2f-f^2b), \end{aligned}$$

and the constant term is

$$\begin{aligned}
& \left(-(d-p)(b-f)(ae-ac+bf-bd) + (b-p)(f-d)(ca-ce+db-df) \right) p \\
& + \left((d-p)(b-f)(ade-acf+f-d) - (b-p)(f-d)(cfa-ceb+b-f) \right) p \\
& = \left((abe+cda+efc-abc-cde-efa+b^2f+d^2b+f^2d-b^2d-d^2f-f^2b)p^2 \right. \\
& \quad + 2(abcf+cdeb+efad-abde-cdfa-efbc)p \\
& \quad + (b^2cef+d^2eab+f^2acd-b^2cde-d^2efa-f^2abc) \\
& \quad \left. + (b^2d+d^2f+f^2b-bd^2-df^2-fb^2) \right) p.
\end{aligned}$$

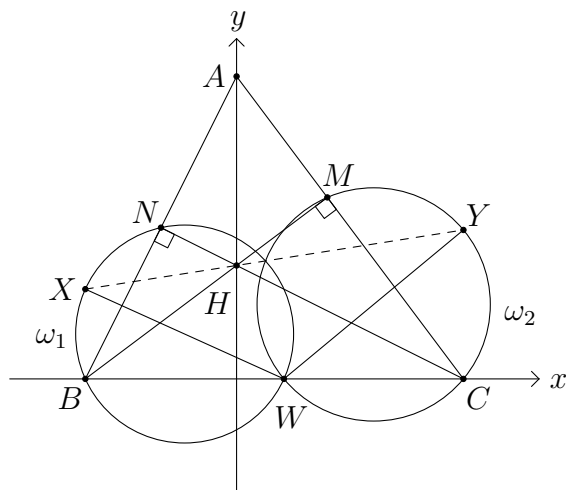
Since every term is symmetric in the variables (a, b) , (c, d) , (e, f) , any two of (AXP) , (BYP) , (CZP) have the same radical axis. As they have a common point P , the result follows readily. $\square$

Problem 65. (IMO Shortlist 2013 G1/Problem 4) Let ABC be an acute-angled triangle with orthocentre H , and let W be a point on the side BC , lying strictly between B and C . The points M and N are the feet of the altitudes from B and C , respectively. Denote by ω_1 the circumcircle of BWN , and let X be the point on ω_1 such that WX is a diameter of ω_1 . Analogously, denote by ω_2 the circumcircle of CWM , and let Y be the point on ω_2 such that WY is a diameter of ω_2 . Prove that X, Y and H are collinear.



Solution. Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$, $W(w, 0)$. Then $N\left(\frac{b(a^2+bc)}{a^2+b^2}, \frac{ab(b-c)}{a^2+b^2}\right)$ and $H\left(0, -\frac{bc}{a}\right)$. Since WX is a diameter of (BWN) , we have $BX \perp BW$ and $NX \perp NW$. The first condition implies the x -coordinate of X is b . The second condition yields

$$\frac{y - \frac{ab(b-c)}{a^2+b^2}}{b - \frac{b(a^2+bc)}{a^2+b^2}} \times \frac{\frac{ab(b-c)}{a^2+b^2}}{\frac{b(a^2+bc)}{a^2+b^2} - w} = -1.$$

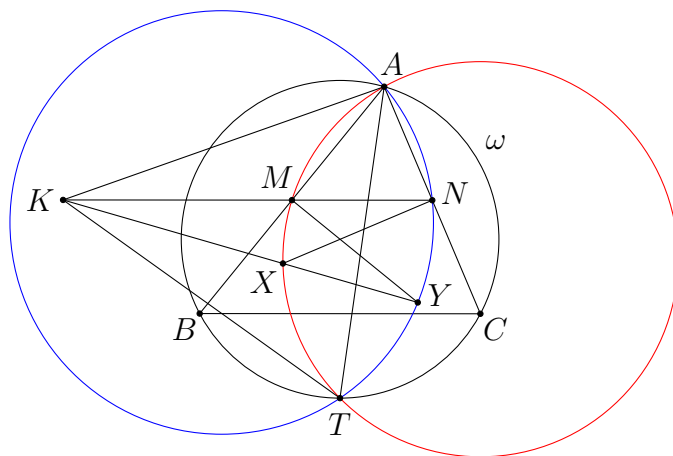


This gives

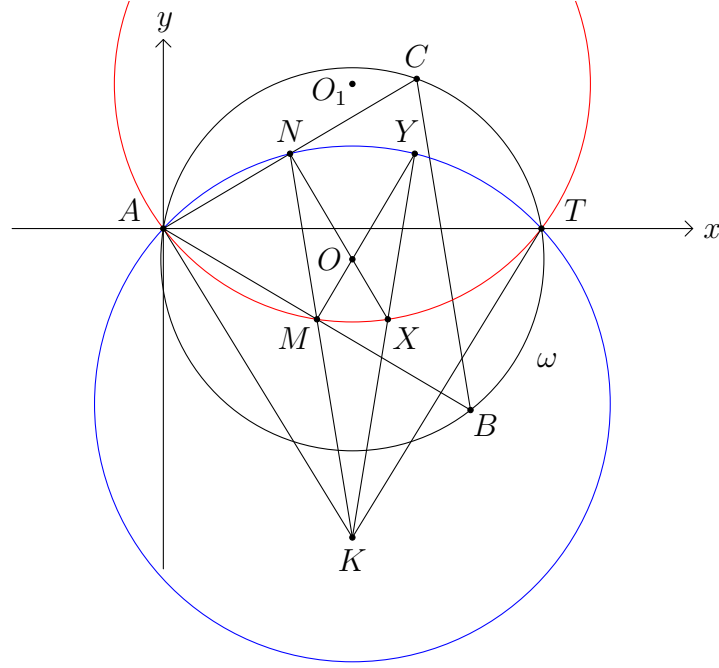
$$y = -\frac{b}{a} \left(\frac{b(a^2 + bc)}{a^2 + b^2} - w \right) + \frac{ab(b - c)}{a^2 + b^2} = \frac{b}{a}w - \frac{bc}{a}.$$

It follows that $X \left(b, \frac{b}{a}w - \frac{bc}{a} \right)$. Thus, $m_{HX} = \frac{\frac{b}{a}w - \frac{bc}{a} + \frac{bc}{a}}{\frac{b}{a}} = \frac{w}{a}$. By symmetry, $m_{HY} = \frac{w}{a} = m_{HX}$. So X, H, Y are collinear. $\square$

Problem 66. (IMO Shortlist 2013 G2) Let ω be the circumcircle of a triangle ABC . Denote by M and N the midpoints of the sides AB and AC , respectively, and denote by T the midpoint of the arc BC of ω not containing A . The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y , respectively; assume that X and Y lie inside the triangle ABC . The lines MN and XY intersect at K . Prove that $KA = KT$.



Solution.



We set the origin at A and set AT as the x -axis. Thus, if AB has equation $y = mx$, then AC has equation $y = -mx$. Let $B(2b, 2mb)$ and $C(2c, -2mc)$ where $b, c > 0$. Then $M(b, mb)$ and $N(c, -mc)$. Note that

$$m_{MY} = \frac{-1}{m_{AB}} = -\frac{1}{m}.$$

This gives

$$MY: \quad y = -\frac{1}{m}x + \frac{b}{m} + mb,$$

i.e.

$$MY: \quad y = -\frac{1}{m}x + \frac{(m^2 + 1)b}{m}. \quad (1)$$

Changing m to $-m$ and changing b to c , we have

$$NX: \quad y = \frac{1}{m}x - \frac{(m^2 + 1)c}{m}. \quad (2)$$

We solve (1) and (2). Then

$$x = \frac{\frac{(m^2 + 1)b}{m} + \frac{(m^2 + 1)c}{m}}{\frac{1}{m} + \frac{1}{m}} = \frac{(m^2 + 1)(b + c)}{2}$$

and

$$y = -\frac{1}{m} \cdot \frac{(m^2 + 1)(b + c)}{2} + \frac{(m^2 + 1)b}{m} = \frac{(m^2 + 1)(b - c)}{2m}.$$

Therefore, by letting O be the circumcentre of $\triangle ABC$, we obtain its coordinates $O\left(\frac{(m^2+1)(b+c)}{2}, \frac{(m^2+1)(b-c)}{2m}\right)$.

Next, let O_1 be the circumcentre of $\triangle AMT$. As OO_1 is the perpendicular bisector of AT , the x -coordinate of O_1 is the same as that of O , which is $\frac{(m^2+1)(b+c)}{2}$. The slope of the perpendicular bisector of AM is

$$\frac{-1}{m_{AM}} = \frac{-1}{m_{AB}} = -\frac{1}{m}.$$

As this perpendicular bisector passes through the midpoint $\left(\frac{b}{2}, \frac{mb}{2}\right)$ of AM , its equation is

$$y = -\frac{1}{m}x + \frac{(m^2+1)b}{2m}.$$

Putting $x = \frac{(m^2+1)(b+c)}{2}$, we obtain

$$y = -\frac{1}{m} \cdot \frac{(m^2+1)(b+c)}{2} + \frac{(m^2+1)b}{2m} = -\frac{(m^2+1)c}{2m}.$$

Therefore, we have $O_1\left(\frac{(m^2+1)(b+c)}{2}, -\frac{(m^2+1)c}{2m}\right)$. Thus,

$$(AMT): \quad x^2 + y^2 - (m^2+1)(b+c)x + \frac{(m^2+1)c}{m}y = 0. \quad (3)$$

Putting (2) in (3), we solve

$$\begin{aligned} \left(1 + \frac{1}{m^2}\right)x^2 + \left(-\frac{2(m^2+1)c}{m^2} - (m^2+1)(b+c) + \frac{(m^2+1)c}{m^2}\right)x \\ + \left(\frac{(m^2+1)^2c^2}{m^2} - \frac{(m^2+1)^2c^2}{m^2}\right) = 0. \end{aligned}$$

Removing the factor m^2+1 , this becomes

$$x^2 - (m^2(b+c) + c)x = 0.$$

From the configuration, we must have $x > 0$. Therefore, we choose the larger root $x = m^2(b+c) + c$. Then we have

$$y = \frac{1}{m}(m^2(b+c) + c) - \frac{(m^2+1)c}{m} = mb,$$

and hence $X(m^2(b+c) + c, mb)$. Changing m to $-m$ and swapping b and c , we find that $Y(m^2(b+c) + b, -mc)$. As

$$m_{XY} = \frac{mb + mc}{(m^2(b+c) + c) - (m^2(b+c) + b)} = \frac{m(b+c)}{c-b},$$

we obtain

$$XY: \quad y = \frac{m(b+c)}{c-b}x - \frac{m(b+c)}{c-b} \cdot (m^2(b+c) + c) + mb,$$

i.e.

$$XY: \quad y = \frac{m(b+c)}{c-b}x - \frac{m(m^2(b+c)^2 + b^2 + c^2)}{c-b}. \quad (4)$$

Also, it is easy to deduce

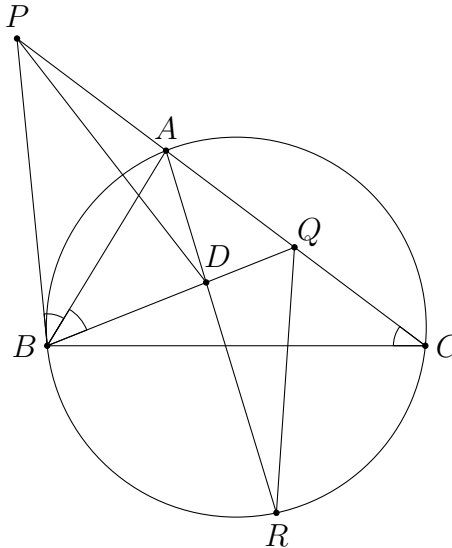
$$MN: \quad y = \frac{m(b+c)}{b-c}x - \frac{2mbc}{b-c}. \quad (5)$$

We solve (4) and (5). Note that

$$x = \frac{\frac{m(m^2(b+c)^2 + b^2 + c^2)}{c-b} + \frac{2mbc}{c-b}}{\frac{m(b+c)}{c-b} + \frac{m(b+c)}{c-b}} = \frac{m(m^2(b+c)^2 + (b+c)^2)}{2m(b+c)} = \frac{(m^2+1)(b+c)}{2}.$$

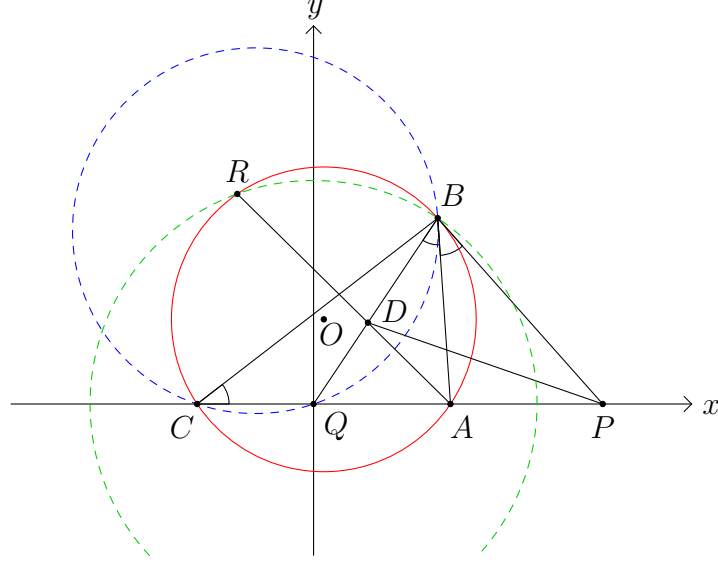
This is the x -coordinate of K , which is the same as that of O . This shows KO is the perpendicular bisector of AT , and hence $KA = KT$. $\square$

Problem 67. (IMO Shortlist 2013 G4) Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C . Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.



Solution. Let $Q(0,0)$, $C(c,0)$ and $B(a,b)$. From $\angle QBA = \angle QCB$, we know that AB is a tangent to (BCQ) , and hence $AB^2 = AQ \cdot AC$. Let x_A be the x -coordinate

of A . Then this implies $(x_A - a)^2 + b^2 = x_A(x_A - c)$. Solving this linear equation in x_A , we have $A\left(\frac{a^2 + b^2}{2a - c}, 0\right)$.



Similarly, from $\angle ABP = \angle ACB$, the line PB is a tangent to (BCA) . Then we have the relation $PB^2 = PA \cdot PC$. Let x_P be the x -coordinate of P . We get

$$(x_P - a)^2 + b^2 = \left(x_P - \frac{a^2 + b^2}{2a - c}\right)(x_P - c),$$

so that $P\left(\frac{2(a - c)(a^2 + b^2)}{(2a - c)^2 - (a^2 + b^2)}, 0\right)$. Next,

$$QB: \quad y = \frac{b}{a}x. \quad (1)$$

The circle with centre P passing through B has equation

$$x^2 + y^2 - \frac{4(a - c)(a^2 + b^2)}{(2a - c)^2 - (a^2 + b^2)}x + \frac{(a^2 + b^2 - c^2)(a^2 + b^2)}{(2a - c)^2 - (a^2 + b^2)} = 0. \quad (2)$$

Putting (1) into (2), we have to solve

$$\left(1 + \frac{b^2}{a^2}\right)x^2 - \frac{4(a - c)(a^2 + b^2)}{(2a - c)^2 - (a^2 + b^2)}x + \frac{(a^2 + b^2 - c^2)(a^2 + b^2)}{(2a - c)^2 - (a^2 + b^2)} = 0.$$

One of the roots corresponds to point B and hence should be $x = a$. Using product of roots, we find that the other root is $x = \frac{a(a^2 + b^2 - c^2)}{(2a - c)^2 - (a^2 + b^2)}$. By (1), we get

$$D\left(\frac{a(a^2 + b^2 - c^2)}{(2a - c)^2 - (a^2 + b^2)}, \frac{b(a^2 + b^2 - c^2)}{(2a - c)^2 - (a^2 + b^2)}\right).$$

Next, we find $m_{AD} = \frac{b(a^2 + b^2 - c^2)(2a - c)}{-a(a - c)^3 + (3a - c)cb^2 + b^4}$ and

$$AD: \quad y = \frac{b(a^2 + b^2 - c^2)(2a - c)}{-a(a - c)^3 + (3a - c)cb^2 + b^4}x - \frac{b(a^2 + b^2 - c^2)(a^2 + b^2)}{-a(a - c)^3 + (3a - c)cb^2 + b^4}. \quad (3)$$

For (ABC) , we first note that its centre O has x -coordinate $\frac{a^2 + 2ac - c^2 + b^2}{2(2a - c)}$. Next,

$$\begin{aligned} OB &= OC \\ \Rightarrow \quad \left(\frac{a^2 + 2ac - c^2 + b^2}{2(2a - c)} - a \right)^2 + (y - b)^2 &= \left(\frac{a^2 + 2ac - c^2 + b^2}{2(2a - c)} - c \right)^2 + y^2 \\ \Rightarrow \quad a^2 - \frac{a(a^2 + 2ac - c^2 + b^2)}{2a - c} + b^2 - 2by &= c^2 - \frac{c(a^2 + 2ac - c^2 + b^2)}{2a - c} \\ \Rightarrow \quad y &= \frac{a(a^2 - 2ac + c^2 + b^2)}{2b(2a - c)}. \end{aligned}$$

Therefore, we get $O \left(\frac{a^2 + 2ac - c^2 + b^2}{2(2a - c)}, \frac{a(a^2 - 2ac + c^2 + b^2)}{2b(2a - c)} \right)$ and

$$(ABC): \quad x^2 + y^2 - \frac{a^2 + 2ac - c^2 + b^2}{2a - c}x - \frac{a(a^2 - 2ac + c^2 + b^2)}{b(2a - c)}y + \frac{c(a^2 + b^2)}{2a - c} = 0. \quad (4)$$

If we want to find R from its definition, then we need to solve (3) and (4). This may not be a simple task as both equations contain bulky terms. It is still possible to do in this way since we need not solve the quadratic equation directly as we know that one of the roots corresponds to A . Alternatively, we may try another path that arises from the result. Indeed, we take the point R' which lies on (ABC) and satisfies $QR' = QB$ and then prove that R' lies on AD . This certainly concludes $R = R'$ and completes the proof.

Let Γ be the circle with centre Q that passes through B . Then

$$\Gamma: \quad x^2 + y^2 - (a^2 + b^2) = 0. \quad (5)$$

The radical axis BR' of Γ and (ABC) is given by the difference of (4) and (5). After simplification, this is

$$y = -\frac{b(a^2 + 2ac - c^2 + b^2)}{a(a^2 - 2ac + c^2 + b^2)}x + \frac{2b(a^2 + b^2)}{a^2 - 2ac + c^2 + b^2}. \quad (6)$$

To find R' , we first find the midpoint M of BR' . Note that $QM \perp BR'$. Hence, $m_{QM} = \frac{a(a^2 - 2ac + c^2 + b^2)}{b(a^2 + 2ac - c^2 + b^2)}$ and

$$QM: \quad y = \frac{a(a^2 - 2ac + c^2 + b^2)}{b(a^2 + 2ac - c^2 + b^2)}x. \quad (7)$$

Solving (6) and (7), we have

$$M \left(\frac{2ab^2(a^2 + 2ac - c^2 + b^2)}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4}, \frac{2a^2b(a^2 - 2ac + c^2 + b^2)}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4} \right).$$

Then the coordinates of R' are

$$\left(\frac{a(-(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + 3b^4)}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4}, \frac{b((a-c)^2(a+c)(3a-c) + 2(a-c)^2b^2 - b^4)}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4} \right).$$

It remains to show that this point satisfies (3). We have

$$\begin{aligned} & (2a-c)x - (a^2 + b^2) \\ &= \frac{a(2a-c)(-(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + 3b^4)}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4} - (a^2 + b^2) \\ &= \frac{-a(3a-c)(a-c)^4 + (a-c)(a^3 + 7a^2c - 5ac^2 + c^3)b^2 + (3a^2 - 7ac + 2c^2)b^4 - b^6}{(a-c)^4 + 2(a^2 + 2ac - c^2)b^2 + b^4}. \end{aligned}$$

In view of the coordinates of R' and (3), we need to show $\alpha\beta = \gamma\delta$ where

$$\begin{aligned} \alpha &= -a(3a-c)(a-c)^4 + (a-c)(a^3 + 7a^2c - 5ac^2 + c^3)b^2 + (3a^2 - 7ac + 2c^2)b^4 - b^6, \\ \beta &= a^2 + b^2 - c^2, \\ \gamma &= (a-c)^2(a+c)(3a-c) + 2(a-c)^2b^2 - b^4, \\ \delta &= -a(a-c)^3 + (3a-c)cb^2 + b^4. \end{aligned}$$

Again, the simplest way is to factorize these expressions. To this end, we simply need

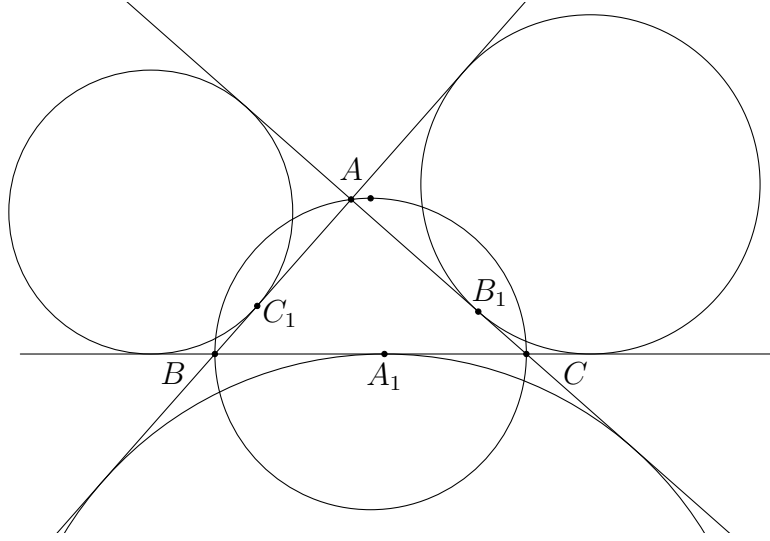
$$\gamma = \beta((a-c)(3a-c) - b^2)$$

and

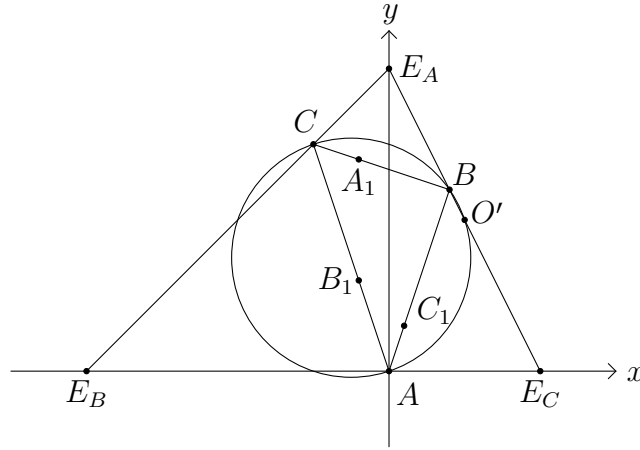
$$\alpha = \delta((a-c)(3a-c) - b^2).$$

The proof is complete. □

Problem 68. (IMO Shortlist 2013 G6/Problem 3) Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Suppose that the circumcentre of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC . Prove that triangle ABC is right-angled.



Solution.



Let E_A, E_B, E_C be the A, B, C -excentres respectively, and let $E_A(0, a)$, $E_B(b, 0)$, $E_C(c, 0)$, $A(0, 0)$. Then $B\left(\frac{c(a^2 + bc)}{a^2 + c^2}, \frac{ac(c - b)}{a^2 + c^2}\right)$ and $C\left(\frac{b(a^2 + bc)}{a^2 + b^2}, \frac{ab(b - c)}{a^2 + b^2}\right)$.

Next, we have

$$\begin{aligned} m_{BC} &= \frac{ac(c - b)(a^2 + b^2) - ab(b - c)(a^2 + c^2)}{c(a^2 + bc)(a^2 + b^2) - b(a^2 + bc)(a^2 + c^2)} \\ &= \frac{a(c - b)(b + c)(a^2 + bc)}{(a^2 + bc)(c - b)(a^2 - bc)} = \frac{a(b + c)}{a^2 - bc}. \end{aligned}$$

Thus, the equation of BC is

$$y = \frac{a(b + c)}{a^2 - bc}x - \frac{a(b + c)}{a^2 - bc} \cdot \frac{c(a^2 + bc)}{a^2 + c^2} + \frac{ac(c - b)}{a^2 + c^2},$$

which means

$$BC: \quad y = \frac{a(b+c)}{a^2-bc}x - \frac{2abc}{a^2-bc}.$$

The equation of E_AA_1 is

$$E_AA_1: \quad y = -\frac{a^2-bc}{a(b+c)}x + a.$$

Solving these, we get

$$x = \frac{a(b+c)(2abc + a(a^2-bc))}{a^2(b+c)^2 + (a^2-bc)^2} = \frac{a^2(b+c)(a^2+bc)}{(a^2+b^2)(a^2+c^2)}$$

and

$$\begin{aligned} y &= -\frac{a^2-bc}{a(b+c)} \cdot \frac{a^2(b+c)(a^2+bc)}{(a^2+b^2)(a^2+c^2)} + a \\ &= \frac{-a(a^2-bc)(a^2+bc)}{(a^2+b^2)(a^2+c^2)} + a = \frac{a(a^2b^2 + a^2c^2 + 2b^2c^2)}{(a^2+b^2)(a^2+c^2)}. \end{aligned}$$

This yields $A_1 \left(\frac{a^2(b+c)(a^2+bc)}{(a^2+b^2)(a^2+c^2)}, \frac{a(a^2b^2 + a^2c^2 + 2b^2c^2)}{(a^2+b^2)(a^2+c^2)} \right)$.

Next, the equation of CA is

$$y = \frac{a(b-c)}{a^2+bc}x.$$

The equation of E_BB_1 is

$$y = -\frac{a^2+bc}{a(b-c)}x + \frac{b(a^2+bc)}{a(b-c)}.$$

Solving these, we have

$$x = \frac{b(a^2+bc)^2}{a^2(b-c)^2 + (a^2+bc)^2} = \frac{b(a^2+bc)^2}{(a^2+b^2)(a^2+c^2)}$$

and

$$y = \frac{a(b-c)}{a^2+bc} \cdot \frac{b(a^2+bc)^2}{(a^2+b^2)(a^2+c^2)} = \frac{ab(b-c)(a^2+bc)}{(a^2+b^2)(a^2+c^2)}.$$

This implies $B_1 \left(\frac{b(a^2+bc)^2}{(a^2+b^2)(a^2+c^2)}, \frac{ab(b-c)(a^2+bc)}{(a^2+b^2)(a^2+c^2)} \right)$. We now find the perpendicular bisector of A_1B_1 . Its slope is

$$\begin{aligned} \frac{-1}{m_{A_1B_1}} &= -\frac{a^2(b+c)(a^2+bc) - b(a^2+bc)^2}{a(a^2b^2 + a^2c^2 + 2b^2c^2) - ab(b-c)(a^2+bc)} \\ &= -\frac{(a^2+bc)(a^2c - b^2c)}{a(a^2bc + a^2c^2 - b^3c + 3b^2c^2)} = -\frac{(a^2+bc)(a^2-b^2)}{a(a^2b + a^2c - b^3 + 3b^2c)} \end{aligned}$$

and it passes through the midpoint of A_1B_1 , whose coordinates are

$$\left(\frac{(2a^2b + a^2c + b^2c)(a^2 + bc)}{2(a^2 + b^2)(a^2 + c^2)}, \frac{a(2a^2b^2 - a^2bc + a^2c^2 + b^3c + b^2c^2)}{2(a^2 + b^2)(a^2 + c^2)} \right).$$

So the equation of the perpendicular bisector of A_1B_1 is

$$y = -\frac{(a^2 + bc)(a^2 - b^2)}{a(a^2b + a^2c - b^3 + 3b^2c)}x + \frac{(a^2 + bc)(a^2 - b^2)}{a(a^2b + a^2c - b^3 + 3b^2c)} \cdot \frac{(2a^2b + a^2c + b^2c)(a^2 + bc)}{2(a^2 + b^2)(a^2 + c^2)} + \frac{a(2a^2b^2 - a^2bc + a^2c^2 + b^3c + b^2c^2)}{2(a^2 + b^2)(a^2 + c^2)},$$

which can be simplified as

$$y = -\frac{(a^2 + bc)(a^2 - b^2)}{a(a^2b + a^2c - b^3 + 3b^2c)}x + \frac{2a^4b + a^4c - 2a^2b^3 + 4a^2b^2c - b^4c}{2a(a^2b + a^2c - b^3 + 3b^2c)}.$$

By symmetry, the perpendicular bisector of A_1C_1 has equation

$$y = -\frac{(a^2 + bc)(a^2 - c^2)}{a(a^2b + a^2c + 3bc^2 - c^3)}x + \frac{a^4b + 2a^4c + 4a^2bc^2 - 2a^2c^3 - bc^4}{2a(a^2b + a^2c + 3bc^2 - c^3)}.$$

We solve these equations to get

$$\begin{aligned} x &= \frac{(2a^4b + a^4c - 2a^2b^3 + 4a^2b^2c - b^4c)(a^2b + a^2c + 3bc^2 - c^3) - (a^4b + 2a^4c + 4a^2bc^2 - 2a^2c^3 - bc^4)(a^2b + a^2c - b^3 + 3b^2c)}{2(a^2 + bc)((a^2 - b^2)(a^2b + a^2c + 3bc^2 - c^3) - (a^2 - c^2)(a^2b + a^2c - b^3 + 3b^2c))} \\ &= \frac{a^6b^2 - a^6c^2 - a^4b^4 + a^4b^3c - a^4bc^3 + a^4c^4 - a^2b^5c - 3a^2b^4c^2 + 3a^2b^2c^4 + a^2bc^5 - 3b^5c^3 + 3b^3c^5}{2(a^2 + bc)(-4a^2b^2c + 4a^2bc^2 - 4b^3c^2 + 4b^2c^3)} \\ &= \frac{(b^2 - c^2)(a^2 + bc)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{-8bc(a^2 + bc)(b - c)(a^2 + bc)} \\ &= -\frac{(b + c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{(a^2 + bc)(a^2 - b^2)}{a(a^2b + a^2c - b^3 + 3b^2c)} \cdot \left(-\frac{(b + c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)} \right) \\ &\quad + \frac{2a^4b + a^4c - 2a^2b^3 + 4a^2b^2c - b^4c}{2a(a^2b + a^2c - b^3 + 3b^2c)} \\ &= \frac{(b + c)(a^2 - b^2)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2) + 4bc(2a^4b + a^4c - 2a^2b^3 + 4a^2b^2c - b^4c)}{8abc(a^2b + a^2c - b^3 + 3b^2c)} \\ &\quad + \frac{(b + c)a^6 + (-2b^3 + 6b^2c + 3bc^2 - c^3)a^4 + (b^5 - 7b^4c + 14b^3c^2 - 2b^2c^3)a^2 + (-b^5c^2 + 3b^4c^3)}{8abc(a^2b + a^2c - b^3 + 3b^2c)} \\ &= \frac{a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2}{8abc}. \end{aligned}$$

Thus, the circumcentre O' of $\triangle A_1B_1C_1$ has coordinates

$$\left(-\frac{(b+c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)}, \frac{a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2}{8abc} \right). \quad (1)$$

Recall that the coordinates of the orthocentre and circumcentre of $\triangle E_AE_BE_C$ are $\left(0, -\frac{bc}{a}\right)$ and $\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$ respectively. So their midpoint O , which is the nine-point centre of $\triangle E_AE_BE_C$, has coordinates $\left(\frac{b+c}{4}, \frac{a^2-bc}{4a}\right)$. Note that O is the circumcentre of $\triangle ABC$ since (ABC) is the nine-point circle of $\triangle E_AE_BE_C$. It follows that

$$(ABC): \quad x^2 + y^2 - \frac{b+c}{2}x - \frac{a^2-bc}{2a}y = 0.$$

We rewrite this as

$$x\left(x - \frac{b+c}{2}\right) + y\left(y - \frac{a^2-bc}{2a}\right) = 0. \quad (2)$$

As O' lies on this circle, we put (1) into (2). This yields

$$\begin{aligned} & \left(\frac{(b+c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)} \right) \left(\frac{(b+c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)} + \frac{b+c}{2} \right) \\ & + \left(\frac{a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2}{8abc} \right) \left(\frac{a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2}{8abc} - \frac{a^2-bc}{2a} \right) = 0. \end{aligned}$$

This means

$$\begin{aligned} & \left(\frac{(b+c)(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2)}{8bc(a^2 + bc)} \right) \left(\frac{(b+c)(a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2)}{8bc(a^2 + bc)} \right) \\ & + \left(\frac{a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2}{8abc} \right) \left(\frac{a^4 - a^2b^2 - a^2c^2 + 5b^2c^2}{8abc} \right) = 0. \end{aligned}$$

Clearing denominators and taking out the common factor, we obtain

$$\begin{aligned} & (a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2)(a^2(b+c)^2(a^4 - a^2b^2 - a^2c^2 - 3b^2c^2) \\ & + (a^2 + bc)^2(a^4 - a^2b^2 - a^2c^2 + 5b^2c^2)) = 0, \end{aligned}$$

i.e.

$$\begin{aligned} & (a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2)(a^8 + 4a^6bc - a^4b^4 - 4a^4b^3c + 4a^4b^2c^2 \\ & - 4a^4bc^3 - a^4c^4 - 4a^2b^4c^2 + 4a^2b^3c^3 - 4a^2b^2c^4 + 5b^4c^4) = 0. \end{aligned}$$

We test for some factors and deduce

$$(a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2)(a^2 - b^2)(a^2 - c^2)(a^4 + a^2b^2 + 4a^2bc + a^2c^2 + 5b^2c^2) = 0. \quad (3)$$

Note that $\angle ABC = 90^\circ$ iff O lies on AC . This is the same as $m_{AO} = m_{AC}$, i.e.

$$\frac{a^2 - bc}{a(b+c)} = \frac{a(b-c)}{a^2 + bc}.$$

This holds iff $a^4 - b^2c^2 = a^2(b^2 - c^2)$, i.e. $(a^2 - b^2)(a^2 + c^2) = 0$. As $a, c \neq 0$, we find that $\angle ABC = 90^\circ$ iff $a^2 - b^2 = 0$. Similarly, $\angle ACB = 90^\circ$ iff $a^2 - c^2 = 0$.

The last factor in (3) cannot be 0 since

$$a^4 + a^2b^2 + 4a^2bc + a^2c^2 + 5b^2c^2 = (a^2 + 2bc)^2 + a^2b^2 + a^2c^2 + b^2c^2 > 0.$$

For the first factor, it is natural to guess it is equivalent to $\angle BAC = 90^\circ$. Indeed, $\angle BAC = 90^\circ$ iff O lies on BC . Recall that

$$BC: \quad y = \frac{a(b+c)}{a^2 - bc}x - \frac{2abc}{a^2 - bc}.$$

Therefore, $\angle BAC = 90^\circ$ iff

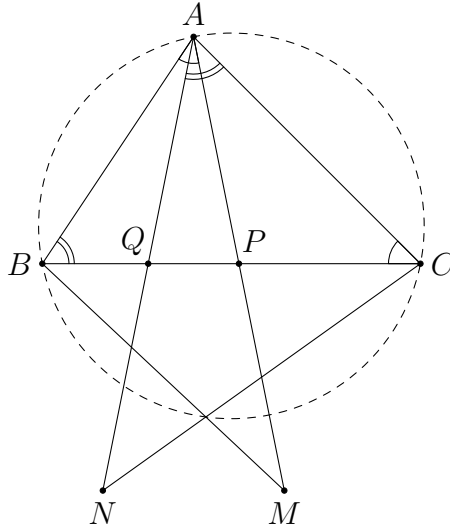
$$\frac{a^2 - bc}{4a} = \frac{a(b+c)}{a^2 - bc} \cdot \frac{b+c}{4} - \frac{2abc}{a^2 - bc}.$$

After expansion, one finds that it is exactly the same as

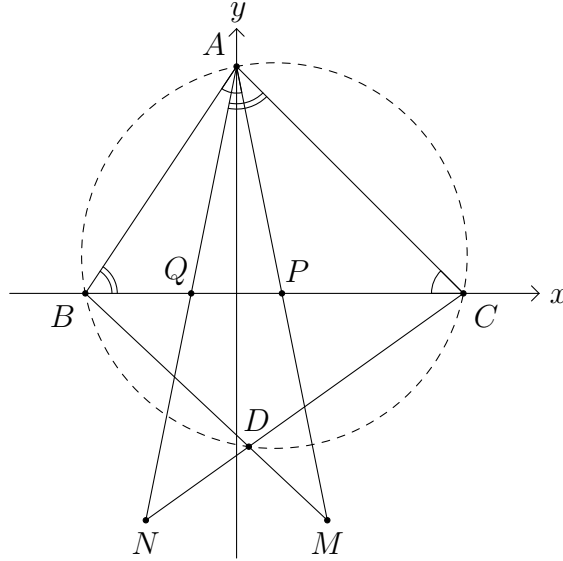
$$a^4 - a^2b^2 + 4a^2bc - a^2c^2 + b^2c^2 = 0.$$

Hence, from (3), we know that one of $\angle ABC$, $\angle ACB$ and $\angle BAC$ is 90° . $\square$

Problem 69. (IMO Shortlist 2014 G1/Problem 4) Points P and Q lie on side BC of acute-angled triangle ABC so that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Points M and N lie on lines AP and AQ , respectively, such that P is the midpoint of AM , and Q is the midpoint of AN . Prove that lines BM and CN intersect on the circumcircle of triangle ABC .



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$ where $b < 0 < c$. From $\angle PAB = \angle BCA$, we know that BA is tangent to (ACP) . So $BA^2 = BP \times BC$. This gives $BP = \frac{a^2 + b^2}{c - b}$ and hence $P\left(\frac{a^2 + bc}{c - b}, 0\right)$. Thus, $M = 2P - A = \left(\frac{2(a^2 + bc)}{c - b}, -a\right)$. This yields

$$m_{BM} = -\frac{a(c - b)}{2a^2 + b^2 + bc}.$$

By symmetry,

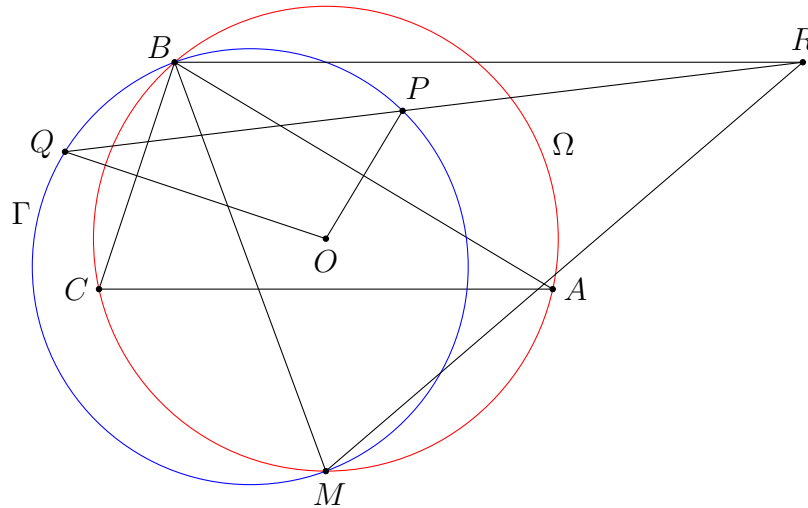
$$m_{CN} = -\frac{a(b - c)}{2a^2 + c^2 + bc}.$$

Let D be the intersection of BM and CN . Then

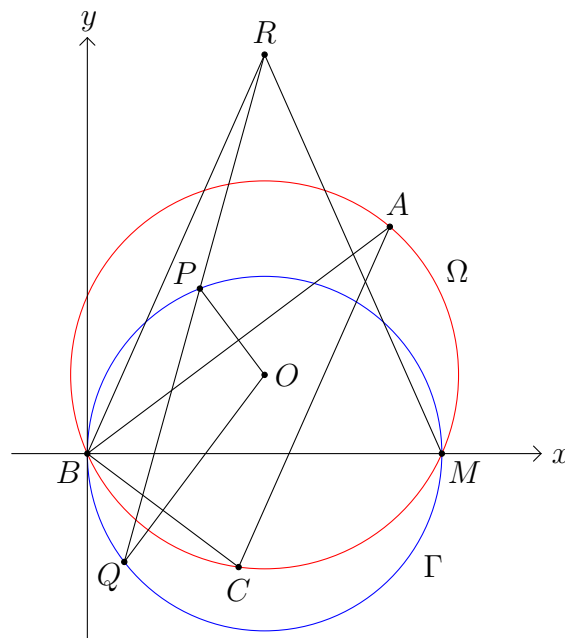
$$\begin{aligned} & A, B, D, C \text{ are concyclic} \\ \Leftrightarrow & \angle CDB + \angle BAC = 180^\circ \\ \Leftrightarrow & \tan \angle CDB = -\tan \angle BAC \\ \Leftrightarrow & \frac{m_{BM} - m_{CN}}{1 + m_{BM} \cdot m_{CN}} = -\frac{m_{AC} - m_{AB}}{1 + m_{AC} \cdot m_{AB}} \\ \Leftrightarrow & \frac{-a(c - b)(2a^2 + c^2 + bc) + a(b - c)(2a^2 + b^2 + bc)}{(2a^2 + b^2 + bc)(2a^2 + c^2 + bc) - a^2(c - b)^2} = -\frac{-ab + ac}{bc + a^2} \\ \Leftrightarrow & \frac{-a(c - b)(4a^2 + b^2 + 2bc + c^2)}{4a^4 + (b^2 + 6bc + c^2)a^2 + bc(b + c)^2} = -\frac{a(c - b)}{a^2 + bc} \\ \Leftrightarrow & \frac{-a(c - b)(4a^2 + b^2 + 2bc + c^2)}{(a^2 + bc)(4a^2 + b^2 + 2bc + c^2)} = -\frac{a(c - b)}{a^2 + bc}. \end{aligned}$$

This is obvious. □

Problem 70. (IMO Shortlist 2014 G3) Let Ω and O be the circumcircle and the circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM . The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q , respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that $BR \parallel AC$. (Here we always assume that an angle bisector is a ray.)



Solution.



We set the origin at B and set BM as the x -axis. Thus, if BA has equation $y = mx$ (with $m > 0$), then BC has equation $y = -mx$. Let $A(2a, 2ma)$ and $C(2c, -2mc)$

where $a, c > 0$. Note that OP is the perpendicular bisector of AB . We easily obtain

$$m_{OP} = \frac{-1}{m_{AB}} = -\frac{1}{m}$$

and

$$OP: \quad y = -\frac{1}{m}x + \frac{(m^2 + 1)a}{m}. \quad (1)$$

Changing m to $-m$ and changing a to c , we have

$$OQ: \quad y = \frac{1}{m}x - \frac{(m^2 + 1)c}{m}. \quad (2)$$

We solve (1) and (2). Then

$$x = \frac{\frac{(m^2 + 1)a}{m} + \frac{(m^2 + 1)c}{m}}{\frac{1}{m} + \frac{1}{m}} = \frac{(m^2 + 1)(a + c)}{2}$$

and

$$y = -\frac{1}{m} \cdot \frac{(m^2 + 1)(a + c)}{2} + \frac{(m^2 + 1)a}{m} = \frac{(m^2 + 1)(a - c)}{2m}.$$

This gives $O\left(\frac{(m^2 + 1)(a + c)}{2}, \frac{(m^2 + 1)(a - c)}{2m}\right)$. As O lies on the perpendicular bisector of BM , we find that $M((m^2 + 1)(a + c), 0)$. Thus,

$$\Gamma: \quad x^2 + y^2 - (m^2 + 1)(a + c)x = 0. \quad (3)$$

Putting (1) in (3), we solve

$$\left(1 + \frac{1}{m^2}\right)x^2 - \left(\frac{2(m^2 + 1)a}{m^2} + (m^2 + 1)(a + c)\right)x + \frac{(m^2 + 1)^2 a^2}{m^2} = 0.$$

Dividing both sides by $\frac{m^2 + 1}{m^2}$, this becomes

$$x^2 - (m^2(a + c) + 2a)x + (m^2 + 1)a^2 = 0.$$

From the configuration, we should choose the smaller root for the x -coordinate of P . Thus, we have

$$\begin{aligned} x &= \frac{m^2(a + c) + 2a - \sqrt{(m^2(a + c) + 2a)^2 - 4(m^2 + 1)a^2}}{2} \\ &= \frac{m^2(a + c) + 2a - \sqrt{m^4(a + c)^2 + 4m^2ac}}{2} \\ &= \frac{m^2(a + c) + 2a - m\sqrt{m^2(a + c)^2 + 4ac}}{2} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{1}{m} \cdot \frac{m^2(a+c) + 2a - m\sqrt{m^2(a+c)^2 + 4ac}}{2} + \frac{(m^2+1)a}{m} \\ &= \frac{m(a-c) + \sqrt{m^2(a+c)^2 + 4ac}}{2}. \end{aligned}$$

So $P\left(\frac{m^2(a+c) + 2a - m\sqrt{m^2(a+c)^2 + 4ac}}{2}, \frac{m(a-c) + \sqrt{m^2(a+c)^2 + 4ac}}{2}\right)$. To find

Q , we can change m to $-m$ and swap a and c . However, we should do this before we take out m from the square root. This means

$$x = \frac{m^2(a+c) + 2c - \sqrt{m^2(a+c)^2 + 4ac}}{2} = \frac{m^2(a+c) + 2c - m\sqrt{m^2(a+c)^2 + 4ac}}{2}$$

and similarly

$$\begin{aligned} y &= \frac{1}{m} \cdot \frac{m^2(a+c) + 2c - m\sqrt{m^2(a+c)^2 + 4ac}}{2} - \frac{(m^2+1)c}{m} \\ &= \frac{m(a-c) - \sqrt{m^2(a+c)^2 + 4ac}}{2}. \end{aligned}$$

Thus, $Q\left(\frac{m^2(a+c) + 2c - m\sqrt{m^2(a+c)^2 + 4ac}}{2}, \frac{m(a-c) - \sqrt{m^2(a+c)^2 + 4ac}}{2}\right)$. For

convenience, let $\Delta = \sqrt{m^2(a+c)^2 + 4ac}$. Then we have $m_{PQ} = \frac{\Delta}{a-c}$, and hence

$$PQ: \quad y = \frac{\Delta}{a-c}x - \frac{\Delta}{a-c} \cdot \frac{m^2(a+c) + 2a - m\Delta}{2} + \frac{m(a-c) + \Delta}{2},$$

i.e.

$$PQ: \quad y = \frac{\Delta}{a-c}x + \frac{m(m^2+1)(a+c)^2 - (m^2+1)(a+c)\Delta}{2(a-c)}.$$

As R lies on the perpendicular bisector of BM , its x -coordinate is the same as that of O , which is $\frac{(m^2+1)(a+c)}{2}$. Its y -coordinate is

$$\frac{\Delta}{a-c} \cdot \frac{(m^2+1)(a+c)}{2} + \frac{m(m^2+1)(a+c)^2 - (m^2+1)(a+c)\Delta}{2(a-c)} = \frac{m(m^2+1)(a+c)^2}{2(a-c)}.$$

This means $R\left(\frac{(m^2+1)(a+c)}{2}, \frac{m(m^2+1)(a+c)^2}{2(a-c)}\right)$. Finally, we check that

$$m_{BR} = \frac{m(a+c)}{a-c}$$

and

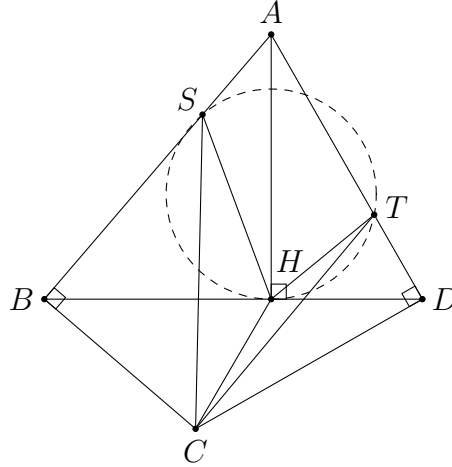
$$m_{AC} = \frac{2ma + 2mc}{2a - 2c} = \frac{m(a+c)}{a-c} = m_{BR}.$$

This proves $BR \parallel AC$ as desired. $\square$

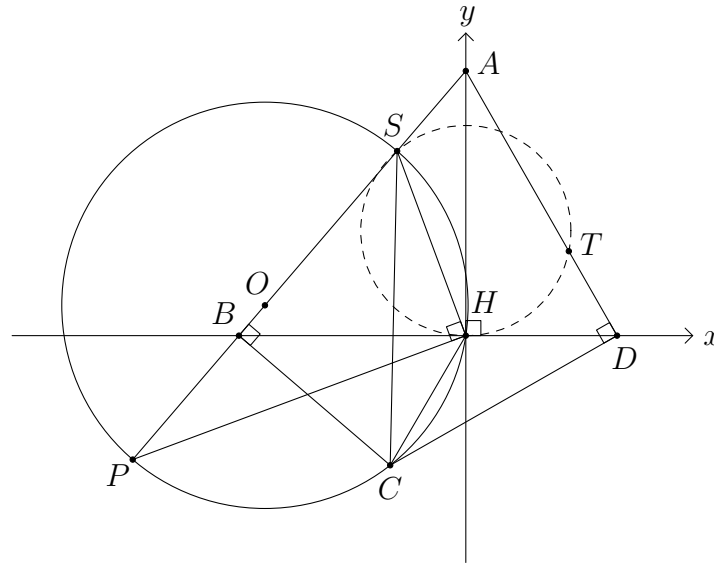
Problem 71. (IMO Shortlist 2014 G5/Problem 3) Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Prove that line BD is tangent to the circumcircle of triangle TSH .



Solution.



Define P on the ray AB such that $\angle SHP = 90^\circ$. Then

$$\angle PHC = \angle SHC - 90^\circ = \angle BSC.$$

This shows P, C, H, S are concyclic. Conversely, if we define O to be the intersection of AB and the perpendicular bisector of CH , then the circle with centre O passing through H meets AB at S and P .

Now, let $H(0,0)$, $A(0,a)$, $B(b,0)$, $D(d,0)$ where $b < 0 < d$ and $a > 0$. We have $m_{BC} = \frac{-1}{m_{AB}} = \frac{b}{a}$. So

$$BC: \quad y = \frac{b}{a}x - \frac{b^2}{a}$$

and similarly,

$$DC: \quad y = \frac{d}{a}x - \frac{d^2}{a}.$$

Solving these, we get $C\left(b+d, \frac{bd}{a}\right)$. Then $m_{CH} = \frac{bd}{a(b+d)}$ and the midpoint of CH is $\left(\frac{b+d}{2}, \frac{bd}{2a}\right)$. Thus, the perpendicular bisector of CH has equation

$$y = -\frac{a(b+d)}{bd}x + \frac{a^2(b+d)^2 + b^2d^2}{2abd}.$$

Together with

$$AB: \quad y = -\frac{a}{b}x + a, \tag{1}$$

we obtain

$$x = \frac{\frac{a^2(b+d)^2 + b^2d^2}{2abd} - a}{\frac{a(b+d)}{bd} - \frac{a}{b}} = \frac{a^2b^2 + a^2d^2 + b^2d^2}{2a^2b}$$

and

$$y = \frac{a}{b} \cdot \left(b - \frac{a^2b^2 + a^2d^2 + b^2d^2}{2a^2b}\right) = \frac{a^2b^2 - a^2d^2 - b^2d^2}{2ab^2}.$$

This yields $O\left(\frac{a^2b^2 + a^2d^2 + b^2d^2}{2a^2b}, \frac{a^2b^2 - a^2d^2 - b^2d^2}{2ab^2}\right)$. The equation with centre O passing through H is

$$x^2 + y^2 - \frac{a^2b^2 + a^2d^2 + b^2d^2}{a^2b}x - \frac{a^2b^2 - a^2d^2 - b^2d^2}{ab^2}y = 0. \tag{2}$$

Combining (1) and (2), we solve

$$\begin{aligned} \left(1 + \frac{a^2}{b^2}\right)x^2 - \left(\frac{2a^2}{b} + \frac{a^2b^2 + a^2d^2 + b^2d^2}{a^2b} - \frac{a^2b^2 - a^2d^2 - b^2d^2}{b^3}\right)x \\ + \left(a^2 - \frac{a^2b^2 - a^2d^2 - b^2d^2}{b^2}\right) = 0, \end{aligned}$$

i.e.

$$a^2bx^2 - (a^2b^2 + a^2d^2 + b^2d^2)x + a^2bd^2 = 0.$$

The larger root

$$x = \frac{a^2b^2 + a^2d^2 + b^2d^2 - \sqrt{(a^2b^2 + a^2d^2 + b^2d^2)^2 - 4a^4b^2d^2}}{2a^2b}$$

is the x -coordinate of S (note that the denominator is negative). Let

$$\Delta = (a^2b^2 + a^2d^2 + b^2d^2)^2 - 4a^4b^2d^2.$$

Then $S\left(\frac{a^2b^2 + a^2d^2 + b^2d^2 - \sqrt{\Delta}}{2a^2b}, \frac{a^2b^2 - a^2d^2 - b^2d^2 + \sqrt{\Delta}}{2ab^2}\right)$. The perpendicular bisector of SH has slope

$$\begin{aligned} \frac{-1}{m_{SH}} &= -\frac{b(a^2b^2 + a^2d^2 + b^2d^2 - \sqrt{\Delta})}{a(a^2b^2 - a^2d^2 - b^2d^2 + \sqrt{\Delta})} \\ &= -\frac{b(a^2b^2 + a^2d^2 + b^2d^2 - \sqrt{\Delta})}{a(a^2b^2 - a^2d^2 - b^2d^2 + \sqrt{\Delta})} \times \frac{a^2b^2 - a^2d^2 - b^2d^2 - \sqrt{\Delta}}{a^2b^2 - a^2d^2 - b^2d^2 - \sqrt{\Delta}} \\ &= \frac{a^2b^2 - a^2d^2 + b^2d^2 - \sqrt{\Delta}}{2abd^2}. \end{aligned}$$

Since this line passes through O , we compute

$$\begin{aligned} &\frac{a^2b^2 - a^2d^2 - b^2d^2}{2ab^2} - \frac{a^2b^2 - a^2d^2 + b^2d^2 - \sqrt{\Delta}}{2abd^2} \cdot \frac{a^2b^2 + a^2d^2 + b^2d^2}{2a^2b} \\ &= \frac{-(a^4(b^2 - d^2)^2 + 2a^2b^2d^2(b^2 + d^2) + b^4d^4) + (a^2b^2 + a^2d^2 + b^2d^2)\sqrt{\Delta}}{4a^3b^2d^2}. \end{aligned}$$

Thus, the equation of the perpendicular bisector of SH is

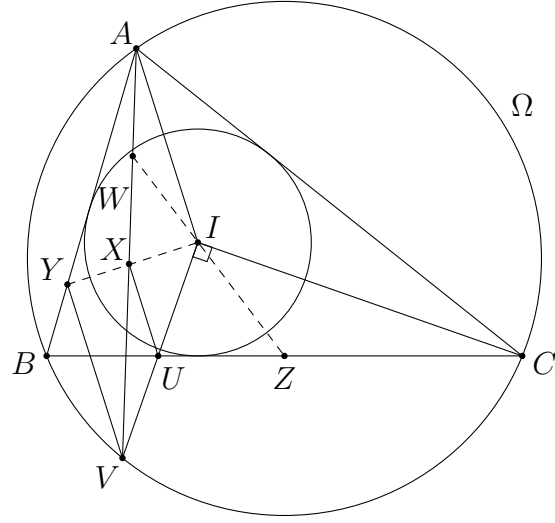
$$\begin{aligned} y &= \frac{a^2b^2 - a^2d^2 + b^2d^2 - \sqrt{\Delta}}{2abd^2}x \\ &+ \frac{-(a^4(b^2 - d^2)^2 + 2a^2b^2d^2(b^2 + d^2) + b^4d^4) + (a^2b^2 + a^2d^2 + b^2d^2)\sqrt{\Delta}}{4a^3b^2d^2}. \end{aligned}$$

Its y -intercept is

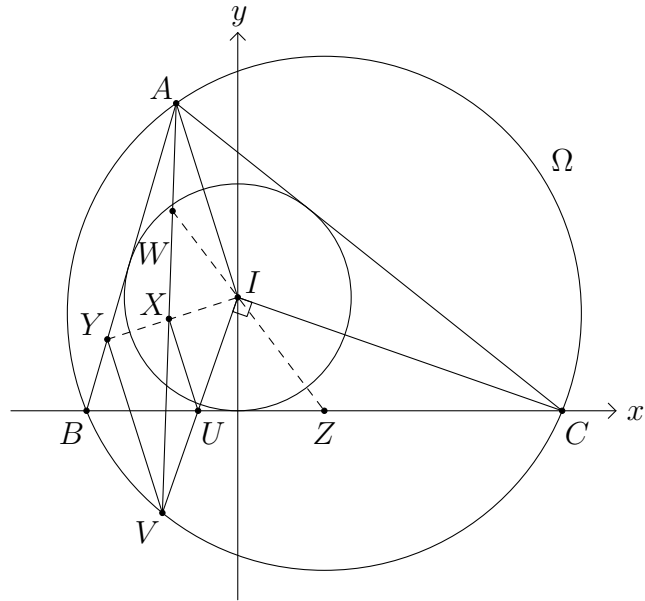
$$\left(0, \frac{-(a^4(b^2 - d^2)^2 + 2a^2b^2d^2(b^2 + d^2) + b^4d^4) + (a^2b^2 + a^2d^2 + b^2d^2)\sqrt{\Delta}}{4a^3b^2d^2}\right).$$

Note that the y -coordinate is symmetric in b and d . Therefore, the perpendicular bisector of TH also passes through this point. Hence, it must be the circumcentre of $\triangle HST$. As this circumcentre lies on the y -axis, which is perpendicular to BD , we know that BD is tangent to (HST) . $\square$

Problem 72. (IMO Shortlist 2014 G7) Let ABC be a triangle with circumcircle Ω and incentre I . Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at points U and V , respectively. Let the line passing through U and parallel to AI intersect AV at X , and let the line passing through V and parallel to AI intersect AB at Y . Let W and Z be the midpoints of AX and BC , respectively. Prove that if the points I, X , and Y are collinear, then the points I, W , and Z are also collinear.



Solution.



Let $I(0,1)$, $B(b,0)$, $C(c,0)$, $Z\left(\frac{b+c}{2}, 0\right)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$ and

$$AB: \quad y = -\frac{2b}{b^2-1}x + \frac{2b^2}{b^2-1}. \quad (1)$$

Next, from $m_{CI} = -\frac{1}{c}$, we get

$$IU: \quad y = cx + 1.$$

Thus, $U\left(-\frac{1}{c}, 0\right)$. Let $V(d, cd+1)$. Since $m_{AI} = \frac{bc-1}{b+c}$, we easily get

$$UX: \quad y = \frac{bc-1}{b+c}x + \frac{bc-1}{c(b+c)} \quad (2)$$

and

$$VY: \quad y = \frac{bc-1}{b+c}x + \frac{(c^2+1)d + (b+c)}{b+c}. \quad (3)$$

Also, we have

$$m_{AV} = \frac{(cd+1)(bc+1) - 2bc}{d(bc+1) - (b+c)} = \frac{cd(bc+1) + (1-bc)}{d(bc+1) - (b+c)}$$

so that

$$AV: \quad y = \frac{cd(bc+1) + (1-bc)}{d(bc+1) - (b+c)}x + \frac{cd(b-c) - (b+c)}{d(bc+1) - (b+c)}. \quad (4)$$

Solving (2) and (4), we get

$$\begin{aligned} x &= \frac{c(b+c)(cd(b-c) - (b+c)) - (bc-1)(d(bc+1) - (b+c))}{c((bc-1)(d(bc+1) - (b+c)) - (b+c)(cd(bc+1) + (1-bc)))} \\ &= \frac{d(c^2-1) + (b+c)}{cd(bc+1)} \end{aligned}$$

and

$$y = \frac{bc-1}{b+c} \cdot \left(\frac{d(c^2-1) + (b+c)}{cd(bc+1)} + \frac{1}{c} \right) = \frac{(dc+1)(bc-1)}{cd(bc+1)}.$$

So

$$X\left(\frac{d(c^2-1) + (b+c)}{cd(bc+1)}, \frac{(dc+1)(bc-1)}{cd(bc+1)}\right)$$

and hence

$$W\left(\frac{d(2c^2+bc-1) + (b+c)}{2cd(bc+1)}, \frac{cd(3bc-1) + (bc-1)}{2cd(bc+1)}\right).$$

Next, solving (1) and (3), we have

$$x = \frac{2b^2(b+c) - (b^2-1)((c^2+1)d + (b+c))}{(bc-1)(b^2-1) + 2b(b+c)} = \frac{-(b^2-1)(c^2+1)d + (b+c)(b^2+1)}{(b^2+1)(bc+1)}$$

and

$$y = \frac{2b}{b^2-1} \cdot \left(b - \frac{-(b^2-1)(c^2+1)d + (b+c)(b^2+1)}{(b^2+1)(bc+1)} \right) = \frac{2b((c^2+1)d + (b^2+1)c)}{(b^2+1)(bc+1)}.$$

This yields

$$Y\left(\frac{-(b^2-1)(c^2+1)d + (b+c)(b^2+1)}{(b^2+1)(bc+1)}, \frac{2b((c^2+1)d + (b^2+1)c)}{(b^2+1)(bc+1)}\right).$$

We now use the condition that I, X, Y are collinear. This implies $m_{IX} = m_{IY}$ so that

$$\frac{-2cd + (bc - 1)}{d(c^2 - 1) + (b + c)} = \frac{2b(c^2 + 1)d + (b^2 + 1)(bc - 1)}{-(b^2 - 1)(c^2 + 1)d + (b + c)(b^2 + 1)}.$$

This yields

$$\begin{aligned} & (2c(b^2 - 1)(c^2 + 1) - 2b(c^2 + 1)(c^2 - 1))d^2 + (- (bc - 1)(b^2 - 1)(c^2 + 1) \\ & - 2c(b + c)(b^2 + 1) - (b^2 + 1)(c^2 - 1)(bc - 1) - 2b(b + c)(c^2 + 1))d \\ & + ((bc - 1)(b + c)(b^2 + 1) - (b + c)(b^2 + 1)(bc - 1)) = 0, \end{aligned}$$

which simplifies to

$$2(c^2 + 1)(bc + 1)(b - c)d^2 - 2(c^2 + 1)(bc + 1)(b^2 + 1)d = 0.$$

The root $d = 0$ is rejected since it yields the impossible case that $V = I$. Therefore, we get

$$d = \frac{b^2 + 1}{b - c}.$$

Recall that the equation of Ω is

$$x^2 + y^2 - (b + c)x - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc + 1)}y + bc = 0.$$

Note that $V(d, cd + 1) = \left(\frac{b^2 + 1}{b - c}, \frac{b(bc + 1)}{b - c}\right)$ lies on this circle. So we have

$$\left(\frac{b^2 + 1}{b - c}\right)^2 + \left(\frac{b(bc + 1)}{b - c}\right)^2 - (b + c) \cdot \frac{b^2 + 1}{b - c} - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc + 1)} \cdot \frac{b(bc + 1)}{b - c} + bc = 0.$$

Now,

$$\begin{aligned} & 2(b^2 + 1)^2 + 2b^2(bc + 1)^2 - 2(b - c)(b + c)(b^2 + 1) \\ & - b(b - c)(b^2c^2 - b^2 + 4bc - c^2 + 1) + 2bc(b - c)^2 = 0 \\ \Rightarrow & 2(b^2 + 1)^2 - 2(b^2 - c^2)(b^2 + 1) + 2b^2(bc + 1)^2 \\ & - b(b - c)(b^2c^2 - b^2 + 4bc - c^2 + 1 - 2c(b - c)) = 0 \\ \Rightarrow & 2(b^2 + 1)(c^2 + 1) + 2b^2(bc + 1)^2 - b(b - c)(b^2c^2 - b^2 + 2bc + c^2 + 1) = 0 \\ \Rightarrow & 2(b^2 + 1)(c^2 + 1) + 2b^2(bc + 1)^2 + 2b^2(b - c)^2 - b(b - c)(b^2 + 1)(c^2 + 1) = 0 \\ \Rightarrow & 2(b^2 + 1)(c^2 + 1) + 2b^2(b^2 + 1)(c^2 + 1) - b(b - c)(b^2 + 1)(c^2 + 1) = 0 \\ \Rightarrow & b^2 + bc + 2 = 0. \end{aligned}$$

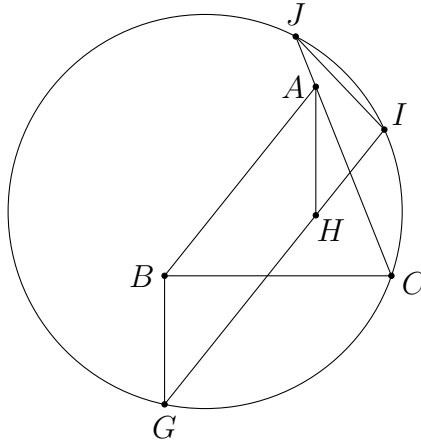
This is the condition we have.

Lastly, we have to check I, W, Z are collinear, i.e. $m_{IW} = m_{IZ}$.

$$\begin{aligned}
& m_{IW} = m_{IZ} \\
\Leftrightarrow & \frac{cd(bc-3) + (bc-1)}{d(2c^2 + bc - 1) + (b+c)} = -\frac{2}{b+c} \\
\Leftrightarrow & d(b^2c^2 + bc^3 + c^2 - bc - 2) + (b+c)(bc+1) = 0 \\
\Leftrightarrow & (bc+1)(d(bc+c^2-2) + (b+c)) = 0 \\
\Leftrightarrow & (bc+1)\left(\frac{b^2+1}{b-c} \cdot (bc+c^2-2) + (b+c)\right) = 0 \\
\Leftrightarrow & (bc+1)(b^3c + b^2c^2 - b^2 + bc - 2) = 0 \\
\Leftrightarrow & (bc+1)(bc-1)(b^2 + bc + 2) = 0.
\end{aligned}$$

This holds since $b^2 + bc + 2 = 0$ from above. So we are done. $\square$

Problem 73. (IMO Shortlist 2015 G1) Let ABC be an acute triangle with orthocentre H . Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI . Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J . Prove that $IJ = AH$.



Solution. Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $H\left(0, -\frac{bc}{a}\right)$ and

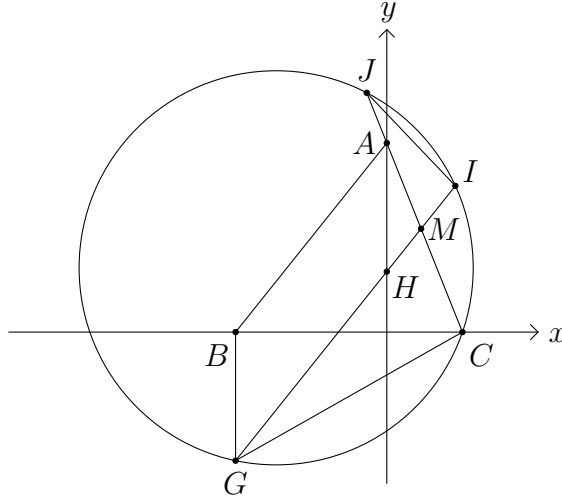
$$G = B + H - A = \left(b, -\frac{a^2 + bc}{a}\right).$$

We have

$$GH: \quad y = -\frac{a}{b}x - \frac{bc}{a}.$$

Let M be the intersection point of GH and AC . Since

$$AC: \quad y = -\frac{a}{c}x + a,$$



we find that $M\left(\frac{bc(a^2 + bc)}{a^2(b - c)}, -\frac{c(a^2 + b^2)}{a(b - c)}\right)$. Note that M is the midpoint of HI . So we have $I\left(\frac{2bc(a^2 + bc)}{a^2(b - c)}, -\frac{c(2a^2 + b^2 + bc)}{a(b - c)}\right)$.

Next, we use the fact that $\triangle MIJ \sim \triangle MCG$. This yields $\frac{MI}{IJ} = \frac{MC}{CG}$ so that

$$IJ = \frac{MI \times CG}{MC}.$$

We now compute the lengths one by one. We have

$$\begin{aligned} MI^2 &= \left(\frac{bc(a^2 + bc)}{a^2(b - c)}\right)^2 + \left(\frac{c(a^2 + b^2)}{a(b - c)}\right)^2 = \frac{c^2(a^2 + bc)^2(a^2 + b^2)}{a^4(b - c)^2}, \\ CG^2 &= (b - c)^2 + \left(\frac{a^2 + bc}{a}\right)^2 = \frac{(a^2 + b^2)(a^2 + c^2)}{a^2}, \\ MC^2 &= \left(\frac{c^2(a^2 + b^2)}{a^2(b - c)}\right)^2 + \left(\frac{c(a^2 + b^2)}{a(b - c)}\right)^2 = \frac{c^2(a^2 + b^2)^2(a^2 + c^2)}{a^4(b - c)^2}. \end{aligned}$$

Combining these, we get

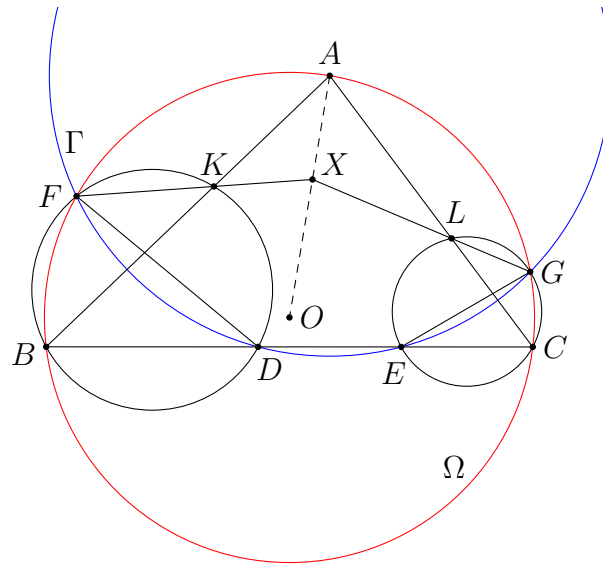
$$IJ^2 = \frac{(a^2 + bc)^2}{a^2} = AH^2.$$

Therefore, $IJ = AH$. □

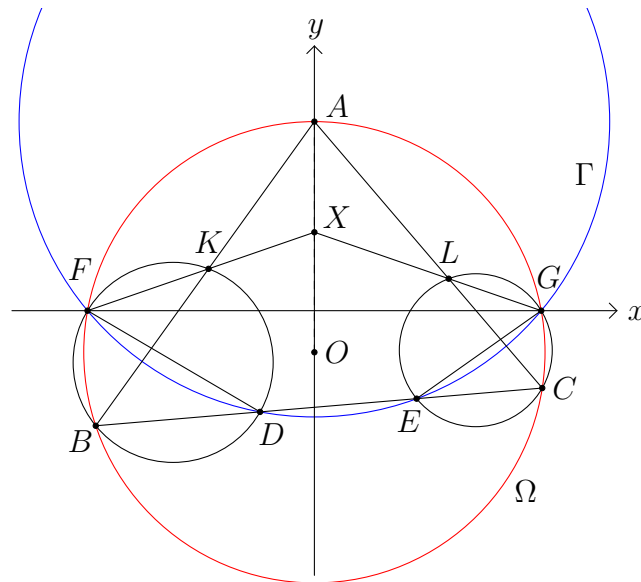
Problem 74. (IMO Shortlist 2015 G2/Problem 4) Triangle ABC has circumcircle Ω and circumcentre O . A circle Γ with centre A intersects the segment BC at points D and E , such that B, D, E and C are all different and lie on line BC in this order. Let F and G be the points of intersection of Γ and Ω , such that A, F, B, C and G lie

on Ω in this order. Let K be the second point of intersection of the circumcircle of triangle BDF and the segment AB . Let L be the second point of intersection of the circumcircle of triangle CGE and the segment CA .

Suppose that the lines FK and GL are different and intersect at the point X . Prove that X lies on the line AO .



Solution.



Note that $AF = AG$. Let $A(0, a)$, $F(-1, 0)$, $G(1, 0)$. Clearly, O lies on the y -axis.

From

$$\begin{aligned}
& OA = OF \\
\Rightarrow & (a - y)^2 = y^2 + 1 \\
\Rightarrow & y = \frac{a^2 - 1}{2a},
\end{aligned}$$

we get $O\left(0, \frac{a^2 - 1}{2a}\right)$. We easily deduce

$$\Omega: \quad x^2 + y^2 - \frac{a^2 - 1}{a}y - 1 = 0, \quad (1)$$

$$\Gamma: \quad x^2 + y^2 - 2ay - 1 = 0. \quad (2)$$

Let $y = mx + k$ be the equation of BC . Let $B(b, mb + k)$, $C(c, mc + k)$, $D(d, md + k)$, $E(e, me + k)$. Putting $y = mx + k$ in (1), we get

$$(1 + m^2)x^2 + \left(2mk - \frac{a^2 - 1}{a} \cdot m\right)x + \left(k^2 - \frac{a^2 - 1}{a} \cdot k - 1\right) = 0.$$

Since b and c are the roots of this quadratic equation, we find that

$$\begin{aligned}
b + c &= \frac{m(-2ka + a^2 - 1)}{(m^2 + 1)a}, \\
bc &= \frac{(k - a)(ka + 1)}{(m^2 + 1)a}.
\end{aligned} \quad (3)$$

Similarly, putting $y = mx + k$ in (2), we get

$$(1 + m^2)x^2 + (2mk - 2ma)x + (k^2 - 2ka - 1) = 0.$$

The roots are d and e . We obtain

$$\begin{aligned}
d + e &= \frac{-2m(k - a)}{m^2 + 1}, \\
de &= \frac{k^2 - 2ka - 1}{m^2 + 1}.
\end{aligned} \quad (4)$$

Now, to show that A, X, O are collinear, we need to show X lies on the y -axis. Since F and G are symmetric point in the y -axis, this is the same as showing $m_{FK} = -m_{GL}$. Therefore, we only need to know the slope of FK while the coordinates of K are not important. We have

$$\begin{aligned}
& F, B, D, K \text{ are concyclic} \\
\Rightarrow & \angle KFD = \angle KBD \\
\Rightarrow & \tan \angle KFD = \tan \angle KBD \\
\Rightarrow & \frac{m_{FK} - m_{DF}}{1 + m_{FK} \cdot m_{DF}} = \frac{m_{BK} - m_{BD}}{1 + m_{BK} \cdot m_{BD}}.
\end{aligned}$$

Using $m_{DF} = \frac{md+k}{d+1}$, $m_{BK} = m_{AB} = \frac{mb+k-a}{b}$ and $m_{BD} = m$, we have

$$\frac{(d+1)m_{FK} - (md+k)}{d+1 + (md+k)m_{FK}} = \frac{mb+k-a-mb}{b+m(mb+k-a)}.$$

From this, we obtain

$$\begin{aligned} m_{FK} &= \frac{(k-a)(d+1) + (md+k)(m^2b+mk-ma+b)}{(d+1)(m^2b+mk-ma+b) - (md+k)(k-a)} \\ &= \frac{m^3bd + m^2k(b+d) + mk^2 - m^2ad - mka + mbd + k(b+d+1) - (ad+a)}{m^2(bd+b) + mk - k^2 - ma + ka + (bd+b)}. \end{aligned}$$

To obtain m_{GL} , the simplest way is to add a negative sign to the above expression (which corresponds to a reflection in the y -axis), and make the following changes: replace b by $-c$; replace d by $-e$; replace m by $-m$. This yields

$$m_{GL} = \frac{m^3ce + m^2k(c+e) + mk^2 - m^2ae - mka + mce + k(c+e-1) - (ae-a)}{m^2(ce-c) - mk - k^2 + ma + ka + (ce-c)}.$$

Recall that we need to prove $m_{FK} = -m_{GL}$. Clearing denominators and collecting like terms, it is equivalent to

$$\begin{aligned} &m^5(2bcde - bcd + bce) + m^4k(bcd + bce + bde + cde + be - cd - bd + ce) \\ &\quad + m^3k^2(e-d) + m^2k^3(-b-c-d-e) - 2mk^4 \\ &\quad + m^4a(-bde - cde - be + cd + bd - ce) + 2m^3ka(d-e) \\ &\quad + m^2k^2a(b+c+2d+2e) + 4mk^3a + m^3(4bcde - 2bcd + 2bce - a^2d + a^2e) \\ &\quad + m^2k(-a^2d - a^2e + 2bcd + 2bce + 2bde + 2cde + 2be - 2cd - 2bd + 2ce - b - c) \\ &\quad + mk^2(-2a^2 - d + e - 2) + k^3(-b-c-d-e) \\ &\quad + m^2a(-2bde - 2cde - 2be + 2cd + 2bd - 2ce + b + c) + 2mka(d-e+2) \\ &\quad + k^2a(b+c+2d+2e) + m(-a^2d + a^2e + 2bcde - bcd + bce - 2a^2) \\ &\quad + k(-a^2d - a^2e + bcd + bce + bde + cde + be - cd - bd + ce - b - c) \\ &\quad + a(-bde - cde - be + cd + bd - ce + b + c) = 0. \end{aligned}$$

This can be rewritten as

$$\begin{aligned} &2bcdem(m^2+1)^2 + bc(d+e)(m^2+1)^2k + bc(e-d)m(m^2+1)^2 \\ &\quad + de(b+c)(m^2+1)^2(k-a) + (b+c)(e-d)(m^2+1)^2(k-a) \\ &\quad - (b+c)(m^2+1)(k^2+1)(k-a) - (d+e)(m^2+1)k(k-a)^2 \\ &\quad + (e-d)m(m^2+1)(k-a)^2 - 2m(k^2+1)(k-a)^2 = 0. \end{aligned}$$

Using (3) and (4), this becomes

$$\begin{aligned}
& \frac{2m(k-a)(ka+1)(k^2-2ka-1)}{a} - \frac{2mk(k-a)^2(ka+1)}{a} \\
& + \frac{m(k^2-2ka-1)(-2ka+a^2-1)(k-a)}{a} - \frac{m(-2ka+a^2-1)(k^2+1)(k-a)}{a} \\
& + 2mk(k-a)^3 - 2m(k^2+1)(k-a)^2 + (e-d) \left(\frac{m(m^2+1)(k-a)(ka+1)}{a} \right. \\
& \left. + \frac{m(m^2+1)(-2ka+a^2-1)(k-a)}{a} + m(m^2+1)(k-a)^2 \right) = 0.
\end{aligned}$$

Removing the common factor $\frac{m(k-a)}{a}$, the terms without the factor $e-d$ are

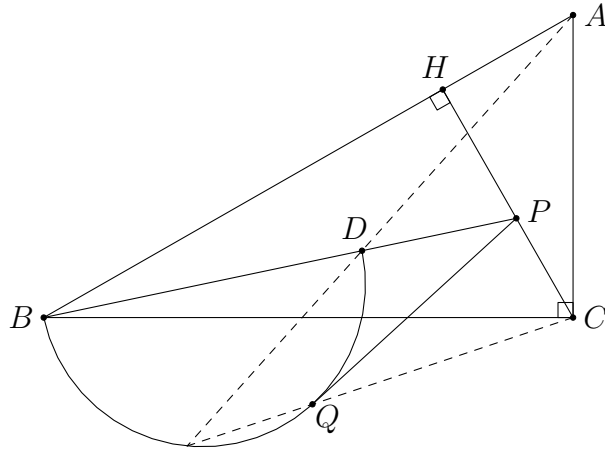
$$\begin{aligned}
& 2(ka+1)(k^2-2ka-1) - 2k(k-a)(ka+1) + (k^2-2ka-1)(-2ka+a^2-1) \\
& - (-2ka+a^2-1)(k^2+1) + 2ka(k-a)^2 - 2a(k^2+1)(k-a) \\
& = -2(ka+1)^2 - 2(ka+1)(-2ka+a^2-1) - 2a(k-a)(ka+1) \\
& = 2(ka+1)(-ka-1+2ka-a^2+1-ka+a^2) = 0,
\end{aligned}$$

while the terms multiplied to $e-d$ are

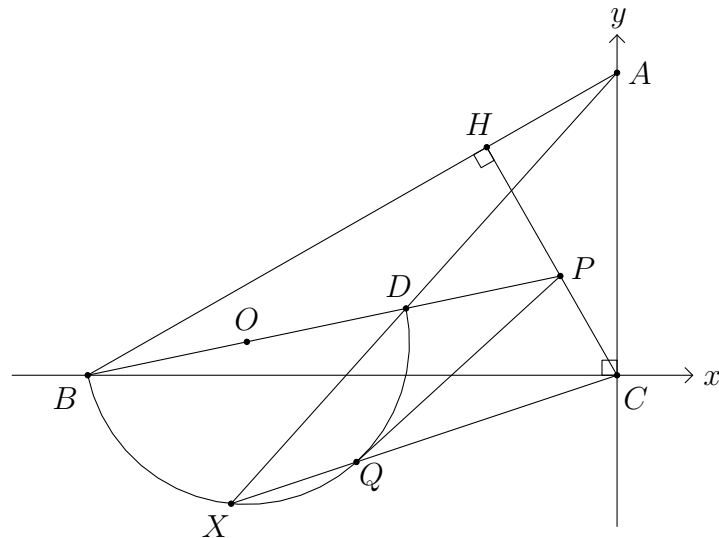
$$(m^2+1)((ka+1) + (-2ka+a^2-1) + a(k-a)) = 0.$$

So the identity is proved and we are done. $\square$

Problem 75. (IMO Shortlist 2015 G3) Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .



Solution.



Let $C(0,0)$, $A(0,a)$, $B(b,0)$. Solving

$$AB: \quad y = -\frac{a}{b}x + a$$

and

$$CH: \quad y = \frac{b}{a}x, \quad (1)$$

we get $H\left(\frac{a^2b}{a^2+b^2}, \frac{ab^2}{a^2+b^2}\right)$. The midpoint of AD lies on CH . Let $\left(d, \frac{bd}{a}\right)$ be its coordinates. Then we get $D\left(2d, \frac{2bd-a^2}{a}\right)$. Then

$$BD: \quad y = \frac{2bd - a^2}{a(2d - b)}x - \frac{b(2bd - a^2)}{a(2d - b)}. \quad (2)$$

Solving (1) and (2), we get

$$P\left(\frac{b(2bd-a^2)}{b^2-a^2}, \frac{b^2(2bd-a^2)}{a(b^2-a^2)}\right).$$

Let AD meet ω again at X . Firstly, we find that

$$AD: \quad y = \frac{bd - a^2}{ad}x + a. \quad (3)$$

Secondly, the centre O of ω is the midpoint $\left(\frac{b+2d}{2}, \frac{2bd-a^2}{2a}\right)$ of BD . So we have

$$\omega: \quad x^2 + y^2 - (b + 2d)x - \frac{2bd - a^2}{a}y + 2bd = 0. \quad (4)$$

(Strictly speaking, this is the equation for the whole circle. But everything involved on the circle in our calculation lies on the semicircle ω from the configuration.) Combining (3) and (4), we solve

$$\left(1 + \frac{(bd - a^2)^2}{a^2d^2}\right)x^2 + \left(\frac{2(bd - a^2)}{d} - (b + 2d) - \frac{(bd - a^2)(2bd - a^2)}{a^2d}\right)x + \left(a^2 - \frac{2bd - a^2}{a} \cdot a + 2bd\right) = 0.$$

Noting that $x = 2d$ is a solution (corresponding to the point D), this is the same as

$$(x - 2d) \left(\left(1 + \frac{(bd - a^2)^2}{a^2d^2}\right)x - \frac{a^2}{d} \right) = 0.$$

From this, we easily obtain

$$X \left(\frac{a^4d}{a^2d^2 + (a^2 - bd)^2}, \frac{ad(-a^2b + a^2d + b^2d)}{a^2d^2 + (a^2 - bd)^2} \right).$$

This yields

$$XC: \quad y = \frac{-a^2b + a^2d + b^2d}{a^3}x. \quad (5)$$

Putting (5) in (4), we solve

$$\left(1 + \frac{(-a^2b + a^2d + b^2d)^2}{a^6}\right)x^2 - \left((b + 2d) + \frac{(2bd - a^2)(-a^2b + a^2d + b^2d)}{a^4}\right)x + 2bd = 0.$$

Again, since $x = \frac{a^4d}{a^2d^2 + (a^2 - bd)^2}$ is a solution, we easily get the factorization

$$((a^2d^2 + (a^2 - bd)^2)x - a^4d)((a^2 + b^2)x - 2a^2b) = 0.$$

Let Q' be the second intersection of CX and ω . Then the x -coordinate of Q' is $\frac{2a^2b}{a^2 + b^2}$ and hence

$$Q' \left(\frac{2a^2b}{a^2 + b^2}, \frac{2b(-a^2b + a^2d + b^2d)}{a(a^2 + b^2)} \right).$$

It suffices to show PQ' is a tangent to ω so that $Q = Q'$. This means we need to check $m_{PQ'} \times m_{OQ'} = -1$. Recall that

$$O \left(\frac{b + 2d}{2}, \frac{2bd - a^2}{2a} \right).$$

So we find that

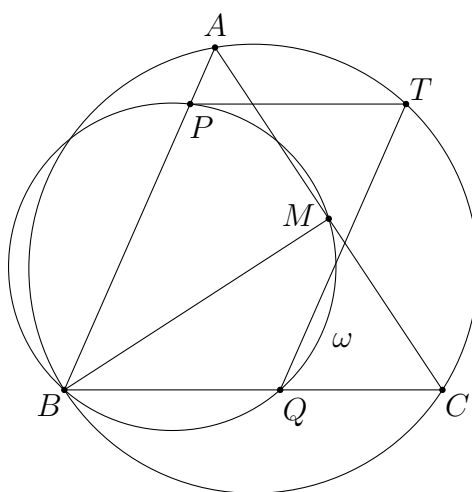
$$\begin{aligned} m_{PQ'} &= \frac{b^2(2bd - a^2)(a^2 + b^2) - 2b(b^2 - a^2)(-a^2b + a^2d + b^2d)}{ab(2bd - a^2)(a^2 + b^2) - 2a^3b(b^2 - a^2)} \\ &= \frac{a(-3a^2b + 2a^2d + 2b^2d + b^3)}{a^4 - 3a^2b^2 + 2a^2bd + 2b^3d} \end{aligned}$$

and

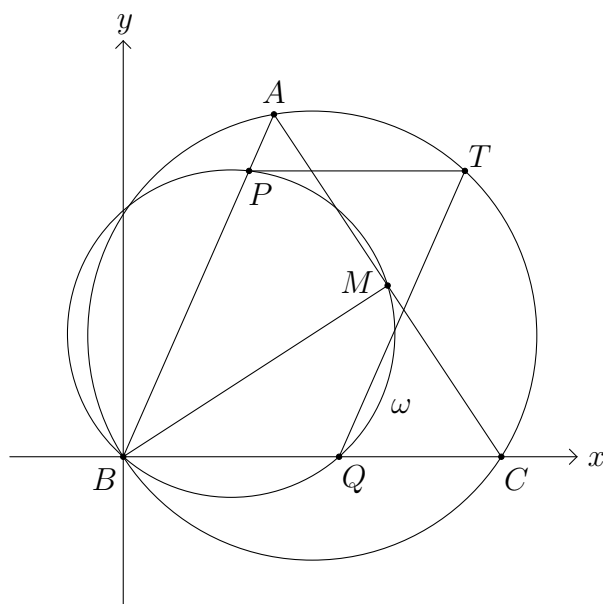
$$m_{OQ'} = \frac{4b(-a^2b + a^2d + b^2d) - (2bd - a^2)(a^2 + b^2)}{4a^3b - a(b + 2d)(a^2 + b^2)} = \frac{a^4 - 3a^2b^2 + 2a^2bd + 2b^3d}{a(3a^2b - 2a^2d - 2b^2d - b^3)}.$$

The result follows readily. $\square$

Problem 76. (IMO Shortlist 2015 G4) Let ABC be an acute triangle, and let M be the midpoint of AC . A circle ω passing through B and M meets the sides AB and BC again at P and Q , respectively. Let T be the point such that the quadrilateral $BPTQ$ is a parallelogram. Suppose that T lies on the circumcircle of the triangle ABC . Determine all possible values of $\frac{BT}{BM}$.



Solution.



Let $B(0,0)$, $C(2c,0)$, $A(2a,2b)$ so that $M(a+c,b)$. The x -coordinate of the circumcentre O of $\triangle ABC$ is c , and O lies on the perpendicular bisector of AB , which has equation

$$y = -\frac{a}{b}x + \frac{a^2 + b^2}{b}.$$

So $O\left(c, \frac{a^2 + b^2 - ac}{b}\right)$. This yields

$$(ABC): \quad x^2 + y^2 - 2cx - \frac{2(a^2 + b^2 - ac)}{b}y = 0.$$

Let $T(s, t)$ where s and t satisfy

$$s^2 + t^2 - 2cs - \frac{2(a^2 + b^2 - ac)}{b}t = 0. \quad (1)$$

The equation of TQ is

$$y = \frac{b}{a}x - \frac{bs}{a} + t.$$

Thus, $Q\left(\frac{bs - at}{b}, 0\right)$. As the equation of AB is $y = \frac{b}{a}x$, we easily get $P\left(\frac{at}{b}, t\right)$.

Now, let O' be the centre of ω . The x -coordinate of O' is $\frac{bs - at}{2b}$. Next, we easily get the equation of the perpendicular bisector of BP , which is

$$y = -\frac{a}{b}x + \frac{(a^2 + b^2)t}{2b^2}.$$

Therefore, we get $O'\left(\frac{bs - at}{2b}, \frac{(2a^2 + b^2)t - abs}{2b^2}\right)$. The equation of ω is

$$x^2 + y^2 - \frac{bs - at}{b}x - \frac{(2a^2 + b^2)t - abs}{b^2}y = 0.$$

As M lies on this circle, we need

$$(a + c)^2 + b^2 - \frac{bs - at}{b} \cdot (a + c) - \frac{(2a^2 + b^2)t - abs}{b^2} \cdot b = 0.$$

After simplification, this becomes

$$-cs - \frac{a^2 + b^2 - ac}{b}t + (a + c)^2 + b^2 = 0. \quad (2)$$

Computing $(1) - 2 \times (2)$, we obtain

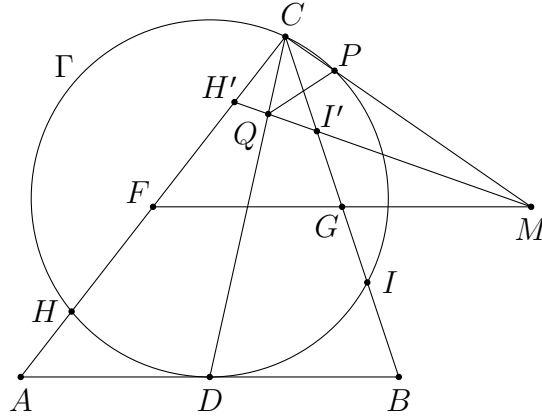
$$s^2 + t^2 = 2((a + c)^2 + b^2).$$

Using this, we find that

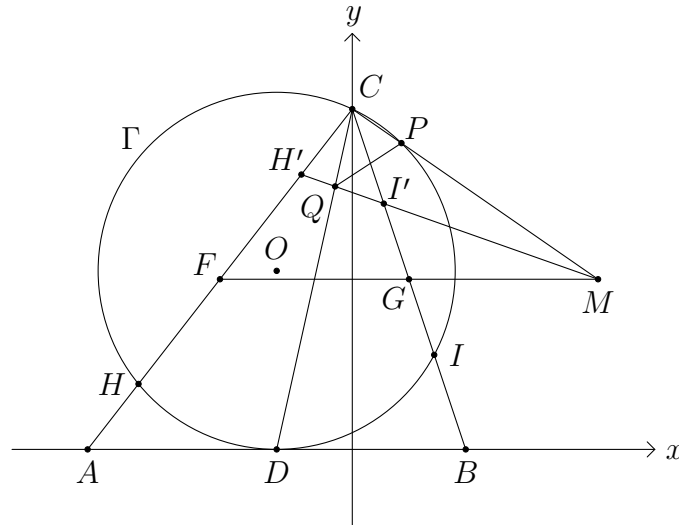
$$\frac{BT^2}{BM^2} = \frac{s^2 + t^2}{(a + c)^2 + b^2} = \frac{2((a + c)^2 + b^2)}{(a + c)^2 + b^2} = 2.$$

Therefore, $\frac{BT}{BM}$ can only be $\sqrt{2}$. □

Problem 77. (IMO Shortlist 2015 G5) Let ABC be a triangle with $CA \neq CB$. Let D, F , and G be the midpoints of the sides AB, AC , and BC , respectively. A circle Γ passing through C and tangent to AB at D meets the segments AF and BG at H and I , respectively. The points H' and I' are symmetric to H and I about F and G , respectively. The line $H'I'$ meets CD and FG at Q and M , respectively. The line CM meets Γ again at P . Prove that $CQ = QP$.



Solution.



Let $A(a, 0)$, $B(b, 0)$, $C(0, c)$. Then $D\left(\frac{a+b}{2}, 0\right)$, $F\left(\frac{a}{2}, \frac{c}{2}\right)$, $G\left(\frac{b}{2}, \frac{c}{2}\right)$. Let O be the centre of Γ . Firstly, the x -coordinate of O is $\frac{a+b}{2}$ since $OD \perp AB$. Also, O lies on the perpendicular bisector of CD , whose equation is

$$y = \frac{a+b}{2c}x - \frac{(a+b)^2 - 4c^2}{8c}.$$

So $O\left(\frac{a+b}{2}, \frac{(a+b)^2+4c^2}{8c}\right)$. Then

$$\Gamma: \quad x^2 + y^2 - (a+b)x - \frac{(a+b)^2+4c^2}{4c}y + \frac{(a+b)^2}{4} = 0. \quad (1)$$

The equation of AC is

$$y = -\frac{c}{a}x + c.$$

Putting this in (1), we solve

$$\left(1 + \frac{c^2}{a^2}\right)x^2 - \left(\frac{2c^2}{a} + (a+b) - \frac{(a+b)^2+4c^2}{4a}\right)x + \left(c^2 - \frac{(a+b)^2+4c^2}{4} + \frac{(a+b)^2}{4}\right) = 0.$$

This means

$$4(a^2 + c^2)x^2 - a(3a^2 + 2ab - b^2 + 4c^2)x = 0.$$

The root $x = 0$ is rejected since it corresponds to C . Thus, we get

$$H\left(\frac{a(3a^2 + 2ab - b^2 + 4c^2)}{4(a^2 + c^2)}, \frac{c(a-b)^2}{4(a^2 + c^2)}\right).$$

Hence, $H'\left(\frac{a(a-b)^2}{4(a^2 + c^2)}, \frac{c(3a^2 + 2ab - b^2 + 4c^2)}{4(a^2 + c^2)}\right)$. Swapping the variables a and b , we

get $I'\left(\frac{b(a-b)^2}{4(b^2 + c^2)}, \frac{c(-a^2 + 2ab + 3b^2 + 4c^2)}{4(b^2 + c^2)}\right)$. This yields

$$H'I': \quad y = \frac{c(a+b)}{c^2 - ab}x + \frac{c(4c^2 - (a+b)^2)}{4(c^2 - ab)}. \quad (2)$$

Solving with

$$CD: \quad y = -\frac{2c}{a+b}x + c,$$

we obtain $Q\left(\frac{(a-b)^2(a+b)}{4(2c^2 + a^2 + b^2)}, \frac{c(4c^2 + (a+b)^2)}{2(2c^2 + a^2 + b^2)}\right)$. Solving (2) with

$$FG: \quad y = \frac{c}{2},$$

we find that $M\left(\frac{-2c^2 + a^2 + b^2}{4(a+b)}, \frac{c}{2}\right)$. The equation of CM is

$$y = \frac{2c(a+b)}{2c^2 - a^2 - b^2}x + c.$$

Putting this in (1), we solve

$$\begin{aligned} \left(1 + \frac{4c^2(a+b)^2}{(2c^2 - a^2 - b^2)^2}\right)x^2 + \left(\frac{4c^2(a+b)}{2c^2 - a^2 - b^2} - (a+b) - \frac{(a+b)((a+b)^2 + 4c^2)}{2(2c^2 - a^2 - b^2)}\right)x \\ + \left(c^2 - \frac{(a+b)^2 + 4c^2}{4} + \frac{(a+b)^2}{4}\right) = 0, \end{aligned}$$

i.e.

$$2(4c^4 + 8abc^2 + (a^2 + b^2)^2)x^2 + (2c^2 - a^2 - b^2)(a + b)(a - b)^2x = 0.$$

Rejecting the solution $x = 0$, we get $x = -\frac{(a + b)(a - b)^2(2c^2 - a^2 - b^2)}{2(4c^4 + 8abc^2 + (a^2 + b^2)^2)}$ and hence

$P\left(-\frac{(a + b)(a - b)^2(2c^2 - a^2 - b^2)}{2(4c^4 + 8abc^2 + (a^2 + b^2)^2)}, \frac{c(4c^4 + 8abc^2 + 4a^2b^2)}{4c^4 + 8abc^2 + (a^2 + b^2)^2}\right)$. Now, we check the result $CQ = QP$. Indeed, we have

$$\begin{aligned} CQ^2 &= \left(\frac{(a - b)^2(a + b)}{4(2c^2 + a^2 + b^2)}\right)^2 + \left(\frac{c(4c^2 + (a + b)^2)}{2(2c^2 + a^2 + b^2)} - c\right)^2 \\ &= \frac{(a - b)^4(a + b)^2}{16(2c^2 + a^2 + b^2)^2} + \frac{c^2(a - b)^4}{4(2c^2 + a^2 + b^2)^2} \\ &= \frac{(a - b)^4(4c^2 + (a + b)^2)}{16(2c^2 + a^2 + b^2)^2} \end{aligned}$$

and

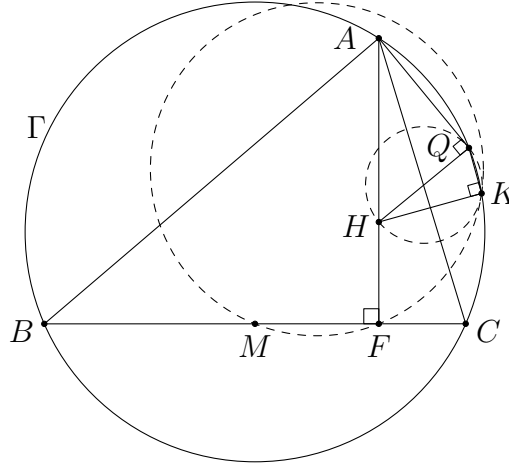
$$\begin{aligned} QP^2 &= \left(\frac{(a - b)^2(a + b)}{4(2c^2 + a^2 + b^2)} + \frac{(a + b)(a - b)^2(2c^2 - a^2 - b^2)}{2(4c^4 + 8abc^2 + (a^2 + b^2)^2)}\right)^2 \\ &\quad + \left(\frac{c(4c^2 + (a + b)^2)}{2(2c^2 + a^2 + b^2)} - \frac{c(4c^4 + 8abc^2 + 4a^2b^2)}{4c^4 + 8abc^2 + (a^2 + b^2)^2}\right)^2 \\ &= \frac{(a - b)^4(a + b)^2(12c^4 + 8abc^2 - (a^2 + b^2)^2)^2}{16(2c^2 + a^2 + b^2)^2(4c^4 + 8abc^2 + (a^2 + b^2)^2)^2} \\ &\quad + \frac{c^2(a - b)^4(4c^4 - 4(a^2 + b^2)c^2 - (a^2 + b^2)(a^2 + 4ab + b^2))^2}{4(2c^2 + a^2 + b^2)^2(4c^4 + 8abc^2 + (a^2 + b^2)^2)^2} \\ &= \frac{(a - b)^4}{16(2c^2 + a^2 + b^2)^2(4c^4 + 8abc^2 + (a^2 + b^2)^2)^2} \\ &\quad \times \left((a + b)^2(144c^8 + 192abc^6 + (-24a^4 + 16a^2b^2 - 24b^4)c^4 - 16ab(a^2 + b^2)^2c^2 \right. \\ &\quad \left. + (a^2 + b^2)^4) + (64c^{10} - 128(a^2 + b^2)c^8 + 32(a^4 - 4a^3b + 2a^2b^2 - 4ab^3 + b^4)c^6 \right. \\ &\quad \left. + 32(a^2 + b^2)^2(a^2 + 4ab + b^2)c^4 + 4(a^2 + b^2)^2(a^2 + 4ab + b^2)^2c^2)\right) \\ &= \frac{(a - b)^4}{16(2c^2 + a^2 + b^2)^2(4c^4 + 8abc^2 + (a^2 + b^2)^2)^2} \\ &\quad \times (64c^{10} + 16(a^2 + 18ab + b^2)c^8 + 32(a^4 + 2a^3b + 14a^2b^2 + 2ab^3 + b^4)c^6 \\ &\quad + 8(a^6 + 10a^5b + 11a^4b^2 + 36a^3b^3 + 11a^2b^4 + 10ab^5 + b^6)c^4 \\ &\quad + 4(a^2 + b^2)^2(a^4 + 4a^3b + 10a^2b^2 + 4ab^3 + b^4)c^2 + (a + b)^2(a^2 + b^2)^4) \end{aligned}$$

$$= \frac{(a-b)^4 (4c^2 + (a+b)^2) (4c^4 + 8abc^2 + (a^2 + b^2)^2)^2}{16(2c^2 + a^2 + b^2)^2 (4c^4 + 8abc^2 + (a^2 + b^2)^2)^2} = CQ^2.$$

So $QP = CQ$. □

Problem 78. (IMO Shortlist 2015 G6/Problem 3) Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocentre, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$, and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different, and lie on Γ in this order.

Prove that the circumcircles of triangles KQH and FKM are tangent to each other.



Solution. Let $K(0,0)$, $Q(a,0)$, $H(0,1)$. From $m_{HQ} = -\frac{1}{a}$, we have $m_{AQ} = \frac{-1}{m_{HQ}} = a$.

The equation of AQ is

$$y = ax - a^2.$$

So we let $A(b, a(b-a))$. The centre O of Γ has x -coordinate $\frac{a}{2}$. Also, it lies on the perpendicular bisector of AK , whose equation is

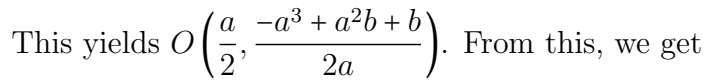
$$y = -\frac{b}{a(b-a)}x + \frac{b^2}{2a(b-a)} + \frac{a(b-a)}{2},$$

i.e.

$$y = -\frac{b}{a(b-a)}x + \frac{a^2(b-a)^2 + b^2}{2a(b-a)}.$$

Thus, we find that

$$y = -\frac{b}{a(b-a)} \cdot \frac{a}{2} + \frac{a^2(b-a)^2 + b^2}{2a(b-a)} = \frac{a^2(b-a)^2 + b^2 - ab}{2a(b-a)} = \frac{-a^3 + a^2b + b}{2a}.$$



Let AH meet Γ again at H' . Recall that H' is the reflection of H in the line BC . From $m_{AH} = \frac{-a^2 + ab - 1}{b}$, we easily get

$$\left(1 + \frac{(-a^2 + ab - 1)^2}{b^2}\right)x^2 + (\dots)x + \left(1 - \frac{-a^3 + a^2b + b}{a}\right) = 0.$$

$$\frac{1 - \frac{-a^3 + a^2b + b}{a}}{1 + \frac{(-a^2 + ab - 1)^2}{b^2}} = \frac{b^2(a^3 - a^2b + a - b)}{a((a^2 - ab + 1)^2 + b^2)}.$$
$$x = \frac{b(a^3 - a^2b + a - b)}{a((a^2 - ab + 1)^2 + b^2)} = \frac{b(a-b)(a^2+1)}{a(a^2+1)(a^2+1-2ab+b^2)} = \frac{b(a-b)}{a(a^2-2ab+b^2+1)}.$$

Then we have

$$\begin{aligned}
y &= \frac{-a^2 + ab - 1}{b} \cdot \frac{b(a-b)}{a(a^2 - 2ab + b^2 + 1)} + 1 \\
&= \frac{-a^3 + 2a^2b - ab^2 - a + b + a^3 - 2a^2b + ab^2 + a}{a(a^2 - 2ab + b^2 + 1)} \\
&= \frac{b}{a(a^2 - 2ab + b^2 + 1)}.
\end{aligned}$$

This gives $H' \left(\frac{b(a-b)}{a(a^2 - 2ab + b^2 + 1)}, \frac{b}{a(a^2 - 2ab + b^2 + 1)} \right)$. As F is the midpoint of HH' , we easily get

$$F \left(\frac{b(a-b)}{2a(a^2 - 2ab + b^2 + 1)}, \frac{b + a(a^2 - 2ab + b^2 + 1)}{2a(a^2 - 2ab + b^2 + 1)} \right).$$

We now find the circumcentre O' of $\triangle FKM$. Firstly, it lies on the perpendicular bisector of FK , whose equation is

$$\begin{aligned}
y &= -\frac{b(a-b)}{b + a(a^2 - 2ab + b^2 + 1)}x + \frac{b(a-b)}{b + a(a^2 - 2ab + b^2 + 1)} \cdot \frac{b(a-b)}{4a(a^2 - 2ab + b^2 + 1)} \\
&\quad + \frac{b + a(a^2 - 2ab + b^2 + 1)}{4a(a^2 - 2ab + b^2 + 1)}.
\end{aligned}$$

This is the same as

$$y = -\frac{b(a-b)}{b + a(a^2 - 2ab + b^2 + 1)}x + \frac{b^2(a-b)^2 + (b + a(a^2 - 2ab + b^2 + 1))^2}{4a(a^2 - 2ab + b^2 + 1)(b + a(a^2 - 2ab + b^2 + 1))}. \quad (3)$$

For the perpendicular bisector of MF , note that it is parallel to AH . Thus, its slope is $\frac{-a^2 + ab - 1}{b}$. Also, it passes through the midpoint of OH (the nine-point centre of $\triangle ABC$), whose coordinates are $\left(\frac{a}{4}, \frac{-a^3 + a^2b + 2a + b}{4a} \right)$. Hence, the equation of the perpendicular bisector of MF is

$$y = \frac{-a^2 + ab - 1}{b}x - \frac{-a^2 + ab - 1}{b} \cdot \frac{a}{4} + \frac{-a^3 + a^2b + 2a + b}{4a}.$$

This means

$$y = \frac{-a^2 + ab - 1}{b}x + \frac{a^4 - 2a^3b + a^2b^2 + a^2 + 2ab + b^2}{4ab}. \quad (4)$$

By solving (3) and (4), we can find the coordinates of O' . Our goal is to show that O' , K and the circumcentre of $\triangle KQH$ are collinear. The circumcentre of $\triangle KQH$ is just the midpoint $\left(\frac{a}{2}, \frac{1}{2} \right)$ of HQ since $\angle HKQ = 90^\circ$. Thus, we need to show that O' lies on the line with equation

$$y = \frac{1}{a}x. \quad (5)$$

In other words, we need to show that (3), (4) and (5) have a common root. Since the equation (5) is much simpler, we put this into the other equations and solve for x . If the value of x is the same, then the lines are concurrent.

Combining (4) and (5), we get

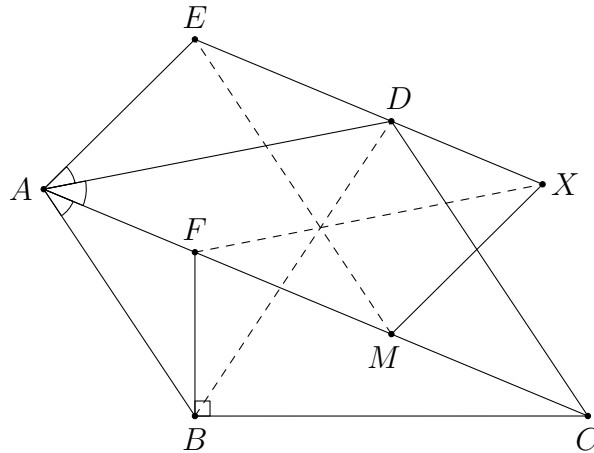
$$x = \frac{a^4 - 2a^3b + a^2b^2 + a^2 + 2ab + b^2}{4(b - a(-a^2 + ab - 1))} = \frac{a^4 - 2a^3b + a^2b^2 + a^2 + 2ab + b^2}{4(a^3 - a^2b + a + b)}.$$

Combining (3) and (5), we get

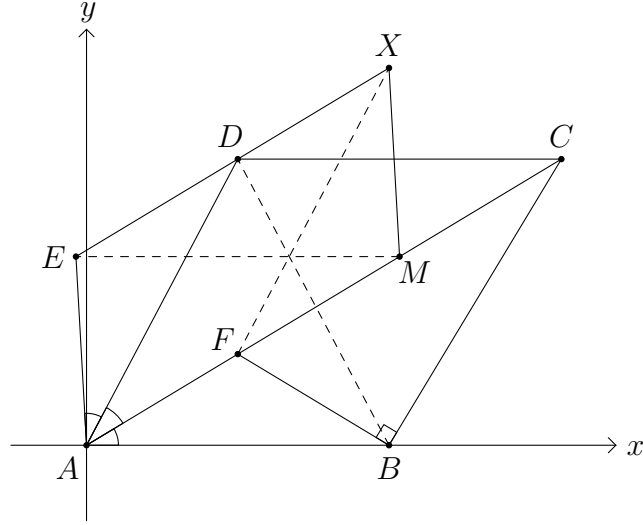
$$\begin{aligned} x &= \frac{b^2(a-b)^2 + (b + a(a^2 - 2ab + b^2 + 1))^2}{4(a^2 - 2ab + b^2 + 1)((b + a(a^2 - 2ab + b^2 + 1)) + ab(a-b))} \\ &= \frac{(b^2(a-b)^2 + b^2) + 2ab(a^2 - 2ab + b^2 + 1) + a^2(a^2 - 2ab + b^2 + 1)^2}{4(a^2 - 2ab + b^2 + 1)(a^3 - 2a^2b + ab^2 + a + b + a^2b - ab^2)} \\ &= \frac{(a^2 - 2ab + b^2 + 1)(b^2 + 2ab + a^2(a^2 - 2ab + b^2 + 1))}{4(a^2 - 2ab + b^2 + 1)(a^3 - a^2b + a + b)} \\ &= \frac{a^4 - 2a^3b + a^2b^2 + a^2 + 2ab + b^2}{4(a^3 - a^2b + a + b)}. \end{aligned}$$

Since the two results are the same, the equations (3), (4) and (5) have a common root. This completes the proof from the above discussion. $\square$

Problem 79. (IMO Shortlist 2016 G1/Problem 1) Triangle BCF has a right angle at B . Let A be the point on line CF such that $FA = FB$ and F lies between A and C . Point D is chosen such that $DA = DC$ and AC is the bisector of $\angle DAB$. Point E is chosen such that $EA = ED$ and AD is the bisector of $\angle EAC$. Let M be the midpoint of CF . Let X be the point such that $AMXE$ is a parallelogram (where $AM \parallel EX$ and $AE \parallel MX$). Prove that lines BD , FX , and ME are concurrent.



Solution.



Let $A(0,0)$, $B(2b,0)$. As F lies on the perpendicular bisector of AB , we let $F(b,a)$. Since $m_{BF} = -\frac{a}{b}$, the equation of BC is

$$y = \frac{b}{a}x - \frac{2b^2}{a}.$$

The equation of AF is

$$y = \frac{a}{b}x.$$

Solving these, we get $C\left(\frac{2b^3}{b^2 - a^2}, \frac{2ab^2}{b^2 - a^2}\right)$ and hence $M\left(\frac{b(3b^2 - a^2)}{2(b^2 - a^2)}, \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}\right)$.

Next, from $\angle DAB = 2\angle CAB$, we have

$$m_{AD} = \frac{2m_{AC}}{1 - m_{AC}^2} = \frac{2ab}{b^2 - a^2}.$$

Therefore, we get

$$AD: y = \frac{2ab}{b^2 - a^2}x.$$

The perpendicular bisector of AC has equation

$$y = -\frac{b}{a}x + \frac{b^2(a^2 + b^2)}{a(b^2 - a^2)}.$$

These yield $D\left(b, \frac{2ab^2}{b^2 - a^2}\right)$. Similarly, from $\angle EAB = \angle CAB + \angle DAB$, we have

$$m_{AE} = \frac{m_{AC} + m_{AD}}{1 - m_{AC} \cdot m_{AD}} = \frac{a(3b^2 - a^2)}{b(b^2 - 3a^2)}.$$

It follows that

$$AE: \quad y = \frac{a(3b^2 - a^2)}{b(b^2 - 3a^2)}x.$$

The perpendicular bisector of AD has equation

$$y = -\frac{b^2 - a^2}{2ab}x + \frac{(a^2 + b^2)^2}{4a(b^2 - a^2)}.$$

These yield $E\left(\frac{b(b^2 - 3a^2)}{2(b^2 - a^2)}, \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}\right)$. Then we get

$$X = E + M - A = \left(2b, \frac{a(3b^2 - a^2)}{b^2 - a^2}\right).$$

It is easy to find the equations of the following straight lines:

$$BD: \quad y = -\frac{2ab}{b^2 - a^2}x + \frac{4ab^2}{b^2 - a^2},$$

$$FX: \quad y = \frac{2ab}{b^2 - a^2}x - \frac{ab^2 + a^3}{b^2 - a^2},$$

$$ME: \quad y = \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}.$$

From BD and ME , we get

$$x = \frac{\frac{4ab^2}{b^2 - a^2} - \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}}{\frac{2ab}{b^2 - a^2}} = \frac{a^2 + 5b^2}{4b}.$$

Thus, the coordinates of the intersection point of BD and ME is

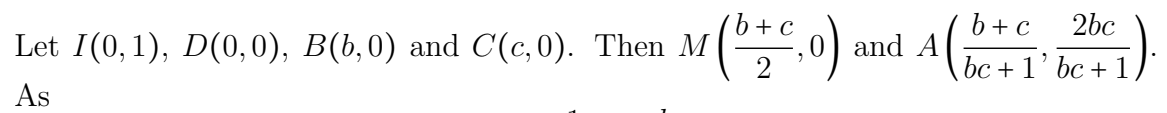
$$\left(\frac{a^2 + 5b^2}{4b}, \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}\right).$$

It remains to check that it satisfies the equation of FX . Indeed, we have

$$\frac{2ab}{b^2 - a^2} \cdot \frac{a^2 + 5b^2}{4b} - \frac{ab^2 + a^3}{b^2 - a^2} = \frac{a(3b^2 - a^2)}{2(b^2 - a^2)}.$$

So BD, FX, ME are concurrent. □

Problem 80. (IMO Shortlist 2016 G2) Let ABC be a triangle with circumcircle Γ and incentre I . Let M be the midpoint of side BC . Denote by D the foot of perpendicular from I to side BC . The line through I perpendicular to AI meets sides AB and AC at F and E respectively. Suppose the circumcircle of triangle AEF intersects Γ at a point X other than A . Prove that lines XD and AM meet on Γ .



we have

Solving with

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we obtain

$$x = \frac{1 - \frac{2b^2}{b^2 - 1}}{\frac{b+c}{bc-1} - \frac{2b}{b^2-1}} = \frac{-(bc-1)(b^2+1)}{b^3 - b^2c + b - c} = \frac{bc-1}{c-b}$$

and

$$y = -\frac{b+c}{bc-1} \cdot \frac{bc-1}{c-b} + 1 = -\frac{2b}{c-b}.$$

This yields $F\left(\frac{bc-1}{c-b}, -\frac{2b}{c-b}\right)$.

Next, let P be the second intersection point of AM and Γ . Putting

$$AM: \quad y = \frac{4bc}{(b+c)(1-bc)}x - \frac{2bc}{1-bc}$$

into

$$\Gamma: \quad x^2 + y^2 - (b+c)x - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc+1)}y + bc = 0, \quad (1)$$

we need to solve

$$\left(1 + \frac{16b^2c^2}{(b+c)^2(1-bc)^2}\right)x^2 + (\dots)x + \left(\frac{4b^2c^2}{(1-bc)^2} + \frac{bc(b^2c^2 - b^2 + 4bc - c^2 + 1)}{(bc+1)(1-bc)} + bc\right) = 0.$$

The constant term is

$$\begin{aligned} & \frac{bc(4bc(bc+1) + (1-bc)(b^2c^2 - b^2 + 4bc - c^2 + 1) + (1-bc)^2(bc+1))}{(1-bc)^2(bc+1)} \\ &= \frac{bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{(1-bc)^2(bc+1)}. \end{aligned}$$

As one of the roots of the quadratic equation is $\frac{b+c}{bc+1}$ (the x -coordinate of A), the x -coordinate of P is

$$\frac{\frac{bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{(1-bc)^2(bc+1)}}{\left(1 + \frac{16b^2c^2}{(b+c)^2(1-bc)^2}\right)} \cdot \frac{b+c}{bc+1} = \frac{bc(b+c)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{16b^2c^2 + (b+c)^2(1-bc)^2}.$$

Then we have

$$\begin{aligned} y &= \frac{4bc}{(b+c)(1-bc)} \cdot \frac{bc(b+c)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{16b^2c^2 + (b+c)^2(1-bc)^2} - \frac{2bc}{1-bc} \\ &= -\frac{2bc(b-c)^2(bc+1)}{16b^2c^2 + (b+c)^2(1-bc)^2}. \end{aligned}$$

Thus, $P\left(\frac{bc(b+c)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{16b^2c^2 + (b+c)^2(1-bc)^2}, -\frac{2bc(b-c)^2(bc+1)}{16b^2c^2 + (b+c)^2(1-bc)^2}\right)$.

Let X' be the second intersection point of DP and Γ . Putting

$$DP: \quad y = -\frac{2(b-c)^2(bc+1)}{(b+c)(b^3c+bc^3-b^2+6bc-c^2+2)}x$$

into (1), we solve

$$\left(1 + \frac{4(b-c)^4(bc+1)^2}{(b+c)^2(b^3c+bc^3-b^2+6bc-c^2+2)^2}\right)x^2 + (\dots)x + bc = 0.$$

One of the roots is $\frac{bc(b+c)(b^3c+bc^3-b^2+6bc-c^2+2)}{16b^2c^2+(b+c)^2(1-bc)^2}$, while the other root is

$$\begin{aligned} & \frac{bc}{1 + \frac{4(b-c)^4(bc+1)^2}{(b+c)^2(b^3c+bc^3-b^2+6bc-c^2+2)^2}} \div \frac{bc(b+c)(b^3c+bc^3-b^2+6bc-c^2+2)}{16b^2c^2+(b+c)^2(1-bc)^2} \\ &= \frac{(b+c)(b^3c+bc^3-b^2+6bc-c^2+2)(16b^2c^2+(b+c)^2(1-bc)^2)}{4(b-c)^4(bc+1)^2+(b+c)^2(b^3c+bc^3-b^2+6bc-c^2+2)^2}. \end{aligned}$$

This implies

$$\begin{aligned} X' & \left(\frac{(b+c)(b^3c+bc^3-b^2+6bc-c^2+2)(16b^2c^2+(b+c)^2(1-bc)^2)}{4(b-c)^4(bc+1)^2+(b+c)^2(b^3c+bc^3-b^2+6bc-c^2+2)^2}, \right. \\ & \quad \left. - \frac{2(b-c)^2(bc+1)(16b^2c^2+(b+c)^2(1-bc)^2)}{4(b-c)^4(bc+1)^2+(b+c)^2(b^3c+bc^3-b^2+6bc-c^2+2)^2} \right). \end{aligned}$$

Let O and O' be the circumcentres of $\triangle ABC$ and $\triangle AEF$ respectively. Recall that $O\left(\frac{b+c}{2}, \frac{b^2c^2-b^2+4bc-c^2+1}{4(bc+1)}\right)$. Also, since $AE = AF$ and $AI \perp EF$, O' lies on AI ,

and so its coordinates are of the form $\left(x, \frac{bc-1}{b+c}x+1\right)$. From

$$O'A = O'F$$

$$\begin{aligned} \Rightarrow & \left(x - \frac{b+c}{bc+1}\right)^2 + \left(\frac{bc-1}{b+c}x + \frac{1-bc}{bc+1}\right)^2 = \left(x - \frac{bc-1}{c-b}\right)^2 + \left(\frac{bc-1}{b+c}x + \frac{c+b}{c-b}\right)^2 \\ \Rightarrow & \left(-\frac{2(b+c)}{bc+1} - \frac{2(bc-1)^2}{(b+c)(bc+1)}\right)x = \left(\frac{bc-1}{c-b}\right)^2 + \left(\frac{c+b}{c-b}\right)^2 - \left(\frac{b+c}{bc+1}\right)^2 - \left(\frac{1-bc}{bc+1}\right)^2 \\ \Rightarrow & -\frac{2(b^2c^2+b^2+c^2+1)}{(b+c)(bc+1)}x = (b^2c^2+b^2+c^2+1)\left(\frac{1}{(c-b)^2} - \frac{1}{(bc+1)^2}\right), \end{aligned}$$

we obtain $x = -\frac{(b^2c^2-b^2+4bc-c^2+1)(b+c)}{2(c-b)^2(bc+1)}$. This yields

$$\begin{aligned} y &= \frac{bc-1}{b+c} \left(-\frac{(b^2c^2-b^2+4bc-c^2+1)(b+c)}{2(c-b)^2(bc+1)} \right) + 1 \\ &= \frac{-b^3c^3+3b^3c-7b^2c^2+3bc^3+b^2-bc+c^2+1}{2(c-b)^2(bc+1)}. \end{aligned}$$

In other words,

$$O' \left(-\frac{(b^2c^2 - b^2 + 4bc - c^2 + 1)(b + c)}{2(c - b)^2(bc + 1)}, \frac{-b^3c^3 + 3b^3c - 7b^2c^2 + 3bc^3 + b^2 - bc + c^2 + 1}{2(c - b)^2(bc + 1)} \right).$$

To show that X' lies on (AEF) , it suffices to show $AX' \perp OO'$. We now compute

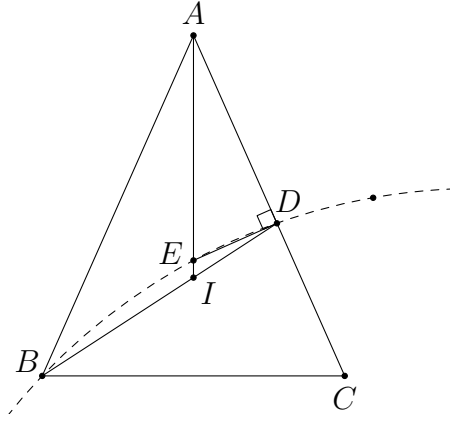
$$\begin{aligned} m_{OO'} &= \frac{(c - b)^2(b^2c^2 - b^2 + 4bc - c^2 + 1) - 2(-b^3c^3 + 3b^3c - 7b^2c^2 + 3bc^3 + b^2 - bc + c^2 + 1)}{2(b + c)((c - b)^2(bc + 1) + (b^2c^2 - b^2 + 4bc - c^2 + 1))} \\ &= \frac{b^4c^2 + b^2c^4 - b^4 + 4b^2c^2 - c^4 - b^2 - c^2 - 2}{2(b + c)(b^3c - b^2c^2 + bc^3 + 2bc + 1)} \end{aligned}$$

and

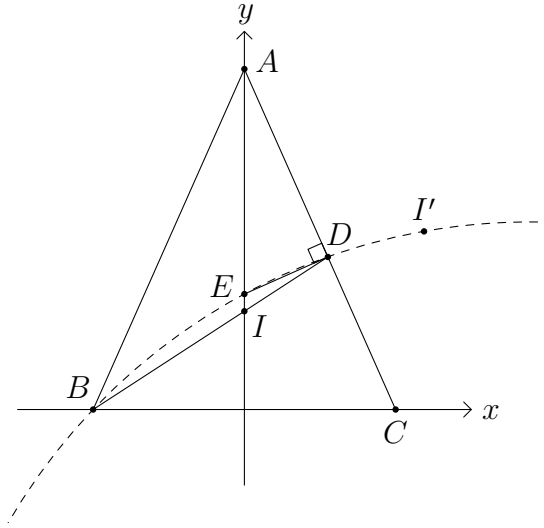
$$\begin{aligned} m_{AX'} &= \frac{2bc(4(b - c)^4(bc + 1)^2 + (b + c)^2(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2) + 2(b - c)^2(bc + 1)^2(16b^2c^2 + (b + c)^2(1 - bc)^2)}{(b + c)(4(b - c)^4(bc + 1)^2 + (b + c)^2(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2 - (bc + 1)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)(16b^2c^2 + (b + c)^2(1 - bc)^2))} \\ &= \frac{2(b + c)^2(bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2 + (b - c)^2(bc + 1)^4)}{(b + c)(4(b - c)^4(bc + 1)^2 + (b + c)^2(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2 - (bc + 1)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)(16b^2c^2 + (b + c)^2(1 - bc)^2))} \\ &= \frac{2(b + c)(bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2 + (b - c)^2(bc + 1)^4)}{4(b - c)^4(bc + 1)^2 + (b + c)^2(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)^2 - (bc + 1)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)(16b^2c^2 + (b + c)^2(1 - bc)^2)} \\ &= \frac{2(b + c)(b^7c^3 + b^6c^4 + b^4c^6 + b^3c^7 - 2b^6c^2 + 16b^5c^3 - 12b^4c^4 + 16b^3c^5 - 2b^2c^6 + b^5c - 2b^4c^2 + 26b^3c^3 - 2b^2c^4 + bc^5 + 16b^2c^2 + b^2 + 2bc + c^2)}{-b^8c^4 - 2b^7c^5 - 2b^6c^6 - 2b^5c^7 - b^4c^8 + b^8c^2 + 4b^7c^3 - 15b^6c^4 - 4b^5c^5 - 15b^4c^6 + 4b^3c^7 + b^2c^8 - 2b^7c + 12b^6c^2 + 6b^5c^3 - 48b^4c^4 + 6b^3c^5 + 12b^2c^6 - 2bc^7 + b^6 + 11b^4c^2 - 8b^3c^3 + 11b^2c^4 + c^6 + b^4 - 2b^3c + 26b^2c^2 - 2bc^3 + c^4 + 2b^2 + 4bc + 2c^2} \\ &= \frac{2(b + c)(b^3c - b^2c^2 + bc^3 + 2bc + 1) \times (b^4c^2 + 2b^3c^3 + b^2c^4 - 2b^3c + 12b^2c^2 - 2bc^3 + b^2 + 2bc + c^2)}{-(b^4c^2 + b^2c^4 - b^4 + 4b^2c^2 - c^4 - b^2 - c^2 - 2) \times (b^4c^2 + 2b^3c^3 + b^2c^4 - 2b^3c + 12b^2c^2 - 2bc^3 + b^2 + 2bc + c^2)} \\ &= -\frac{2(b + c)(b^3c - b^2c^2 + bc^3 + 2bc + 1)}{b^4c^2 + b^2c^4 - b^4 + 4b^2c^2 - c^4 - b^2 - c^2 - 2}. \end{aligned}$$

Thus, $m_{OO'} \cdot m_{AX'} = -1$, and hence $OO' \perp AX'$ as desired. $\square$

Problem 81. (IMO Shortlist 2016 G4) Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D , and the line through D perpendicular to AC meets AI at E . Prove that the reflection of I in AC lies on the circumcircle of triangle BDE .



Solution.



Let $I(0, 1)$, $B(-a, 0)$, $C(a, 0)$. Then $A\left(0, \frac{2a^2}{a^2 - 1}\right)$. Solving

$$BI: \quad y = \frac{1}{a}x + 1$$

and

$$AC: \quad y = -\frac{2a}{a^2 - 1}x + \frac{2a^2}{a^2 - 1}, \quad (1)$$

we have $D\left(\frac{a(a^2 + 1)}{3a^2 - 1}, \frac{4a^2}{3a^2 - 1}\right)$. Then

$$DE: \quad y = \frac{a^2 - 1}{2a}x + \frac{-a^4 + 8a^2 + 1}{2(3a^2 - 1)}.$$

This yields $E\left(0, \frac{-a^4 + 8a^2 + 1}{2(3a^2 - 1)}\right)$.

Next, let I' be the reflection of I in AC . We have

$$II' : y = \frac{a^2 - 1}{2a}x + 1. \quad (2)$$

Combining (1) and (2), we get

$$x = \frac{\frac{2a^2}{a^2 - 1} - 1}{\frac{a^2 - 1}{2a} + \frac{2a}{a^2 - 1}} = \frac{2a}{a^2 + 1}$$

and

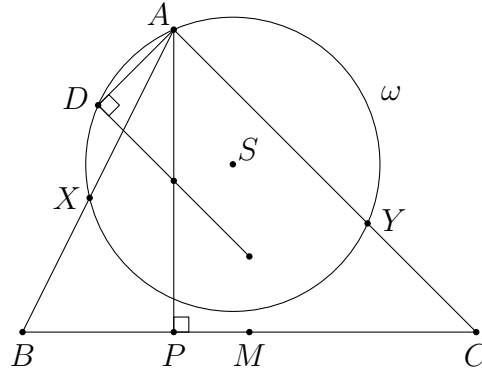
$$y = \frac{a^2 - 1}{2a} \cdot \frac{2a}{a^2 + 1} + 1 = \frac{2a^2}{a^2 + 1}.$$

This means $\left(\frac{2a}{a^2 + 1}, \frac{2a^2}{a^2 + 1}\right)$ are the coordinates of the intersection point of AC and II' . Note that this is the midpoint of II' . So $I' \left(\frac{4a}{a^2 + 1}, \frac{3a^2 - 1}{a^2 + 1}\right)$. Now,

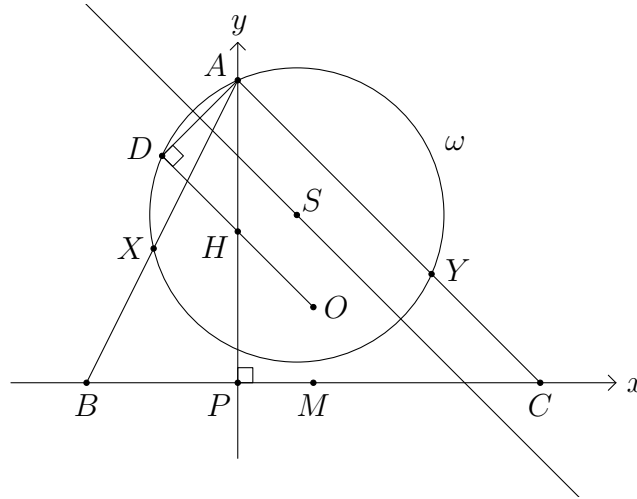
$$\begin{aligned} & D, E, B, I' \text{ are concyclic} \\ \Leftrightarrow & \angle DEI' = \angle DBI' \\ \Leftrightarrow & \tan \angle DEI' = \tan \angle DBI' \\ \Leftrightarrow & \frac{m_{DE} - m_{EI'}}{1 + m_{DE} \cdot m_{EI'}} = \frac{m_{DB} - m_{BI'}}{1 + m_{DB} \cdot m_{BI'}} \\ \Leftrightarrow & \frac{\frac{a^2 - 1}{2a} - \frac{a^6 + 11a^4 - 21a^2 + 1}{8a(3a^2 - 1)}}{1 + \frac{a^2 - 1}{2a} \cdot \frac{a^6 + 11a^4 - 21a^2 + 1}{8a(3a^2 - 1)}} = \frac{\frac{1}{a} - \frac{3a^2 - 1}{a(a^2 + 5)}}{1 + \frac{1}{a} \cdot \frac{3a^2 - 1}{a(a^2 + 5)}} \\ \Leftrightarrow & \frac{2a(-a^6 + a^4 + 5a^2 + 3)}{a^8 + 10a^6 + 16a^4 + 6a^2 - 1} = \frac{a(-2a^2 + 6)}{a^4 + 8a^2 - 1} \\ \Leftrightarrow & \frac{2a(-a^2 + 3)(a^2 + 1)^2}{(a^2 + 1)^2(a^4 + 8a^2 - 1)} = \frac{2a(-a^2 + 3)}{a^4 + 8a^2 - 1}. \end{aligned}$$

This is true. So we are done. $\square$

Problem 82. (IMO Shortlist 2016 G5) Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC . A circle ω with centre S passes through A and D , and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC , and let M be the midpoint of BC . Prove that the circumcentre of triangle $XS Y$ is equidistant from P and M .



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then the orthocentre is $H\left(0, -\frac{bc}{a}\right)$ and the circumcentre is $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$ (alternatively, one can use the centroid of $\triangle ABC$ as it also lies on the Euler line). This gives

$$m_{OH} = \frac{a^2 + bc + 2bc}{a(b+c)} = \frac{a^2 + 3bc}{a(b+c)}.$$

The only restriction on S is that $SA = SD$. This means S lies on the perpendicular bisector of AD . Since $AD \perp OH$, this perpendicular bisector has slope $\frac{a^2 + 3bc}{a(b+c)}$ and it passes through the midpoint $\left(0, \frac{a^2 - bc}{2a}\right)$ of AH . Thus, the equation of this line is

$$y = \frac{a^2 + 3bc}{a(b+c)}x + \frac{a^2 - bc}{2a}.$$

Let $S\left(s, \frac{2(a^2 + 3bc)s + (b+c)(a^2 - bc)}{2a(b+c)}\right)$. As (AXY) has centre S and passes through

A, its equation is

$$x^2 + y^2 - 2sx - \frac{2(a^2 + 3bc)s + (b+c)(a^2 - bc)}{a(b+c)}y + \frac{2(a^2 + 3bc)s - bc(b+c)}{b+c} = 0.$$

We put the equation

$$AB: \quad y = -\frac{a}{b}x + a$$

into the equation of (AXY). Then we need to solve

$$\begin{aligned} & \left(1 + \frac{a^2}{b^2}\right)x^2 - \left(\frac{2a^2}{b} + 2s - \frac{2(a^2 + 3bc)s + (b+c)(a^2 - bc)}{b(b+c)}\right)x \\ & + \left(a^2 - \frac{2(a^2 + 3bc)s + (b+c)(a^2 - bc)}{b+c} + \frac{2(a^2 + 3bc)s - bc(b+c)}{b+c}\right) = 0. \end{aligned}$$

This is the same as

$$(b+c)(a^2 + b^2)x^2 - b(2(b^2 - a^2 - 2bc)s + (b+c)(a^2 + bc))x = 0.$$

The solution $x = 0$ is rejected. Hence, $x = \frac{b(2(b^2 - a^2 - 2bc)s + (b+c)(a^2 + bc))}{(b+c)(a^2 + b^2)}$ and

$$\begin{aligned} y &= -\frac{a}{b} \cdot \frac{b(2(b^2 - a^2 - 2bc)s + (b+c)(a^2 + bc))}{(b+c)(a^2 + b^2)} + a \\ &= \frac{a(-2(b^2 - a^2 - 2bc)s + b(b+c)(b-c))}{(b+c)(a^2 + b^2)}. \end{aligned}$$

This yields

$$X \left(\frac{b(2(b^2 - a^2 - 2bc)s + (b+c)(a^2 + bc))}{(b+c)(a^2 + b^2)}, \frac{a(-2(b^2 - a^2 - 2bc)s + b(b+c)(b-c))}{(b+c)(a^2 + b^2)} \right).$$

Now we find the equation of the perpendicular bisector of XS . Recall that the coordinates of S are $\left(s, \frac{2(a^2 + 3bc)s + (b+c)(a^2 - bc)}{2a(b+c)}\right)$. So the slope of the perpendicular bisector is

$$\begin{aligned} \frac{-1}{m_{XS}} &= \frac{-2a\left((b+c)(a^2 + b^2)s - b(2(b^2 - a^2 - 2bc)s + (b+c)(a^2 + bc))\right)}{(2(a^2 + 3bc)s + (b+c)(a^2 - bc))(a^2 + b^2) - 2a^2(-2(b^2 - a^2 - 2bc)s + b(b+c)(b-c))} \\ &= -\frac{2a\left((3a^2b + a^2c - b^3 + 5b^2c)s - b(b+c)(a^2 + bc)\right)}{(-2a^4 + 6a^2b^2 - 2a^2bc + 6b^3c)s + (b+c)(a^4 - a^2b^2 + a^2bc - b^3c)} \\ &= -\frac{2a\left((3a^2b + a^2c - b^3 + 5b^2c)s - b(b+c)(a^2 + bc)\right)}{(a^2 + bc)((-2a^2 + 6b^2)s + (b+c)(a^2 - b^2))}. \end{aligned}$$

Also, it passes through the midpoint of XS , whose coordinates are

$$\left(\frac{(c-b)(a^2-3b^2)s + b(b+c)(a^2+bc)}{2(b+c)(a^2+b^2)}, \frac{(6a^4-2a^2b^2+14a^2bc+6b^3c)s + (b+c)(a^4+3a^2b^2-3a^2bc-b^3c)}{4a(b+c)(a^2+b^2)} \right).$$

So we compute

$$\begin{aligned} & \frac{(6a^4-2a^2b^2+14a^2bc+6b^3c)s + (b+c)(a^4+3a^2b^2-3a^2bc-b^3c)}{4a(b+c)(a^2+b^2)} \\ & + \frac{2a((3a^2b+a^2c-b^3+5b^2c)s - b(b+c)(a^2+bc))}{(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & \cdot \frac{(c-b)(a^2-3b^2)s + b(b+c)(a^2+bc)}{2(b+c)(a^2+b^2)} \\ & = \frac{4(a^2-3b^2)(-3a^6-2a^4b^2-8a^4bc+a^4c^2+a^2b^4-8a^2b^3c-2a^2b^2c^2-3b^4c^2)s^2}{4a(b+c)(a^2+b^2)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{4(b+c)(a^2+bc)(a^6+2a^4b^2+5a^4bc+a^2b^4+2a^2b^3c-3b^5c)s}{4a(b+c)(a^2+b^2)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{(a^2+bc)(b+c)^2(a^6-2a^4b^2-3a^4bc-3a^2b^4-2a^2b^3c+b^5c)}{4a(b+c)(a^2+b^2)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & = \frac{4(a^2-3b^2)(-3a^4+a^2b^2-8a^2bc+a^2c^2-3b^2c^2)s^2}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{4(b+c)(a^2+bc)(a^4+a^2b^2+5a^2bc-3b^3c)s}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{(a^2+bc)(b+c)^2(a^4-3a^2b^2-3a^2bc+b^3c)}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))}. \end{aligned}$$

Thus, the equation of the perpendicular bisector of XS is

$$\begin{aligned} y = & -\frac{2a((3a^2b+a^2c-b^3+5b^2c)s - b(b+c)(a^2+bc))}{(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))}x \\ & + \frac{4(a^2-3b^2)(-3a^4+a^2b^2-8a^2bc+a^2c^2-3b^2c^2)s^2}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{4(b+c)(a^2+bc)(a^4+a^2b^2+5a^2bc-3b^3c)s}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{(a^2+bc)(b+c)^2(a^4-3a^2b^2-3a^2bc+b^3c)}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))}. \end{aligned}$$

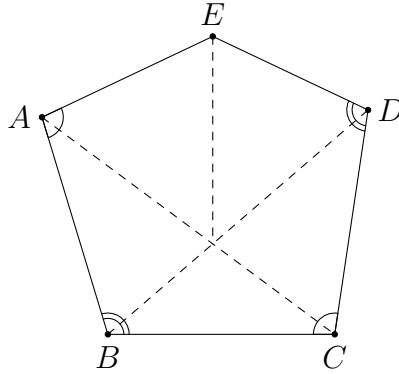
We easily get $P(0,0)$ and $M\left(\frac{b+c}{2}, 0\right)$. The perpendicular bisector of PM has equation $x = \frac{b+c}{4}$. Putting this in the above equation, we find

$$\begin{aligned} y = & -\frac{2a((3a^2b+a^2c-b^3+5b^2c)s - b(b+c)(a^2+bc))}{(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \cdot \frac{b+c}{4} \\ & + \frac{4(a^2-3b^2)(-3a^4+a^2b^2-8a^2bc+a^2c^2-3b^2c^2)s^2}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{4(b+c)(a^2+bc)(a^4+a^2b^2+5a^2bc-3b^3c)s}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \\ & + \frac{(a^2+bc)(b+c)^2(a^4-3a^2b^2-3a^2bc+b^3c)}{4a(b+c)(a^2+bc)((-2a^2+6b^2)s + (b+c)(a^2-b^2))} \end{aligned}$$

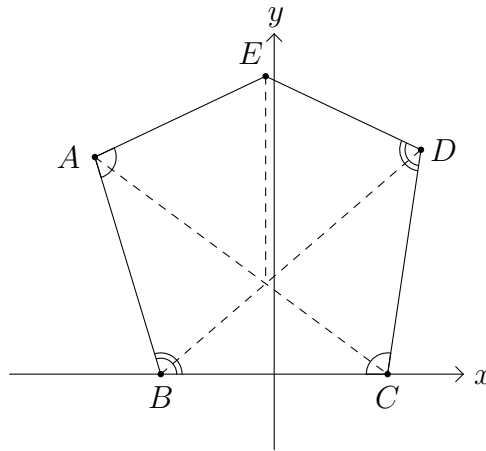
$$\begin{aligned}
& \frac{4(a^2 - 3b^2)(-3a^4 + a^2b^2 - 8a^2bc + a^2c^2 - 3b^2c^2)s^2}{+2(b+c)(2a^6 - a^4b^2 + 8a^4bc - a^4c^2 + a^2b^4 - 8a^2b^3c + 5a^2b^2c^2 - 6b^4c^2)s} \\
& + \frac{(b+c)^2(a^2 + bc)(a^2 - b^2)(a^2 - bc)}{4a(b+c)(a^2 + bc)((-2a^2 + 6b^2)s + (b+c)(a^2 - b^2))} \\
& = \frac{2(3a^4 - a^2b^2 + 8a^2bc - a^2c^2 + 3b^2c^2)s + (b+c)(a^2 + bc)(a^2 - bc)}{4a(b+c)(a^2 + bc)}.
\end{aligned}$$

Since this expression is symmetric in b and c , the intersection of the perpendicular bisectors of XS and PM also lies on the perpendicular bisector of YS . Therefore, it must be the circumcentre of $\triangle XYS$. As its x -coordinate is $\frac{b+c}{4}$, it is equidistant from P and M . $\square$

Problem 83. (IMO Shortlist 2017 G1) Let $ABCDE$ be a convex pentagon such that $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that the perpendicular line from E to BC and the line segments AC and BD are concurrent.



Solution.



Let $B(-1, 0)$, $C(1, 0)$, $A(a, b)$, $D(c, d)$. From $AB = BC = CD$, we have

$$(a + 1)^2 + b^2 = 4, \quad (1)$$

$$(c - 1)^2 + d^2 = 4. \quad (2)$$

Firstly,

$$\begin{aligned} & \angle EAB = \angle BCD \\ \Rightarrow & \tan \angle EAB = \tan \angle BCD \\ \Rightarrow & \frac{m_{AE} - m_{AB}}{1 + m_{AE} \cdot m_{AB}} = -m_{CD} \\ \Rightarrow & m_{AE} = \frac{m_{AB} - m_{CD}}{1 + m_{AB} \cdot m_{CD}}. \end{aligned}$$

Thus,

$$m_{AE} = \frac{\frac{b}{a+1} - \frac{d}{c-1}}{1 + \frac{b}{a+1} \cdot \frac{d}{c-1}} = \frac{b(c-1) - d(a+1)}{(a+1)(c-1) + bd}.$$

This yields

$$AE: \quad y = \frac{b(c-1) - d(a+1)}{(a+1)(c-1) + bd}x + \frac{a^2d + b^2d + bc + ad - b}{(a+1)(c-1) + bd}.$$

Using (1) and expanding, this becomes

$$AE: \quad y = \frac{bc - ad - b - d}{ac + bd - a + c - 1}x + \frac{bc - ad - b + 3d}{ac + bd - a + c - 1}. \quad (3)$$

Swapping b and d ; a and $-c$ (because we need to take a reflection in the y -axis to interchange the points $(-1, 0)$ and $(1, 0)$), and replacing x by $-x$, we get

$$DE: \quad y = \frac{-bc + ad + b + d}{ac + bd - a + c - 1}x + \frac{bc - ad + 3b - d}{ac + bd - a + c - 1}. \quad (4)$$

We solve (3) and (4) to get

$$x = \frac{(bc - ad + 3b - d) - (bc - ad - b + 3d)}{(bc - ad - b - d) - (-bc + ad + b + d)} = \frac{2(b - d)}{bc - ad - b - d}. \quad (5)$$

This is the x -coordinate of E , and is what we need. Solving

$$AC: \quad y = \frac{b}{a-1}x - \frac{b}{a-1}$$

and

$$BD: \quad y = \frac{d}{c+1}x + \frac{d}{c+1},$$

we get

$$x = \frac{d(a-1) + b(c+1)}{b(c+1) - d(a-1)} = \frac{bc + ad + b - d}{bc - ad + b + d}.$$

This is the x -coordinate of the intersection point of AC and BD . We need to show that it is the same as (5). Indeed,

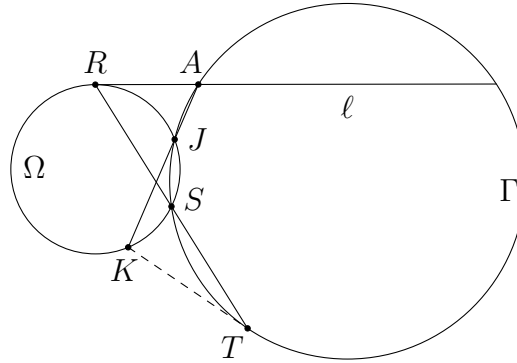
$$\begin{aligned} & \frac{2(b-d)}{bc-ad-b-d} = \frac{bc+ad+b-d}{bc-ad+b+d} \\ \Leftrightarrow & 2(b^2c+d^2a-bcd-abd+b^2-d^2) = b^2c^2-a^2d^2-2bcd-2abd-b^2+d^2 \\ \Leftrightarrow & b^2(-c^2+2c+3) = d^2(-a^2-2a+3). \end{aligned}$$

Using (1) and (2), we get exactly

$$b^2(-c^2+2c+3) = b^2d^2 = d^2(-a^2-2a+3).$$

This completes the proof. $\square$

Problem 84. (IMO Shortlist 2017 G2/Problem 4) Let R and S be different points on a circle Ω such that RS is not a diameter. Let ℓ be the tangent line to Ω at R . Point T is such that S is the midpoint of the line segment RT . Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects ℓ at two distinct points. Let A be the common point of Γ and ℓ that is closer to R . Line AJ meets Ω again at K . Prove that the line KT is tangent to Γ .



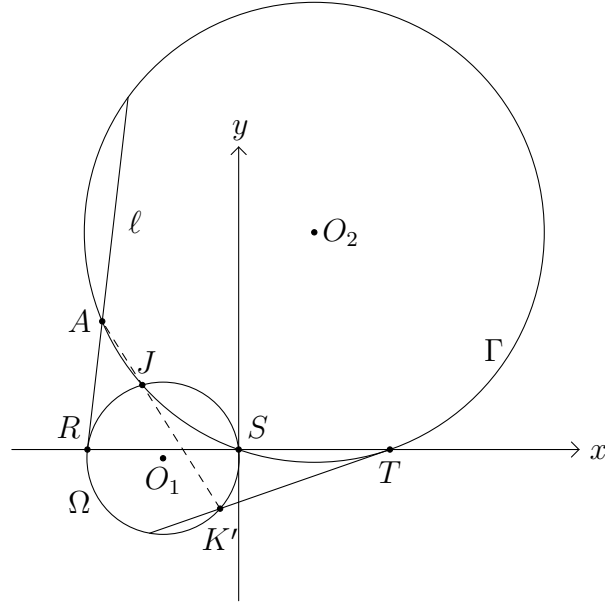
Solution. Let O_1 and O_2 be the centres of Ω and Γ respectively. Let $S(0,0)$, $R(-1,0)$, $T(1,0)$, $O_1\left(-\frac{1}{2},a\right)$, $O_2\left(\frac{1}{2},b\right)$. WLOG assume $a < 0 < b$. Then we have

$$\Omega: \quad x^2 + y^2 + x - 2ay = 0, \quad (1)$$

$$\Gamma: \quad x^2 + y^2 - x - 2by = 0. \quad (2)$$

The difference of (1) and (2) is $x + (b-a)y = 0$, which is the equation of JS . Putting $y = \frac{1}{a-b}x$ in (1), we solve

$$\left(1 + \frac{1}{(a-b)^2}\right)x^2 + \left(1 - \frac{2a}{a-b}\right)x = 0.$$



Rejecting the solution $x = 0$, we get

$$x = \frac{a^2 - b^2}{(a - b)^2 + 1}$$

and

$$y = \frac{1}{a - b} \cdot \frac{a^2 - b^2}{(a - b)^2 + 1} = \frac{a + b}{(a - b)^2 + 1}.$$

Thus, we have $J\left(\frac{a^2 - b^2}{(a - b)^2 + 1}, \frac{a + b}{(a - b)^2 + 1}\right)$.

Consider the tangent at R to Ω . Its slope is $\frac{-1}{m_{RO_1}} = -\frac{1}{2a}$. So the equation of RA is

$$y = -\frac{1}{2a}x - \frac{1}{2a}.$$

Putting this in (2), we solve

$$\left(1 + \frac{1}{4a^2}\right)x^2 + \left(\frac{1}{2a^2} - 1 + \frac{b}{a}\right)x + \left(\frac{1}{4a^2} + \frac{b}{a}\right) = 0.$$

This is the same as

$$(4a^2 + 1)x^2 + 2(-2a^2 + 2ab + 1)x + (4ab + 1) = 0.$$

The discriminant is

$$4(-2a^2 + 2ab + 1)^2 - 4(4a^2 + 1)(4ab + 1) = 16a^2(a^2 - 6ab + b^2 - 2).$$

From the configuration, we choose the smaller root of the quadratic equation. Also, note that $a < 0$. This gives

$$x = \frac{2a^2 - 2ab - 1 - 2(-a)\sqrt{a^2 - 6ab + b^2 - 2}}{4a^2 + 1}$$

and

$$y = -\frac{1}{2a} \cdot \left(\frac{2a^2 - 2ab - 1 + 2a\sqrt{a^2 - 6ab + b^2 - 2}}{4a^2 + 1} + 1 \right) = \frac{-3a + b - \sqrt{a^2 - 6ab + b^2 - 2}}{4a^2 + 1}.$$

Hence, we have $A\left(\frac{2a^2 - 2ab - 1 + 2a\Delta}{4a^2 + 1}, \frac{-3a + b - \Delta}{4a^2 + 1}\right)$ where

$$\Delta = \sqrt{a^2 - 6ab + b^2 - 2}.$$

It follows that

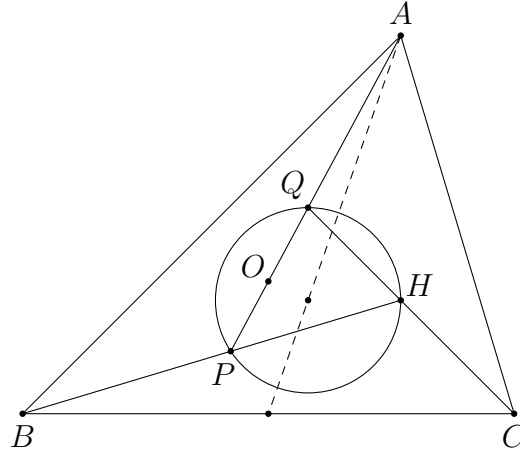
$$\begin{aligned} m_{AJ} &= \frac{(-3a + b - \Delta)((a - b)^2 + 1) - (a + b)(4a^2 + 1)}{(2a^2 - 2ab - 1 + 2a\Delta)((a - b)^2 + 1) - (a^2 - b^2)(4a^2 + 1)} \\ &= \frac{(-7a^3 + 3a^2b - 5ab^2 + b^3 - 4a) - ((a - b)^2 + 1)\Delta}{(-2a^4 - 6a^3b + 10a^2b^2 - 2ab^3 - 1) + 2a((a - b)^2 + 1)\Delta} \\ &\quad \times \left((-7a^3 + 3a^2b - 5ab^2 + b^3 - 4a) - ((a - b)^2 + 1)\Delta \right) \\ &= \frac{(-2a^4 - 6a^3b + 10a^2b^2 - 2ab^3 - 1)^2 - 4a^2((a - b)^2 + 1)^2\Delta^2}{(-2a^4 - 6a^3b + 10a^2b^2 - 2ab^3 - 1)^2 - 4a^2((a - b)^2 + 1)^2\Delta^2} \\ &= \frac{(a + b)^2(a - b)(4a^2 + 1)^2 + ((a - b)^2 + 1)(4a^2 + 1)^2\Delta}{(4a^2 + 1)^2(4ab(a - b)^2 + 1)} \\ &= \frac{(a + b)^2(a - b) + ((a - b)^2 + 1)\Delta}{4ab(a - b)^2 + 1}. \end{aligned}$$

On the other hand, consider the intersection of Ω and the tangent at T to Γ . With some trial and error, we consider one of the intersection points K' which is closer to T . Our goal is to show that $K = K'$. This means J lies on AK' , or equivalently, $m_{AJ} = m_{K'J}$. Due to symmetry, we can obtain $m_{K'J}$ from m_{AJ} by swapping a and $-b$. This easily gives

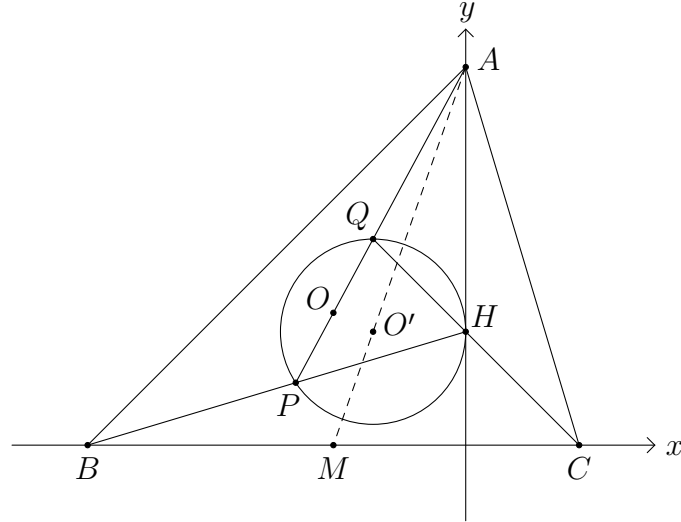
$$m_{K'J} = \frac{(-b - a)^2(-b + a) + ((-b + a)^2 + 1)\Delta}{4(-b)(-a)(-b + a)^2 + 1} = m_{AJ}.$$

The proof is complete. □

Problem 85. (IMO Shortlist 2017 G3) Let O be the circumcentre of an acute scalene triangle ABC . Line OA intersects the altitudes of ABC through B and C at P and Q , respectively. The altitudes meet at H . Prove that the circumcentre of triangle PQH lies on a median of triangle ABC .



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$. Then $H\left(0, -\frac{bc}{a}\right)$, $O\left(\frac{b+c}{2}, \frac{a^2+bc}{2a}\right)$ and

$$BH: \quad y = \frac{c}{a}x - \frac{bc}{a}. \quad (1)$$

We easily obtain

$$AO: \quad y = -\frac{a^2-bc}{a(b+c)}x + a. \quad (2)$$

Solving (1) and (2), we get

$$x = \frac{a + \frac{bc}{a}}{\frac{c}{a} + \frac{a^2-bc}{a(b+c)}} = \frac{(b+c)(a^2+bc)}{a^2+c^2}$$

and

$$y = \frac{c}{a} \cdot \left(\frac{(b+c)(a^2+bc)}{a^2+c^2} - b \right) = \frac{c^2(a^2+b^2)}{a(a^2+c^2)}.$$

This means $P \left(\frac{(b+c)(a^2+bc)}{a^2+c^2}, \frac{c^2(a^2+b^2)}{a(a^2+c^2)} \right)$. From $m_{PH} = m_{BH} = \frac{c}{a}$, the perpendicular bisector of PH has slope $-\frac{a}{c}$. Also, it passes through the midpoint of PH , whose coordinates are $\left(\frac{(b+c)(a^2+bc)}{2(a^2+c^2)}, \frac{c(c-b)(a^2-bc)}{2a(a^2+c^2)} \right)$. Therefore, its equation is

$$y = -\frac{a}{c}x + \frac{a^2b + a^2c + b^2c - bc^2}{2ac}.$$

By symmetry, the equation of the perpendicular bisector of QH is

$$y = -\frac{a}{b}x + \frac{a^2c + a^2b + bc^2 - b^2c}{2ab}.$$

We now solve these equations as follows:

$$x = \frac{c(a^2c + a^2b + bc^2 - b^2c) - b(a^2b + a^2c + b^2c - bc^2)}{2a^2(c-b)} = \frac{(b+c)(a^2+bc)}{2a^2},$$

$$y = -\frac{a}{c} \cdot \frac{(b+c)(a^2+bc)}{2a^2} + \frac{a^2b + a^2c + b^2c - bc^2}{2ac} = -\frac{bc}{a}.$$

This gives the coordinates $\left(\frac{(b+c)(a^2+bc)}{2a^2}, -\frac{bc}{a} \right)$ of the circumcentre O' of $\triangle PQH$.

Let $M \left(\frac{b+c}{2}, 0 \right)$ be the midpoint of BC . The equation of AM is

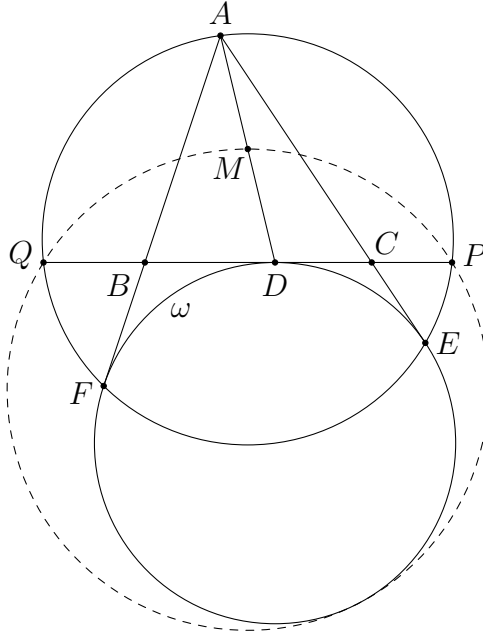
$$y = -\frac{2a}{b+c}x + a.$$

We check that

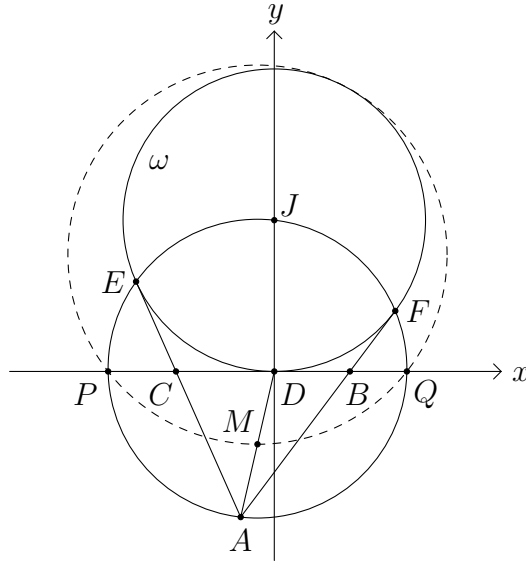
$$-\frac{2a}{b+c} \cdot \frac{(b+c)(a^2+bc)}{2a^2} + a = -\frac{bc}{a}.$$

So O' lies on the median AM as desired. $\square$

Problem 86. (IMO Shortlist 2017 G4) In triangle ABC , let ω be the excircle opposite A . Let D, E , and F be the points where ω is tangent to lines BC, CA , and AB , respectively. The circle AEF intersects line BC at P and Q . Let M be the midpoint of AD . Prove that the circle MPQ is tangent to ω .



Solution.



Let J be the A -excentre of $\triangle ABC$. Let $D(0,0)$, $J(0,1)$, $B(b,0)$, $C(c,0)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$ and hence $M\left(\frac{b+c}{2(bc+1)}, \frac{bc}{bc+1}\right)$. Since $\angle AEJ = \angle AFJ = 90^\circ$, the circumcircle of $\triangle AEF$ is just the circle with AJ as diameter. Its centre is the midpoint $\left(\frac{b+c}{2(bc+1)}, \frac{3bc+1}{2(bc+1)}\right)$ of AJ . So

$$(AEF): \quad x^2 + y^2 - \frac{b+c}{bc+1}x - \frac{3bc+1}{bc+1}y + \frac{2bc}{bc+1} = 0.$$

Putting $y = 0$, we solve

$$x^2 - \frac{b+c}{bc+1}x + \frac{2bc}{bc+1} = 0.$$

Let $P(p, 0)$, $Q(q, 0)$. Then the roots p and q satisfy

$$p+q = \frac{b+c}{bc+1}, \quad pq = \frac{2bc}{bc+1}. \quad (1)$$

Now, the perpendicular bisector of PM has slope $-\frac{b+c-2(bc+1)p}{2bc}$. It passes through $\left(\frac{b+c+2(bc+1)p}{4(bc+1)}, \frac{bc}{2(bc+1)}\right)$. So its equation is

$$y = -\frac{b+c-2(bc+1)p}{2bc}x + \frac{4b^2c^2 + (b+c)^2 - 4(bc+1)^2p^2}{8bc(bc+1)}.$$

Putting $x = \frac{p+q}{2}$, we get the circumcentre of $\triangle MPQ$, which is

$$\left(\frac{p+q}{2}, \frac{4b^2c^2 + (b+c)^2 - 2(b+c)(bc+1)(p+q) + 4(bc+1)^2pq}{8bc(bc+1)}\right).$$

Using (1), this becomes

$$\left(\frac{b+c}{2(bc+1)}, \frac{12b^2c^2 - b^2 + 6bc - c^2}{8bc(bc+1)}\right).$$

Hence, the circumcircle of $\triangle MPQ$ has equation

$$\begin{aligned} x^2 + y^2 - \frac{b+c}{bc+1}x - \frac{12b^2c^2 - b^2 + 6bc - c^2}{4bc(bc+1)}y - \left(\frac{b+c}{2(bc+1)}\right)^2 \\ - \left(\frac{bc}{bc+1}\right)^2 + \frac{b+c}{bc+1} \cdot \frac{b+c}{2(bc+1)} + \frac{12b^2c^2 - b^2 + 6bc - c^2}{4bc(bc+1)} \cdot \frac{bc}{bc+1} = 0, \end{aligned}$$

i.e.

$$x^2 + y^2 - \frac{b+c}{bc+1}x - \frac{12b^2c^2 - b^2 + 6bc - c^2}{4bc(bc+1)}y + \frac{2bc}{bc+1} = 0. \quad (2)$$

On the other hand, the A -excircle has equation

$$x^2 + (y-1)^2 = 1. \quad (3)$$

Taking the difference of (2) and (3), we get the equation of the radical axis of the two circles, which is

$$\frac{b+c}{bc+1}x + \frac{4b^2c^2 - b^2 - 2bc - c^2}{4bc(bc+1)}y - \frac{2bc}{bc+1} = 0.$$

This can be rewritten as

$$x = -\frac{4b^2c^2 - b^2 - 2bc - c^2}{4bc(b+c)}y + \frac{2bc}{b+c}. \quad (4)$$

It remains to show that this straight line is tangent to the A -excircle, which would then imply the two circles are tangent to each other. Putting (4) in (3), we get

$$\left(\frac{(4b^2c^2 - b^2 - 2bc - c^2)^2}{16b^2c^2(b+c)^2} + 1 \right) y^2 - \left(\frac{4b^2c^2 - b^2 - 2bc - c^2}{(b+c)^2} + 2 \right) y + \frac{4b^2c^2}{(b+c)^2} = 0.$$

Clearing denominators and simplifying, we get the quadratic equation

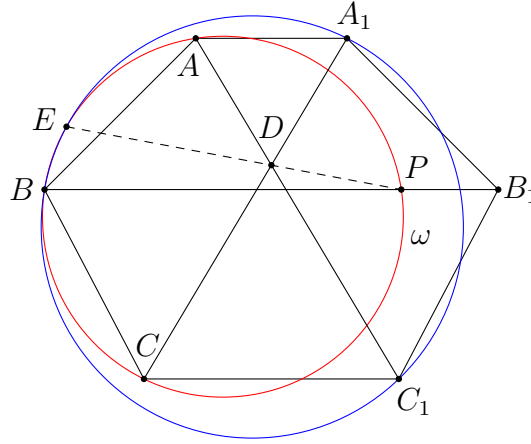
$$(4b^2c^2 + (b+c)^2)^2 y^2 - 16b^2c^2(4b^2c^2 + (b+c)^2)y + 64b^4c^4 = 0.$$

This becomes

$$\left((4b^2c^2 + (b+c)^2)y - 8b^2c^2 \right)^2 = 0.$$

As there is a unique solution, the radical axis is indeed a tangent to the A -excircle. The proof is complete. $\square$

Problem 87. (IMO Shortlist 2017 G5) Let $ABCC_1B_1A_1$ be a convex hexagon such that $AB = BC$, and suppose that the line segments AA_1, BB_1 , and CC_1 have the same perpendicular bisector. Let the diagonals AC_1 and A_1C meet at D , and denote by ω the circle ABC . Let ω intersect the circle A_1BC_1 again at $E \neq B$. Prove that the lines BB_1 and DE intersect on ω .



Solution. Let $B(-1,0)$, $B_1(1,0)$, $A(-a,s)$, $A_1(a,s)$, $C(-c,t)$, $C_1(c,t)$. The condition $AB = BC$ means

$$(a-1)^2 + s^2 = (c-1)^2 + t^2,$$

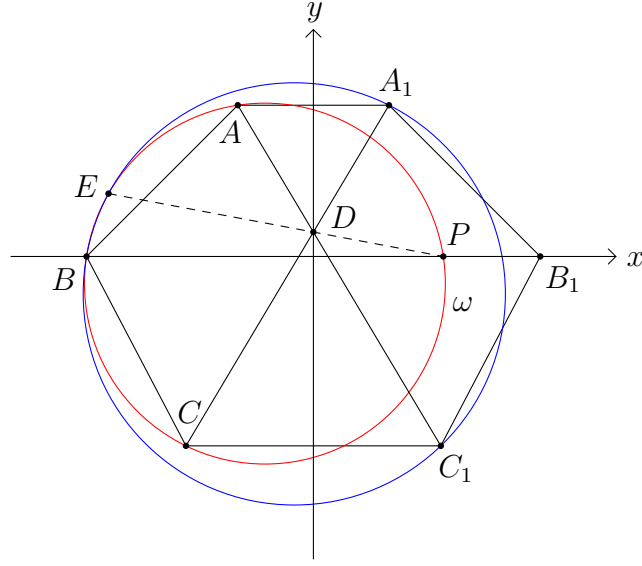
i.e.

$$s^2 - t^2 = c^2 - a^2 + 2a - 2c. \quad (1)$$

The equation of AC_1 is

$$y = \frac{t-s}{a+c}x + \frac{at+cs}{a+c}.$$

By symmetry, D lies on the y -axis. So $D\left(0, \frac{at+cs}{a+c}\right)$.



The slope of the perpendicular bisector of AB is $\frac{-1}{m_{AB}} = \frac{a-1}{s}$. So its equation is

$$y = \frac{a-1}{s}x + \frac{a-1}{s} \cdot \frac{a+1}{2} + \frac{s}{2},$$

i.e.

$$y = \frac{a-1}{s}x + \frac{s^2 + a^2 - 1}{2s}.$$

Changing variables, we get the equation of the perpendicular bisector of BC , which is

$$y = \frac{c-1}{t}x + \frac{t^2 + c^2 - 1}{2t}.$$

Solving these, we get

$$x = \frac{t(s^2 + a^2 - 1) - s(t^2 + c^2 - 1)}{2(s(c-1) - t(a-1))}$$

and

$$\begin{aligned} y &= \frac{a-1}{s} \cdot \frac{t(s^2 + a^2 - 1) - s(t^2 + c^2 - 1)}{2(s(c-1) - t(a-1))} + \frac{s^2 + a^2 - 1}{2s} \\ &= \frac{-s(a-1)(t^2 + c^2 - 1) + s(c-1)(s^2 + a^2 - 1)}{2s(s(c-1) - t(a-1))} \\ &= \frac{(c-1)(s^2 + a^2 - 1) - (a-1)(t^2 + c^2 - 1)}{2(s(c-1) - t(a-1))}. \end{aligned}$$

Thus, $\left(\frac{t(s^2 + a^2 - 1) - s(t^2 + c^2 - 1)}{2(s(c-1) - t(a-1))}, \frac{(c-1)(s^2 + a^2 - 1) - (a-1)(t^2 + c^2 - 1)}{2(s(c-1) - t(a-1))} \right)$ is the

centre of ω . Since ω passes through B , we easily get the equation of ω as follows:

$$\begin{aligned} x^2 + y^2 - \frac{t(s^2 + a^2 - 1) - s(t^2 + c^2 - 1)}{s(c-1) - t(a-1)}x \\ - \frac{(c-1)(s^2 + a^2 - 1) - (a-1)(t^2 + c^2 - 1)}{s(c-1) - t(a-1)}y - \frac{t(s^2 + a^2 - a) - s(t^2 + c^2 - c)}{s(c-1) - t(a-1)} = 0. \end{aligned} \quad (2)$$

Changing a to $-a$ and c to $-c$, we get the equation

$$\begin{aligned} x^2 + y^2 - \frac{t(s^2 + a^2 - 1) - s(t^2 + c^2 - 1)}{-s(c+1) + t(a+1)}x \\ - \frac{-(c+1)(s^2 + a^2 - 1) + (a+1)(t^2 + c^2 - 1)}{-s(c+1) + t(a+1)}y - \frac{t(s^2 + a^2 + a) - s(t^2 + c^2 + c)}{-s(c+1) + t(a+1)} = 0. \end{aligned}$$

of (A_1BC_1) . If we take the difference of the two equations of the circles, we get the equation of the radical axis BE . After expansion and simplification, this equation becomes

$$\begin{aligned} & (-s(cs - at)(t^2 + c^2 - 1) + t(cs - at)(s^2 + a^2 - 1))x \\ & + (s(c - a)(t^2 + c^2 - 1) - t(c - a)(s^2 + a^2 - 1))y \\ & + (-s(cs - at)(t^2 + c^2 - 1) + t(cs - at)(s^2 + a^2 - 1)) = 0. \end{aligned}$$

(When we do the simplification, it is more convenient to consider the terms $t^2 + c^2 - 1$ and $s^2 + a^2 - 1$ as a whole.) A careful observation suggests there is a common factor $s(t^2 + c^2 - 1) - t(s^2 + a^2 - 1)$ for the coefficients. So this can be simplified to

$$y = \frac{cs - at}{c - a}x + \frac{cs - at}{c - a}.$$

We substitute this into (2) to get a quadratic polynomial of x . One of the root is -1 and the other is the x -coordinate of E . The coefficient of x^2 is

$$1 + \left(\frac{cs - at}{c - a} \right)^2 = \frac{(c - a)^2 + (cs - at)^2}{(c - a)^2}$$

and the constant term is

$$\begin{aligned} & \left(\frac{cs - at}{c - a} \right)^2 - \frac{(cs - at)((c-1)(s^2 + a^2 - 1) - (a-1)(t^2 + c^2 - 1))}{(c - a)(s(c-1) - t(a-1))} \\ & - \frac{t(s^2 + a^2 - a) - s(t^2 + c^2 - c)}{s(c-1) - t(a-1)} \\ & = \frac{ac((c-1)s^3 + (-a - 2c + 3)s^2t + (2a + c - 3)st^2 + (-a + 1)t^3 \\ & \quad + (a^2c - 2ac^2 + c^3 - a^2 + 2ac - c^2)s + (-a^3 + 2a^2c - ac^2 + a^2 - 2ac + c^2)t)}{(c - a)^2(s(c-1) - t(a-1))} \\ & = \frac{ac(s^2 - 2st + t^2 + a^2 - 2ac + c^2)}{(c - a)^2} \\ & = \frac{ac((s - t)^2 + (a - c)^2)}{(c - a)^2}. \end{aligned}$$

Thus, from the product of roots, we find that the root other than -1 is

$$-\frac{ac((s-t)^2 + (a-c)^2)}{(c-a)^2 + (cs-at)^2}.$$

Then we have

$$\begin{aligned} y &= \frac{cs-at}{c-a} \cdot \left(-\frac{ac((s-t)^2 + (a-c)^2)}{(c-a)^2 + (cs-at)^2} + 1 \right) \\ &= \frac{cs-at}{c-a} \cdot \frac{(c-a)(cs^2 - at^2 - ac^2 + a^2c + c - a)}{(c-a)^2 + (cs-at)^2} \\ &= \frac{(cs-at)(cs^2 - at^2 - ac^2 + a^2c + c - a)}{(c-a)^2 + (cs-at)^2}. \end{aligned}$$

This yields $E \left(-\frac{ac((s-t)^2 + (a-c)^2)}{(c-a)^2 + (cs-at)^2}, \frac{(cs-at)(cs^2 - at^2 - ac^2 + a^2c + c - a)}{(c-a)^2 + (cs-at)^2} \right)$. Recall that $D \left(0, \frac{at+cs}{a+c} \right)$. So

$$\begin{aligned} m_{DE} &= \frac{(at+cs)((c-a)^2 + (cs-at)^2) - (a+c)(cs-at)(cs^2 - at^2 - ac^2 + a^2c + c - a)}{ac(a+c)((s-t)^2 + (a-c)^2)} \\ &= \frac{-cs^3 + as^2t + cst^2 - at^3 + (-a^2c + c^3 + 2a - 2c)s + (a^3 - ac^2 - 2a + 2c)t}{(a+c)((s-t)^2 + (a-c)^2)}. \end{aligned}$$

We use the condition (1) to rewrite the numerator. We have

$$\begin{aligned} &-cs^3 + as^2t + cst^2 - at^3 + (-a^2c + c^3 + 2a - 2c)s + (a^3 - ac^2 - 2a + 2c)t \\ &= -cs(s^2 - t^2) + as^2t - at^3 + (-a^2c + c^3 + 2a - 2c)s + (a^3 - ac^2 - 2a + 2c)t \\ &= at(s^2 - t^2) + (-2ac + 2c^2 + 2a - 2c)s + (a^3 - ac^2 - 2a + 2c)t \\ &= (-2ac + 2c^2 + 2a - 2c)s + (2a^2 - 2ac - 2a + 2c)t \\ &= 2(c-a)(s(c-1) - t(a-1)). \end{aligned}$$

Therefore, the equation of DE is

$$y = \frac{2(c-a)(s(c-1) - t(a-1))}{(a+c)((s-t)^2 + (a-c)^2)}x + \frac{at+cs}{a+c}.$$

Let P be the second intersection of ω and BB_1 . We put $y = 0$ in (2). The roots to the quadratic equation in x are -1 and the x -coordinate of P . So we get

$$x = \frac{t(s^2 + a^2 - a) - s(t^2 + c^2 - c)}{s(c-1) - t(a-1)}.$$

Hence, $P \left(\frac{t(s^2 + a^2 - a) - s(t^2 + c^2 - c)}{s(c-1) - t(a-1)}, 0 \right)$. We need to prove P lies on DE , i.e.

$$\frac{2(c-a)(s(c-1) - t(a-1))}{(a+c)((s-t)^2 + (a-c)^2)} \cdot \frac{t(s^2 + a^2 - a) - s(t^2 + c^2 - c)}{s(c-1) - t(a-1)} + \frac{at + cs}{a+c} = 0.$$

Cancelling the common factors and expanding, we need to prove

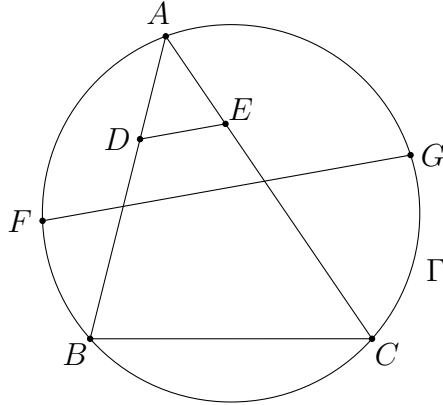
$$cs^3 - as^2t - cst^2 + at^3 + c(a-c)(a+c-2)s - a(a-c)(a+c-2)t = 0.$$

This can be factorized as

$$(s^2 - t^2 + a^2 - 2a + 2c - c^2)(cs - at) = 0.$$

The first factor is 0 from (1). So we are done. $\square$

Problem 88. (IMO Shortlist 2018 G1/Problem 1) Let Γ be the circumcircle of acute-angled triangle ABC . Points D and E lie on segments AB and AC , respectively, such that $AD = AE$. The perpendicular bisectors of BD and CE intersect the minor arcs AB and AC of Γ at points F and G , respectively. Prove that the lines DE and FG are parallel (or are the same line).



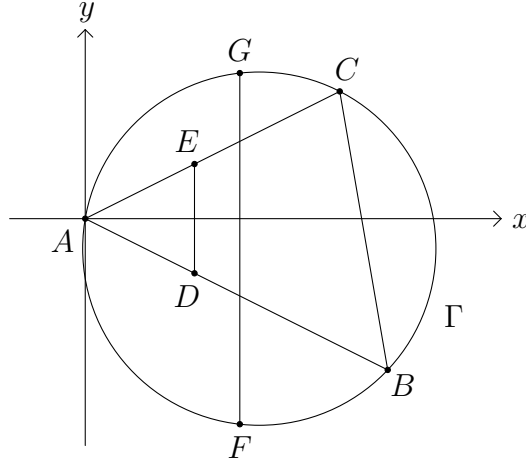
Solution. Let $A(0,0)$ and let the internal angle bisector of A be the x -axis. Then AB and AC are symmetric in the x -axis. Let $B(b, -mb)$, $C(c, mc)$ where $b, c > 0$ so that the equations of AB and AC are $y = -mx$ and $y = mx$ respectively. From $AD = AE$, we know that DE is perpendicular to the x -axis. So we let $D(d, -md)$, $E(d, md)$.

The perpendicular bisector of AB is

$$y = \frac{1}{m}x - \frac{b(1+m^2)}{2m}$$

while that of AC is

$$y = -\frac{1}{m}x + \frac{c(1+m^2)}{2m}.$$



Solving these, we get the circumcentre $\left(\frac{(b+c)(1+m^2)}{4}, \frac{(c-b)(1+m^2)}{4m} \right)$ of $\triangle ABC$. Thus,

$$(ABC): \quad x^2 + y^2 - \frac{(b+c)(1+m^2)}{2}x - \frac{(c-b)(1+m^2)}{2m}y = 0. \quad (1)$$

Next, the perpendicular bisector of BD is

$$y = \frac{1}{m}x - \frac{(b+d)(1+m^2)}{2m}.$$

Putting this in (1), we solve

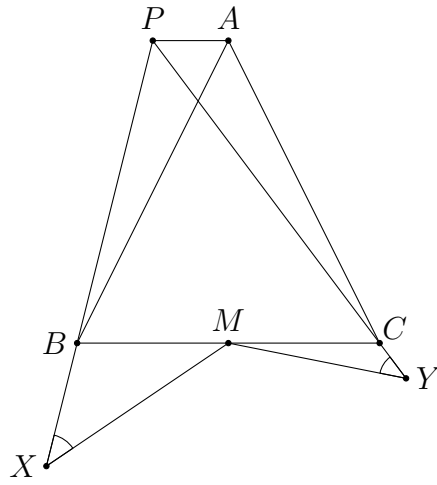
$$\begin{aligned} \left(1 + \frac{1}{m^2}\right)x^2 - \left(\frac{(b+d)(1+m^2)}{m^2} + \frac{(b+c)(1+m^2)}{2} + \frac{(c-b)(1+m^2)}{2m^2}\right)x \\ + \left(\frac{(b+d)^2(1+m^2)^2}{4m^2} + \frac{(b+d)(c-b)(1+m^2)^2}{4m^2}\right) = 0. \end{aligned}$$

This means

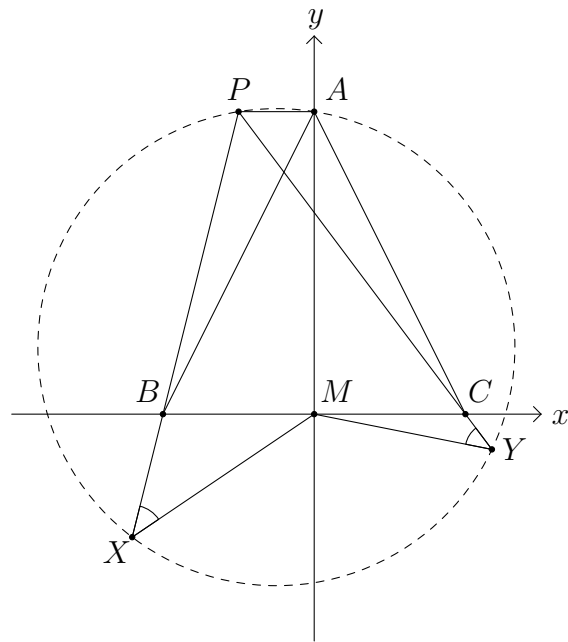
$$4x^2 - 2((b+c)(1+m^2) + 2d)x + (1+m^2)(b+d)(c+d) = 0.$$

The smaller root of this equation is the x -coordinate of F . If we replace m by $-m$ and then swap b and c , we get the corresponding quadratic equation whose smaller root is the x -coordinate of G . That equation is indeed the same as the above one. Hence, the x -coordinates of F and G are the same. This implies FG is perpendicular to the x -axis, and hence $FG \parallel DE$. $\square$

Problem 89. (IMO Shortlist 2018 G2) Let ABC be a triangle with $AB = AC$, and let M be the midpoint of BC . Let P be a point such that $PB < PC$ and PA is parallel to BC . Let X and Y be points on the lines PB and PC , respectively, so that B lies on the segment PX , C lies on the segment PY , and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic.



Solution.



Let $M(0,0)$, $B(-1,0)$, $C(1,0)$, $A(0,a)$, $P(p,a)$. Then

$$PB: \quad y = \frac{a}{p+1}x + \frac{a}{p+1}.$$

Similarly, we have

$$PC: \quad y = \frac{a}{p-1}x - \frac{a}{p-1}.$$

Let $X\left(b, \frac{a(b+1)}{p+1}\right)$ and $Y\left(c, \frac{a(c-1)}{p-1}\right)$. Then

$$\begin{aligned}
& \angle PXM = \angle PYM \\
\Rightarrow & \tan \angle PXM = \tan \angle PYM \\
\Rightarrow & \frac{m_{PX} - m_{MX}}{1 + m_{PX} \cdot m_{MX}} = \frac{m_{MY} - m_{PY}}{1 + m_{MY} \cdot m_{PY}} \\
\Rightarrow & \frac{\frac{a}{p+1} - \frac{a(b+1)}{b(p+1)}}{1 + \frac{a}{p+1} \cdot \frac{a(b+1)}{b(p+1)}} = \frac{\frac{a(c-1)}{c(p-1)} - \frac{a}{p-1}}{1 + \frac{a(c-1)}{c(p-1)} \cdot \frac{a}{p-1}} \\
\Rightarrow & \frac{-a(p+1)}{b(p+1)^2 + a^2(b+1)} = \frac{-a(p-1)}{c(p-1)^2 + a^2(c-1)} \\
\Rightarrow & (p+1)(c(p-1)^2 + a^2(c-1)) = (p-1)(b(p+1)^2 + a^2(b+1)). \tag{1}
\end{aligned}$$

Next, the circumcentre O of $\triangle APX$ has x -coordinate $\frac{p}{2}$. From

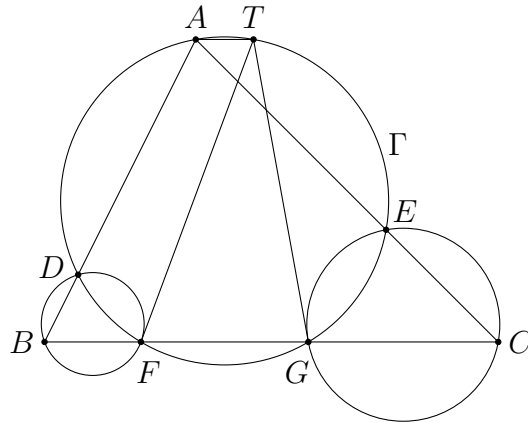
$$\begin{aligned}
& OA = OX \\
\Rightarrow & \left(\frac{p}{2}\right)^2 + (y-a)^2 = \left(\frac{p}{2} - b\right)^2 + \left(y - \frac{a(b+1)}{p+1}\right)^2 \\
\Rightarrow & a^2 - 2ay = b^2 - pb + \frac{a^2(b+1)^2}{(p+1)^2} - \frac{2a(b+1)}{p+1}y \\
\Rightarrow & y = \frac{a^2(p+b+2) + b(p+1)^2}{2a(p+1)},
\end{aligned}$$

we get $O\left(\frac{p}{2}, \frac{a^2(p+b+2) + b(p+1)^2}{2a(p+1)}\right)$. Adding a negative sign to the x -coordinate (corresponding to a reflection in the y -axis), swapping b and $-c$, and replacing p by $-p$, we get the circumcentre $O'\left(\frac{p}{2}, \frac{a^2(p+c-2) + c(p-1)^2}{2a(p-1)}\right)$ of $\triangle APY$. Then

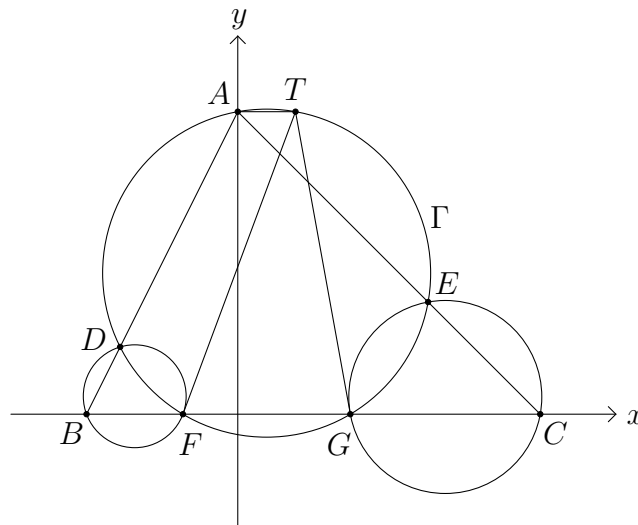
$$\begin{aligned}
& A, P, X, Y \text{ are concyclic} \\
\Leftrightarrow & O = O' \\
\Leftrightarrow & \frac{a^2(p+b+2) + b(p+1)^2}{2a(p+1)} = \frac{a^2(p+c-2) + c(p-1)^2}{2a(p-1)} \\
\Leftrightarrow & \frac{a^2(b+1) + b(p+1)^2}{2a(p+1)} + \frac{a}{2} = \frac{a^2(c-1) + c(p-1)^2}{2a(p-1)} + \frac{a}{2} \\
\Leftrightarrow & (p-1)(a^2(b+1) + b(p+1)^2) = (p+1)(a^2(c-1) + c(p-1)^2).
\end{aligned}$$

This is indeed the same as our condition (1). So we are done. $\square$

Problem 90. (IMO Shortlist 2019 G1) Let ABC be a triangle. Circle Γ passes through A , meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G . The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T . Suppose that points A and T are distinct. Prove that line AT is parallel to BC .



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(c, 0)$, $F(f, 0)$ and $G(g, 0)$. Then

$$AB: \quad y = -\frac{a}{b}x + a.$$

Let $D(bd, a(1 - d))$ where $0 < d < 1$.

Note that

$$\begin{aligned}
& A, D, F, G \text{ are concyclic} \\
\Rightarrow & BA \times BD = BF \times BG \\
\Rightarrow & \sqrt{(a^2 + b^2)(b^2(d-1)^2 + a^2(d-1)^2)} = (f-b)(g-b) \\
\Rightarrow & (1-d)(a^2 + b^2) = (f-b)(g-b). \tag{1}
\end{aligned}$$

Let O be the circumcentre of $\triangle BDF$. Its x -coordinate is $\frac{b+f}{2}$. From

$$\begin{aligned}
& OB = OD \\
\Rightarrow & \left(\frac{b-f}{2}\right)^2 + y^2 = \left(\frac{b+f}{2} - bd\right)^2 + (y - a(1-d))^2 \\
\Rightarrow & -\frac{bf}{2} = \frac{bf}{2} - bd(b+f) + b^2d^2 - 2a(1-d)y + a^2(1-d)^2 \\
\Rightarrow & y = \frac{a^2(1-d)^2 + b^2d^2 - bd(b+f) + bf}{2a(1-d)} \\
\Rightarrow & y = \frac{a^2(1-d) - b^2d + bf}{2a}.
\end{aligned}$$

Using (1) to rewrite $a^2(1-d) - b^2d = (f-b)(g-b) - b^2$, we obtain

$$y = \frac{(f-b)(g-b) - b^2 + bf}{2a} = \frac{g(f-b)}{2a}.$$

Thus, $O\left(\frac{b+f}{2}, \frac{g(f-b)}{2a}\right)$. Then

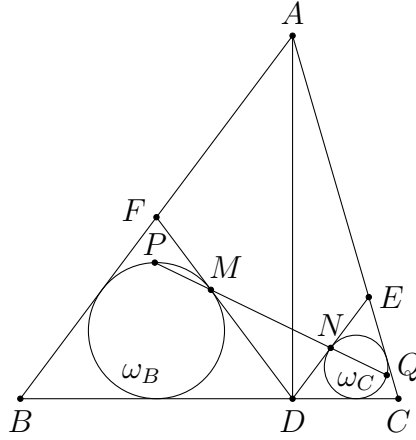
$$m_{FT} = \frac{-1}{m_{OF}} = \frac{a}{g}$$

and

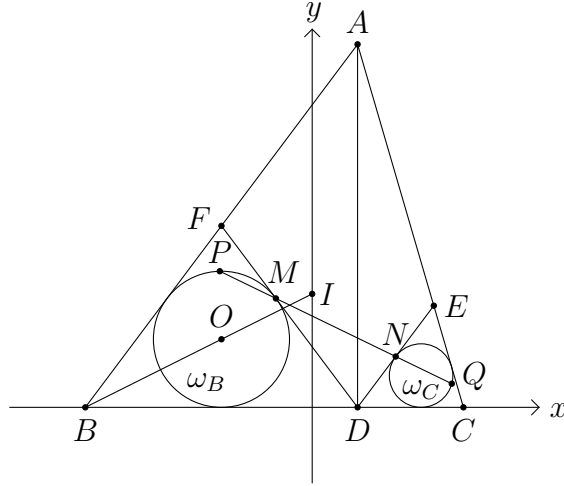
$$FT: \quad y = \frac{a}{g}x - \frac{af}{g}.$$

Putting $y = a$, we obtain $x = f + g$. This means $(f + g, a)$ is the intersection point of FT and the line passing through A and parallel to BC . By symmetry, it also lies on GT . Therefore, this is the point T , and the result follows immediately. $\square$

Problem 91. (IMO Shortlist 2019 G2) Let ABC be an acute-angled triangle and let D, E , and F be the feet of altitudes from A, B , and C to sides BC, CA , and AB , respectively. Denote by ω_B and ω_C the incircles of triangles BDF and CDE , and let these circles be tangent to segments DF and DE at M and N , respectively. Let line MN meet circles ω_B and ω_C again at $P \neq M$ and $Q \neq N$, respectively. Prove that $MP = NQ$.



Solution.



Let I be the incentre of $\triangle ABC$, and let $I(0,1)$, $B(b,0)$, $C(c,0)$. Then we have $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$ and $D\left(\frac{b+c}{bc+1}, 0\right)$. Solving

$$\begin{aligned} AB: \quad y &= -\frac{2b}{b^2-1}x + \frac{2b^2}{b^2-1}, \\ CF: \quad y &= \frac{b^2-1}{2b}x - \frac{(b^2-1)c}{2b}, \end{aligned}$$

we get

$$x = \frac{4b^3 + (b^2-1)^2c}{4b^2 + (b^2-1)^2} = \frac{4b^3 + (b^2-1)^2c}{(b^2+1)^2}$$

and

$$y = \frac{2b}{b^2-1} \left(b - \frac{4b^3 + (b^2-1)^2c}{(b^2+1)^2} \right) = \frac{2b}{b^2-1} \cdot \frac{b(b^2-1)^2 - (b^2-1)^2c}{(b^2+1)^2} = \frac{2b(b-c)(b^2-1)}{(b^2+1)^2}.$$

Therefore, we have $F\left(\frac{4b^3 + (b^2 - 1)^2c}{(b^2 + 1)^2}, \frac{2b(b - c)(b^2 - 1)}{(b^2 + 1)^2}\right)$.

Let O be the centre of ω_B . Note that $\triangle BDF \sim \triangle BAC$, and the ratio of similarity is $\frac{BD}{BA}$. This implies O lies on BI and

$$\frac{BO}{BI} = \frac{BD}{BA} = \frac{\frac{b+c}{bc+1} - b}{-\frac{c(1+b^2)}{bc+1}} = \frac{b^2 - 1}{b^2 + 1}.$$

Using directed lengths, this yields $BO : OI = (b^2 - 1) : 2$, and hence $O\left(\frac{2b}{b^2 + 1}, \frac{b^2 - 1}{b^2 + 1}\right)$. Thus, the length of the tangent from D to ω_B is

$$\frac{b+c}{bc+1} - \frac{2b}{b^2+1} = \frac{(b+c)(b^2+1) - 2b(bc+1)}{(bc+1)(b^2+1)} = \frac{(b-c)(b^2-1)}{(bc+1)(b^2+1)}.$$

This is also the length of DM . Note that

$$\begin{aligned} m_{DF} &= \frac{2b(b-c)(b^2-1)(bc+1)}{(4b^3 + (b^2-1)^2c)(bc+1) - (b+c)(b^2+1)^2} \\ &= \frac{2b(b-c)(b^2-1)(bc+1)}{b(b^2-1)^2c^2 + 4b^2(b^2-1)c - b(b^2-1)^2} \\ &= \frac{2(b-c)(bc+1)}{(b^2-1)c^2 + 4bc - (b^2-1)}. \end{aligned}$$

As $m_{DF} < 0$, $b - c < 0$ and $bc + 1 < 0$, the denominator is negative. We compute

$$\begin{aligned} m_{DF}^2 + 1 &= \frac{4(b-c)^2(bc+1)^2 + ((b^2-1)c^2 + 4bc - (b^2-1))^2}{((b^2-1)c^2 + 4bc - (b^2-1))^2} \\ &= \frac{(b^2+1)^2c^4 + 2(b^2+1)^2c^2 + (b^2+1)^2}{((b^2-1)c^2 + 4bc - (b^2-1))^2} \\ &= \left(\frac{(b^2+1)(c^2+1)}{(b^2-1)c^2 + 4bc - (b^2-1)}\right)^2. \end{aligned}$$

Therefore, the x -coordinate of M is

$$\begin{aligned} x_D - \frac{DM}{\sqrt{m_{DF}^2 + 1}} &= \frac{b+c}{bc+1} - \frac{\frac{(b-c)(b^2-1)}{(bc+1)(b^2+1)}}{\frac{(b^2+1)(c^2+1)}{-(b^2-1)c^2 - 4bc + (b^2-1)}} \\ &= \frac{(b+c)(b^2+1)^2(c^2+1) - (b-c)(b^2-1)(-(b^2-1)c^2 - 4bc + (b^2-1))}{(bc+1)(b^2+1)^2(c^2+1)} \end{aligned}$$

$$\begin{aligned}
&= \frac{4b^2c^3 + 2b(b^4 - 2b^2 + 3)c^2 + 2(3b^4 - 2b^2 + 1)c + 4b^3}{(bc + 1)(b^2 + 1)^2(c^2 + 1)} \\
&= \frac{4bc^2 + 2(b^2 - 1)^2c + 4b^3}{(b^2 + 1)^2(c^2 + 1)}
\end{aligned}$$

and its y -coordinate is

$$\begin{aligned}
\frac{DM \cdot |m_{DF}|}{\sqrt{m_{DF}^2 + 1}} &= \frac{\frac{(b-c)(b^2-1)}{(bc+1)(b^2+1)} \cdot \frac{2(b-c)(bc+1)}{-(b^2-1)c^2 - 4bc + (b^2-1)}}{\frac{(b^2+1)(c^2+1)}{-(b^2-1)c^2 - 4bc + (b^2-1)}} \\
&= \frac{2(b-c)^2(b^2-1)}{(b^2+1)^2(c^2+1)}.
\end{aligned}$$

This means $M \left(\frac{4bc^2 + 2(b^2-1)^2c + 4b^3}{(b^2+1)^2(c^2+1)}, \frac{2(b-c)^2(b^2-1)}{(b^2+1)^2(c^2+1)} \right)$. By symmetry, we have $N \left(\frac{4cb^2 + 2(c^2-1)^2b + 4c^3}{(c^2+1)^2(b^2+1)}, \frac{2(c-b)^2(c^2-1)}{(c^2+1)^2(b^2+1)} \right)$. Then we compute

$$\begin{aligned}
m_{MN} &= \frac{2(b-c)^2(b^2-1)(c^2+1) - 2(c-b)^2(c^2-1)(b^2+1)}{(4bc^2 + 2(b^2-1)^2c + 4b^3)(c^2+1) - (4cb^2 + 2(c^2-1)^2b + 4c^3)(b^2+1)} \\
&= \frac{4(b-c)^2(b^2-c^2)}{2(b-c)(b^3c^3 - b^3c + 3b^2c^2 - bc^3 + b^2 - 3bc + c^2 - 1)} \\
&= \frac{2(b-c)^2(b+c)}{(bc-1)(b^2c^2 - b^2 + 4bc - c^2 + 1)}.
\end{aligned}$$

This gives

$$\begin{aligned}
MN: \quad y &= \frac{2(b-c)^2(b+c)}{(bc-1)(b^2c^2 - b^2 + 4bc - c^2 + 1)} \cdot \left(x - \frac{4bc^2 + 2(b^2-1)^2c + 4b^3}{(b^2+1)^2(c^2+1)} \right) \\
&\quad + \frac{2(b-c)^2(b^2-1)}{(b^2+1)^2(c^2+1)}.
\end{aligned}$$

After simplification, this gives

$$\begin{aligned}
MN: \quad y &= \frac{2(b-c)^2(b+c)}{(bc-1)(b^2c^2 - b^2 + 4bc - c^2 + 1)}x \\
&\quad + \frac{2(b-c)^2(bc+1)(b^2c^2 - 3b^2 - 3c^2 + 1)}{(b^2+1)(c^2+1)(bc-1)(b^2c^2 - b^2 + 4bc - c^2 + 1)}.
\end{aligned}$$

We can rewrite this in the form

$$MN: \quad x = \frac{(bc-1)(b^2c^2 - b^2 + 4bc - c^2 + 1)}{2(b-c)^2(b+c)}y - \frac{(bc+1)(b^2c^2 - 3b^2 - 3c^2 + 1)}{(b+c)(b^2+1)(c^2+1)}. \quad (1)$$

Recall that the centre of ω_B is $O\left(\frac{2b}{b^2+1}, \frac{b^2-1}{b^2+1}\right)$. So the radius of ω_B is $\frac{b^2-1}{b^2+1}$, and

$$\omega_B: \quad x^2 + y^2 - \frac{4b}{b^2+1}x - \frac{2(b^2-1)}{b^2+1}y + \frac{4b^2}{(b^2+1)^2} = 0. \quad (2)$$

Putting (1) into (2), we obtain a quadratic equation in y , whose roots are the y -coordinates of M and P . The leading coefficient is

$$1 + \frac{(bc-1)^2(b^2c^2 - b^2 + 4bc - c^2 + 1)^2}{4(b-c)^4(b+c)^2}$$

while the constant term is

$$\begin{aligned} & \frac{(bc+1)^2(b^2c^2 - 3b^2 - 3c^2 + 1)^2}{(b+c)^2(b^2+1)^2(c^2+1)^2} + \frac{4b(bc+1)(b^2c^2 - 3b^2 - 3c^2 + 1)}{(b+c)(b^2+1)^2(c^2+1)} + \frac{4b^2}{(b^2+1)^2} \\ &= \left(\frac{(bc+1)(b^2c^2 - 3b^2 - 3c^2 + 1)}{(b+c)(b^2+1)(c^2+1)} + \frac{2b}{b^2+1} \right)^2 \\ &= \left(\frac{(bc+1)(b^2c^2 - 3b^2 - 3c^2 + 1) + 2b(b+c)(c^2+1)}{(b+c)(b^2+1)(c^2+1)} \right)^2 \\ &= \left(\frac{(b^2-1)(bc^3 - 3bc + 3c^2 - 1)}{(b+c)(b^2+1)(c^2+1)} \right)^2. \end{aligned}$$

Thus, by the product of roots, the y -coordinate of P is

$$\begin{aligned} & \frac{\frac{(b^2-1)^2(bc^3 - 3bc + 3c^2 - 1)^2}{(b+c)^2(b^2+1)^2(c^2+1)^2}}{1 + \frac{(bc-1)^2(b^2c^2 - b^2 + 4bc - c^2 + 1)^2}{4(b-c)^4(b+c)^2}} \div \frac{2(b-c)^2(b^2-1)}{(b^2+1)^2(c^2+1)} \\ &= \frac{2(b-c)^2(b^2-1)(bc^3 - 3bc + 3c^2 - 1)^2}{(c^2+1)((bc-1)^2(b^2c^2 - b^2 + 4bc - c^2 + 1)^2 + 4(b-c)^4(b+c)^2)} \\ &= \frac{2(b-c)^2(b^2-1)(bc^3 - 3bc + 3c^2 - 1)^2}{(b^2+1)(c^2+1)((b^4 - 3b^2 + 4)c^6 + 2b(3b^2 - 5)c^5 + (-2b^4 + 7b^2 + 1)c^4 - 4b(b^2+1)c^3 + (b^4 + 7b^2 - 2)c^2 - 2b(5b^2 - 3)c + (4b^4 - 3b^2 + 1))} \end{aligned}$$

It follows that

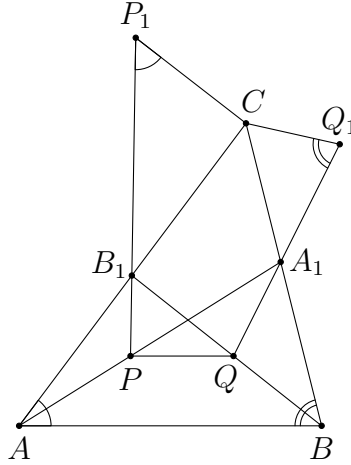
$$\begin{aligned} & y_M - y_P \\ &= \frac{2(b-c)^2(b^2-1)}{(b^2+1)^2(c^2+1)} - \frac{2(b-c)^2(b^2-1)(bc^3 - 3bc + 3c^2 - 1)^2}{(b^2+1)(c^2+1)((b^4 - 3b^2 + 4)c^6 + 2b(3b^2 - 5)c^5 + (-2b^4 + 7b^2 + 1)c^4 - 4b(b^2+1)c^3 + (b^4 + 7b^2 - 2)c^2 - 2b(5b^2 - 3)c + (4b^4 - 3b^2 + 1))} \end{aligned}$$

$$\begin{aligned}
& \frac{2(b-c)^2(b^2-1)(-4(b^2-1)c^6 - 16bc^5 + 4(b^4+b^2-2)c^4 + 16b(b^2+1)c^3 \\
& + 4(-2b^4+b^2+1)c^2 - 16b^3c + 4b^2(b^2-1))}{(b^2+1)^2(c^2+1)((b^4-3b^2+4)c^6 + 2b(3b^2-5)c^5 + (-2b^4+7b^2+1)c^4 \\
& - 4b(b^2+1)c^3 + (b^4+7b^2-2)c^2 - 2b(5b^2-3)c + (4b^4-3b^2+1))} \\
& = \frac{2(b-c)^2(b^2-1)(c^2-1)(c^2-b^2)(-4b^2c^2 + 4c^2 - 16bc + 4b^2 - 4)}{(b^2+1)^2(c^2+1)^2((b^4-3b^2+4)c^4 + 2b(3b^2-5)c^3 + (-3b^4+10b^2-3)c^2 \\
& + 2b(-5b^2+3)c + (4b^4-3b^2+1))}.
\end{aligned}$$

Note that the denominator is symmetric in b and c . Therefore, by symmetry, we have $y_N - y_Q = -(y_M - y_P)$. As $m_{MP} = m_{NQ}$, this implies $MP = NQ$. $\square$

Problem 92. (IMO Shortlist 2019 G3/Problem 2) In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be a point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$.

Prove that points P, Q, P_1 , and Q_1 are concyclic.



Solution. Let AB meet PB_1 and QA_1 at D and E respectively. Let $A(a, 0)$, $B(b, 0)$, $C(0, c)$, $D(d, 0)$ and $E(e, 0)$. Let

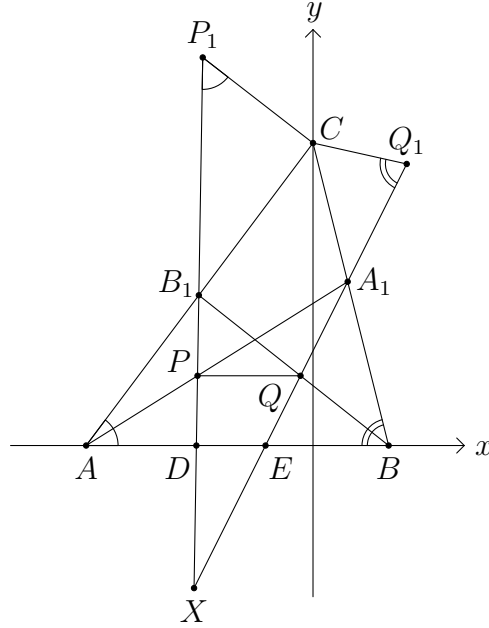
$$DB_1: y = mx - md, \quad (1)$$

$$EA_1: y = nx - ne. \quad (2)$$

Solving (1) and

$$AC: y = -\frac{c}{a}x + c,$$

we easily obtain $B_1\left(\frac{a(md+c)}{ma+c}, \frac{mc(a-d)}{ma+c}\right)$.



It follows that

$$BB_1: y = \frac{mc(a-d)}{ma(d-b) + c(a-b)}x - \frac{mbc(a-d)}{ma(d-b) + c(a-b)}.$$

Combining with (2), we get

$$x = \frac{\frac{mbc(a-d)}{ma(d-b) + c(a-b)} - ne}{\frac{mc(a-d)}{ma(d-b) + c(a-b)} - n} = \frac{mnae(b-d) + mbc(a-d) + nce(b-a)}{mna(b-d) + mc(a-d) + nc(b-a)}$$

and

$$\begin{aligned} y &= n \left(\frac{mnae(b-d) + mbc(a-d) + nce(b-a)}{mna(b-d) + mc(a-d) + nc(b-a)} - e \right) \\ &= \frac{mnc(a-d)(b-e)}{mna(b-d) + mc(a-d) + nc(b-a)}. \end{aligned}$$

$$\text{So } Q \left(\frac{mnae(b-d) + mbc(a-d) + nce(b-a)}{mna(b-d) + mc(a-d) + nc(b-a)}, \frac{mnc(a-d)(b-e)}{mna(b-d) + mc(a-d) + nc(b-a)} \right).$$

Swapping the pairs (a, b) , (d, e) and (m, n) , we find that the y -coordinate of P is

$$\frac{nmc(b-e)(a-d)}{nmb(a-e) + nc(b-e) + mc(a-b)}.$$

As $PQ \parallel AB$, the y -coordinates of P and Q are the same. This gives

$$\frac{mnc(a-d)(b-e)}{mna(b-d) + mc(a-d) + nc(b-a)} = \frac{nmc(b-e)(a-d)}{nmb(a-e) + nc(b-e) + mc(a-b)},$$

i.e. $mn(ad - be) - mc(b - d) + nc(a - e) = 0$. This implies

$$c = \frac{mn(ad - be)}{m(b - d) - n(a - e)}. \quad (3)$$

Next, by the condition $\angle CBA = \angle CQ_1Q = \angle CQ_1E$, we see that Q_1, C, E, B are concyclic. Recall that

$$(CEB): \quad x^2 + y^2 - (b + e)x - \frac{c^2 + be}{c}y + be = 0.$$

Putting (2) into this equation, we solve

$$(1 + n^2)x^2 + (\dots)x + \left(n^2e^2 + \frac{ne(c^2 + be)}{c} + be\right) = 0.$$

One of the roots to this quadratic equation is e (the x -coordinate of E), while the other root is

$$x = \frac{n^2ce^2 + ne(c^2 + be) + bce}{c(n^2 + 1)} \div e = \frac{n^2ce + n(c^2 + be) + bc}{c(n^2 + 1)}.$$

This yields

$$y = n \left(\frac{n^2ce + n(c^2 + be) + bc}{c(n^2 + 1)} - e \right) = \frac{n^2(c^2 + be) + nc(b - e)}{c(n^2 + 1)}.$$

Thus, $Q_1 \left(\frac{n^2ce + n(c^2 + be) + bc}{c(n^2 + 1)}, \frac{n^2(c^2 + be) + nc(b - e)}{c(n^2 + 1)} \right)$. By symmetry, we have

$$P_1 \left(\frac{m^2cd + m(c^2 + ad) + ac}{c(m^2 + 1)}, \frac{m^2(c^2 + ad) + mc(a - d)}{c(m^2 + 1)} \right).$$

We first solve the general case where PP_1 intersects QQ_1 at a point X . Solving (1) and (2), we easily obtain $X \left(\frac{md - ne}{m - n}, \frac{mn(d - e)}{m - n} \right)$. Using power, it suffices to show $XQ \cdot XQ_1$ is a constant (up to symmetry). As the slope of QQ_1 is n , we have

$$XQ \cdot XQ_1 = \sqrt{n^2 + 1}(x_X - x_Q) \cdot \sqrt{n^2 + 1}(x_X - x_{Q_1}).$$

Now,

$$\begin{aligned} x_X - x_Q &= \frac{md - ne}{m - n} - \frac{mnae(b - d) + mbc(a - d) + nce(b - a)}{mna(b - d) + mc(a - d) + nc(b - a)} \\ &= \frac{m(b - d)(mna(d - e) - mc(a - d) + nc(a - e))}{(m - n)(mna(b - d) + mc(a - d) + nc(b - a))}. \end{aligned}$$

Using (3), we replace c in the numerator by the other variables. This gives

$$\begin{aligned}
x_X - x_Q &= \frac{m(b-d)(m^2na(b-d)(d-e) - mn^2a(a-e)(d-e) \\
&\quad - m^2n(a-d)(ad-be) + mn^2(a-e)(ad-be))}{(m-n)(m(b-d) - n(a-e))(mna(b-d) + mc(a-d) + nc(b-a))} \\
&= \frac{m(b-d)(-m^2nd(a-b)(a-e) + mn^2e(a-e)(a-b))}{(m-n)(m(b-d) - n(a-e))(mna(b-d) + mc(a-d) + nc(b-a))} \\
&= \frac{m^2n(a-b)(a-e)(b-d)(-md + ne)}{(m-n)(m(b-d) - n(a-e))(mna(b-d) + mc(a-d) + nc(b-a))}.
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
&x_X - x_{Q_1} \\
&= \frac{md - ne}{m - n} - \frac{n^2ce + n(c^2 + be) + bc}{c(n^2 + 1)} \\
&= \frac{-n(m-n)c^2 + (mn^2(d-e) - m(b-d) + n(b-e))c - n(m-n)be}{c(m-n)(n^2 + 1)} \\
&= \frac{-m^2n^3(m-n)(ad-be)^2 - n(m-n)be(m(b-d) - n(a-e))^2 \\
&\quad + mn(mn^2(d-e) - m(b-d) + n(b-e))(ad-be)(m(b-d) - n(a-e))}{c(m-n)(n^2 + 1)(m(b-d) - n(a-e))^2} \\
&= \frac{n(m^3n^2d(ad-be)(-a+b-d+e) + m^2n^3e(ad-be)(a-b+d-e) \\
&\quad - m^3ad(b-d)^2 + n^3be(a-e)^2 + m^2n(b-d)(a^2d + abd + abe - bde - 2ade) \\
&\quad + mn^2(a-e)(-b^2e - abd - abe + ade + 2bde))}{c(m-n)(n^2 + 1)(m(b-d) - n(a-e))^2}.
\end{aligned}$$

It follows that

$$\begin{aligned}
&XQ \cdot XQ_1 \\
&= \frac{m^2n^2(a-b)(a-e)(b-d)(-md + ne) \\
&\quad \times (m^3n^2d(ad-be)(-a+b-d+e) + m^2n^3e(ad-be)(a-b+d-e) \\
&\quad - m^3ad(b-d)^2 + n^3be(a-e)^2 + m^2n(b-d)(a^2d + abd + abe - bde - 2ade) \\
&\quad + mn^2(a-e)(-b^2e - abd - abe + ade + 2bde))}{c(m-n)^2(m(b-d) - n(a-e))^3(mna(b-d) + mc(a-d) + nc(b-a))}.
\end{aligned}$$

It is now routine to check that each term is symmetric by swapping the pairs (a, b) , (d, e) and (m, n) . In particular,

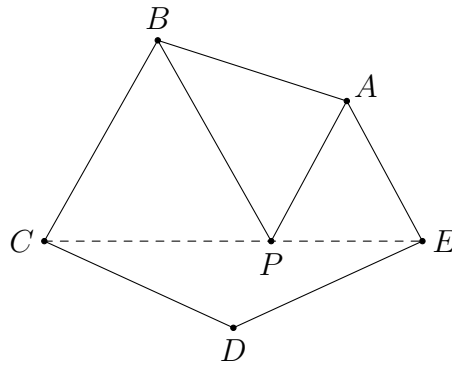
$$mna(b-d) + mc(a-d) + nc(b-a) = nmb(a-e) + nc(b-e) + mc(a-b)$$

from above. This proves the general case.

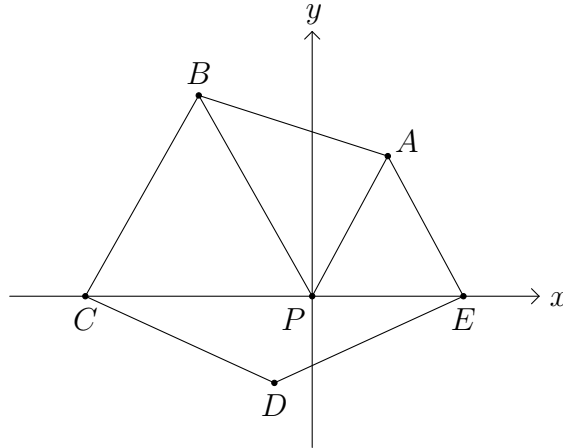
We are required to consider the special case $PP_1//QQ_1$ (or $m = n$) due to the marking scheme. From the configuration, PQ_1 and QP_1 always meet at a point Y . In the

general case, we must have $YP \times YQ_1 = YQ \times YP_1$ since P, Q, Q_1, P_1 are concyclic. This identity is a rational function in the variables a, b, d, e, m, n and it holds for all cases except possibly $m = n$. Also, since $m - n$ is not a factor of the denominators (as the lengths are finite when $m = n$), the identity must also hold when $m = n$. This proves P, Q, Q_1, P_1 are concyclic in any case. $\square$

Problem 93. (IMO Shortlist 2019 G5) Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2\angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $\text{area}(BCD) + \text{area}(ADE) = \text{area}(ABD) + \text{area}(ABP)$.



Solution.



Firstly, suppose P lies on CE . Let $P(0, 0)$, $C(2a, 0)$, $E(2b, 0)$, $B(a, c)$, $A(b, d)$ and $D(a + b, t)$. Then

$$[BCD] = \frac{1}{2}(2at + (a + b)c - 2ac - at) = \frac{1}{2}(at - ac + bc),$$

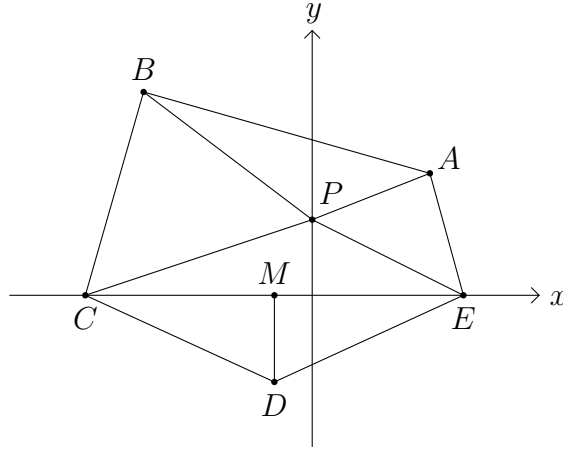
$$[ADE] = \frac{1}{2}(bt + 2bd - (a + b)d - 2bt) = \frac{1}{2}(-bt - ad + bd),$$

$$[ABD] = \frac{1}{2}(bc + at + (a+b)d - ad - (a+b)c - bt) = \frac{1}{2}(at - bt - ac + bd),$$

$$[ABP] = \frac{1}{2}(bc - ad).$$

It is routine to check

$$[BCD] + [ADE] = \frac{1}{2}(at - bt - ac + bc - ad + bd) = [ABD] + [ABP].$$



Secondly, suppose $[BCD] + [ADE] = [ABD] + [ABP]$. Suppose on the contrary that P does not lie on CE . We shall derive the contradiction that $\angle EDC = 2\angle ADB$. Let $C(2a, 0)$, $E(2b, 0)$ and $P(0, 2c)$ with $c \neq 0$. Then the perpendicular bisector of CP has equation

$$y = \frac{a}{c}x + \frac{c^2 - a^2}{c}.$$

Let $B(p, s)$ where

$$s = \frac{ap + c^2 - a^2}{c}.$$

Similarly, let $A(q, t)$ where

$$t = \frac{bq + c^2 - b^2}{c}.$$

Let $D(a+b, d)$. We have

$$[BCD] = \frac{1}{2}(2ad + (a+b)s - 2as - pd) = \frac{1}{2}(-pd + 2ad - as + bs),$$

$$[ADE] = \frac{1}{2}(qd + 2bt - (a+b)t - 2bd) = \frac{1}{2}(qd - 2bd - at + bt),$$

$$[ABD] = \frac{1}{2}(qs + pd + (a+b)t - pt - (a+b)s - qd),$$

$$[ABP] = \frac{1}{2}(qs + 2cp - pt - 2cq).$$

Then

$$\begin{aligned} & [BCD] + [ADE] = [ABD] + [ABP] \\ \Rightarrow & (-p + q + a - b)d + (-q + b)s + (p - a)t + (-p + q)c = 0. \end{aligned} \quad (1)$$

Let $M(a + b, 0)$ be the midpoint of CE . As $DC = DE$, we have $\angle EDC = 2\angle EDM$. Therefore, it suffices to prove $\angle EDM = \angle ADB$. Indeed,

$$\begin{aligned} & \angle EDM = \angle ADB \\ \Leftrightarrow & \frac{1}{m_{DE}} = \frac{m_{BD} - m_{AD}}{1 + m_{BD} \cdot m_{AD}} \\ \Leftrightarrow & \frac{a - b}{d} = \frac{\frac{s - d}{p - a - b} - \frac{t - d}{q - a - b}}{1 + \frac{s - d}{p - a - b} \cdot \frac{t - d}{q - a - b}} \\ \Leftrightarrow & \frac{a - b}{d} = \frac{(p - q)d + (q - a - b)s - (p - a - b)t}{d^2 - (s + t)d + st + (p - a - b)(q - a - b)}. \end{aligned}$$

After expansion, this is equivalent to

$$(-p + q + a - b)d^2 + (-qs + pt + 2bs - 2at)d + (a - b)(st + (p - a - b)(q - a - b)) = 0.$$

Using (1), the left-hand side becomes

$$\begin{aligned} & ((q - b)s + (-p + a)t + (p - q)c)d + (-qs + pt + 2bs - 2at)d \\ & + (a - b)(st + (p - a - b)(q - a - b)) \\ = & (bs - at + (p - q)c)d + (a - b)(st + (p - a - b)(q - a - b)). \end{aligned}$$

Recall that $s = \frac{ap + c^2 - a^2}{c}$ and $t = \frac{bq + c^2 - b^2}{c}$. This becomes

$$\begin{aligned} & \left(\frac{b(ap + c^2 - a^2)}{c} - \frac{a(bq + c^2 - b^2)}{c} + (p - q)c \right) d \\ & + (a - b) \left(\frac{ap + c^2 - a^2}{c} \cdot \frac{bq + c^2 - b^2}{c} + (p - a - b)(q - a - b) \right) \\ = & \frac{(c^2 + ab)(p - q - a + b)}{c} d \\ & + \frac{a - b}{c^2} ((ap + c^2 - a^2)(bq + c^2 - b^2) + (p - a - b)(q - a - b)c^2). \end{aligned}$$

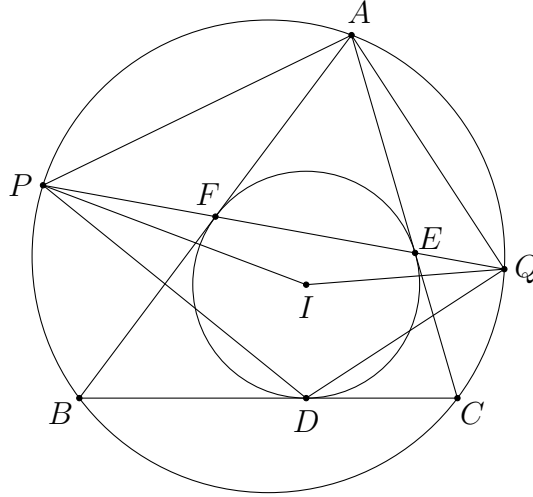
Using (1) again, this further becomes

$$\begin{aligned} & \frac{c^2 + ab}{c} \left(\frac{(-q + b)(ap + c^2 - a^2)}{c} + \frac{(p - a)(bq + c^2 - b^2)}{c} + (-p + q)c \right) \\ & + \frac{a - b}{c^2} ((ap + c^2 - a^2)(bq + c^2 - b^2) + (p - a - b)(q - a - b)c^2) \end{aligned}$$

$$\begin{aligned}
&= \frac{c^2 + ab}{c^2}((-a + b)c^2 + a(p - a)(-q + b) + b(p - a)(q - b)) \\
&\quad + \frac{a - b}{c^2}(c^4 + (pq - bp - aq + 2ab)c^2 + ab(p - a)(q - b)) \\
&= -\frac{c^2 + ab}{c^2}(a - b)(c^2 + (p - a)(q - b)) + \frac{a - b}{c^2}(c^2 + ab)(c^2 + (p - a)(q - b)) \\
&= 0.
\end{aligned}$$

This proves the angle relation, which gives a contradiction. Therefore, we are done. $\square$

Problem 94. (IMO Shortlist 2019 G6) Let I be the incentre of acute-angled triangle ABC . Let the incircle meet BC, CA , and AB at D, E , and F respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.

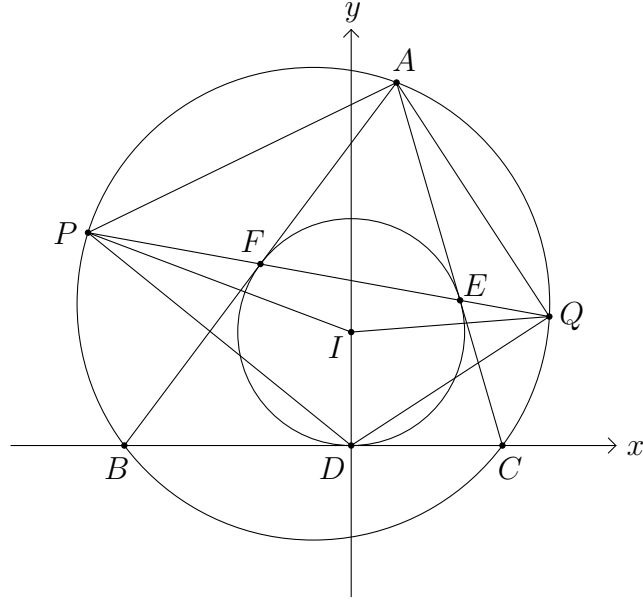


Solution. Let $D(0, 0)$, $I(0, 1)$, $B(b, 0)$ and $C(c, 0)$. Then we have $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$, $E\left(\frac{2c}{c^2+1}, \frac{2c^2}{c^2+1}\right)$ and $F\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right)$. Then it is easy to deduce

$$EF: \quad y = -\frac{b+c}{bc-1}x + \frac{2bc}{bc-1}.$$

Putting this into

$$(ABC): \quad x^2 + y^2 - (b+c)x - \frac{b^2c^2 - b^2 + 4bc - c^2 + 1}{2(bc+1)}y + bc = 0,$$



we obtain

$$\left(1 + \frac{(b+c)^2}{(bc-1)^2}\right)x^2 - \left(\frac{4bc(b+c)}{(bc-1)^2} + (b+c) - \frac{(b+c)(b^2c^2 - b^2 + 4bc - c^2 + 1)}{2(bc+1)(bc-1)}\right)x + \left(\frac{4b^2c^2}{(bc-1)^2} - \frac{bc(b^2c^2 - b^2 + 4bc - c^2 + 1)}{(bc+1)(bc-1)} + bc\right) = 0.$$

After expansion, this means

$$2(b^2+1)(c^2+1)(bc+1)x^2 - (b+c)(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)x + 2bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2) = 0.$$

Let $P\left(\alpha, \frac{-(b+c)\alpha + 2bc}{bc-1}\right)$ and $Q\left(\beta, \frac{-(b+c)\beta + 2bc}{bc-1}\right)$. Then α, β are the roots of the above quadratic equation. This implies

$$\alpha + \beta = \frac{(b+c)(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)}{2(b^2+1)(c^2+1)(bc+1)}, \quad (1)$$

$$\alpha\beta = \frac{bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{(b^2+1)(c^2+1)(bc+1)}. \quad (2)$$

Now, note that

$$\begin{aligned} & \angle DPA + \angle AQD = \angle QIP \\ \Leftrightarrow & 360^\circ - \angle QDP - \angle PAQ = \angle QIP \\ \Leftrightarrow & \tan(\angle QDP + \angle PAQ) = -\tan \angle QIP. \end{aligned}$$

The last equivalence holds since

$$\begin{aligned} 180^\circ &< \angle QDP + \angle PAQ \\ &< (\angle QDP + \angle PAQ) + \angle QIP \\ &< 360^\circ + 180^\circ = 540^\circ. \end{aligned}$$

The remaining tasks are some crazy computations. Firstly,

$$\begin{aligned} \tan \angle QIP &= \frac{m_{IP} - m_{IQ}}{1 + m_{IP} \cdot m_{IQ}} \\ &= \frac{\frac{-(b+c)\alpha + (bc+1)}{(bc-1)\alpha} - \frac{-(b+c)\beta + (bc+1)}{(bc-1)\beta}}{1 + \frac{-(b+c)\alpha + (bc+1)}{(bc-1)\alpha} \cdot \frac{-(b+c)\beta + (bc+1)}{(bc-1)\beta}} \\ &= \frac{(bc-1)(bc+1)(\beta-\alpha)}{(b^2+1)(c^2+1)\alpha\beta - (b+c)(bc+1)(\alpha+\beta) + (bc+1)^2}. \end{aligned}$$

Using (1) and (2), the denominator becomes

$$\begin{aligned} &\frac{bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{bc+1} - \frac{(b+c)^2(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)}{2(b^2+1)(c^2+1)} \\ &+ (bc+1)^2 \\ &\frac{b^6c^4 + b^4c^6 + b^6c^2 - 6b^5c^3 + 12b^4c^4 - 6b^3c^5 + b^2c^6 - 2b^5c + 8b^4c^2 - 16b^3c^3}{+8b^2c^4 - 2bc^5 + b^4 - 2b^3c - 2b^2c^2 - 2bc^3 + c^4 - b^2 + 4bc - c^2 + 2} \\ &= \frac{2(bc+1)(b^2+1)(c^2+1)}{2(bc+1)(b^2+1)(c^2+1)}. \end{aligned}$$

Therefore,

$$\tan \angle QIP = \frac{2(bc-1)(bc+1)^2(b^2+1)(c^2+1)(\beta-\alpha)}{b^6c^4 + b^4c^6 + b^6c^2 - 6b^5c^3 + 12b^4c^4 - 6b^3c^5 + b^2c^6 - 2b^5c + 8b^4c^2 - 16b^3c^3 + 8b^2c^4 - 2bc^5 + b^4 - 2b^3c - 2b^2c^2 - 2bc^3 + c^4 - b^2 + 4bc - c^2 + 2}.$$

Secondly,

$$\begin{aligned} \tan \angle QDP &= \frac{m_{DP} - m_{DQ}}{1 + m_{DP} \cdot m_{DQ}} \\ &= \frac{\frac{-(b+c)\alpha + 2bc}{(bc-1)\alpha} - \frac{-(b+c)\beta + 2bc}{(bc-1)\beta}}{1 + \frac{-(b+c)\alpha + 2bc}{(bc-1)\alpha} \cdot \frac{-(b+c)\beta + 2bc}{(bc-1)\beta}} \\ &= \frac{2bc(bc-1)(\beta-\alpha)}{(b^2+1)(c^2+1)\alpha\beta - 2bc(b+c)(\alpha+\beta) + 4b^2c^2}. \end{aligned}$$

The denominator is equal to

$$\begin{aligned}
& \frac{bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{bc + 1} \\
& - \frac{bc(b + c)^2(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)}{(b^2 + 1)(c^2 + 1)(bc + 1)} + 4b^2c^2 \\
& = \frac{2bc(b^4c^4 - b^4c^2 + 2b^3c^3 - b^2c^4 + 2b^3c - 6b^2c^2 + 2bc^3 - b^2 + 2bc - c^2 + 1)}{(b^2 + 1)(c^2 + 1)(bc + 1)},
\end{aligned}$$

hence

$$\tan \angle QDP = \frac{(bc - 1)(bc + 1)(b^2 + 1)(c^2 + 1)(\beta - \alpha)}{b^4c^4 - b^4c^2 + 2b^3c^3 - b^2c^4 + 2b^3c - 6b^2c^2 + 2bc^3 - b^2 + 2bc - c^2 + 1}.$$

Thirdly,

$$\begin{aligned}
& \tan \angle PAQ \\
& = \frac{m_{AQ} - m_{AP}}{1 + m_{AQ} \cdot m_{AP}} \\
& = \frac{\frac{-(b + c)(bc + 1)\beta + 4bc}{(bc - 1)((bc + 1)\beta - (b + c))} - \frac{-(b + c)(bc + 1)\alpha + 4bc}{(bc - 1)((bc + 1)\alpha - (b + c))}}{1 + \frac{-(b + c)(bc + 1)\beta + 4bc}{(bc - 1)((bc + 1)\beta - (b + c))} \cdot \frac{-(b + c)(bc + 1)\alpha + 4bc}{(bc - 1)((bc + 1)\alpha - (b + c))}} \\
& = \frac{(b - c)^2(bc - 1)(bc + 1)(\beta - \alpha)}{(b^2 + 1)(c^2 + 1)(bc + 1)^2\alpha\beta - (b + c)(bc + 1)^3(\alpha + \beta) + (b + c)^2(bc - 1)^2 + 16b^2c^2}.
\end{aligned}$$

The denominator is equal to

$$\begin{aligned}
& bc(bc + 1)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2) + (b + c)^2(bc - 1)^2 + 16b^2c^2 \\
& - \frac{(b + c)^2(bc + 1)^2(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)}{2(b^2 + 1)(c^2 + 1)} \\
& = \frac{(b - c)^2(b^5c^5 + b^5c^3 - 3b^4c^4 + b^3c^5 + b^4c^2 - 4b^3c^3 + b^2c^4 - 5b^3c + 12b^2c^2 - 5bc^3 + 3b^2 - 5bc + 3c^2 - 1)}{2(b^2 + 1)(c^2 + 1)},
\end{aligned}$$

hence

$$\tan \angle PAQ = \frac{2(bc - 1)(bc + 1)(b^2 + 1)(c^2 + 1)(\beta - \alpha)}{b^5c^5 + b^5c^3 - 3b^4c^4 + b^3c^5 + b^4c^2 - 4b^3c^3 + b^2c^4 - 5b^3c + 12b^2c^2 - 5bc^3 + 3b^2 - 5bc + 3c^2 - 1}.$$

Lastly,

$$\begin{aligned}
& \tan(\angle QDP + \angle PAQ) = \frac{\tan \angle QDP + \tan \angle PAQ}{1 - \tan \angle QDP \cdot \tan \angle PAQ} \\
& = \frac{(bc - 1)(bc + 1)(b^2 + 1)(c^2 + 1)(\beta - \alpha) \times (b^5c^5 + b^5c^3 - b^4c^4 + b^3c^5 - b^4c^2 - b^2c^4 - b^3c - bc^3 + b^2 - bc + c^2 + 1)}{(b^4c^4 - b^4c^2 + 2b^3c^3 - b^2c^4 + 2b^3c - 6b^2c^2 + 2bc^3 - b^2 + 2bc - c^2 + 1) \times (b^5c^5 + b^5c^3 - 3b^4c^4 + b^3c^5 + b^4c^2 - 4b^3c^3 + b^2c^4 - 5b^3c + 12b^2c^2 - 5bc^3 + 3b^2 - 5bc + 3c^2 - 1) - 2(bc - 1)^2(bc + 1)^2(b^2 + 1)^2(c^2 + 1)^2(\beta - \alpha)^2}.
\end{aligned}$$

The numerator can be factorized as

$$(bc - 1)(bc + 1)^2(b^2 + 1)(c^2 + 1)(\beta - \alpha) \\ \times (b^4c^4 + b^4c^2 - 2b^3c^3 + b^2c^4 - 2b^3c + 2b^2c^2 - 2bc^3 + b^2 - 2bc + c^2 + 1).$$

Using

$$(\beta - \alpha)^2 = (\alpha + \beta)^2 - 4\alpha\beta \\ = \frac{(b + c)^2(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)^2}{4(b^2 + 1)^2(c^2 + 1)^2(bc + 1)^2} \\ - \frac{4bc(b^3c + bc^3 - b^2 + 6bc - c^2 + 2)}{(b^2 + 1)(c^2 + 1)(bc + 1)},$$

the denominator becomes

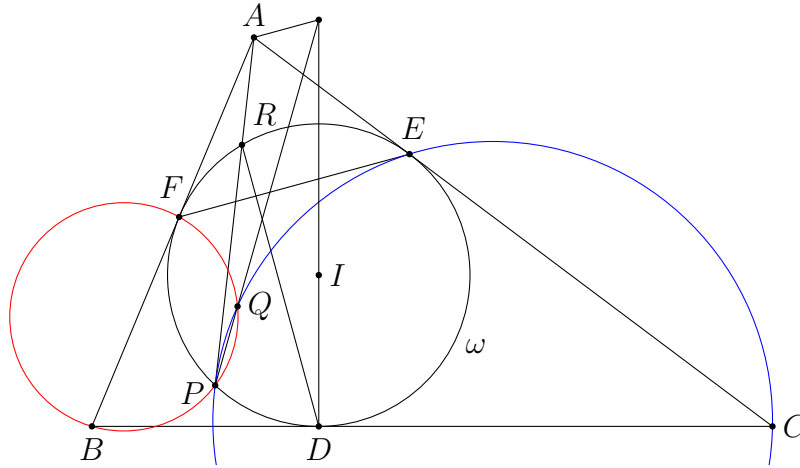
$$(b^4c^4 - b^4c^2 + 2b^3c^3 - b^2c^4 + 2b^3c - 6b^2c^2 + 2bc^3 - b^2 + 2bc - c^2 + 1) \\ \times (b^5c^5 + b^5c^3 - 3b^4c^4 + b^3c^5 + b^4c^2 - 4b^3c^3 + b^2c^4 - 5b^3c + 12b^2c^2 - 5bc^3 + 3b^2 \\ - 5bc + 3c^2 - 1) - \frac{1}{2}(b + c)^2(bc - 1)^2(b^3c^3 + b^3c + 3b^2c^2 + bc^3 - b^2 + 9bc - c^2 + 3)^2 \\ + 8bc(bc - 1)^2(bc + 1)(b^2 + 1)(c^2 + 1)(b^3c + bc^3 - b^2 + 6bc - c^2 + 2) \\ = \frac{1}{2}(-b^{10}c^8 - b^8c^{10} - 2b^{10}c^6 + 8b^9c^7 - 14b^8c^8 + 8b^7c^9 - 2b^6c^{10} - b^{10}c^4 + 12b^9c^5 - 35b^8c^6 \\ + 56b^7c^7 - 35b^6c^8 + 12b^5c^9 - b^4c^{10} + 4b^9c^3 - 28b^8c^4 + 76b^7c^5 - 96b^6c^6 + 76b^5c^7 \\ - 28b^4c^8 + 4b^3c^9 - 6b^8c^2 + 32b^7c^3 - 80b^6c^4 + 96b^5c^5 - 80b^4c^6 + 32b^3c^7 - 6b^2c^8 \\ + 4b^7c - 18b^6c^2 + 36b^5c^3 - 56b^4c^4 + 36b^3c^5 - 18b^2c^6 + 4bc^7 - b^6 + 4b^5c - 3b^4c^2 \\ + 8b^3c^3 - 3b^2c^4 + 4bc^5 - c^6 + 8b^2c^2 - b^2 - c^2 - 2) \\ = -\frac{1}{2}(b^4c^4 + b^4c^2 - 2b^3c^3 + b^2c^4 - 2b^3c + 2b^2c^2 - 2bc^3 + b^2 - 2bc + c^2 + 1) \\ \times (b^6c^4 + b^4c^6 + b^6c^2 - 6b^5c^3 + 12b^4c^4 - 6b^3c^5 + b^2c^6 - 2b^5c + 8b^4c^2 - 16b^3c^3 \\ + 8b^2c^4 - 2bc^5 + b^4 - 2b^3c - 2b^2c^2 - 2bc^3 + c^4 - b^2 + 4bc - c^2 + 2).$$

Therefore, we find that

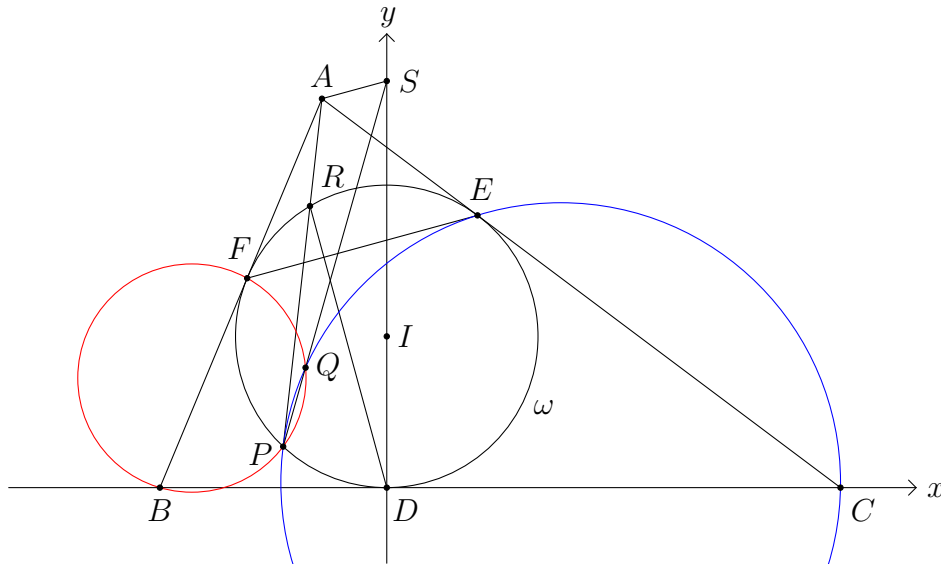
$$\tan(\angle QDP + \angle PAQ) \\ = -\frac{2(bc - 1)(bc + 1)^2(b^2 + 1)(c^2 + 1)(\beta - \alpha)}{b^6c^4 + b^4c^6 + b^6c^2 - 6b^5c^3 + 12b^4c^4 - 6b^3c^5 + b^2c^6 - 2b^5c + 8b^4c^2 - 16b^3c^3 \\ + 8b^2c^4 - 2bc^5 + b^4 - 2b^3c - 2b^2c^2 - 2bc^3 + c^4 - b^2 + 4bc - c^2 + 2} \\ = -\tan \angle QIP.$$

This completes the proof. $\square$

Problem 95. (IMO Shortlist 2019 G7) The incircle ω of acute-angled scalene triangle ABC has centre I and meets sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω again at R . Line AR meets ω again at P . The circumcircles of triangles PCE and PBF meet again at $Q \neq P$. Prove that lines DI and PQ meet on the external bisector of angle BAC .



Solution.



Let $D(0,0)$, $I(0,1)$, $B(b,0)$ and $C(c,0)$. Then $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$, $E\left(\frac{2c}{c^2+1}, \frac{2c^2}{c^2+1}\right)$ and $F\left(\frac{2b}{b^2+1}, \frac{2b^2}{b^2+1}\right)$. It is easy to deduce

$$m_{DR} = \frac{-1}{m_{EF}} = \frac{bc-1}{b+c}.$$

Therefore, we have

$$DR: \quad y = \frac{bc-1}{b+c}x.$$

Putting this into

$$\odot I: \quad x^2 + y^2 - 2y = 0, \tag{1}$$

we solve

$$\left(1 + \frac{(bc-1)^2}{(b+c)^2}\right)x^2 - \frac{2(bc-1)}{b+c}x = 0.$$

Rejecting the root $x = 0$, we obtain $x = \frac{2(b+c)(bc-1)}{(b^2+1)(c^2+1)}$, and hence

$$R\left(\frac{2(b+c)(bc-1)}{(b^2+1)(c^2+1)}, \frac{2(bc-1)^2}{(b^2+1)(c^2+1)}\right).$$

Then we have

$$m_{AR} = \frac{2bc(b^2+1)(c^2+1) - 2(bc-1)^2(bc+1)}{(b+c)(b^2+1)(c^2+1) - 2(b+c)(bc-1)(bc+1)} = \frac{2(b^3c + b^2c^2 + bc^3 + 2bc - 1)}{(b+c)(-b^2c^2 + b^2 + c^2 + 3)}.$$

It follows that

$$AR: \quad y = \frac{2(b^3c + b^2c^2 + bc^3 + 2bc - 1)}{(b+c)(-b^2c^2 + b^2 + c^2 + 3)}x + \frac{2(-b^2c^2 + 1)}{-b^2c^2 + b^2 + c^2 + 3}.$$

We first rewrite this as

$$AR: \quad x = \frac{(b+c)(-b^2c^2 + b^2 + c^2 + 3)}{2(b^3c + b^2c^2 + bc^3 + 2bc - 1)}y + \frac{(b+c)(b^2c^2 - 1)}{b^3c + b^2c^2 + bc^3 + 2bc - 1}.$$

Putting this into (1), we solve

$$\left(\frac{(b+c)^2(-b^2c^2 + b^2 + c^2 + 3)^2}{4(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2} + 1\right)y^2 + (\dots)y + \frac{(b+c)^2(b^2c^2 - 1)^2}{(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2} = 0.$$

The roots are the y -coordinates of R and P . This implies

$$\begin{aligned} y_P &= \frac{\frac{(b+c)^2(b^2c^2 - 1)^2}{(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2}}{\frac{(b+c)^2(-b^2c^2 + b^2 + c^2 + 3)^2 + 4(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2}{4(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2}} \div \frac{2(bc-1)^2}{(b^2+1)(c^2+1)} \\ &= \frac{2(b+c)^2(bc+1)^2(b^2+1)(c^2+1)}{(b+c)^2(-b^2c^2 + b^2 + c^2 + 3)^2 + 4(b^3c + b^2c^2 + bc^3 + 2bc - 1)^2} \\ &= \frac{2(b+c)^2(bc+1)^2(b^2+1)(c^2+1)}{b^6c^4 + 2b^5c^5 + b^4c^6 + 2b^6c^2 + 4b^5c^3 + 8b^4c^4 + 4b^3c^5 + 2b^2c^6 + b^6 + 2b^5c + 13b^4c^2 \\ &\quad + 8b^3c^3 + 13b^2c^4 + 2bc^5 + c^6 + 6b^4 + 4b^3c + 20b^2c^2 + 4bc^3 + 6c^4 + 9b^2 + 2bc + 9c^2 + 4} \\ &= \frac{2(b+c)^2(bc+1)^2}{b^4c^2 + 2b^3c^3 + b^2c^4 + b^4 + 2b^3c + 6b^2c^2 + 2bc^3 + c^4 + 5b^2 + 2bc + 5c^2 + 4}. \end{aligned}$$

Then

$$\begin{aligned}
x &= \frac{b+c}{b^3c+b^2c^2+bc^3+2bc-1} \left(\frac{-b^2c^2+b^2+c^2+3}{2} \right. \\
&\quad \times \frac{2(b+c)^2(bc+1)^2}{b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4} + (b^2c^2-1) \Big) \\
&= \frac{(b+c)(bc+1)}{b^3c+b^2c^2+bc^3+2bc-1} \\
&\quad \times \frac{2b^5c+2b^4c^2+4b^3c^3+2b^2c^4+2bc^5+8b^3c+4b^2c^2+8bc^3-2b^2+8bc-2c^2-4}{b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4} \\
&= \frac{2(b+c)(bc+1)(b^2+c^2+2)}{b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4}.
\end{aligned}$$

This means

$$P \left(\frac{2(b+c)(bc+1)(b^2+c^2+2)}{b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4}, \frac{2(b+c)^2(bc+1)^2}{b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4} \right).$$

Next, we find the circumcentre of $\triangle BFP$. The perpendicular bisector of BF has slope $\frac{-1}{m_{BF}} = \frac{b^2-1}{2b}$. It passes through the midpoint $\left(\frac{b(b^2+3)}{2(b^2+1)}, \frac{b^2}{b^2+1} \right)$ of BF . Thus, its equation is

$$y = \frac{b^2-1}{2b}x - \frac{b^2-3}{4}. \quad (2)$$

Next, we have

$$\begin{aligned}
m_{FP} &= \frac{2b^2(b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4) - 2(b+c)^2(bc+1)^2(b^2+1)}{2b(b^4c^2+2b^3c^3+b^2c^4+b^4+2b^3c+6b^2c^2+2bc^3+c^4+5b^2+2bc+5c^2+4) - 2(b+c)(bc+1)(b^2+c^2+2)(b^2+1)} \\
&= \frac{b^6+b^4c^2-2b^3c^3+4b^4-2b^3c-2bc^3+3b^2-2bc-c^2}{-b^6c+b^4c^3-2b^4c+2b^3c^2+2b^3-3b^2c+2bc^2-c^3+2b-2c} \\
&= \frac{(b-c)(b^2+1)(b^3+b^2c+2bc^2+3b+c)}{(b-c)(b^2+1)(-b^3c-b^2c^2-bc+c^2+2)} \\
&= \frac{b^3+b^2c+2bc^2+3b+c}{-b^3c-b^2c^2-bc+c^2+2}.
\end{aligned}$$

The perpendicular bisector of FP has slope $\frac{-1}{m_{FP}}$. Also, I lies on this perpendicular bisector. So its equation is

$$y = \frac{b^3c+b^2c^2+bc-c^2-2}{b^3+b^2c+2bc^2+3b+c}x + 1. \quad (3)$$

Solving (2) and (3), we obtain

$$\begin{aligned}
x &= \frac{1 + \frac{b^2 - 3}{4}}{\frac{b^2 - 1}{2b} - \frac{b^3c + b^2c^2 + bc - c^2 - 2}{b^3 + b^2c + 2bc^2 + 3b + c}} \\
&= \frac{b(b^3 + b^2c + 2bc^2 + 3b + c)(b^2 + 1)}{2(b^5 - b^4c + 2b^3 - 2b^2c + b - c)} \\
&= \frac{b(b^3 + b^2c + 2bc^2 + 3b + c)(b^2 + 1)}{2(b - c)(b^2 + 1)^2} \\
&= \frac{b(b^3 + b^2c + 2bc^2 + 3b + c)}{2(b - c)(b^2 + 1)}
\end{aligned}$$

and

$$\begin{aligned}
y &= \frac{b^3c + b^2c^2 + bc - c^2 - 2}{b^3 + b^2c + 2bc^2 + 3b + c} \cdot \frac{b(b^3 + b^2c + 2bc^2 + 3b + c)}{2(b - c)(b^2 + 1)} + 1 \\
&= \frac{b^4c + b^3c^2 + 2b^3 - b^2c - bc^2 - 2c}{2(b - c)(b^2 + 1)}.
\end{aligned}$$

As (BFP) passes through $B(b, 0)$, we find that

$$\begin{aligned}
(BFP): \quad x^2 + y^2 - \frac{b(b^3 + b^2c + 2bc^2 + 3b + c)}{(b - c)(b^2 + 1)}x - \frac{b^4c + b^3c^2 + 2b^3 - b^2c - bc^2 - 2c}{(b - c)(b^2 + 1)}y \\
+ \frac{2b^2(b + c)(bc + 1)}{(b - c)(b^2 + 1)} = 0.
\end{aligned}$$

By symmetry,

$$\begin{aligned}
(CEP): \quad x^2 + y^2 - \frac{c(c^3 + c^2b + 2cb^2 + 3c + b)}{(c - b)(c^2 + 1)}x - \frac{c^4b + c^3b^2 + 2c^3 - c^2b - cb^2 - 2b}{(c - b)(c^2 + 1)}y \\
+ \frac{2c^2(c + b)(cb + 1)}{(c - b)(c^2 + 1)} = 0.
\end{aligned}$$

Taking their difference and clearing the denominators, we find that

$$\begin{aligned}
PQ: \quad (3b^4c^2 + 2b^3c^3 + 3b^2c^4 + b^4 + 2b^3c + 10b^2c^2 + 2bc^3 + c^4 + 3b^2 + 2bc + 3c^2)x \\
+ (2b^4c^3 + 2b^3c^4 + 2b^3c^2 + 2b^2c^3 - 2b^2c - 2bc^2 - 2b - 2c)y \\
- 2(b + c)(bc + 1)(2b^2c^2 + b^2 + c^2) = 0.
\end{aligned}$$

Let S be the intersection point of PQ and DI . As DI is the y -axis, we substitute $x = 0$ in the above equation. This gives

$$\begin{aligned}
y &= \frac{2(b + c)(bc + 1)(2b^2c^2 + b^2 + c^2)}{2b^4c^3 + 2b^3c^4 + 2b^3c^2 + 2b^2c^3 - 2b^2c - 2bc^2 - 2b - 2c} \\
&= \frac{2(b + c)(bc + 1)(b^2c^2 - 1)}{2(b + c)(bc + 1)(2b^2c^2 + b^2 + c^2)} \\
&= \frac{2b^2c^2 + b^2 + c^2}{b^2c^2 - 1}.
\end{aligned}$$

Thus, $S\left(0, \frac{2b^2c^2 + b^2 + c^2}{b^2c^2 - 1}\right)$. Finally, we check that

$$m_{AS} = \frac{\frac{2bc}{bc+1} - \frac{2b^2c^2 + b^2 + c^2}{b^2c^2 - 1}}{\frac{b+c}{bc+1}} = -\frac{b+c}{bc-1}.$$

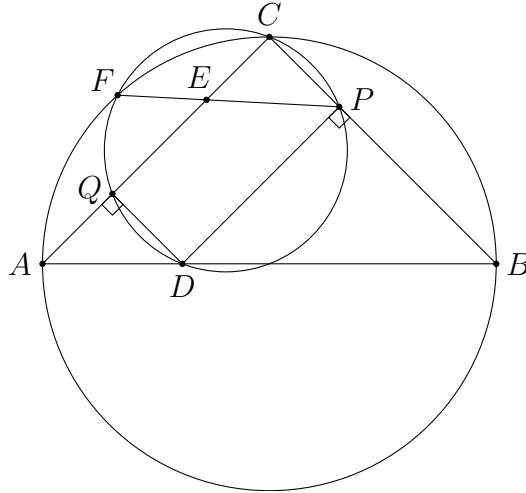
Then

$$m_{AS} \times m_{AI} = -\frac{b+c}{bc-1} \times \frac{bc-1}{b+c} = -1.$$

This implies $AS \perp AI$, and hence AS is the external angle bisector of $\angle BAC$ as desired. $\square$

Problem 96. (IMO Shortlist 2020 G1) Let ABC be an isosceles triangle with $BC = CA$, and let D be a point inside side AB such that $AD < DB$. Let P and Q be two points inside sides BC and CA , respectively, such that $\angle DPB = \angle DQA = 90^\circ$. Let the perpendicular bisector of PQ meet line segment CQ at E , and let the circumcircles of triangles ABC and CPQ meet again at point F , different from C .

Suppose that P, E, F are collinear. Prove that $\angle ACB = 90^\circ$.



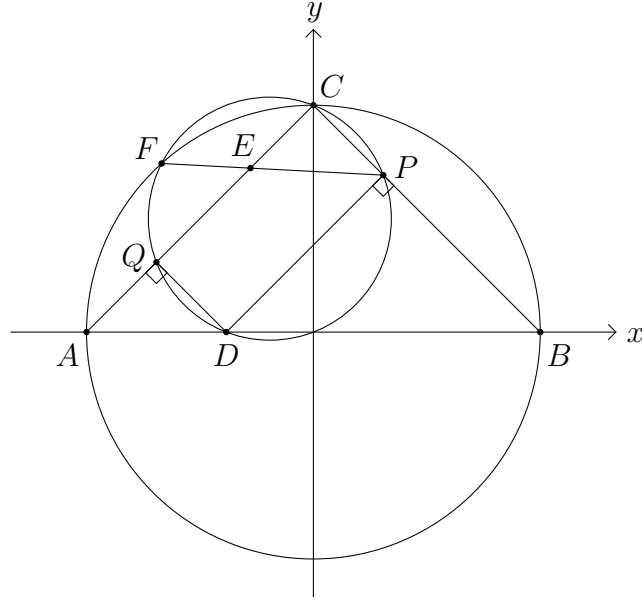
Solution. Let $A(-a, 0)$, $B(a, 0)$, $C(0, c)$ and $D(d, 0)$. Solving

$$BC: \quad y = -\frac{c}{a}x + c,$$

$$DP: \quad y = \frac{a}{c}x - \frac{ad}{c},$$

we obtain $P\left(\frac{a(ad + c^2)}{a^2 + c^2}, \frac{ac(a - d)}{a^2 + c^2}\right)$. By symmetry, changing a to $-a$, we obtain

$$Q\left(\frac{a(ad - c^2)}{a^2 + c^2}, \frac{ac(a + d)}{a^2 + c^2}\right).$$



Recall that

$$(ABC): \quad x^2 + y^2 + \frac{a^2 - c^2}{c}y - a^2 = 0. \quad (1)$$

Note that C, Q, D, P are concyclic as $\angle CPD + \angle CQD = 180^\circ$. Also, the centre of (CPQ) is the midpoint $\left(\frac{d}{2}, \frac{c}{2}\right)$ of CD . This gives

$$(CPQ): \quad x^2 + y^2 - dx - cy = 0. \quad (2)$$

As E is the intersection of the diagonals of the cyclic quadrilateral $CFQP$, and $EP = EQ$, it is obvious that $CFQP$ is an isosceles trapezoid with $CF \parallel PQ$ (for example, $\angle CFE = \angle PQE = \angle QPE = \angle FCE$). For CF , it is the radical axis of (ABC) and (CPQ) , which can be found by taking the difference of (1) and (2). This gives

$$CF: \quad dx + \frac{a^2}{c}y - a^2 = 0.$$

Thus, $m_{CF} = -\frac{cd}{a^2}$. Also, it is easy to deduce

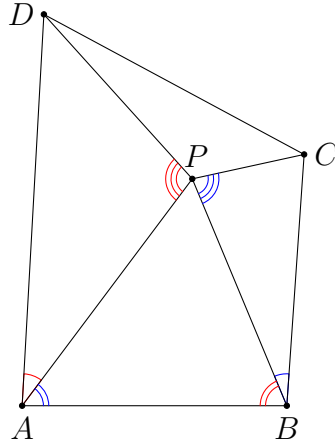
$$m_{PQ} = \frac{\frac{ac(a+d)}{a^2+c^2} - \frac{ac(a-d)}{a^2+c^2}}{\frac{a(ad-c^2)}{a^2+c^2} - \frac{a(ad+c^2)}{a^2+c^2}} = \frac{2acd}{-2ac^2} = -\frac{d}{c}.$$

This implies $-\frac{cd}{a^2} = -\frac{d}{c}$, and hence $c = \pm a$. This means $\angle ACB = 90^\circ$. $\square$

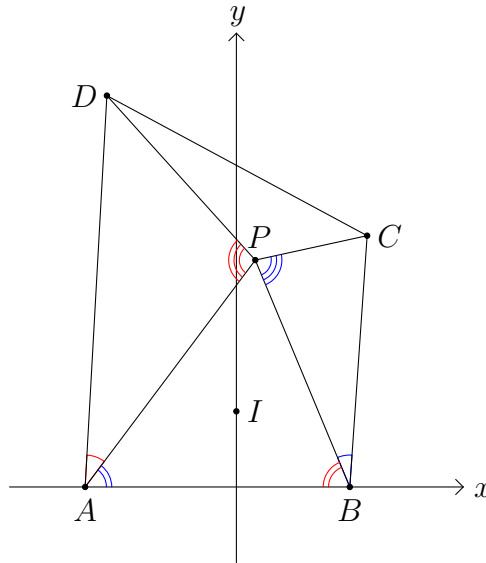
Problem 97. (IMO Shortlist 2020 G2/Problem 1) Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB .



Solution.



Let $I(0, 1)$ be the incentre of $\triangle PAB$, and let $A(a, 0)$, $B(b, 0)$. Then we have $P\left(\frac{a+b}{ab+1}, \frac{2ab}{ab+1}\right)$. Let $\alpha = \angle PAD$ and $\beta = \angle CBP$. Note that

$$\tan \alpha = \tan \angle IBA = \frac{1}{b}$$

and

$$\tan \beta = \tan \angle BAI = -\frac{1}{a}.$$

This implies

$$m_{AD} = \tan(\alpha + 2\beta) = \frac{\tan \alpha + \tan 2\beta}{1 - \tan \alpha \tan 2\beta} = \frac{\frac{1}{b} - \frac{2a}{a^2 - 1}}{1 + \frac{2a}{b(a^2 - 1)}} = \frac{a^2 - 2ab - 1}{a^2b + 2a - b},$$

hence

$$AD: \quad y = \frac{a^2 - 2ab - 1}{a^2b + 2a - b}x - \frac{a(a^2 - 2ab - 1)}{a^2b + 2a - b}. \quad (1)$$

Also, we compute

$$\tan 3\alpha = \frac{\tan \alpha + \tan 2\alpha}{1 - \tan \alpha \tan 2\alpha} = \frac{\frac{1}{b} + \frac{2b}{b^2 - 1}}{1 - \frac{2}{b^2 - 1}} = \frac{3b^2 - 1}{b(b^2 - 3)}.$$

It follows that

$$\begin{aligned} m_{DP} &= \tan(2\beta - 3\alpha) = \frac{\tan 2\beta - \tan 3\alpha}{1 + \tan 2\beta \tan 3\alpha} \\ &= \frac{-\frac{2a}{a^2 - 1} - \frac{3b^2 - 1}{b(b^2 - 3)}}{1 - \frac{2a}{a^2 - 1} \cdot \frac{3b^2 - 1}{b(b^2 - 3)}} \\ &= \frac{-3a^2b^2 - 2ab^3 + a^2 + 6ab + 3b^2 - 1}{a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b}. \end{aligned}$$

In view of the coordinates of P , we compute

$$\begin{aligned} &\frac{2ab}{ab + 1} - \frac{-3a^2b^2 - 2ab^3 + a^2 + 6ab + 3b^2 - 1}{a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b} \cdot \frac{a + b}{ab + 1} \\ &= \frac{2a^3b^4 - 3a^3b^2 - 7a^2b^3 - a^3 - 3a^2b - 3ab^2 - 3b^3 + a + b}{(ab + 1)(a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b)}. \end{aligned}$$

This gives

$$\begin{aligned} DP: \quad y &= \frac{-3a^2b^2 - 2ab^3 + a^2 + 6ab + 3b^2 - 1}{a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b}x \\ &\quad + \frac{2a^3b^4 - 3a^3b^2 - 7a^2b^3 - a^3 - 3a^2b - 3ab^2 - 3b^3 + a + b}{(ab + 1)(a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b)}. \end{aligned} \quad (2)$$

Then we solve (1) and (2). We have

$$\begin{aligned} x &= \frac{a(a^2 - 2ab - 1)(ab + 1)(a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b) \\ &\quad + (a^2b + 2a - b)(2a^3b^4 - 3a^3b^2 - 7a^2b^3 - a^3 - 3a^2b - 3ab^2 - 3b^3 + a + b)}{(ab + 1)((a^2 - 2ab - 1)(a^2b^3 - 3a^2b - 6ab^2 - b^3 + 2a + 3b) \\ &\quad - (a^2b + 2a - b)(-3a^2b^2 - 2ab^3 + a^2 + 6ab + 3b^2 - 1))} \end{aligned}$$

$$\begin{aligned}
& \frac{a^6b^4 - 3a^6b^2 - 2a^5b^3 + 5a^4b^4 - 2a^5b - 7a^4b^2 - 4a^3b^3 + 7a^2b^4 - 4a^3b - 5a^2b^2 - 2ab^3 + 3b^4 - 2ab - b^2}{(ab+1)(4a^4b^3 - 4a^4b + 8a^2b^3 - 8a^2b + 4b^3 - 4b)} \\
&= \frac{b(a^2+1)^2(a^2b^3 - 3a^2b - 2ab^2 + 3b^3 - 2a - b)}{4(ab+1)b(b^2-1)(a^2+1)^2} \\
&= \frac{a^2b^3 - 3a^2b - 2ab^2 + 3b^3 - 2a - b}{4(ab+1)(b^2-1)}
\end{aligned}$$

and

$$\begin{aligned}
y &= \frac{a^2 - 2ab - 1}{a^2b + 2a - b} \left(\frac{a^2b^3 - 3a^2b - 2ab^2 + 3b^3 - 2a - b}{4(ab+1)(b^2-1)} - a \right) \\
&= \frac{a^2 - 2ab - 1}{a^2b + 2a - b} \cdot \frac{-3a^2b^3 + a^2b - 6ab^2 + 3b^3 + 2a - b}{4(ab+1)(b^2-1)} \\
&= \frac{(a^2 - 2ab - 1)(-3b^2 + 1)}{4(ab+1)(b^2-1)}.
\end{aligned}$$

Thus, $D \left(\frac{a^2b^3 - 3a^2b - 2ab^2 + 3b^3 - 2a - b}{4(ab+1)(b^2-1)}, \frac{(a^2 - 2ab - 1)(-3b^2 + 1)}{4(ab+1)(b^2-1)} \right)$. Let ℓ be the internal angle bisector of $\angle ADP$. As $\frac{1}{2}\angle ADP = 90^\circ - 2\alpha$, we have

$$\begin{aligned}
m_\ell &= \tan((\alpha + 2\beta) + (90^\circ - 2\alpha)) = \frac{\tan(\alpha + 2\beta) + \cot 2\alpha}{1 - \tan(\alpha + 2\beta) \cot 2\alpha} \\
&= \frac{\frac{a^2 - 2ab - 1}{a^2b + 2a - b} + \frac{b^2 - 1}{2b}}{1 - \frac{a^2 - 2ab - 1}{a^2b + 2a - b} \cdot \frac{b^2 - 1}{2b}} = \frac{a^2b^3 + a^2b - 2ab^2 - b^3 - 2a - b}{a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1}.
\end{aligned}$$

This implies

$$\begin{aligned}
\ell: \quad y &= \frac{a^2b^3 + a^2b - 2ab^2 - b^3 - 2a - b}{a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1}x + \frac{(a^2 - 2ab - 1)(-3b^2 + 1)}{4(ab+1)(b^2-1)} \\
&\quad - \frac{a^2b^3 + a^2b - 2ab^2 - b^3 - 2a - b}{a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1} \cdot \frac{a^2b^3 - 3a^2b - 2ab^2 + 3b^3 - 2a - b}{4(ab+1)(b^2-1)},
\end{aligned}$$

i.e.

$$\ell: \quad y = \frac{a^2b^3 + a^2b - 2ab^2 - b^3 - 2a - b}{a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1}x + \frac{-a^4b^4 - 2a^4b^2 + 4a^3b^3 + 10a^2b^4 - a^4 + 4a^3b + 16a^2b^2 + 4ab^3 + 3b^4 + 6a^2 + 4ab + 2b^2 - 1}{4(ab+1)(a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1)}.$$

Putting $x = \frac{a+b}{2}$, we obtain the y -coordinate of the intersection point of ℓ and the perpendicular bisector of AB .

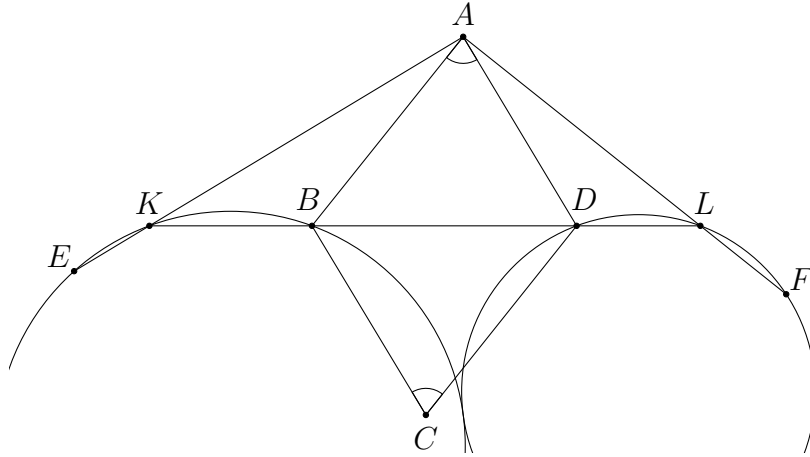
The result is

$$y = \frac{a^4b^4 + 2a^3b^5 + 4a^3b^3 + 6a^2b^4 - 2ab^5 - a^4 + 2a^3b + 8a^2b^2 - 4ab^3 + b^4 + 2a^2 - 2ab - 1}{4(ab+1)(a^2b^2 + 2ab^3 + a^2 + 2ab - b^2 - 1)}$$

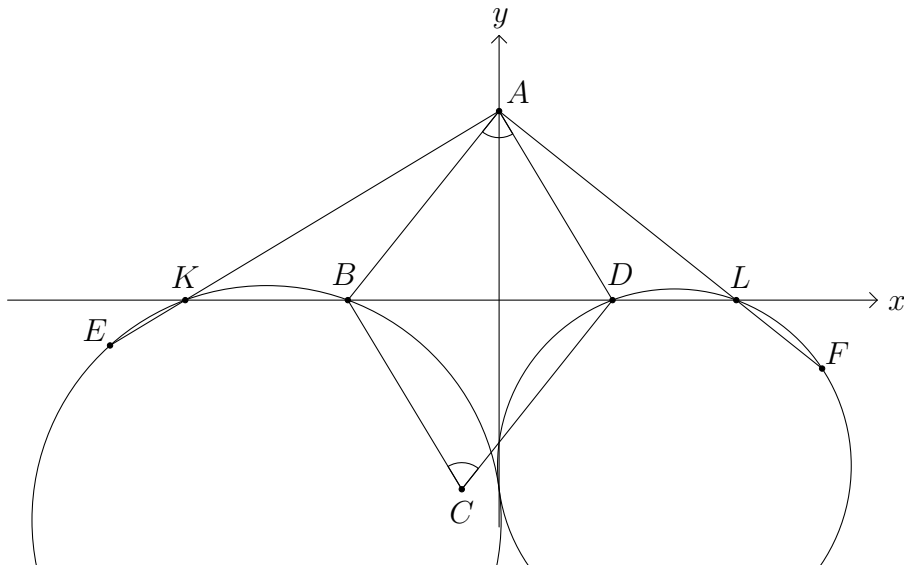
$$= \frac{a^2b^2 - a^2 + 4ab - b^2 + 1}{4(ab+1)}.$$

As this is symmetric in a and b , the internal angle bisector of $\angle PCB$ also passes through the same point. This completes the proof. $\square$

Problem 98. (IMO Shortlist 2020 G3) Let $ABCD$ be a convex quadrilateral with $\angle ABC > 90^\circ$, $\angle CDA > 90^\circ$, and $\angle DAB = \angle BCD$. Denote by E and F the reflections of A in lines BC and CD , respectively. Suppose that the segments AE and AF meet the line BD at K and L , respectively. Prove that the circumcircles of triangles BEK and DFL are tangent to each other.



Solution.



Let $A(0, a)$, $B(b, 0)$, $C(s, t)$ and $D(d, 0)$. Firstly,

$$\begin{aligned}
& \angle DCB = \angle BAD \\
\Rightarrow & \frac{m_{CB} - m_{CD}}{1 + m_{CB} \cdot m_{CD}} = \frac{m_{AD} - m_{AB}}{1 + m_{AD} \cdot m_{AB}} \\
\Rightarrow & \frac{\frac{t}{s-b} - \frac{t}{s-d}}{1 + \frac{t^2}{(s-b)(s-d)}} = \frac{-\frac{a}{d} + \frac{a}{b}}{1 + \frac{a^2}{bd}} \\
\Rightarrow & \frac{t(b-d)}{(s-b)(s-d) + t^2} = \frac{a(d-b)}{bd + a^2} \\
\Rightarrow & a(s-b)(s-d) + at^2 + a^2t + bdt = 0.
\end{aligned} \tag{1}$$

Next, we have

$$BC: \quad y = \frac{t}{s-b}x - \frac{bt}{s-b}$$

and

$$AE: \quad y = -\frac{s-b}{t}x + a. \tag{2}$$

Solving these, we obtain

$$x = \frac{t(a(s-b) + bt)}{(s-b)^2 + t^2}$$

and

$$y = \frac{t}{s-b} \left(\frac{t(a(s-b) + bt)}{(s-b)^2 + t^2} - b \right) = \frac{t}{s-b} \cdot \frac{at(s-b) - b(s-b)^2}{(s-b)^2 + t^2} = \frac{t(at - bs + b^2)}{(s-b)^2 + t^2}.$$

This means $\left(\frac{t(as + bt - ab)}{(s-b)^2 + t^2}, \frac{t(at - bs + b^2)}{(s-b)^2 + t^2} \right)$ are the coordinates of the intersection point of AE and BC . This is the midpoint of AE . Thus,

$$E \left(\frac{2t(as + bt - ab)}{(s-b)^2 + t^2}, \frac{-as^2 + at^2 - 2bst + 2abs + 2b^2t - ab^2}{(s-b)^2 + t^2} \right).$$

Putting $y = 0$ in (2), we find that $K \left(\frac{at}{s-b}, 0 \right)$. Thus, the centre of (BEK) lies on the line $x = \frac{at + bs - b^2}{2(s-b)}$. The slope of the perpendicular bisector of EK is $\frac{-1}{m_{AE}} = \frac{t}{s-b}$. Its equation is

$$\begin{aligned}
y = & \frac{t}{s-b}x + \frac{-as^2 + at^2 - 2bst + 2abs + 2b^2t - ab^2}{2((s-b)^2 + t^2)} \\
& - \frac{t}{s-b} \cdot \frac{2t(s-b)(as + bt - ab) + at(s-b)^2 + at^3}{2(s-b)((s-b)^2 + t^2)}.
\end{aligned}$$

Putting $x = \frac{at + bs - b^2}{2(s - b)}$, we have

$$\begin{aligned}
y &= \frac{\begin{aligned} &\textcolor{red}{t}(at + bs - b^2)((s - b)^2 + \textcolor{red}{t}^2) + (s - b)^2(-as^2 + at^2 - 2bst + 2abs + 2b^2t - ab^2) \\ &\textcolor{red}{-t}(2t(s - b)(as + bt - ab) + at(s - b)^2 + \textcolor{red}{at}^3) \end{aligned}}{2(s - b)^2((s - b)^2 + t^2)} \\
&= \frac{\begin{aligned} &t(at + bs - b^2)(s - b)^2 + (s - b)^2(-as^2 + at^2 - 2bst + 2abs + 2b^2t - ab^2) \\ &- 2t^2(s - b)(as + bt - ab) - at^2(s - b)^2 + \textcolor{red}{bt}^3(s - b) \end{aligned}}{2(s - b)^2((s - b)^2 + t^2)} \\
&= \frac{\begin{aligned} &t(at + bs - b^2)(s - b) + (s - b)(-as^2 + at^2 - 2bst + 2abs + 2b^2t - ab^2) \\ &- 2t^2(as + bt - ab) - at^2(s - b) + bt^3 \end{aligned}}{2(s - b)((s - b)^2 + t^2)} \\
&= \frac{-as^3 - ast^2 - bs^2t - bt^3 + 3abs^2 + abt^2 + 2b^2st - 3ab^2s - b^3t + ab^3}{2(s - b)((s - b)^2 + t^2)} \\
&= \frac{((s - b)^2 + t^2)(-as - bt + ab)}{2(s - b)((s - b)^2 + t^2)} \\
&= \frac{-as - bt + ab}{2(s - b)}.
\end{aligned}$$

As (BEK) passes through $B(b, 0)$, it follows that

$$(BEK): \quad x^2 + y^2 - \frac{at + bs - b^2}{s - b}x + \frac{as + bt - ab}{s - b}y + \frac{abt}{s - b} = 0. \quad (3)$$

By symmetry,

$$(DFL): \quad x^2 + y^2 - \frac{at + ds - d^2}{s - d}x + \frac{as + dt - ad}{s - d}y + \frac{adt}{s - d} = 0.$$

Taking their difference, we obtain the equation of the radical axis

$$\left(\frac{at + ds - d^2}{s - d} - \frac{at + bs - b^2}{s - b} \right)x + \left(\frac{as + bt - ab}{s - b} - \frac{as + dt - ad}{s - d} \right)y + \left(\frac{abt}{s - b} - \frac{adt}{s - d} \right) = 0.$$

This is the same as

$$(d - b)(s^2 + at - bs - ds + bd)x + (b - d)sty + ast(b - d) = 0,$$

i.e.

$$y = \frac{s^2 + at - bs - ds + bd}{st}x - a.$$

Using (1), we can rewrite

$$s^2 + at - bs - ds + bd = \frac{a(s - b)(s - d) + a^2t}{a} = -\frac{at^2 + bdt}{a}.$$

Thus, the above equation can be rewritten as

$$y = -\frac{at + bd}{as}x - a.$$

We have to show that this is tangent to (BEK) (which implies it is a common tangent to both circles). Thus, we substitute this into (3) and solve the quadratic equation in x . The constant term is

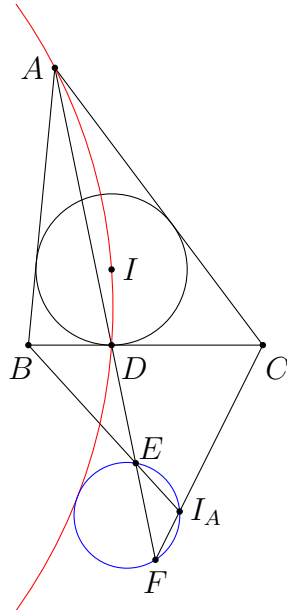
$$a^2 + \frac{as + bt - ab}{s - b} \cdot (-a) + \frac{abt}{s - b} = 0,$$

while the coefficient of x is

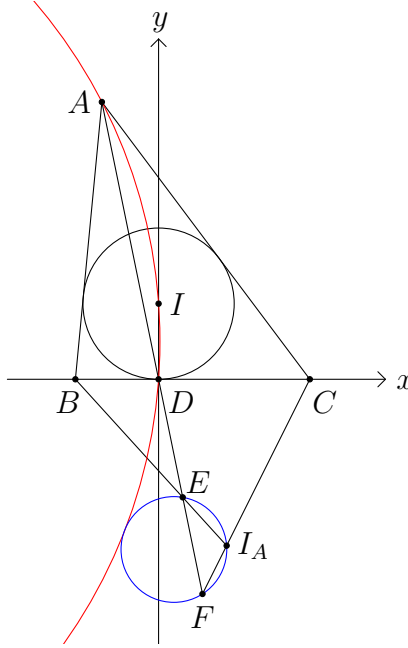
$$\begin{aligned} & \frac{2a(at + bd)}{as} - \frac{at + bs - b^2}{s - b} - \frac{(as + bt - ab)(at + bd)}{as(s - b)} \\ &= \frac{2a(s - b)(at + bd) - as(at + bs - b^2) - (as + bt - ab)(at + bd)}{as(s - b)} \\ &= \frac{-abs^2 - abt^2 + ab^2s + abds - a^2bt - b^2dt - ab^2d}{as(s - b)} \\ &= \frac{-b(a(s - b)(s - d) + at^2 + a^2t + bdt)}{as(s - b)} = 0 \end{aligned}$$

by (1). Therefore, the quadratic equation is essentially $x^2 = 0$, which has a double root. This proves (BEK) and (DFL) are tangent. $\square$

Problem 99. (IMO Shortlist 2020 G6) Let I and I_A be the incentre and the A -excentre of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D . The line AD meets BI_A and CI_A at E and F , respectively. Prove that the circumcircles of triangles AID and I_AEF are tangent to each other.



Solution.



Let $D(0,0)$, $I(0,1)$, $B(b,0)$ and $C(c,0)$. Then we have $A\left(\frac{b+c}{bc+1}, \frac{2bc}{bc+1}\right)$. Note that $BI_A \perp BI$. Therefore, $m_{BI_A} = \frac{-1}{m_{BI}} = b$. This implies

$$BI_A: y = bx - b^2.$$

By symmetry,

$$CI_A: y = cx - c^2.$$

Solving these, we easily obtain $I_A(b+c, bc)$. Also, we have

$$AD: y = \frac{2bc}{b+c}x.$$

This yields $E\left(\frac{b(b+c)}{b-c}, \frac{2b^2c}{b-c}\right)$ and $F\left(\frac{c(c+b)}{c-b}, \frac{2c^2b}{c-b}\right)$. Next, as the perpendicular bisector of DI is $y = \frac{1}{2}$, we have

$$(ADI): x^2 + y^2 + \alpha x - y = 0$$

for some constant α . Putting $x = \frac{b+c}{bc+1}$ and $y = \frac{2bc}{bc+1}$, we obtain

$$\alpha = \frac{bc+1}{b+c} \left(\frac{2bc}{bc+1} - \frac{(b+c)^2}{(bc+1)^2} - \frac{4b^2c^2}{(bc+1)^2} \right) = -\frac{2b^2c^2 + b^2 + c^2}{(bc+1)(b+c)}.$$

In other words,

$$(ADI): x^2 + y^2 - \frac{2b^2c^2 + b^2 + c^2}{(bc+1)(b+c)}x - y = 0. \quad (1)$$

On the other hand, the perpendicular bisector of EI_A has slope $\frac{-1}{m_{EI_A}} = -\frac{1}{b}$. Its equation is

$$y = -\frac{1}{b}x + \frac{1}{b} \cdot \frac{1}{2} \left(\frac{b(b+c)}{b-c} + (b+c) \right) + \frac{1}{2} \left(\frac{2b^2c}{b-c} + bc \right),$$

i.e.

$$y = -\frac{1}{b}x + \frac{3b^3c - b^2c^2 + 2b^2 + bc - c^2}{2b(b-c)}.$$

By symmetry, the perpendicular bisector of FI_A has equation

$$y = -\frac{1}{c}x + \frac{3c^3b - b^2c^2 + 2c^2 + bc - b^2}{2c(c-b)}.$$

Solving these two equations, we obtain

$$\begin{aligned} x &= \frac{c(3b^3c - b^2c^2 + 2b^2 + bc - c^2) + b(3c^3b - b^2c^2 + 2c^2 + bc - b^2)}{2(b-c)(c-b)} \\ &= \frac{-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3}{2(b-c)^2} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{1}{b} \cdot \frac{-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3}{2(b-c)^2} + \frac{3b^3c - b^2c^2 + 2b^2 + bc - c^2}{2b(b-c)} \\ &= \frac{-(-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3) + (b-c)(3b^3c - b^2c^2 + 2b^2 + bc - c^2)}{2b(b-c)^2} \\ &= \frac{3b^4c - 2b^3c^2 + 3b^2c^3 + b^3 + 2b^2c + bc^2}{2b(b-c)^2} \\ &= \frac{3b^3c - 2b^2c^2 + 3bc^3 + b^2 + 2bc + c^2}{2(b-c)^2}. \end{aligned}$$

These are the coordinates of the centre of (EFI_A) . Therefore,

$$\begin{aligned} (EFI_A) : \quad & x^2 + y^2 - \frac{-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3}{(b-c)^2}x \\ & - \frac{3b^3c - 2b^2c^2 + 3bc^3 + b^2 + 2bc + c^2}{(b-c)^2}y - (b+c)^2 - (bc)^2 \\ & + \frac{-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3}{(b-c)^2}(b+c) + \frac{3b^3c - 2b^2c^2 + 3bc^3 + b^2 + 2bc + c^2}{(b-c)^2}bc = 0, \end{aligned}$$

i.e.

$$\begin{aligned} (EFI_A) : \quad & x^2 + y^2 - \frac{-2b^3c^2 - 2b^2c^3 + b^3 - 3b^2c - 3bc^2 + c^3}{(b-c)^2}x \\ & - \frac{3b^3c - 2b^2c^2 + 3bc^3 + b^2 + 2bc + c^2}{(b-c)^2}y - \frac{bc(4b^2c^2 + b^2 + 2bc + c^2)}{(b-c)^2} = 0 \end{aligned} \quad (2)$$

Taking the difference of (1) and (2), and clearing denominators, we obtain

$$bc(2b^4c^2 + 4b^3c^3 + 2b^2c^4 - b^4 + 6b^3c + 6b^2c^2 + 6bc^3 - c^4 + 8bc)x - bc(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)y - bc(b+c)(bc+1)(4b^2c^2 + b^2 + 2bc + c^2) = 0.$$

This means

$$y = \frac{2b^4c^2 + 4b^3c^3 + 2b^2c^4 - b^4 + 6b^3c + 6b^2c^2 + 6bc^3 - c^4 + 8bc}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)}x - \frac{4b^2c^2 + b^2 + 2bc + c^2}{3b^2 - 2bc + 3c^2 + 4}.$$

We have to show that this radical axis is tangent to (ADI). So we put it into (1) and obtain a quadratic equation in x . Firstly, the constant term is

$$\begin{aligned}\delta &= \left(\frac{4b^2c^2 + b^2 + 2bc + c^2}{3b^2 - 2bc + 3c^2 + 4} \right)^2 + \frac{4b^2c^2 + b^2 + 2bc + c^2}{3b^2 - 2bc + 3c^2 + 4} \\ &= \frac{(4b^2c^2 + b^2 + 2bc + c^2)(4b^2c^2 + b^2 + 2bc + c^2)}{(3b^2 - 2bc + 3c^2 + 4)^2} \\ &= \frac{4(4b^2c^2 + b^2 + 2bc + c^2)(b^2 + 1)(c^2 + 1)}{(3b^2 - 2bc + 3c^2 + 4)^2}.\end{aligned}$$

Secondly, the coefficient of x^2 is

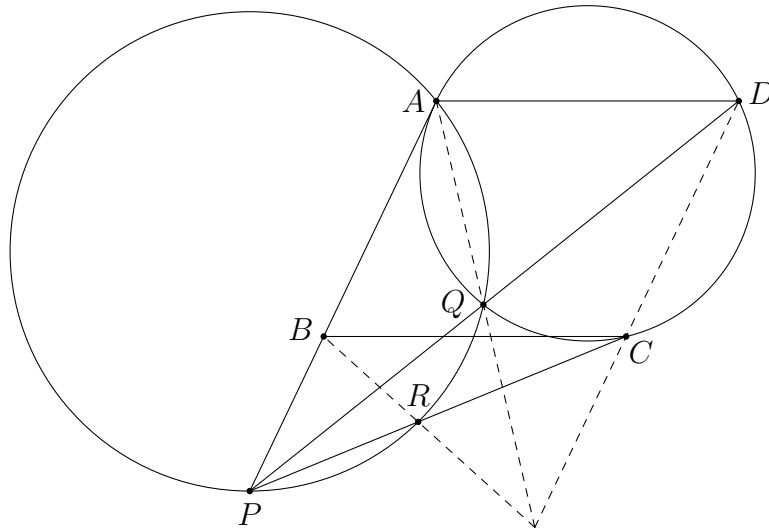
$$\begin{aligned}\beta &= \left(\frac{2b^4c^2 + 4b^3c^3 + 2b^2c^4 - b^4 + 6b^3c + 6b^2c^2 + 6bc^3 - c^4 + 8bc}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)} \right)^2 + 1 \\ &\quad \frac{4b^4c^4(b+c)^4 + b^2c^2(5b^6 + 22b^5c + 75b^4c^2 + 116b^3c^3 + 75b^2c^4 + 22bc^5 + 5c^6) \\ &\quad + (b^8 + 6b^7c + 60b^6c^2 + 138b^5c^3 + 230b^4c^4 + 138b^3c^5 + 60b^2c^6 + 6bc^7 + c^8) \\ &\quad + (9b^6 + 38b^5c + 183b^4c^2 + 180b^3c^3 + 183b^2c^4 + 38bc^5 + 9c^6) \\ &\quad + 8(3b^4 + 8b^3c + 18b^2c^2 + 8bc^3 + 3c^4) + 16(b+c)^2}{(b+c)^2(bc+1)^2(3b^2 - 2bc + 3c^2 + 4)^2} \\ &= \frac{(4b^2c^2 + b^2 + 2bc + c^2)(b^2 + 1)(c^2 + 1)((b+c)^2 + 4)^2}{(b+c)^2(bc+1)^2(3b^2 - 2bc + 3c^2 + 4)^2}.\end{aligned}$$

(The first three factors can be guessed from the factors of δ .) Thirdly, the coefficient of x is

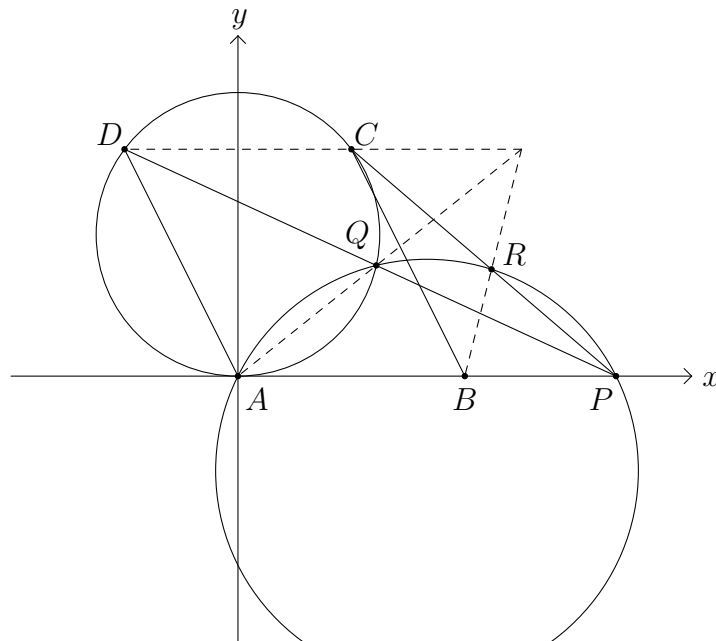
$$\begin{aligned}\gamma &= -\frac{2(4b^2c^2 + b^2 + 2bc + c^2)(2b^4c^2 + 4b^3c^3 + 2b^2c^4 - b^4 + 6b^3c + 6b^2c^2 + 6bc^3 - c^4 + 8bc)}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)^2} \\ &\quad - \frac{2b^2c^2 + b^2 + c^2}{(bc+1)(b+c)} - \frac{2b^4c^2 + 4b^3c^3 + 2b^2c^4 - b^4 + 6b^3c + 6b^2c^2 + 6bc^3 - c^4 + 8bc}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)} \\ &\quad \frac{-16b^4c^4(b+c)^2 - 4b^2c^2(5b^4 + 12b^3c + 30b^2c^2 + 12bc^3 + 5c^4) \\ &\quad - 4(b^6 + 4b^5c + 31b^4c^2 + 24b^3c^3 + 31b^2c^4 + 4bc^5 + c^6) \\ &\quad - 4(5b^4 + 12b^3c + 30b^2c^2 + 12bc^3 + 5c^4) - 16(b+c)^2}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)^2} \\ &= \frac{-4(4b^2c^2 + b^2 + 2bc + c^2)(b^2 + 1)(c^2 + 1)((b+c)^2 + 4)}{(b+c)(bc+1)(3b^2 - 2bc + 3c^2 + 4)^2}.\end{aligned}$$

It is now obvious that $\gamma^2 = 4\beta\delta$. Therefore, the quadratic equation has a unique solution, and we are done. $\square$

Problem 100. (IMO Shortlist 2021 G1) Let $ABCD$ be a parallelogram such that $AC = BC$. A point P is chosen on the extension of the segment AB beyond B . The circumcircle of the triangle ACD meets the segment PD again at Q , and the circumcircle of the triangle APQ meets the segment PC again at R . Prove that the lines CD , AQ , and BR are concurrent.



Solution.



Let $A(0,0)$, $B(2a,0)$, $C(a,c)$ and $P(b,0)$. Then $D = A + C - B = (-a, c)$. Clearly, the centre of (ACD) lies on the y -axis. As it passes through A , its equation is of the form

$$x^2 + y^2 + \alpha y = 0$$

for some constant α . Putting $x = a$ and $y = c$, we obtain $\alpha = -\frac{c^2 + a^2}{c}$. Thus,

$$(ACD): \quad x^2 + y^2 - \frac{c^2 + a^2}{c}y = 0.$$

Putting

$$DP: \quad y = -\frac{c}{a+b}x + \frac{bc}{a+b}$$

into this equation, we solve

$$\left(1 + \frac{c^2}{(a+b)^2}\right)x^2 + (\dots)x + \left(\frac{b^2c^2}{(a+b)^2} - \frac{b(c^2 + a^2)}{a+b}\right) = 0.$$

One of the roots is $-a$, while the other root is

$$\frac{\frac{b^2c^2}{(a+b)^2} - \frac{b(c^2 + a^2)}{a+b}}{1 + \frac{c^2}{(a+b)^2}} \div (-a) = \frac{b(c^2 + a^2 + ab)}{c^2 + (a+b)^2}.$$

Then we have

$$y = \frac{c}{a+b} \left(b - \frac{b(c^2 + a^2 + ab)}{c^2 + (a+b)^2} \right) = \frac{c}{a+b} \cdot \frac{b^2(a+b)}{c^2 + (a+b)^2} = \frac{b^2c}{c^2 + (a+b)^2}.$$

This means $Q\left(\frac{b(c^2 + a^2 + ab)}{c^2 + (a+b)^2}, \frac{b^2c}{c^2 + (a+b)^2}\right)$. Similarly, the x -coordinate of the centre of (APQ) is $\frac{b}{2}$. So the equation of (APQ) is of the form

$$x^2 + y^2 - bx + \beta y = 0$$

for some constant β . Putting $x = \frac{b(c^2 + a^2 + ab)}{c^2 + (a+b)^2}$ and $y = \frac{b^2c}{c^2 + (a+b)^2}$, we obtain

$$\begin{aligned} \beta &= \frac{c^2 + (a+b)^2}{b^2c} \left(-\frac{b^2(c^2 + a^2 + ab)^2}{(c^2 + (a+b)^2)^2} - \frac{b^4c^2}{(c^2 + (a+b)^2)^2} + \frac{b^2(c^2 + a^2 + ab)}{c^2 + (a+b)^2} \right) \\ &= \frac{c^2 + (a+b)^2}{b^2c} \cdot \frac{ab^3}{c^2 + (a+b)^2} \\ &= \frac{ab}{c}. \end{aligned}$$

Therefore,

$$(APQ): \quad x^2 + y^2 - bx + \frac{ab}{c}y = 0.$$

Putting

$$PC: \quad y = \frac{c}{a-b}x - \frac{bc}{a-b}$$

into this equation, we solve

$$\left(1 + \frac{c^2}{(a-b)^2}\right)x^2 + (\dots) + \left(\frac{b^2c^2}{(a-b)^2} - \frac{ab^2}{a-b}\right) = 0.$$

One of the roots is b , while the other root is

$$\frac{\frac{b^2c^2}{(a-b)^2} - \frac{ab^2}{a-b}}{1 + \frac{c^2}{(a-b)^2}} \div b = \frac{b(c^2 - a^2 + ab)}{c^2 + (a-b)^2}.$$

Then we have

$$y = \frac{c}{a-b} \left(\frac{b(c^2 - a^2 + ab)}{c^2 + (a-b)^2} - b \right) = \frac{c}{a-b} \cdot \frac{b(a-b)(-2a+b)}{c^2 + (a-b)^2} = \frac{bc(-2a+b)}{c^2 + (a-b)^2}.$$

This means $R\left(\frac{b(c^2 - a^2 + ab)}{c^2 + (a-b)^2}, \frac{bc(-2a+b)}{c^2 + (a-b)^2}\right)$. Next, it is obvious that

$$AQ: \quad y = \frac{bc}{c^2 + a^2 + ab}x.$$

Its intersection with

$$CD: \quad y = c$$

is $\left(\frac{c^2 + a^2 + ab}{b}, c\right)$. Also, we have

$$\begin{aligned} m_{BR} &= \frac{bc(-2a+b)}{b(c^2 - a^2 + ab) - 2a(c^2 + (a-b)^2)} \\ &= \frac{bc(-2a+b)}{(-2a+b)c^2 + a(a-b)(-2a+b)} = \frac{bc}{c^2 + a^2 - ab}. \end{aligned}$$

This gives

$$BR: \quad y = \frac{bc}{c^2 + a^2 - ab}x - \frac{2abc}{c^2 + a^2 - ab}.$$

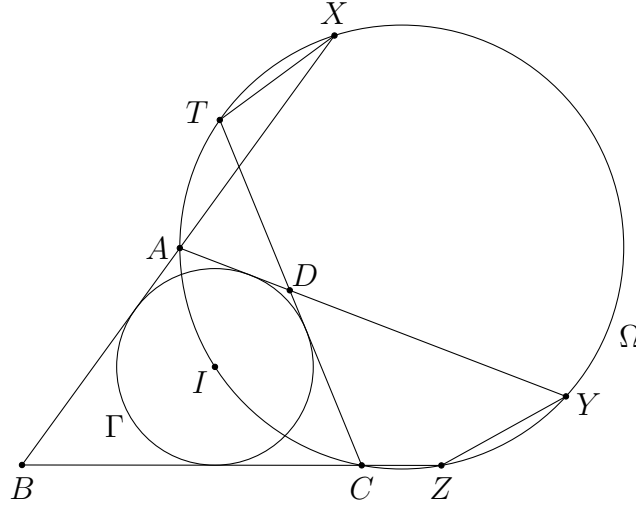
We check that

$$\frac{bc}{c^2 + a^2 - ab} \cdot \frac{c^2 + a^2 + ab}{b} - \frac{2abc}{c^2 + a^2 - ab} = \frac{c(c^2 + a^2 - ab)}{c^2 + a^2 - ab} = c.$$

Therefore, $\left(\frac{c^2 + a^2 + ab}{b}, c\right)$ lies on BR as well. This shows AQ, CD, BR are concurrent. $\square$

Problem 101. (IMO Shortlist 2021 G2/Problem 4) Let Γ be a circle with centre I , and $ABCD$ a convex quadrilateral such that each of the segments AB, BC, CD and DA is tangent to Γ . Let Ω be the circumcircle of the triangle AIC . The extension of BA beyond A meets Ω at X , and the extension of BC beyond C meets Ω at Z . The extensions of AD and CD beyond D meet Ω at Y and T , respectively. Prove that

$$AD + DT + TX + XA = CD + DY + YZ + ZC.$$



Solution. Let $O(0,0)$ be the centre of Ω , and let $I(0,1)$. Let $A(a,b)$ and $C(c,d)$. Note that

$$a^2 + b^2 = c^2 + d^2 = 1.$$

Let

$$\Gamma: x^2 + (y-1)^2 = r^2.$$

Consider a tangent $y = mx - ma + b$ to Γ passing through A . Putting this into the above equation of Γ , we have

$$(m^2 + 1)x^2 - 2m(ma - b + 1)x + (ma - b + 1)^2 - r^2 = 0.$$

There is a unique solution in x . Therefore,

$$\Delta = 4m^2(ma - b + 1)^2 - 4(m^2 + 1)((ma - b + 1)^2 - r^2) = 0.$$

After expansion, this becomes

$$(r^2 - a^2)m^2 + 2a(b-1)m + r^2 - (b-1)^2 = 0. \quad (1)$$

Putting $y = mx - ma + b$ into $x^2 + y^2 = 1$, we solve

$$(1 + m^2)x^2 - 2m(ma - b)x + (ma - b)^2 - 1 = 0.$$

One of the roots is a , and so the other root is

$$\frac{2m(ma - b)}{1 + m^2} - a = \frac{am^2 - 2bm - a}{m^2 + 1}.$$



Using (1) and $a^2 + b^2 = 1$, we can rewrite this as

As this is independent of m , the y -coordinates of X and Y are the same. Since X and Y lie on Ω , we may let $X(\alpha, \beta)$ and $Y(-\alpha, \beta)$. Similarly, let $T(\gamma, \delta)$ and $Z(-\gamma, \delta)$.

Let E, F, G be the contact points of BA, AD, DC and Γ respectively. Then

$$\begin{aligned} AD + DT + TX + XA &= (\textcolor{red}{AF} + \textcolor{blue}{DF}) + (TG - \textcolor{blue}{DG}) + TX + (XE - \textcolor{red}{AE}) \\ &= TX + TG + XE. \end{aligned}$$

Similarly, $CD + DY + YZ + ZC$ is equal to YZ plus the lengths of tangents from Y and Z to Γ . Indeed,

$$TX = \sqrt{(\alpha - \gamma)^2 + (\beta - \delta)^2} = YZ.$$

The tangent from T to Γ has length

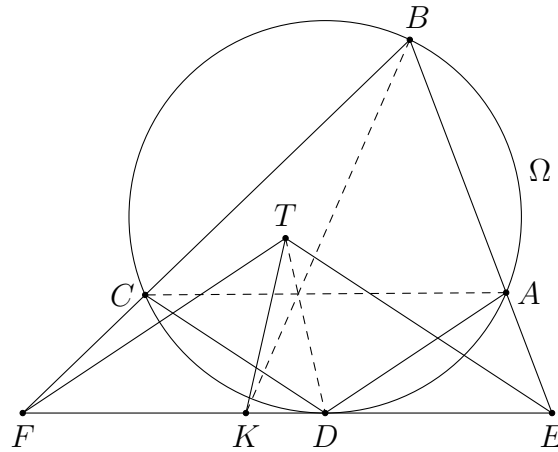
$$\sqrt{IT^2 - IG^2} = \sqrt{\gamma^2 + (\delta - 1)^2 - r^2}.$$

By symmetry, this is equal to the length of the tangent from Z to Γ . Similarly, the tangents from X and Y to Γ have the same length. Therefore, we have proved

$$AD + DT + TX + XA = CD + DY + YZ + ZC.$$

□

Problem 102. (IMO Shortlist 2021 G4) Let $ABCD$ be a quadrilateral inscribed in a circle Ω . Let the tangent to Ω at D intersect the rays BA and BC at points E and F , respectively. A point T is chosen inside the triangle ABC so that $TE \parallel CD$ and $TF \parallel AD$. Let $K \neq D$ be a point on the segment DF such that $TD = TK$. Prove that the lines AC, DT and BK intersect at one point.

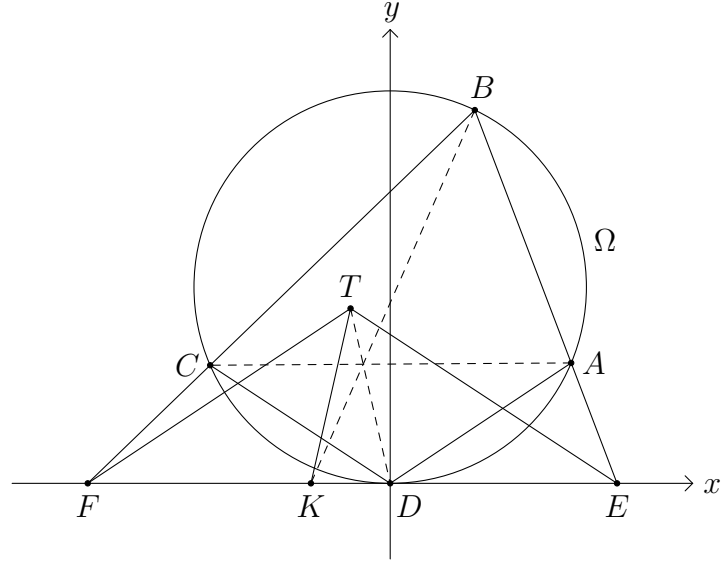


Solution. Let $D(0,0)$, $E(a,0)$, $F(b,0)$ and $B(c,d)$. The centre of Ω lies on the y axis as Ω is tangent to the x -axis. So the equation of Ω is of the form

$$x^2 + y^2 + \alpha y = 0$$

for some constant α . Putting $x = c$ and $y = d$, we obtain $\alpha = -\frac{c^2 + d^2}{d}$. Thus,

$$\Omega: \quad x^2 + y^2 - \frac{c^2 + d^2}{d}y = 0.$$



Putting

$$BE: \quad y = \frac{d}{c-a}x - \frac{ad}{c-a}$$

into this equation, we solve

$$\left(1 + \frac{d^2}{(c-a)^2}\right)x^2 + (\dots)x + \left(\frac{a^2d^2}{(c-a)^2} + \frac{a(c^2+d^2)}{c-a}\right) = 0.$$

One of the roots is c , while the other root is

$$\frac{\frac{a^2d^2}{(c-a)^2} + \frac{a(c^2+d^2)}{c-a}}{1 + \frac{d^2}{(c-a)^2}} \div c = \frac{a(c^2+d^2-ac)}{d^2 + (c-a)^2}.$$

Then we have

$$y = \frac{d}{c-a} \left(\frac{a(c^2+d^2-ac)}{d^2 + (c-a)^2} - a \right) = \frac{d}{c-a} \cdot \frac{a^2(c-a)}{d^2 + (c-a)^2} = \frac{a^2d}{d^2 + (c-a)^2}.$$

Therefore, we obtain $A\left(\frac{a(c^2+d^2-ac)}{d^2 + (c-a)^2}, \frac{a^2d}{d^2 + (c-a)^2}\right)$. It follows that

$$m_{FT} = m_{AD} = \frac{ad}{c^2 + d^2 - ac}.$$

This implies

$$FT: \quad y = \frac{ad}{c^2 + d^2 - ac}x - \frac{abd}{c^2 + d^2 - ac}.$$

By symmetry,

$$ET: \quad y = \frac{bd}{c^2 + d^2 - bc}x - \frac{abd}{c^2 + d^2 - bc}.$$

Solving these, we get

$$x = \frac{abd(c^2 + d^2 - bc) - abd(c^2 + d^2 - ac)}{ad(c^2 + d^2 - bc) - bd(c^2 + d^2 - ac)} = \frac{abcd(a - b)}{d(c^2 + d^2)(a - b)} = \frac{abc}{c^2 + d^2}$$

and

$$y = \frac{ad}{c^2 + d^2 - ac} \left(\frac{abc}{c^2 + d^2} - b \right) = \frac{ad}{c^2 + d^2 - ac} \cdot \frac{-b(c^2 + d^2 - ac)}{c^2 + d^2} = -\frac{abd}{c^2 + d^2}.$$

Thus, $T \left(\frac{abc}{c^2 + d^2}, -\frac{abd}{c^2 + d^2} \right)$. Since T lies on the perpendicular bisector of DK , we easily obtain $K \left(\frac{2abc}{c^2 + d^2}, 0 \right)$. Now, we have

$$\begin{aligned} DT: \quad y &= -\frac{d}{c}x, \\ BK: \quad y &= \frac{d(c^2 + d^2)}{c(c^2 + d^2 - 2ab)}x - \frac{2abd}{c^2 + d^2 - 2ab}. \end{aligned}$$

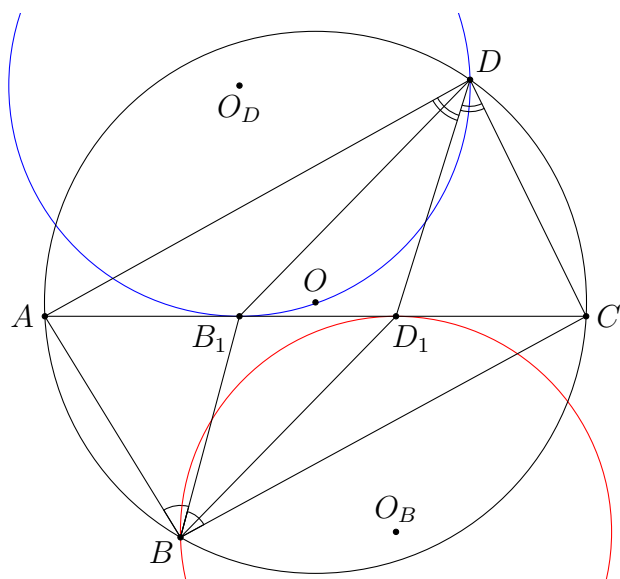
Solving these, we get the intersection point $X \left(\frac{abc}{c^2 + d^2 - ab}, -\frac{abd}{c^2 + d^2 - ab} \right)$. Lastly, we compute

$$\begin{aligned} m_{AX} &= \frac{a^2d(c^2 + d^2 - ab) + abd(d^2 + (c - a)^2)}{a(c^2 + d^2 - ac)(c^2 + d^2 - ab) - abc(d^2 + (c - a)^2)} \\ &= \frac{d(ac^2 + bc^2 + ad^2 + bd^2 - 2abc)}{(c^2 + d^2)(c^2 + d^2 - ab) - ac^3 - bc^3 - acd^2 - bcd^2 + 2abc^2}. \end{aligned}$$

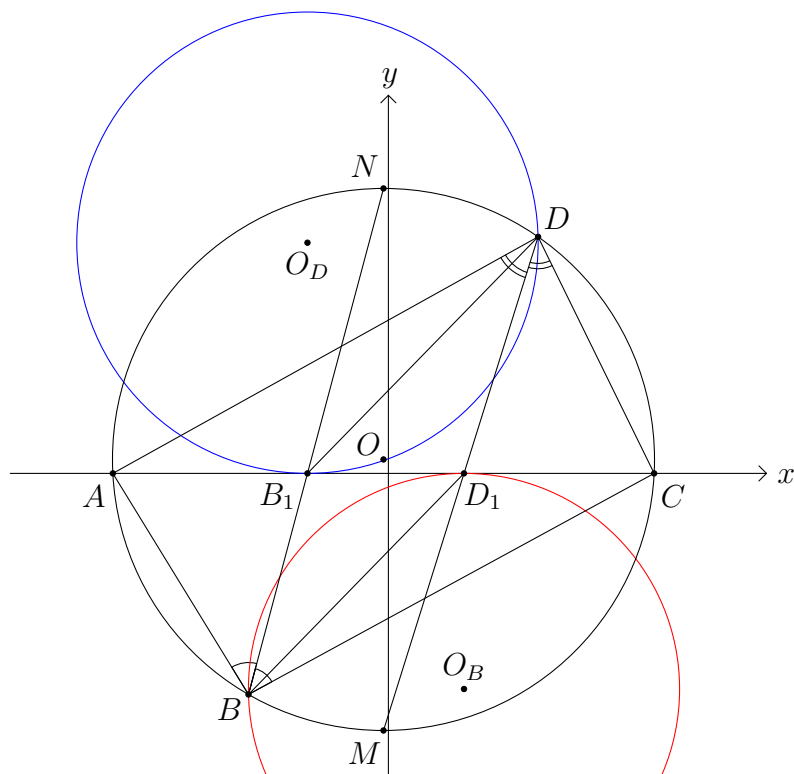
Since this is symmetric in a and b , we have $m_{CX} = m_{AX}$. This means X also lies on AC , and hence AC, DT, BK are concurrent. $\square$

Problem 103. (IMO Shortlist 2021 G5) Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcentre of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at B_1 and D_1 , respectively. Let O_B be the centre of the circle which passes through B and is tangent to AC at D_1 . Similarly, let O_D be the centre of the circle which passes through D and is tangent to AC at B_1 .

Assume that $BD_1 \parallel DB_1$. Prove that O lies on the line O_BO_D .



Solution.



We set the intersection point of B_1D_1 and O_BO_D to be the origin. This is the centre of a homothety (with ratio $k < 0$) that maps $\odot O_B$ to $\odot O_D$. Thus, by letting $D_1(1, 0)$ and $O_B = (1, a)$, we have $B_1(k, 0)$ and $O_D(k, ka)$. Note that

$$\odot O_B: \quad x^2 + y^2 - 2x - 2ay + 1 = 0.$$

Let $B(b, c)$ so that $b^2 + c^2 - 2b - 2ac + 1 = 0$, i.e.

$$a = \frac{c^2 + (b-1)^2}{2c}. \quad (1)$$

Since $BD_1 \parallel DB_1$, D is the image of B under the above homothety (in any case, we can use this parallel condition to find the coordinates of D). This implies $D(kb, kc)$. Let DD_1 and BB_1 meet (ABC) at M and N respectively. From the angle bisectors, we know that M and N are the midpoints of $\widehat{AC}$. In particular, it is a vertical line, it passes through O , and it is a diameter. We find that

$$DD_1: \quad y = \frac{kc}{kb-1}x - \frac{kc}{kb-1}. \quad (2)$$

Since MN is a diameter, $m_{BM} = \frac{-1}{m_{BN}} = \frac{-1}{m_{BB_1}} = -\frac{b-k}{c}$. This easily implies

$$BM: \quad y = -\frac{b-k}{c}x + \frac{c^2 + b^2 - kb}{c}. \quad (3)$$

Solving (2) and (3), we get

$$x = \frac{kc^2 + (kb-1)(c^2 + b^2 - kb)}{kc^2 + (b-k)(kb-1)} = \frac{(kb+k-1)c^2 + b(b-k)(kb-1)}{kc^2 + (b-k)(kb-1)}$$

and

$$\begin{aligned} y &= \frac{kc}{kb-1} \left(\frac{(kb+k-1)c^2 + b(b-k)(kb-1)}{kc^2 + (b-k)(kb-1)} - 1 \right) \\ &= \frac{kc}{\textcolor{red}{kb-1}} \cdot \frac{(\textcolor{red}{kb-1})c^2 + (b-1)(b-k)(\textcolor{red}{kb-1})}{kc^2 + (b-k)(kb-1)} \\ &= \frac{kc(c^2 + (b-1)(b-k))}{kc^2 + (b-k)(kb-1)}. \end{aligned}$$

This means $M \left(\frac{(kb+k-1)c^2 + b(b-k)(kb-1)}{kc^2 + (b-k)(kb-1)}, \frac{kc(c^2 + (b-1)(b-k))}{kc^2 + (b-k)(kb-1)} \right)$. To find the coordinates of N , we can replace k by $\frac{1}{k}$, and multiply both coordinates by k (just think about how to swap the roles of D_1 and B_1). This gives

$$\begin{aligned} N &= \left(k \cdot \frac{\left(\frac{1}{k}b + \frac{1}{k} - 1 \right) c^2 + b \left(b - \frac{1}{k} \right) \left(\frac{1}{k}b - 1 \right)}{\frac{1}{k}c^2 + \left(b - \frac{1}{k} \right) \left(\frac{1}{k}b - 1 \right)}, k \cdot \frac{\frac{1}{k}c \left(c^2 + (b-1) \left(b - \frac{1}{k} \right) \right)}{\frac{1}{k}c^2 + \left(b - \frac{1}{k} \right) \left(\frac{1}{k}b - 1 \right)} \right) \\ &= \left(\frac{k(k(b+1-k)c^2 + b(b-k)(kb-1))}{kc^2 + (b-k)(kb-1)}, \frac{kc(kc^2 + (b-1)(kb-1))}{kc^2 + (b-k)(kb-1)} \right). \end{aligned}$$

As MN is a vertical line, M and N have the same x -coordinates. Equating these, we see that

$$(k-1)((kb-k^2-1)c^2+b(b-k)(kb-1))=0. \quad (4)$$

Since O is the midpoint of MN , we easily obtain

$$O \left(\frac{(k+1)((kb-k^2+2k-1)c^2+b(b-k)(kb-1))}{2(kc^2+(b-k)(kb-1))}, \frac{k(k+1)c(c^2+(b-1)^2)}{2(kc^2+(b-k)(kb-1))} \right).$$

Finally, it is clear that

$$O_BO_D: \quad y = ax.$$

So it suffices to check

$$\frac{k(k+1)c(c^2+(b-1)^2)}{2(kc^2+(b-k)(kb-1))} = a \cdot \frac{(k+1)((kb-k^2+2k-1)c^2+b(b-k)(kb-1))}{2(kc^2+(b-k)(kb-1))}.$$

Indeed, using (1), this is the same as

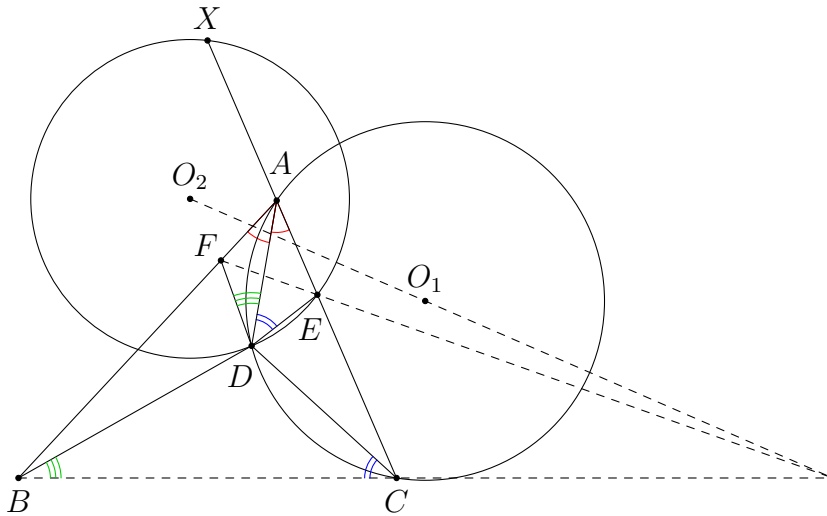
$$kc(c^2+(b-1)^2) = \frac{c^2+(b-1)^2}{2c} \cdot ((kb-k^2+2k-1)c^2+b(b-k)(kb-1)).$$

This is equivalent to

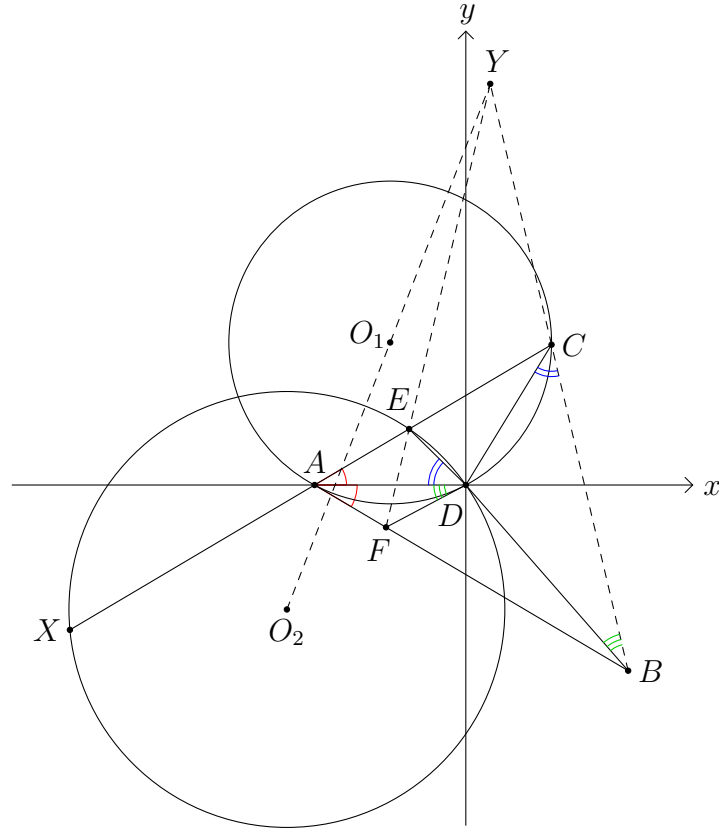
$$0 = (kb-k^2-1)c^2+b(b-k)(kb-1),$$

which follows from (4). So we are done. $\square$

Problem 104. (IMO Shortlist 2021 G7/Problem 3) Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and EXD , respectively. Prove that the lines BC , EF , and O_1O_2 are concurrent.



Solution.



Let $D(0,0)$ and $A(-1,0)$. As AD bisects $\angle BAC$, we let $m_{AB} = -m$ and $m_{AC} = m$. Then

$$\begin{aligned} AB: & y = -mx - m, \\ AC: & y = mx + m. \end{aligned} \tag{1}$$

Let $B(b, -mb - m)$ and $C(c, mc + m)$. Then

$$\begin{aligned} m_{DE} &= -\tan \angle DCB = -\frac{m_{BC} - m_{CD}}{1 + m_{BC} \cdot m_{CD}} \\ &= -\frac{\frac{mb + mc + 2m}{c - b} - \frac{mc + m}{c}}{1 + \frac{mb + mc + 2m}{c - b} \cdot \frac{mc + m}{c}} \\ &= -\frac{m(2bc + b + c)}{m^2(c + 1)(b + c + 2) - c(b - c)}. \end{aligned}$$

This implies

$$DE: \quad y = -\frac{m(2bc + b + c)}{m^2(c + 1)(b + c + 2) - c(b - c)}x. \tag{2}$$

Solving (1) and (2), we obtain

$$E \left(-\frac{m^2(c+1)(b+c+2) - c(b-c)}{m^2(c+1)(b+c+2) + (c+1)(b+c)}, \frac{m(2bc+b+c)}{m^2(c+1)(b+c+2) + (c+1)(b+c)} \right).$$

Changing m to $-m$, and swapping b and c , we deduce

$$F \left(-\frac{m^2(b+1)(b+c+2) + b(b-c)}{m^2(b+1)(b+c+2) + (b+1)(b+c)}, -\frac{m(2bc+b+c)}{m^2(b+1)(b+c+2) + (b+1)(b+c)} \right).$$

Then we compute

$$\begin{aligned} m_{EF} &= \frac{m(2bc+b+c) \left(m^2(b+1)(b+c+2) + (b+1)(b+c) \right. \\ &\quad \left. + m^2(c+1)(b+c+2) + (c+1)(b+c) \right)}{\left(m^2(b+1)(b+c+2) + b(b-c) \right) \left(m^2(c+1)(b+c+2) + (c+1)(b+c) \right) \\ &\quad - \left(m^2(c+1)(b+c+2) - c(b-c) \right) \left(m^2(b+1)(b+c+2) + (b+1)(b+c) \right)} \\ &= \frac{m(2bc+b+c) \left(m^2(b+c+2)^2 + (b+c)(b+c+2) \right)}{m^2(b+c+2)(b-c)(2bc+b+c) + (b-c)(b+c)(2bc+b+c)} \\ &= \frac{m(b+c+2)}{b-c}. \end{aligned}$$

This yields

$$\begin{aligned} EF: \quad y &= \frac{m(b+c+2)}{b-c}x + \frac{m(b+c+2) \left(m^2(c+1)(b+c+2) - c(b-c) \right)}{(b-c) \left(m^2(c+1)(b+c+2) + (c+1)(b+c) \right)} \\ &\quad + \frac{m(2bc+b+c)}{m^2(c+1)(b+c+2) + (c+1)(b+c)}. \end{aligned}$$

This can be simplified as

$$EF: \quad y = \frac{m(b+c+2)}{b-c}x + \frac{m \left(m^2(b+c+2)^2 + (b-c)^2 \right)}{(b-c) \left(m^2(b+c+2) + b+c \right)}.$$

Solving with

$$BC: \quad y = -\frac{m(b+c+2)}{b-c}x + \frac{m(2bc+b+c)}{b-c},$$

we get

$$\begin{aligned} x &= \frac{\color{red}{m}(2bc+b+c) \left(m^2(b+c+2) + b+c \right) - \color{red}{m} \left(m^2(b+c+2)^2 + (b-c)^2 \right)}{2\color{red}{m}(b+c+2) \left(m^2(b+c+2) + b+c \right)} \\ &= \frac{2m^2(b+c+2)(bc-1) + 2b^2c + 2bc^2 + 4bc}{2(b+c+2) \left(m^2(b+c+2) + b+c \right)} \\ &= \frac{m^2(bc-1) + bc}{m^2(b+c+2) + b+c} \end{aligned}$$

and

$$\begin{aligned} y &= -\frac{m(b+c+2)}{b-c} \cdot \frac{m^2(bc-1)+bc}{m^2(b+c+2)+b+c} + \frac{m(2bc+b+c)}{b-c} \\ &= \frac{m(m^2(b+c+2)(b+1)(c+1)+b^2c+bc^2+b^2+c^2)}{(b-c)(m^2(b+c+2)+b+c)}. \end{aligned}$$

Thus, the intersection point Y of BC and EF has coordinates

$$Y \left(\frac{m^2(bc-1)+bc}{m^2(b+c+2)+b+c}, \frac{m(m^2(b+c+2)(b+1)(c+1)+b^2c+bc^2+b^2+c^2)}{(b-c)(m^2(b+c+2)+b+c)} \right).$$

Next, O_1 has x -coordinate $-\frac{1}{2}$. Also,

$$\begin{aligned} O_1D &= O_1C \\ \Rightarrow \left(-\frac{1}{2}\right)^2 + y^2 &= \left(-\frac{1}{2}-c\right)^2 + (y-mc-m)^2 \\ \Rightarrow 0 &= c^2 + c - 2m(c+1)y + m^2(c+1)^2 \\ \Rightarrow y &= \frac{m^2(c+1)+c}{2m}. \end{aligned}$$

This gives $O_1 \left(-\frac{1}{2}, \frac{m^2(c+1)+c}{2m}\right)$. Then we compute

$$\begin{aligned} m_{O_1Y} &= \frac{2m^2(m^2(b+c+2)(b+1)(c+1)+b^2c+bc^2+b^2+c^2) - (m^2(c+1)+c)(b-c)(m^2(b+c+2)+b+c)}{m(b-c)(2m^2(bc-1)+2bc+m^2(b+c+2)+b+c)} \\ &= \frac{m^4(c+1)(b+c+2)^2 + m^2(2bc^2+2c^3+b^2-2bc+5c^2) - c(b-c)(b+c)}{m(b-c)(2bc+b+c)(m^2+1)}. \end{aligned}$$

The perpendicular bisector of BC has slope $\frac{-1}{m_{BC}} = \frac{b-c}{m(b+c+2)}$. It passes through the midpoint $\left(\frac{b+c}{2}, \frac{m(-b+c)}{2}\right)$ of BC . So its equation is

$$y = \frac{b-c}{m(b+c+2)}x + \frac{m^2(-b^2+c^2-2b+2c)-b^2+c^2}{2m(b+c+2)}.$$

Combining with (1), we obtain

$$x = \frac{m^2(-b^2+c^2-2b+2c)-b^2+c^2-2m^2(b+c+2)}{2(m^2(b+c+2)-b+c)} = \frac{m^2(-b^2+c^2-4b-4)-b^2+c^2}{2(m^2(b+c+2)-b+c)}$$

and

$$y = m \left(\frac{m^2(-b^2 + c^2 - 4b - 4) - b^2 + c^2}{2(m^2(b + c + 2) - b + c)} + 1 \right) \\ = \frac{m(m^2(-b^2 + c^2 - 2b + 2c) - b^2 + c^2 - 2b + 2c)}{2(m^2(b + c + 2) - b + c)}.$$

So $X \left(\frac{m^2(b + c + 2)(c - b - 2) - (b - c)(b + c)}{2(m^2(b + c + 2) - b + c)}, \frac{m(b + c + 2)(c - b)(m^2 + 1)}{2(m^2(b + c + 2) - b + c)} \right)$. To find O_2 , we use the perpendicular bisectors of DE and EX . Using $D(0, 0)$ and

$$E \left(-\frac{m^2(c + 1)(b + c + 2) - c(b - c)}{m^2(c + 1)(b + c + 2) + (c + 1)(b + c)}, \frac{m(2bc + b + c)}{m^2(c + 1)(b + c + 2) + (c + 1)(b + c)} \right),$$

we find that the slope of the perpendicular bisector of DE is

$$\frac{-1}{m_{DE}} = \frac{m^2(c + 1)(b + c + 2) - c(b - c)}{m(2bc + b + c)}.$$

This line passes through the midpoint of DE , and so its equation is

$$y = \frac{m^2(c + 1)(b + c + 2) - c(b - c)}{m(2bc + b + c)}x + \frac{m(2bc + b + c)}{2(m^2(c + 1)(b + c + 2) + (c + 1)(b + c))} \\ + \frac{m^2(c + 1)(b + c + 2) - c(b - c)}{m(2bc + b + c)} \cdot \frac{m^2(c + 1)(b + c + 2) - c(b - c)}{2(m^2(c + 1)(b + c + 2) + (c + 1)(b + c))}.$$

This means

$$y = \frac{m^2(c + 1)(b + c + 2) - c(b - c)}{m(2bc + b + c)}x \\ + \frac{m^4(c + 1)^2(b + c + 2)^2 + m^2(2b^2c^2 + 2c^4 + 2b^2c + 6c^3 + b^2 - 2bc + 5c^2) + c^2(b - c)^2}{2m(c + 1)(2bc + b + c)(m^2(b + c + 2) + b + c)}. \quad (3)$$

For the perpendicular bisector of EX , we first note that its slope is $\frac{-1}{m_{EX}} = -\frac{1}{m}$. Secondly, we find the coordinates of the midpoint of EX . We have

$$x = \frac{(m^2(b + c + 2)(c - b - 2) - (b - c)(b + c))(c + 1)(m^2(b + c + 2) + b + c) - 2(m^2(c + 1)(b + c + 2) - c(b - c))(m^2(b + c + 2) - b + c)}{4(c + 1)(m^2(b + c + 2) + b + c)(m^2(b + c + 2) - b + c)} \\ = \frac{m^4(c + 1)(b + c + 2)^2(c - b - 4) + 2m^2(b + c + 2)(-b^2c + c^3 - b^2 + bc - 2c^2 - 2c) - (b - c)(b^2c + 2bc^2 + c^3 + b^2 + 4bc - c^2)}{4(c + 1)(m^2(b + c + 2) + b + c)(m^2(b + c + 2) - b + c)}$$

and

$$y = \frac{2m(2bc+b+c)(m^2(b+c+2)-b+c)}{+m(b+c+2)(c-b)(m^2+1)(c+1)(m^2(b+c+2)+b+c)} \\ = \frac{4(c+1)(m^2(b+c+2)+b+c)(m^2(b+c+2)-b+c)}{m(m^4(c+1)(b+c+2)^2(c-b)+2m^2(b+c+2)(-b^2c+c^3-b^2+bc+2c^2+2c) \\ -(b-c)(b^2c+2bc^2+c^3+b^2+8bc+3c^2+4b+4c))}.$$

Then the equation of the perpendicular bisector of EX is

$$y = -\frac{1}{m}x + \frac{m^6(c+1)(b+c+2)^2(c-b) + m^4(b+c+2)(-3b^2c+3c^3-3b^2-4bc+3c^2-6b-6c-8) \\ + m^2(-3b^3c-3b^2c^2+3bc^3+3c^4-3b^3-11b^2c+3bc^2+3c^3-8b^2-8c^2-8c) \\ -(b-c)(b^2c+2bc^2+c^3+b^2+4bc-c^2)}{4m(c+1)(m^2(b+c+2)+b+c)(m^2(b+c+2)-b+c)}.$$

Combining this with (3), we obtain

$$x = \frac{(2bc+b+c)(m^6(c+1)(b+c+2)^2(c-b) + m^4(b+c+2)(-3b^2c+3c^3-3b^2-4bc+3c^2-6b-6c-8) \\ + m^2(-3b^3c-3b^2c^2+3bc^3+3c^4-3b^3-11b^2c+3bc^2+3c^3-8b^2-8c^2-8c) \\ -(b-c)(b^2c+2bc^2+c^3+b^2+4bc-c^2)) \\ -2(m^2(b+c+2)-b+c)(m^4(c+1)^2(b+c+2)^2 \\ + m^2(2b^2c^2+2c^4+2b^2c+6c^3+b^2-2bc+5c^2)+c^2(b-c)^2)}{4(c+1)(m^2(b+c+2)+b+c)(m^2(b+c+2)-b+c) \\ \times (m^2(c+1)(b+c+2)-c(b-c)+2bc+b+c)} \\ = \frac{m^6(c+1)(b+c+2)^2(-2b^2c+2bc^2-b^2-2bc-c^2-2b-6c-4) \\ + m^4(b+c+2)(-6b^3c^2+6bc^4-9b^3c-13b^2c^2+9bc^3-3c^4-3b^3-19b^2c-9bc^2 \\ -17c^3-6b^2-16bc-26c^2-4b-12c) \\ + m^2(-6b^4c^2-6b^3c^3+6b^2c^4+6bc^5-9b^4c-26b^3c^2+4b^2c^3+18bc^4-3c^5 \\ -3b^4-26b^3c-16b^2c^2+10bc^3-13c^4-6b^3-14b^2c-10bc^2-18c^3-8bc-8c^2) \\ + (b-c)(-2b^3c^2-4b^2c^3-2bc^4-3b^3c-9b^2c^2-5bc^3+c^4-b^3-5b^2c-3bc^2+c^3)}{4(c+1)^2(m^2(b+c+2)+b+c)^2(m^2(b+c+2)-b+c)} \\ = \frac{m^4(b+c+2)(-2b^2c+2bc^2-b^2-2bc-c^2-2b-6c-4) \\ + 2m^2(b+c+2)(-2b^2c+2bc^2-b^2-c^2-2c) \\ + (-2b^3c+2bc^3-b^3-3b^2c+5bc^2-c^3)}{4(c+1)(m^2(b+c+2)+b+c)(m^2(b+c+2)-b+c)}.$$

Afterwards, using the equation of the perpendicular bisector of EX , we deduce

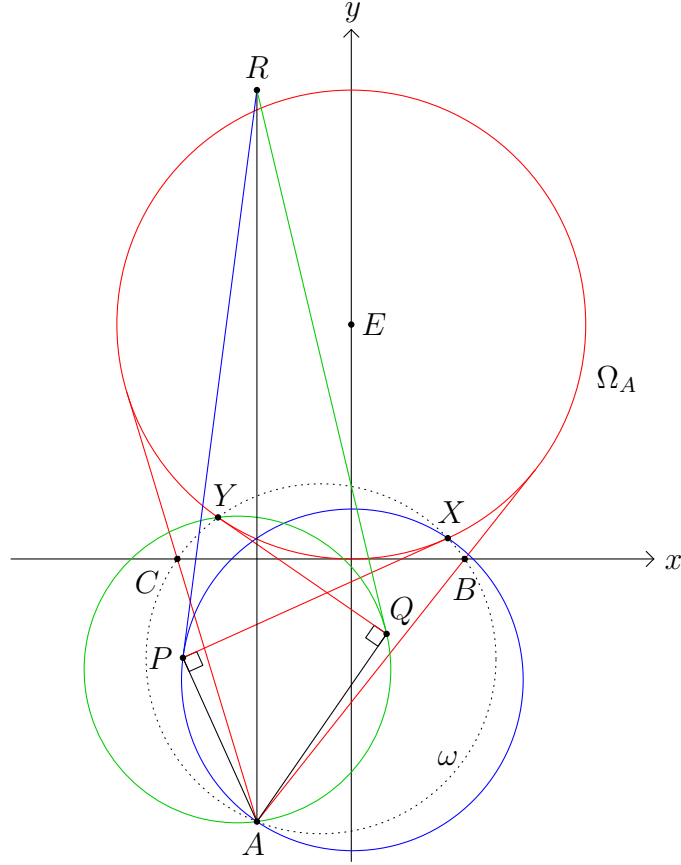
$$y = \frac{m^6(c+1)(b+c+2)^2(c-b) + m^4(b+c+2)(-b^2c - 2bc^2 + 3c^3 - 2b^2 - 2bc + 4c^2 - 4b - 4) + m^2(b^3c - 3b^2c^2 - bc^3 + 3c^4 - b^3 - b^2c - 3bc^2 + 5c^3 - 4b^2 + 4bc) + c(b-c)^2(b+c)}{4m(c+1)(m^2(b+c+2) + b+c)(m^2(b+c+2) - b+c)}.$$

These are the coordinates of O_2 . Finally, as $O_1\left(-\frac{1}{2}, \frac{m^2(c+1)+c}{2m}\right)$, we compute

$$\begin{aligned} & m_{O_1O_2} \\ & \frac{m^6(c+1)(b+c+2)^2(c-b) + m^4(b+c+2)(-b^2c - 2bc^2 + 3c^3 - 2b^2 - 2bc + 4c^2 - 4b - 4) + m^2(b^3c - 3b^2c^2 - bc^3 + 3c^4 - b^3 - b^2c - 3bc^2 + 5c^3 - 4b^2 + 4bc) + c(b-c)^2(b+c)}{4m(c+1)(m^2(b+c+2) + b+c)(m^2(b+c+2) - b+c)} \\ & = \frac{m(m^4(b+c+2)(-2b^2c + 2bc^2 - b^2 - 2bc - c^2 - 2b - 6c - 4) + 2m^2(b+c+2)(-2b^2c + 2bc^2 - b^2 - c^2 - 2c) + (-2b^3c + 2bc^3 - b^3 - 3b^2c + 5bc^2 - c^3) + 2(c+1)(m^2(b+c+2) + b+c)(m^2(b+c+2) - b+c))}{-m^6(c+1)(b+c+2)^3 + m^4(b+c+2)(-b^2c - 4bc^2 - 3c^3 - 2b^2 - 4bc - 10c^2 - 4b - 8c - 4) + m^2(b^3c - b^2c^2 - 5bc^3 - 3c^4 - b^3 + 3b^2c - 7bc^2 - 11c^3 - 2b^2 + 4bc - 10c^2) + c(b^2 - c^2)(b+c+2)} \\ & = \frac{m(m^4(b+c+2)(-2b^2c + 2bc^2 - b^2 + c^2) + 2m^2(b+c+2)(-2b^2c + 2bc^2 - b^2 + c^2) + (-2b^3c + 2bc^3 - b^3 - 5b^2c + 5bc^2 + c^3 - 2b^2 + 2c^2))}{-(b+c+2)(m^2+1) \times (m^4(c+1)(b+c+2)^2 + m^2(2bc^2 + 2c^3 + b^2 - 2bc + 5c^2) - c(b-c)(b+c))} \\ & = \frac{m(b+c+2)(c-b)(2bc+b+c)(m^2+1)^2}{m^4(c+1)(b+c+2)^2 + m^2(2bc^2 + 2c^3 + b^2 - 2bc + 5c^2) - c(b-c)(b+c)} \\ & = \frac{m(b-c)(2bc+b+c)(m^2+1)}{m(b-c)(2bc+b+c)(m^2+1)} \\ & = m_{O_1Y}. \end{aligned}$$

This shows Y, O_1, O_2 are collinear. So the proof is complete. $\square$

Problem 105. (IMO Shortlist 2021 G8) Let ω be the circumcircle of a triangle ABC , and let Ω_A be its excircle which is tangent to the segment BC . Let X and Y be the intersection points of ω and Ω_A . Let P and Q be the projections of A onto the tangent lines to Ω_A at X and Y , respectively. The tangent line at P to the circumcircle of the triangle APX intersects the tangent line at Q to the circumcircle of the triangle AQY at a point R . Prove that $AR \perp BC$.



Also, as $AP \perp PX$, we have $m_{AP} = \frac{\beta - 1}{\alpha}$. This implies

$$AP: \quad y = \frac{\beta - 1}{\alpha}x + \frac{2bc\alpha - (b + c)(\beta - 1)}{(bc + 1)\alpha}.$$

Combining these two equations, we get

$$x = \frac{(bc + 1)\alpha\beta - (2bc\alpha - (b + c)(\beta - 1))(\beta - 1)}{(bc + 1)(\alpha^2 + (\beta - 1)^2)}.$$

Again, using (1), this becomes

$$x = \frac{(-bc + 1)\alpha\beta + (b + c)(\beta - 1)^2 + 2bc\alpha}{bc + 1}.$$

Then

$$\begin{aligned} y &= -\frac{\alpha}{\beta - 1} \cdot \frac{(-bc + 1)\alpha\beta + (b + c)(\beta - 1)^2 + 2bc\alpha}{bc + 1} + \frac{\beta}{\beta - 1} \\ &= \frac{(bc - 1)\alpha^2\beta - (b + c)(\beta - 1)^2\alpha - 2bc\alpha^2 + (bc + 1)\beta}{(bc + 1)(\beta - 1)} \end{aligned}$$

$$\begin{aligned}
&= \frac{(bc-1)(-\beta^2+2\beta)\beta - (b+c)(\beta-1)^2\alpha - 2bc(-\beta^2+2\beta) + (bc+1)\beta}{(bc+1)(\beta-1)} \\
&= \frac{(-bc+1)\beta^3 + (4bc-2)\beta^2 + (-3bc+1)\beta - (b+c)(\beta-1)^2\alpha}{(bc+1)(\beta-1)} \\
&= \frac{-(b+c)\alpha\beta + (-bc+1)\beta^2 + (b+c)\alpha + (3bc-1)\beta}{bc+1}.
\end{aligned}$$

Thus,

$$P \left(\frac{(-bc+1)\alpha\beta + (b+c)(\beta-1)^2 + 2bc\alpha}{bc+1}, \frac{-(b+c)\alpha\beta + (-bc+1)\beta^2 + (b+c)\alpha + (3bc-1)\beta}{bc+1} \right).$$

Next, let O be the centre of (APX) . Since $\angle APX = 90^\circ$, O is the midpoint of AX , which is $\left(\frac{(bc+1)\alpha + (b+c)}{2(bc+1)}, \frac{(bc+1)\beta + 2bc}{2(bc+1)} \right)$. As $OP \perp PR$, we have

$$m_{PR} = \frac{-1}{m_{OP}} = \frac{2(-bc+1)\alpha\beta + 2(b+c)\beta^2 + (3bc-1)\alpha - 4(b+c)\beta + (b+c)}{2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc}.$$

It follows that

$$\begin{aligned}
PR: \quad y &= \frac{2(-bc+1)\alpha\beta + 2(b+c)\beta^2 + (3bc-1)\alpha - 4(b+c)\beta + (b+c)}{2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc} x \\
&\quad - \frac{2(-bc+1)\alpha\beta + 2(b+c)\beta^2 + (3bc-1)\alpha - 4(b+c)\beta + (b+c)}{2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc} \\
&\quad \times \frac{(-bc+1)\alpha\beta + (b+c)(\beta-1)^2 + 2bc\alpha}{bc+1} \\
&\quad + \frac{-(b+c)\alpha\beta + (-bc+1)\beta^2 + (b+c)\alpha + (3bc-1)\beta}{bc+1}.
\end{aligned}$$

To find the intersection point of PR and the perpendicular line from A to BC , we substitute $x = \frac{b+c}{bc+1}$. This gives

$$\begin{aligned}
&(b+c)(2(-bc+1)\alpha\beta + 2(b+c)\beta^2 + (3bc-1)\alpha - 4(b+c)\beta + (b+c)) \\
&- (2(-bc+1)\alpha\beta + 2(b+c)\beta^2 + (3bc-1)\alpha - 4(b+c)\beta + (b+c)) \\
&\times ((-bc+1)\alpha\beta + (b+c)\beta^2 + 2bc\alpha - 2(b+c)\beta + (b+c)) \\
&+ ((-bc+1)\alpha\beta + (b+c)(\beta-1)^2 + 2bc\alpha) \\
&\times (2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc) \\
y &= \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{bc+1} \\
&\quad - \frac{2(b^2+1)(c^2+1)\alpha^2\beta^2 - 2(b^2+1)(c^2+1)\beta^4 + (7b^2c^2 + 4b^2 + 4c^2 + 1)\alpha^2\beta}{bc+1} \\
&\quad + \frac{(11b^2c^2 + 8b^2 + 8c^2 + 5)\beta^3 - 2(3b^2c^2 + b^2 + bc + c^2)\alpha^2 + 2(b+c)(bc+1)\alpha\beta}{bc+1} \\
&\quad - \frac{(17b^2c^2 + 9b^2 + 2bc + 9c^2 + 3)\beta^2 + 2(3b^2c^2 + b^2 + bc + c^2)\beta}{bc+1} \\
&= \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{bc+1}
\end{aligned}$$

$$\begin{aligned}
& -4(b^2+1)(c^2+1)\beta^3 + (7b^2c^2+4b^2+4c^2+1)\alpha^2\beta \\
& + (11b^2c^2+8b^2+8c^2+5)\beta^3 - 2(3b^2c^2+b^2+bc+c^2)\alpha^2 + 2(b+c)(bc+1)\alpha\beta \\
& - (17b^2c^2+9b^2+2bc+9c^2+3)\beta^2 + 2(3b^2c^2+b^2+bc+c^2)\beta \\
= & \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& \frac{(7b^2c^2+4b^2+4c^2+1)\alpha^2\beta + (7b^2c^2+4b^2+4c^2+1)\beta^3}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& - 2(3b^2c^2+b^2+bc+c^2)\alpha^2 + 2(b+c)(bc+1)\alpha\beta \\
& - (17b^2c^2+9b^2+2bc+9c^2+3)\beta^2 + 2(3b^2c^2+b^2+bc+c^2)\beta \\
= & \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& \frac{2(7b^2c^2+4b^2+4c^2+1)\beta^2 - 2(3b^2c^2+b^2+bc+c^2)\alpha^2}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& + 2(b+c)(bc+1)\alpha\beta - (17b^2c^2+9b^2+2bc+9c^2+3)\beta^2 + 2(3b^2c^2+b^2+bc+c^2)\beta \\
= & \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& \frac{-2(3b^2c^2+b^2+bc+c^2)\alpha^2 - 2(3b^2c^2+b^2+bc+c^2)\beta^2}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& + 2(b+c)(bc+1)\alpha\beta + (3b^2c^2+b^2+c^2-1)\beta^2 + 2(3b^2c^2+b^2+bc+c^2)\beta \\
= & \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& \frac{-4(3b^2c^2+b^2+bc+c^2)\beta + 2(b+c)(bc+1)\alpha\beta}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& + (3b^2c^2+b^2+c^2-1)\beta^2 + 2(3b^2c^2+b^2+bc+c^2)\beta \\
= & \frac{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)} \\
& \frac{2(b+c)(bc+1)\alpha\beta + (3b^2c^2+b^2+c^2-1)\beta^2 - 2(3b^2c^2+b^2+bc+c^2)\beta}{(bc+1)(2(b+c)\alpha\beta + 2(bc-1)\beta^2 - 2(b+c)\alpha + (-5bc+3)\beta + 2bc)}.
\end{aligned}$$

Lastly, we use (2) to replace α . This yields

$$\begin{aligned}
& \frac{(-(b^2c^2-b^2-c^2-3)\beta + 2bc(bc+1))\beta + (3b^2c^2+b^2+c^2-1)\beta^2}{-2(3b^2c^2+b^2+bc+c^2)\beta} \\
y = & \frac{(bc+1)(2(bc-1)\beta^2 + (-5bc+3)\beta + 2bc)}{(bc+1)(2(bc-1)\beta^2 + (-5bc+3)\beta + 2bc)} \\
& + \frac{(-(b^2c^2-b^2-c^2-3)\beta + 2bc(bc+1))(\beta-1)}{(bc+1)(2(bc-1)\beta^2 + (-5bc+3)\beta + 2bc)} \\
= & \frac{2(b^2+1)(c^2+1)\beta^2 - 2(2b^2c^2+b^2+c^2)\beta}{(b^2+1)(c^2+1)\beta^2 - (2b^2c^2+b^2+c^2)\beta} \\
= & 2.
\end{aligned}$$

It is now obvious that the same point lies on QR by symmetry. This means it is just the point R , and hence $AR \perp BC$. $\square$

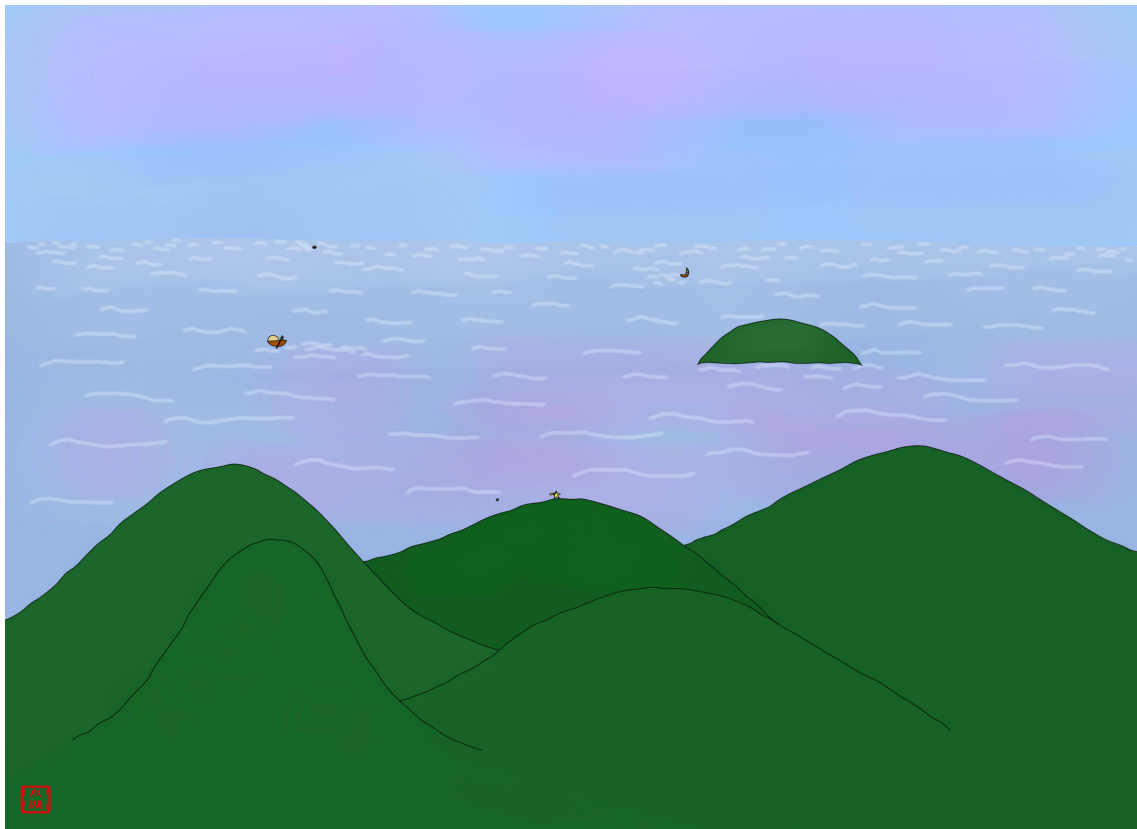
Collection

Here is the list of all geometry problems in the IMO shortlist from 2000 to 2021, together with some (subjective) comments on the suitability of using coordinate geometry to solve these problems. The following gives the meanings of the abbreviations.

- -: there is no such a shortlist problem that year
- number: this problem is solved in this note
- U: this problem is unsolved in this note
- N: this is not a classical geometry problem (e.g. combinatorial geometry)
- I: this is a geometric inequality problem (usually hard)
- P: this is a geometry problem involving a polygon with n sides (impossible)
- ○: this is very easy using coordinate geometry (< 30 minutes is needed)
- ✓: this is easy using coordinate geometry (about 1 hour is needed)
- ≈: this is medium using coordinate geometry (1.5 to 2 hours is needed)
- ×: this is hard using coordinate geometry (3 hours or more is needed)
- !: the solution provided requires some geometric insight

Year	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10
2000	U	1 ○	2 ≈!	N	3 ✓	4 ×	N	5 ≈	-	-
2001	6 ≈	7 ✓	8 ○	I	U	9 ✓	I	10 ✓	-	-
2002	11 ○	12 ≈	13 ✓	14 ≈	I	I	15 ○	16 ≈	-	-
2003	17 ✓	18 ○	19 ○!	20 ✓	21 ✓	U	U	-	-	-
2004	22 ≈	23 ≈	24 ≈	25 ✓	P	P	26 ✓	U	-	-
2005	27 ○	U	28 ≈	U	29 ○	30 ○	I	-	-	-
2006	31 ×	U	32 ○	33 ○	34 ○	35 ✓	U	I	U	P
2007	36 ○	37 ○	38 ○	39 ○	U	I	U	U	-	-
2008	40 ○	41 ○	42 ○	43 ○	N	U	44 ×	-	-	-

Year	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10
2009	45 ○	46 ○	47 ✓	48 ○	P	49 ○	U	U	-	-
2010	50 ✓	51 ≈	P	52 ≈	U	U	U	-	-	-
2011	53 ≈	54 ✓	U	55 ✓	56 ≈	57 ≈	U	U	-	-
2012	58 ○	59 ✓	60 ✓	61 ○	62 ≈	63 ×	U	64 ✓	-	-
2013	65 ○	66 ✓	U	67 ≈	U	68 ×	-	-	-	-
2014	69 ○	I	70 ✓	U	71 ≈	U	72 ≈	-	-	-
2015	73 ○	74 ×	75 ✓	76 ○	77 ≈	78 ≈!	U	P	-	-
2016	79 ○	80 ×	N	81 ○	82 ≈	U	U	I	-	-
2017	83 ○	84 ✓	85 ○	86 ✓	87 ×	N	U	N	-	-
2018	88 ○	89 ○	N	U	U	U	U	-	-	-
2019	90 ○	91 ×	92 ≈	U	93 ≈	94 ×	95 ×	N	-	-
2020	96 ○	97 ≈	98 ✓	N	U	99 ≈	U	U	N	-
2021	100 ○	101 ✓!	N	102 ✓	103 ✓!	N	104 ×	105 ≈	-	-



‘Of course it is happening inside your head, but why on earth should that mean
that it is not real?’

—*Harry Potter and the Deathly Hallows*